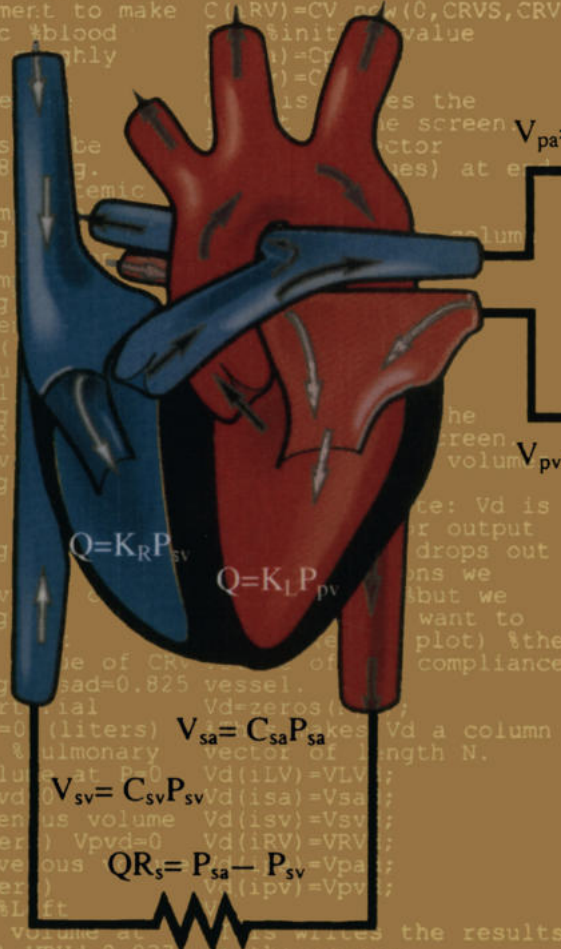


Frank C. Hoppensteadt
Charles S. Peskin

Modeling and Simulation in Medicine and the Life Sciences

Second Edition



Texts in Applied Mathematics **10**

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(continued after index)

Frank C. Hoppensteadt

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Second Edition

With 93 Illustrations



Springer

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Cover illustration: cover art by Charles S. Peskin and Peter H. Carrington.

Mathematics Subject Classification (2000): 92-01, 92B05, 92D99

Library of Congress Cataloging-in-Publication Data
Hoppensteadt, F.C.

Modeling and simulation in medicine and the life sciences / Frank C. Hoppensteadt,
Charles S. Peskin.—2nd ed.

p. cm. — (Texts in applied mathematics ; 10)

Original title: Mathematics in medicine and the life sciences.

Includes bibliographical references and index.

ISBN 978-1-4419-2871-9

1. Biomathematics. 2. Population biology—Mathematics. 3. Physiology—
Mathematics. I. Peskin, Charles S. II. Hoppensteadt, F.C. Mathematics in medicine
and the life sciences. III. Title. IV. Series.

QH323.5 .H67 2001

570'.1'51—dc21

2001032010

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ISBN 978-1-4419-2871-9

ISBN 978-0-387-21571-6 (eBook)

DOI 10.1007/978-0-387-21571-6

© 2002 Springer Science+Business Media New York

Originally published by Springer-Verlag New York, Inc. in 2002

Softcover reprint of the hardcover 2nd edition 2002

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9 8 7 6 5 4 3 2 (Corrected printing, 2004)

SPIN 10965207

Springer-Verlag is part of Springer-Science+Business Media

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This book is dedicated to the memory of
ANNELI LAX,
who helped and inspired us.

Series Preface

Mathematics is playing an ever more important role in the physical and biological sciences, provoking a blurring of boundaries between scientific disciplines and a resurgence of interest in the modern as well as the classical techniques of applied mathematics. This renewal of interest, both in research and teaching, has led to the establishment of the series: *Texts in Applied Mathematics (TAM)*.

The development of new courses is a natural consequence of a high level of excitement on the research frontier as newer techniques, such as numerical and symbolic computer systems, dynamical systems, and chaos, mix with and reinforce the traditional methods of applied mathematics. Thus, the purpose of the textbook series is to meet the current and future needs of these advances and encourage the teaching of new courses.

TAM will publish textbooks suitable for use in advanced undergraduate and beginning graduate courses, and will complement the *Applied Mathematical Sciences (AMS)* series, which will focus on advanced textbooks and research level monographs.

Pasadena, California
Providence, Rhode Island
Houston, Texas
College Park, Maryland

J.E. Marsden
L. Sirovich
M. Golubitsky
S.S. Antman

Preface

Important knowledge in medicine and the life sciences has been created using mathematical models that help predict and interpret what we observe. This book describes several major contributions that have been made to physiology, engineering, and population biology by such theoretical work.

We have tried to keep the presentation brief, and tried to ensure that the topics are presented at a level that is accessible to a wide audience. Each topic could serve as a launching point for more advanced study, and suitable references are suggested to help with this. If the underlying mathematics is understood for these basic examples, then mathematical aspects of more advanced life science problems will be within reach.

The material in each chapter is presented in general order of increasing mathematical difficulty, which ranges up to calculus, matrix theory, and elementary differential equations. Some exercises deal with material in preceding sections, while others are projects that extend preceding material. Some topics in matrices, MATLAB[®] and probability are described in appendices.

Our purpose in this book is not the systematic presentation of mathematical material, although there are important threads that run through several chapters. Instead, we hope to illustrate how mathematics can be used. In particular, our goal is to make available to students having at least one term of calculus topics in the life sciences and medicine that have benefited from mathematical modeling and analysis. In addition to exposing students to current ideas, the material is intended to reinforce their mathematics education by presenting familiar mathematical topics from novel points of view. Finally, enabling students to think in terms of models

early in their academic experience should motivate them to develop and apply modeling skills further.

While hoping this cross-disciplinary book will be useful to a wide variety of individuals, we believe that it can have special significance for the pre-medical student, who will find a mathematical introduction to a host of phenomena that are central to the practice of medicine. These include genetics and epidemics as well as the functions of the heart, lungs, and kidneys. It is our hope that the mathematical study of these topics will give the student a depth of understanding and insight that could not have been achieved through traditional, descriptive education in the medical sciences.

The mix of topics, taken largely from physiology and from population biology, includes important phenomena that are within reach of the students described above. The population part of the book draws its material from the areas of demographics, genetics, epidemics, and molecular biology, while the physiological part surveys cardiovascular, pulmonary, renal, and muscle physiology. One chapter (Chapter 6) is intended to introduce students to models of nerve cells and some neural circuits as a basis for studying how the brain works from the point of view of a neural engineer. We are on the rise of a wave of understanding brain function, and mathematical modeling can be useful in understanding this complex organ.

Anneli Lax maintained an interest in the course that led to this book and provided helpful discussions during the preparation of the lecture notes on which this book is based. We are grateful to her. Besides the authors, the course has been taught by W. Stephen Childress, H. Michael Lacker, Daniel Tranchina, and Tatyana Izhikevich, and we are indebted to them for their comments and advice.

Arizona State University
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May 2001

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Contents

Series Preface	vii
Preface	ix
Introduction	1
1 The Heart and Circulation	5
1.1 Plan of the Circulation	5
1.2 Volume, Flow, and Pressure	7
1.3 Resistance and Compliance Vessels	8
1.4 The Heart as a Pair of Pumps	10
1.5 Mathematical Model of the Uncontrolled Circulation	14
1.6 Balancing the Two Sides of the Heart and the Two Circulations	18
1.7 The Need for External Circulatory Control Mechanisms	20
1.8 Neural Control: The Baroreceptor Loop	21
1.9 Autoregulation	25
1.10 Changes in the Circulation Occurring at Birth	28
1.11 Dynamics of the Arterial Pulse	33
1.12 Computer Simulation of Pulsatile Blood Flow	39
1.13 Suggestions for Computing Projects Concerning the Circulation	63
1.14 Annotated References	68
Exercises	69

2	Gas Exchange in the Lungs	75
2.1	The Ideal Gas Law and the Solubility of Gases	76
2.2	The Equations of Gas Transport in One Alveolus	78
2.3	Gas Transport in the Lung	82
2.4	Optimal Gas Transport	83
2.5	Mean Alveolar and Arterial Partial Pressures	85
2.6	Transport of O_2	87
2.7	Computer Solution of the Equations for O_2 Transport in the Lung	91
2.8	Computing Projects Concerning Oxygen Transport by the Lung	104
2.9	Annotated References	107
	Exercises	107
3	Control of Cell Volume and Electrical Properties of Cell Membranes	109
3.1	Osmotic Pressure and the Work of Concentration	109
3.2	A Simple Model of Cell Volume Control	113
3.3	The Movement of Ions Across Cell Membranes	115
3.4	The Interaction of Electrical and Osmotic Effects	118
3.5	The Hodgkin-Huxley Equations for the Nerve Action Potential	124
3.6	Computer Simulation of the Nerve Action Potential	132
3.7	Suggestions for Computing Projects Concerning the Nerve Impulse	139
3.8	Annotated References	143
	Exercises	144
4	The Renal Countercurrent Mechanism	147
4.1	The Nephron	147
4.2	Dynamics of Na^+ and H_2O : Transport along the Renal Tubules	150
4.3	The Loop of Henle	152
4.4	The Juxtaglomerular Apparatus and the Renin-Angiotensin System	155
4.5	The Distal Tubule and Collecting Duct: Concentrating and Diluting Modes	157
4.6	Remarks on the Significance of the Juxtaglomerular Apparatus	158
4.7	How Nephrons Do Better Than a Factor of e	159
4.8	Computing Project on the Interacting Nephron Population Model	166
4.9	Annotated References	168
	Exercises	169

5	Muscle Mechanics	171
5.1	The Force–Velocity Curve	171
5.2	Crossbridge Dynamics	173
5.3	Computer Simulation of Crossbridge Attachment and Detachment	179
5.4	Suggested Computing Projects on Crossbridge Dynamics	188
5.5	Annotated References	190
	Exercises	190
6	Neural Systems	193
6.1	Guttman's Experiments on Phase Locking	195
6.2	Biological Rhythms	197
6.2.1	Modulating the Clock on the Wall	198
6.2.2	Biological Clocks	199
6.2.3	Electronic Clocks: Voltage–Controlled Oscillators	200
6.3	Model Neural Networks	200
6.3.1	A Thalamocortical Circuit.	201
6.3.2	Hippocampus	203
6.4	Annotated References	207
	Exercises	208
7	Population Dynamics	215
7.1	Bacterial Cultures	216
7.1.1	Batch Culture Growth	217
7.1.2	Least–Squares Estimation of the Growth Rate . .	221
7.2	Age Structures	222
7.2.1	Cohort Analysis	223
7.2.2	Birth Rates	224
7.2.3	Age Distributions	228
7.3	Microbial Ecology	230
7.3.1	Uptake of Nutrients	230
7.3.2	Growth and Quiescence	232
7.3.3	Biofilms and Growth in a Continuous Flow	234
7.3.4	A Solutivory Food Chain	236
7.4	Nonlinear Reproduction Curves	237
7.4.1	Compensatory Growth: Verhulst's Model	238
7.4.2	Chaos	243
7.5	Controlling Populations	248
7.5.1	Cell Synchronization	249
7.5.2	Cohort Dynamics near the Limits of Growth . . .	250
7.5.3	Harvesting: Economics of Populations	252
7.6	Annotated References	256
	Exercises	257

8	Genetics	267
8.1	Population Genetics	269
8.1.1	Natural Selection	269
8.1.2	Slow Selection	272
8.1.3	Genotype Frequencies	273
8.1.4	Random Genetic Drift	275
8.2	Biotechnology	277
8.2.1	Plasmids	277
8.2.2	Control of Plasmid Replication	281
8.2.3	Regulation of Gene Expression	286
8.2.4	Stoichiometry of Metabolic Pathways	290
8.3	Annotated References	291
	Exercises	292
9	A Theory of Epidemics	295
9.1	Spread of Infection Within a Family	296
9.2	The Threshold of an Epidemic	300
9.3	Predicting the Severity of an Epidemic	303
9.4	Annotated References	306
	Exercises	306
10	Patterns of Population Growth and Dispersal	311
10.1	Random Walks and the Process of Diffusion	312
10.1.1	Random Walks	313
10.1.2	Boundary Conditions	314
10.1.3	Sample Paths	316
10.1.4	Diffusion Approximation	317
10.1.5	Two-Dimensional Diffusion	318
10.1.6	Sample Paths in Two Dimensions	319
10.1.7	Numerical Solutions: The Method of Lines	320
10.2	Bacterial Growth on a Petri Plate	321
10.2.1	Diffusion of a Drop of Substrate	321
10.2.2	Bacterial Growth Due to Diffusing Nutrition	323
10.2.3	Growth Patterns Due to Two Nutrients	323
10.3	Concluding Remarks	325
10.4	Annotated References	326
	Exercises	326
	Appendix A Getting Started with Matrices and MATLAB	335
	Appendix B Background on Random Processes	343
	Index	349

Introduction

There are two major parts to this book. The first part (Chapters 1–5) deals with models from physiology. We introduce models of various organs and body systems using elementary ideas from physics. A background in high-school-level physics is sufficient for this material. The biological material need not have been studied before, since the essentials are summarized in the text.

These five chapters present several important physiological systems (heart and circulation, kidney, etc.) while introducing many of the essential terms and ideas used in modeling the physical processes of flow, transport, etc. The goal is to introduce students to the use of mathematical methods in modeling biological systems by studying simple models that have been used decisively in understanding how these systems work. In addition to introducing students to important topics in physiology and steering them toward further study of them, our approach is intended to reinforce mathematical skills by applying them to significant physiological problems.

Each of these five physiological chapters now contains a section on computer simulation followed by a section on suggested computing projects. These are new to this Second Edition. In Chapters 3 and 4, the focus of the computer simulation section is substantial new material on the Hodgkin-Huxley equations for the nerve action potential (in Chapter 3) and on the interaction between nephrons of different lengths in the renal countercurrent mechanism (in Chapter 4). Except in Chapter 4, where the student is asked to write the necessary program from scratch, the sections on computer simulation contain complete MATLAB[®] programs that are also available at the Springer-Verlag website (<http://www.springer-ny.com>).

Because these programs are intended to be modified by the student, they are thoroughly documented with comments, and surrounded by explanatory text. These MATLAB® programs generate graphical output that are not included in the book, thus providing virtual figures that each student can modify by changing parameters or by making more extensive changes in the assumptions underlying any given model. The suggested computing projects are open-ended and can serve as a basis for course projects, independent study, or separate courses on computer simulation in biology and medicine. Computer simulation invites student participation, and these new sections are designed to encourage maximum student involvement in learning the subject matter of this book.

Chapter 6 introduces some ideas and methods from neuroengineering about how neurons might be organized into neural circuits that can accomplish specific tasks. Some basic electronic circuits are described in connection with these models.

The second part of the book deals with problems from population biology and biotechnology. Chapter 7 begins with models used to study bacterial growth. Starting with Malthus's model of geometric population growth and proceeding to models whose solutions are chaotic, we introduce various mathematical methods for studying and simulating model systems. Other topics include age structure and infinitesimal sampling intervals. The eighth, ninth, and tenth chapters are shorter than Chapter 7, and they are separated only because they study quite separate population phenomena. These chapters use ideas of probability theory to model random events in populations. Random models and their nonrandom analogues are developed, and we discover how the random and nonrandom models are related. Problems from genetics are studied in Chapter 8, Chapter 9 is about two problems from epidemiology, and Chapter 10 is about dispersal and spatial pattern formation processes. One intention of the second part is to expose students to uses of probability in modeling, to build random and nonrandom models of similar phenomena, and so to gain insight into how the two are related.

Random sampling occurs in many settings. For example, if a person who is capable of spreading a disease enters a group of people susceptible to it, there may or may not be an effective contact between the infective and a susceptible individual. Contact is a sampling process. If the group is very small and not in close contact, the probability of spread of the disease is small. If the group is large and in close contact, the probability is near one, and random effects can be ignored.

The basic tool developed here for accounting for random sampling is the Markov chain. Markov chain models are derived in all three chapters, and they are compared to analogous nonrandom models.

Both the random and nonrandom models lend themselves nicely to simulation using a computer. Computers are attractive because they make it possible to solve the equations of more realistic models and to vary pa-

rameters at will while observing the effects on model behavior through graphical output. Throughout the book sample programs and simulations are described using MATLAB®. This approach to computing gives concise code that can be easily translated into other computing languages and environments, such as BASIC, FORTRAN, C++, MAPLE® and MATHEMATICA®. In the population part of the book, however, the emphasis is more on simulation results than on simulation methods. Therefore, the figures showing simulation results are included in the text. The programs that generated these figures are available through the Springer-Verlag web-site (<http://www.springer-ny.com>).

The range of material is intended to make possible courses at the undergraduate level ranging from one quarter to a full year without a great deal of editing being required of the instructor and students. The exercises are graded (roughly) in difficulty from those that help beginning students to those that challenge advanced undergraduates. Some exercises are denoted by *. These require more mathematical preparation or a softer touch in teaching. They can be skipped, although we believe that the material in these exercises is important to bringing certain topics to closure.

Using this material we have taught premedical students, biology students, mathematics and physics students, computer science and engineering students. This material makes possible cross-disciplinary courses that can be taught by one instructor or several who come from various backgrounds.

Here are some sample courses:

One quarter: Chapters 1–3; or Chapters 3 and 6.

One semester: Chapters 1–5 or Chapters 7–10.

One year: The entire text.

We have suggested several projects that can be assigned to groups of students working in teams. For example, students can construct a neuron model circuit, derive the mathematical model of the circuit, simulate the model on a computer, and analyze it using methods of differential equations; or students can perform a variety of chemostat experiments; or conduct a project in the economics of exhaustible resources.

Explanations are brief in most cases, and the intention is to provide a wide range of material to the students while still providing enough information to enable them to complete the exercises at the end of each chapter and to start reading in the scientific literature. The exercises are an important part of the text, since it is through them that the students get an opportunity to apply material in the text and to create new applications using those methods.

With additional work by the instructor, it is possible to spend an entire term on one or two chapters. For example, heart and circulation lead to a variety of clinical examples of disease, or beginning with Chapter 6, one

can launch into a term's worth of work on brain modeling. The modular structure of the book allows flexibility in course design.

This book uses mathematical modeling throughout, and the reader should be aware that such models are always based on simplifying assumptions. Simplification in mathematical modeling is both a blessing and a curse. The curse is the partial loss of predictive power that comes from whatever lack of correspondence there may be between the model and the real world. The blessing is the insight that comes from the process of pruning away unnecessary detail and leaving behind only what is essential. When Euclid said, "A line is that which has no breadth," he was not describing anything that exists in the physical world, but history has vindicated his somewhat artificial conception. Because this is an introductory book, the simplifications made herein are even more extensive than usual, so the reader should be warned not to regard the statements or equations of this book as being literal or exact descriptions of biological reality. The models presented here are in the nature of metaphors, and these metaphors will have served their purpose if they have helped the reader to see through the bewildering complexity of living systems to the underlying simplicity of certain fundamental biological processes and functions.

1

The Heart and Circulation

This chapter begins a discussion of mathematics in physiology. The discussion begins with the heart and blood circulation in the body. We first outline the structure of the circulation, and then we derive models of blood flow and pressure and mechanisms for controlling them.

1.1 Plan of the Circulation

The function of the heart is to pump blood. The blood carries oxygen (O_2) from the lungs to the various tissues of the body, and it carries carbon dioxide (CO_2) from these tissues back to the lungs.

Since the circulation forms a closed loop, its description can begin anywhere. We begin on the left side of the heart (see Figure 1.1). The *left heart* receives blood that is rich in O_2 and pumps this blood into the systemic arteries. These form a tree of progressively smaller vessels that supply fully oxygenated (and hence bright red!) blood to all of the organs and tissues of the body. From the smallest of the *systemic arteries* blood flows into the systemic capillaries, which are roughly the diameter of a single red blood cell. It is in the capillaries that the actual exchange of O_2 and CO_2 takes place. The blood that leaves the systemic capillaries carries less O_2 and more CO_2 than the blood that entered. (The loss of O_2 causes a change in the color, so that the blood is now more bluish than before.)

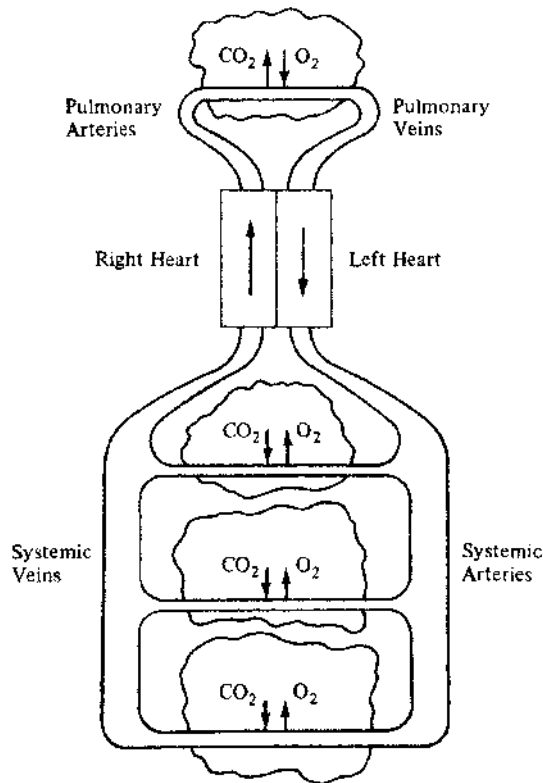


Figure 1.1. Plan of the circulation. Blood flows away from the heart in arteries and back towards the heart in veins. The pulmonary circulation begins on the right side of the heart and ends on the left side, whereas the systemic circulation begins on the left side and ends on the right side. In the pulmonary circulation, oxygen (O_2) is picked up by the blood, and carbon dioxide (CO_2) is removed, as blood flows through the lungs. In the systemic circulation, the reverse happens (O_2 is removed and CO_2 is picked up) as blood flows through the various tissues of the body.

Leaving the systemic capillaries, the blood enters the *systemic veins*, through which it flows in vessels of progressively increasing size toward the right side of the heart.

The *right heart* pumps blood into the *pulmonary arteries*, which form a tree that distributes the blood to the tissues of the lung. The smallest branches of this tree give rise to the *pulmonary capillaries*, where CO_2 leaves the blood stream and O_2 enters from the air space of the lungs. Leaving the pulmonary capillaries, the oxygenated blood is collected in the *pulmonary veins*, through which it flows back to the left heart. This

completes the circulation. The average time required for a red blood cell to complete the circuit that we have described is about 1 minute.

The reader has probably noticed an important symmetry in the plan of the circulation: The blood that leaves the left heart flows through the arteries, capillaries, and veins of the *systemic circulation* before returning to the right heart. Similarly, the blood that leaves the right heart flows through the arteries, capillaries, and veins of the *pulmonary circulation* before returning to the left heart. Of course, these partial “circulations” are not really closed loops, since there is no blood flow directly from one side of the heart to the other (except in the fetal circulation and in certain forms of congenital heart disease).

Because of this symmetry between the systemic and pulmonary circulations, we can expect that the equations of the systemic circulation will have the same form as the equations of the pulmonary circulation. Similarly, the equations of the right and left heart will be the same in form, and the relation of the right heart to the systemic circulation will be the same as the relation of the left heart to the pulmonary circulation. These symmetries in the *form* of the equations are not matched by corresponding symmetries in the *magnitude*, however. That is, the parameters appearing in the equations of the pulmonary circulation have values that are different from the corresponding parameters of the systemic circulation. Because of these differences, the systemic blood volume is about 10 times the pulmonary blood volume, and the systemic arterial pressure is about 6 times the pulmonary arterial pressure (see Table 1.1). One of the themes of this chapter is to study the consequences of this quantitative disparity between the two circulations.

1.2 Volume, Flow, and Pressure

The purpose of this section is to introduce the three physical variables that are needed in a quantitative description of the circulation, to explain the system of units used by physiologists, and to give typical values that occur in the different parts of the circulation.

Since blood is nearly incompressible, the volume of the blood serves as a convenient measure of the amount of blood in any part of the circulation. Volume will be measured in *liters* (1 liter = 1000 cm³) and designated by the symbol V . The total blood volume (V_0) is about 5 liters, partitioned roughly as shown in Table 1.1.

The definition of the *flow* that we use in this book is the volume of blood per unit time passing a point in the circulation. Thus, flow is measured in liters/minute. We designate flow by the symbol Q . The most important flow in the circulation is the *cardiac output*, which is defined as the volume of blood pumped per unit time by either side of the heart. (This definition

Table 1.1. Normal Resting Pressures and Volumes of the Systemic and Pulmonary Arteries and Veins

	$P(\text{mmHg})$	$V(\text{liters})$
sa	100	1.0
sv	2	3.5
pa	15	0.1
pv	5	0.4

(s=systemic, p=pulmonary, a=artery, v=vein)

assumes that the two sides of the heart produce identical outputs. At this point, the reader should be wondering why this is so. The mystery will be cleared up later.) The cardiac output may be calculated as the product of the stroke volume (volume of blood pumped per beat) and the heart rate (number of beats per unit time). Typical values are

$$\begin{aligned}\text{stroke volume} &= 70 \text{ cm}^3/\text{beat} = 0.070 \text{ liters/beat;} \\ \text{heart rate} &= 80 \text{ beats/minute;} \\ \text{cardiac output} &= 5.6 \text{ liters/minute.}\end{aligned}$$

The usual definition of *pressure* (P) is force per unit area, and pressure is expressed in terms of the height of a column of mercury that can be supported by the pressure in question. (Note that this height is independent of the cross-sectional area of the column: If the area is doubled the weight of the column is doubled, but so is the force produced by a given pressure.) Thus, the conventional units of pressure in physiology are mmHg (millimeters of mercury).

When considering pressures, it is important to remember that only pressure *differences* produce observable effects. Thus, any reference pressure can be called zero, and other pressures can be reported as differences from the reference pressure. A particularly convenient reference pressure in physiology is the pressure of the atmosphere, since this is the pressure outside of most of the blood vessels of the circulation. Pressure differences with respect to the atmosphere can be measured directly by the simple device of leaving the far end of the (U-shaped) mercury column open to the atmosphere. Throughout this chapter the symbol P will stand for pressure measured with respect to the atmosphere. As we shall see, this definition of pressure has greater relevance for the circulation than the absolute pressure.

1.3 Resistance and Compliance Vessels

Consider the blood vessel shown in Figure 1.2. The volume of the vessel is V , its inflow is Q_1 at pressure P_1 , and its outflow is Q_2 at pressure P_2 . The external pressure is zero (atmospheric). Suppose that the vessel is in a

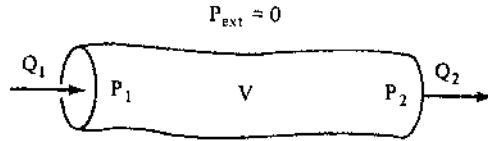


Figure 1.2. A typical blood vessel. V = volume of the vessel, P_1 = upstream pressure, P_2 = downstream pressure, P_{ext} = external pressure (pressure outside the vessel, which is here chosen as our reference pressure, hence $P_{\text{ext}} = 0$), Q_1 = inflow (volume per unit time), Q_2 = outflow.

steady state, which means that none of these quantities are changing with time. Then we must have $Q_1 = Q_2$ (why?), so we shall drop the subscript and just use the symbol Q to designate either the inflow or the outflow.

How are Q , P_1 , P_2 , and V related? This is a complicated question, since it involves two separate properties of the blood vessel: its *resistance* to blood flow and its *compliance* in response to distending pressure. We can isolate these properties by considering special cases.

First, suppose that the vessel is rigid, so that the volume is known and constant. Then we need only a relationship between Q , P_1 , and P_2 . Since only pressure differences matter, it is a safe assumption that Q is determined by $P_1 - P_2$. The simplest relationship of this sort is

$$Q = \frac{P_1 - P_2}{R}, \quad (1.3.1)$$

where R is a constant called the *resistance* of the vessel. If a vessel satisfies (1.3.1), we call it a *resistance vessel*. Next suppose that the vessel is elastic but that it has no resistance to blood flow, so that the pressures at the two ends of the vessel are equal for all Q . Because the vessel is elastic, there is some relation between the distending pressure $P = P_1 = P_2$ and the volume V . The simplest relation of this kind is

$$V = CP, \quad (1.3.2)$$

where C is a constant called the *compliance* of the vessel.

Alternatively, if we want to take into account the nonzero residual volume of the vessel at zero pressure, we could use the relationship

$$V = V_d + CP, \quad (1.3.3)$$

where C is still called the compliance and where V_d is the ("dead") volume at $P = 0$. A vessel satisfying (1.3.2) or (1.3.3) is called a *compliance vessel*.

Clearly, the concepts of resistance vessels and compliance vessels are idealizations. First, a real vessel must exhibit both resistance and compliance properties simultaneously. Second, we have assumed linear relationships that may be too simple to describe real blood vessels.

The reply to these objections is as follows. First, the circulation does seem to produce a clear separation between the resistance and compliance

functions of its different vessels. Thus, the large arteries and veins are primarily compliance vessels in the sense that only small pressure differences are needed to drive the cardiac output through these vessels, while their changes in volume are highly significant. The main site of resistance is in the tissues themselves (primarily at the level of the smallest arteries, the *arterioles*), where volume changes are less important but where large pressure drops are observed.

Second, the approximation of a linear relation between flow and pressure differences (1.3.1) is a good approximation when we allow for changes in resistance. This may appear to be a circular argument, since we can always make (1.3.1) true by introducing the definition $R = (P_1 - P_2)/Q$. In fact, tissues exhibit reasonably constant values of R under conditions where the diameters of their blood vessels remain constant. When R changes, a physiological explanation must be sought. This explanation usually involves a stimulus that leads to contraction or relaxation of the smooth muscles in the walls of the arterioles. Such a stimulus may be generated by the nervous system, by circulating hormones (or other substances circulating in the blood), or by the action of locally produced products of metabolism. Some of these effects will be studied in Section 1.9, on autoregulation.

Finally, there is no such justification for the linear compliance relations that we have assumed. It is an excellent approximation to say that the volume of a blood vessel is determined by the internal pressure, but the relationship between volume and pressure becomes progressively less compliant as it is distended. The linear model is introduced for simplicity.

1.4 The Heart as a Pair of Pumps

A pump is a device that can accept fluid at low pressure (P_1) and transfer it to a region where the pressure is high ($P_2 > P_1$). Thus, the pump performs work on the fluid; the rate at which work is performed by the pump is the product of the volume rate of flow Q and the pressure difference $P_2 - P_1$. To characterize the pump we need to say how Q depends on P_1 and P_2 .

Consider, for example, the left side of the heart (Figure 1.3). The left ventricle is equipped with an inflow (mitral) valve and an outflow (aortic) valve. When the ventricle is relaxed (diastole) the inflow valve is open and the outflow valve is closed. During this period of time, the left ventricle receives blood from the left atrium at a pressure that is essentially that of the pulmonary veins. Thus, for the left ventricle, $P_1 = P_{sv} = 5$ mmHg. When the ventricle contracts (systole), the inflow valve closes and the outflow valve opens. Then the left ventricle actively pumps blood into the systemic arterial tree. Thus, for the left ventricle, $P_2 = P_{sa} = 100$ mmHg.

What determines the output of the left ventricle under these conditions? To answer this question we follow an approach introduced by Sagawa and

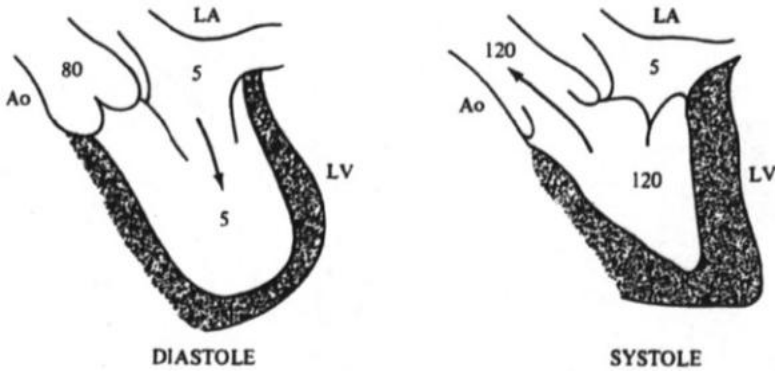


Figure 1.3. Cross section of the left side of the heart with the left ventricle in DIASTOLE (relaxed) and SYSTOLE (contracted). LA = left atrium, LV = left ventricle, Ao = Aorta. The numbers in the chambers are pressures in millimeters of mercury (mmHg). Note that the inflow (mitral) valve is open in diastole and closed in systole, whereas the outflow (aortic) valve is closed in diastole and open in systole. Pressures are approximately equal across an open valve, but can be very unequal, with the downstream pressure higher, across a closed valve. The valves ensure the unidirectional flow of blood from left atrium to left ventricle to aorta.

his collaborators: We regard the ventricle as a compliance vessel whose compliance changes with time. Thus, the ventricle is described by

$$V(t) = V_d + C(t)P(t), \quad (1.4.1)$$

where $C(t)$ is a given function with the qualitative character shown in Figure 1.4. The important point is that $C(t)$ takes on a small value C_{systole} when the ventricle is contracting, and a much larger value C_{diastole} when the ventricle is relaxed. For simplicity, we have taken V_d to be independent of time.

Using (1.4.1), we can construct a pressure-volume diagram of the cardiac cycle (Figure 1.5). For a detailed explanation of this type of diagram, see the paper by Sagawa et al., cited in Section 1.14. For our purposes, it is sufficient to note that the maximum volume achieved by the ventricle (at end-diastole) is given by

$$V_{\text{ED}} = V_d + C_{\text{diastole}}P_v, \quad (1.4.2)$$

while the minimum volume (achieved at end-systole) is given by

$$V_{\text{ES}} = V_d + C_{\text{systole}}P_a, \quad (1.4.3)$$

where P_a is the pressure in the arteries supplied by the ventricle and P_v is the pressure in the veins that fill it. Thus, the stroke volume is given by

$$V_{\text{stroke}} = V_{\text{ED}} - V_{\text{ES}} = C_{\text{diastole}}P_v - C_{\text{systole}}P_a. \quad (1.4.4)$$

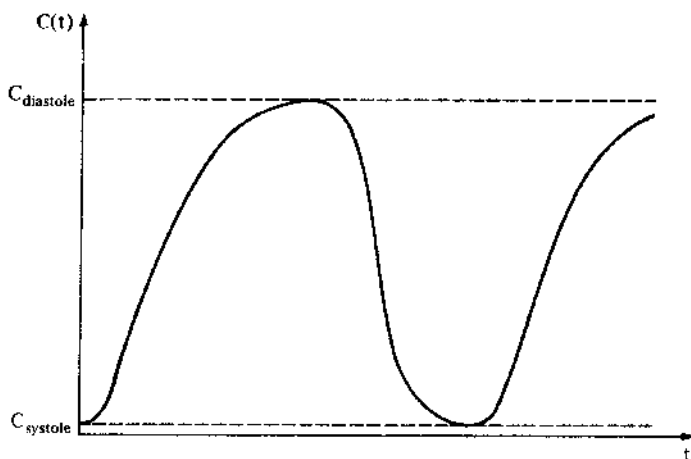


Figure 1.4. Compliance of either ventricle as a function of time. During diastole, the compliance rises from a low value C_{systole} to a high value C_{diastole} as the ventricle relaxes. During systole, it does the reverse as the ventricle contracts.

A particularly simple special case is when $C_{\text{systole}} = 0$, so that the systolic line is vertical in Figure 1.5. This gives a stroke volume

$$V_{\text{stroke}} = C_{\text{diastole}} P_v, \quad (1.4.5)$$

which is the model of the ventricle that we shall use. Finally, if F (frequency) is the heart rate (beats per minute), we get cardiac output

$$Q = F V_{\text{stroke}} = F C_{\text{diastole}} P_v. \quad (1.4.6)$$

For the present, F is taken to be constant, although we shall consider changes in heart rate later in the chapter. We define

$$K = F C_{\text{diastole}}, \quad (1.4.7)$$

so that $Q = K P_v$. We call K the *pump coefficient* of the ventricle. Although F is the same for the two sides of the heart (which are driven by the same pacemaker), the constant C_{diastole} is greater in the thin-walled right ventricle than in the thick-walled left ventricle, so K is greater on the right than on the left. Also, the two sides of the heart are connected to different venous systems. Thus, we have the following expressions for the right and

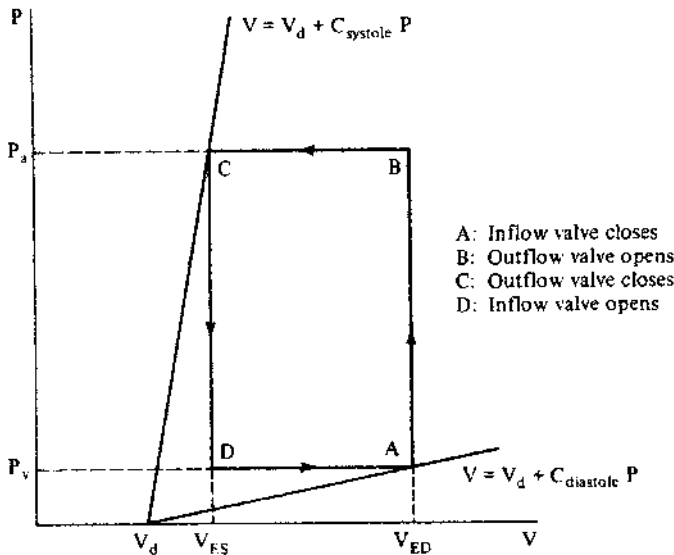


Figure 1.5. Pressure-volume diagram for either ventricle. Landmarks on the volume axis (V) are V_d = "dead volume," i.e., the volume at zero transmural pressure, V_{ES} = end-systolic volume, V_{ED} = end-diastolic volume. Landmarks on the pressure (P) axis are the venous (inflow) pressure P_V , and the arterial (outflow) pressure P_A . Since the venous pressure is essentially the same as the atrial pressure, it may be considered the inflow pressure for the ventricle. Slanting lines radiating from V_d describe the pressure-volume relationships of the ventricle at the end of diastole and at the end of systole; these lines are labeled with their equations. At other times, the pressure-volume relationship is indicated by a similar slanting line (not shown) that lies between the two lines shown. Rectangular loop ABCDA shows the path followed by the ventricle during the cardiac cycle. On the vertical branches (AB and CD), both valves are closed, so the volume of the ventricle is constant. On each of the horizontal branches, one valve is open: the arterial (outflow) valve along BC and the atrioventricular (inflow) valve along DA. The pressure difference across the open valve is here idealized as zero, and the arterial and venous pressures are idealized as constants. The names of the four phases of the cardiac cycle are AB = isovolumetric contraction, BC = ejection, CD = isovolumetric relaxation, DA = filling.

left cardiac outputs:

$$Q_R = K_R P_{sv} \quad (1.4.8)$$

$$Q_L = K_L P_{pv}. \quad (1.4.9)$$

Throughout this section we have tacitly assumed that the pressure outside the heart is zero (atmospheric). If not, then the distending pressures during diastole are not simply P_{sv} and P_{pv} but $P_{sv} - P_{thorax}$ and $P_{pv} - P_{thorax}$, where P_{thorax} is the pressure in the chest. In fact, P_{thorax} is slightly negative (with respect to the atmosphere), and this contributes to increased cardiac output by increasing the end-diastolic volume V_{ED} . This effect was first noticed because it disappears when the chest is opened during surgery. In the model developed below, we assume for simplicity that $P_{thorax} = 0$ so that we can use (1.4.8) and (1.4.9) without modification. Then, effects of $P_{thorax} < 0$ are considered briefly in Exercises 1.8 through 1.9.

1.5 Mathematical Model of the Uncontrolled Circulation

In this section we put together the ideas that have been developed above to construct a mathematical model of the circulation. In the form that we first present it, the model lacks the control mechanisms that regulate the circulation and make it serve the needs of the body. In subsequent sections we will use this model in several ways:

1. to study the *self*-regulating properties of the circulation, independent of external control mechanisms;
2. to explain the need for external control mechanisms;
3. to serve as a foundation on which we can construct a simple model of the control of circulation.

Our model is defined by the following equations (see Figure 1.6): First, we have the equations of the right and left hearts:

$$Q_R = K_R P_{sv}, \quad (1.5.1)$$

$$Q_L = K_L P_{pv}. \quad (1.5.2)$$

Second, we make the assumption that the systemic and pulmonary arteries and veins are compliance vessels. For simplicity, we use (1.3.2) instead of

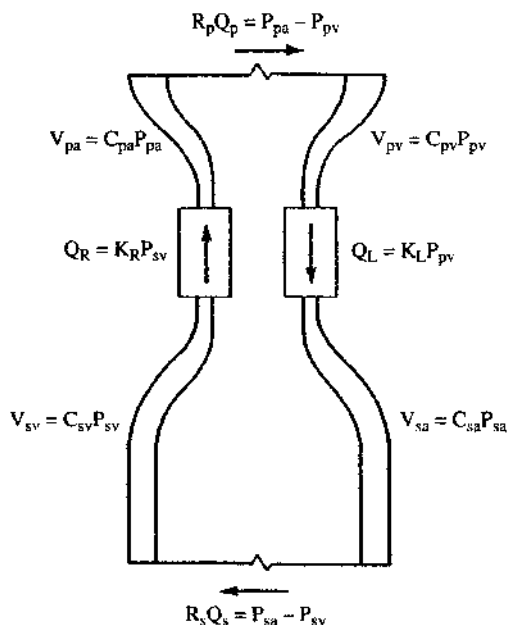


Figure 1.6. Equations of the circulation. One equation is shown for each of the eight principal elements of the circulation. For additional equations that relate these elements to each other, see the text. Each element is characterized by one parameter (K = pump coefficient, C = compliance, R = resistance) and two or three unknowns (Q = flow, P = pressure, V = volume). Subscript notation is s = systemic, p = pulmonary, a = arterial, v = venous, L = left, R = right.

(1.3.3). That is, we neglect V_d in these vessels. This gives the equations

$$V_{sa} = C_{sa} P_{sa}, \quad (1.5.3)$$

$$V_{sv} = C_{sv} P_{sv}, \quad (1.5.4)$$

$$V_{pa} = C_{pa} P_{pa}, \quad (1.5.5)$$

$$V_{pv} = C_{pv} P_{pv}. \quad (1.5.6)$$

Third, we assume that the systemic and pulmonary tissues act like resistance vessels, so that

$$Q_s = \frac{1}{R_s} (P_{sa} - P_{sv}), \quad (1.5.7)$$

$$Q_p = \frac{1}{R_p} (P_{pa} - P_{pv}). \quad (1.5.8)$$

At this point, we have an equation for each element of the circulation. Each equation contains a *parameter* that characterizes that element. These

parameters are the pump coefficients K_R , K_L , the resistances R_s , R_p , and the compliances C_{sa} , C_{sv} , C_{pa} , and C_{pv} . Suppose we are given the values of these parameters. Can we use equations (1.5.1) through (1.5.8) to determine the flows, pressures, and volumes of the model circulation? The answer to this question is negative; we do not yet have enough equations to determine the 12 unknowns

$$Q_R, Q_L, Q_s, Q_p; P_{sa}, P_{sv}, P_{pa}, P_{pv}; V_{sa}, V_{sv}, V_{pa}, V_{pv}.$$

The missing equations refer not to any particular element but to the circulation as a whole and to the way that its elements are connected. (Try to discover the missing equations for yourself before reading further.)

First, it is reasonable to assume that the total blood volume V_0 is given. This gives the equation

$$V_{sa} + V_{sv} + V_{pa} + V_{pv} = V_0, \quad (1.5.9)$$

in which V_0 is an additional parameter.

Next, we assume that the circulation is in a *steady state*, so that the flow into each of the compliance vessels must equal the flow out (why?). It follows that $Q_R = Q_L = Q_s = Q_p$, so we can drop the subscripts and just refer to all of the flows as Q , the cardiac output.

With these additional assumptions, we have nine equations for the nine unknowns $Q, P_{sa}, P_{sv}, P_{pa}, P_{pv}, V_{sa}, V_{sv}, V_{pa}, V_{pv}$. The model is complete.

Our next task is to solve the equations of the model. That is, we want to express each of the unknowns in terms of the parameters. (Try this for yourself before reading further.) An efficient plan of attack is as follows: First, express all of the pressures in terms of the flow Q . Then use the compliance equations to get the volumes in terms of Q . Finally, substitute in the equation for the total blood volume and solve for Q . With Q known (in terms of parameters only) we can go back and express the pressures and then the volumes in terms of parameters.

Here are the details. From the pump equations, we get the venous pressures in terms of Q :

$$P_{sv} = \frac{Q}{K_R}, \quad (1.5.10)$$

$$P_{pv} = \frac{Q}{K_L}. \quad (1.5.11)$$

Substituting this result in the resistance equations, we get the arterial pressures in terms of Q :

$$P_{sa} = \frac{Q}{K_R} + R_s Q, \quad (1.5.12)$$

$$P_{pa} = \frac{Q}{K_L} + R_p Q. \quad (1.5.13)$$

Substituting all four pressures into the compliance equations, we obtain

$$V_{sv} = \frac{C_{sv}}{K_R} Q, \quad (1.5.14)$$

$$V_{pv} = \frac{C_{pv}}{K_L} Q, \quad (1.5.15)$$

$$V_{sa} = \left[\frac{C_{sa}}{K_R} + C_{sa} R_s \right] Q, \quad (1.5.16)$$

$$V_{pa} = \left[\frac{C_{pa}}{K_L} + C_{pa} R_p \right] Q. \quad (1.5.17)$$

To save writing, we introduce the following combinations of parameters

$$T_{sv} = C_{sv}/K_R, \quad (1.5.18)$$

$$T_{pv} = C_{pv}/K_L, \quad (1.5.19)$$

$$T_{sa} = (C_{sa}/K_R) + C_{sa} R_s, \quad (1.5.20)$$

$$T_{pa} = (C_{pa}/K_L) + C_{pa} R_p. \quad (1.5.21)$$

Then (1.5.14) through (1.5.17) can be summarized by the equations

$$V_i = T_i Q, \quad i = sv, pv, sa, pa. \quad (1.5.22)$$

We are now ready to substitute these expressions in the equations for the total blood volume and solve for Q . We get

$$(T_{sa} + T_{sv} + T_{pa} + T_{pv})Q = V_0, \quad (1.5.23)$$

so

$$Q = \frac{V_0}{(T_{sa} + T_{sv} + T_{pa} + T_{pv})}. \quad (1.5.24)$$

The solution is completed using the equations $V_i = T_i Q$ and $P_i = V_i/C_i$. We get

$$V_i = \frac{T_i V_0}{(T_{sa} + T_{sv} + T_{pa} + T_{pv})}, \quad (1.5.25)$$

$$P_i = \frac{T_i V_0}{C_i (T_{sa} + T_{sv} + T_{pa} + T_{pv})}, \quad (1.5.26)$$

where $i = sa, sv, pa$, and pv . Thus, we have a formula for each unknown in terms of parameters only.

The quantitative use of these formulae depends on having numerical values for the parameters. In particular, we need *normal resting values* for the parameters so that we can use the model to predict the effects of parameter changes away from the normal resting state of the circulation. It is easy to determine the parameters from data such as are given in Table 1.1 and Section 1.2 because each equation of our model (1.5.5–1.5.9) contains exactly one of the parameters, so it can be written as a formula

Table 1.2. Normal Resting Parameters of the Model Circulation

	Systemic	Pulmonary
R :	$R_s = 17.5$	$R_p = 1.79 \text{ mmHg}/(\text{liter}/\text{min})$
C :	$C_{sa} = 0.01$	$C_{pa} = 0.00667 \text{ liters}/\text{mmHg}$
	$C_{sv} = 1.75$	$C_{pv} = 0.08 \text{ liters}/\text{mmHg}$
	Right	Left
K :	$K_R = 2.8$	$K_L = 1.12 \text{ (liters}/\text{min})/\text{mmHg}$
V :	$V_0 = 5.0 \text{ liters}$	

for that parameter in terms of the observed pressures, volumes, and flows. The results of this procedure are summarized in Table 1.2.

The procedure that we have just used for *identification* of parameters is based on the assumption that the model is correct. If we improve the model, then the best choice of parameters may change. An example of this is studied in Exercises 1.7 and 1.17.

1.6 Balancing the Two Sides of the Heart and the Two Circulations

The reader has probably noticed that most of the equations of the previous section came in pairs. The reason for this is the symmetry of form between the right and left heart and the systemic and pulmonary circulations. In fact, we can obtain one member of a pair from the other by making the subscript interchanges $s \leftrightarrow p$ and $R \leftrightarrow L$ (try it and see!). In the few equations that stand alone (because they refer to the circulation as a whole) these interchanges give us back the same equation as before.

If the corresponding parameters were quantitatively equal (that is, if we had $K_R = K_L$, $R_s = R_p$, etc.), then the two circulations would be quantitatively symmetrical with $P_{sa} = P_{pa}$, and so on. A glance at Tables 1.1 and 1.2 shows that this is far from being the case.

This raises the question of how the two sides of the heart and the two circulations are coordinated. What keeps the outputs of the right and left hearts equal? What mechanisms control the partition of blood volume between the systemic and pulmonary circulations? These are vital (and closely related) questions. If the left output exceeded the right output by only 10% for 1 minute, this would be enough to empty the vessels of the pulmonary circulation.

In our steady-state model of the circulation, the right and left cardiac outputs are equal by definition. In a time-dependent version of the model, we could see how this equality of output is maintained. Suppose, for ex-

ample, that K_R is suddenly reduced. Temporarily, Q_R will be less than Q_L , so there will be a net transfer of blood volume away from the pulmonary circulation and into the systemic circulation. This will raise the systemic venous pressure and lower the pulmonary venous pressure. The effect of these pressure changes will be to drive the cardiac outputs back toward equality. A net rate of transfer of volume will persist until equality of output of the two sides has been restored. Then a new equilibrium is established with a different partition of the blood volume than before.

In the steady-state model, we compute only the end result of this process. Using (1.5.25) and (1.5.18) through (1.5.21), we see that

$$\begin{aligned} \frac{V_p}{V_s} &= \frac{V_{pa} + V_{pv}}{V_{sa} + V_{sv}} \\ &= \frac{T_{pa} + T_{pv}}{T_{sa} + T_{sv}} \\ &= \left(\frac{C_{pa} + C_{pv}}{K_L} + C_{pa}R_p \right) / \left(\frac{C_{sa} + C_{sv}}{K_R} + C_{sa}R_s \right), \quad (1.6.1) \end{aligned}$$

where V_p is the total pulmonary volume and V_s is the total systemic volume. Thus, the partition of the blood volume between the two circulations is determined by the parameters, and a change in parameters that temporarily produces a disparity in output between the two sides of the heart eventually results in a volume shift that compensates for the parameter change and restores the equality of output.

The key to the success of this intrinsic control mechanism is the dependence of *cardiac output* on *venous pressure*. Suppose instead that the cardiac outputs of the two sides of the heart were given and equal. In that case Q would be a parameter and we would have to drop equations (1.5.1) and (1.5.2). We would have lost two equations but only one unknown, so we would be free to specify one more relationship. In fact, we could then assume that the pulmonary and systemic volumes (V_p, V_s) were separately given. This would lead to the equations

$$V_{sa} + V_{sv} = V_s, \quad (1.6.2)$$

$$V_{pa} + V_{pv} = V_p, \quad (1.6.3)$$

which would replace (1.5.9), increasing the number of equations by one. Thus, we would have the eight equations (1.5.3) through (1.5.8) and (1.6.2) and (1.6.3) for the eight unknowns (V_i, P_i) with Q as a new parameter. With these assumptions the pulmonary and systemic volumes would be arbitrary; there would be no mechanism available to hold them in a reasonable relationship to each other. These considerations show the importance of the dependence of cardiac output on venous pressure, not only for maintaining a balance between the two sides of the heart, but also for establishing a controlled partition of the blood volume between the pulmonary and systemic circulations.

1.7 The Need for External Circulatory Control Mechanisms

The arterioles in an exercising muscle dilate, and the systemic resistance R_s falls. The cardiac output rises, and the systemic arterial pressure is maintained. The increase in cardiac output comes primarily from an increase in heart rate while stroke volume remains fairly constant.

In this section we study the consequences of a change in R_s in our model of the *uncontrolled* circulation. We shall find a predicted response that is very different from the observed response described above. In the uncontrolled circulation a decrease in R_s results in only a modest increase in cardiac output. The most noticeable effect is a substantial fall in systemic arterial pressure. This shows the need for the external circulatory control mechanisms that are outlined in the next section.

We begin with an obvious but important remark. The effects of a change in R_s cannot be predicted solely from the equation of the systemic resistance, even though that is the only equation where R_s appears. If we neglect P_{sv} in equation (1.5.7) (an excellent approximation because P_{sv} is about 2 mmHg, whereas P_{sa} is about 100 mmHg), we get $P_{sa} = QR_s$. From this we might conclude that P_{sa} is proportional to R_s with Q constant or that Q is inversely proportional to R_s with P_{sa} constant. Neither conclusion is correct, since both P_{sa} and Q vary when R_s changes. The actual effects on P_{sa} and Q cannot be predicted without taking all of the other equations into account. That is the essence of a system of *simultaneous* equations.

In fact, we have already taken these equations into account when we solved for the unknowns in terms of the parameters. The formulae that we need are

$$Q = \frac{V_0}{T_{sa} + T_{sv} + T_{pa} + T_{pv}} \quad (1.7.1)$$

and

$$P_{sa} = \frac{V_0}{C_{sa}} \frac{T_{sa}}{T_{sa} + T_{sv} + T_{pa} + T_{pv}}, \quad (1.7.2)$$

where T_{sa} , etc., are given by (1.5.18) through (1.5.21).

Using these formulae and the parameter values given in Table 1.2, we can find the effects on Q and P_{sa} of reducing R_s to 50% of its normal value (while leaving the other parameters unchanged). The results are summarized in Table 1.3.

Note that the increase in cardiac output was only about 10% whereas the drop in arterial pressure was about 40%. This mechanism of adjusting the cardiac output is definitely inadequate to sustain reasonable levels of exercise, where cardiac output must be doubled or even tripled and where blood supply to nonmuscular tissue must be maintained.

Table 1.3. Effect of Changing Systemic Resistance on Cardiac Output and Systemic Arterial Pressure in the Uncontrolled Circulation

	Normal	$R_s = R_s^{\text{normal}}/2$	Change	% Change
Q	5.6	6.2	+0.6 liters/min	+11%
P_{sa}	100.0	57	-43.0 mmHg	-43%

The results that we have just derived can be summarized using the concept of sensitivity. If Y depends on X , and X changes, then the sensitivity of Y to X is defined to be

$$\begin{aligned}\sigma_{YX} &= \frac{\Delta \log Y}{\Delta \log X} = \frac{\log Y' - \log Y}{\log X' - \log X} \\ &= \frac{\log(Y'/Y)}{\log(X'/X)},\end{aligned}\quad (1.7.3)$$

where $X' = X + \Delta X$ and Y' is the value that Y takes on when X is changed to X' . Note that the sensitivity is not influenced by a change of units in X or Y . It also makes no difference what base is used for the logarithms in these formulae. When the changes in X and Y are small, we have approximately

$$\sigma_{YX} = \frac{dY}{Y} / \frac{dX}{X}, \quad (1.7.4)$$

which shows that the sensitivity is roughly the ratio of relative (or %) changes. If $Y = aX^n$, then $\log Y = n \log X$ and $\sigma_{YX} = n$. In particular, if Y is proportional to X , then $\sigma_{YX} = 1$. If Y is inversely proportional to X , then $\sigma_{YX} = -1$.

From the numbers in Table 1.3, we conclude that $\sigma_{QR_s} = -0.15$, while $\sigma_{P_{sa}R_s} = +0.81$. It is not a coincidence that

$$-(\sigma_{QR_s}) + \sigma_{P_{sa}R_s} \approx 1. \quad (1.7.5)$$

This follows from the fact that $P_{sa} \approx QR_s$ as the reader can show by taking logarithms and applying the definition of sensitivity. Because of (1.7.5), we cannot increase the magnitude of the sensitivity of cardiac output to systemic resistance without *decreasing* the sensitivity of systemic arterial pressure to systemic resistance. Any mechanism that accomplishes one will automatically accomplish the other.

1.8 Neural Control: The Baroreceptor Loop

From the results of the previous section, it appears that it would be a good idea to hold P_{sa} constant. In that case, we should have $\sigma_{P_{sa}R_s} = 0$ and $\sigma_{QR_s} = -1$, which would be a tremendous improvement from the

standpoint of the circulatory response to exercise. The improvement would be twofold. First, $\sigma_{QR_s} = -1$ would mean that the cardiac output would double every time the systemic resistance were halved. Second, $\sigma_{P_{sa}R_s} = 0$ would mean that the systemic arterial pressure (and hence the blood flow to the nonexercising tissues and organs) would be maintained.

In the body, P_{sa} is controlled by a feedback mechanism called the baroreceptor loop (see Figure 1.7, in which an arrow indicates a positive, or excitatory, influence, and a bar indicates a negative, or inhibitory, influence). The elements of the baroreceptor loop are as follows:

1. The *baroreceptors* (B) are stretch receptors located in the carotid arteries and in the arch of the aorta. The baroreceptors transmit nerve

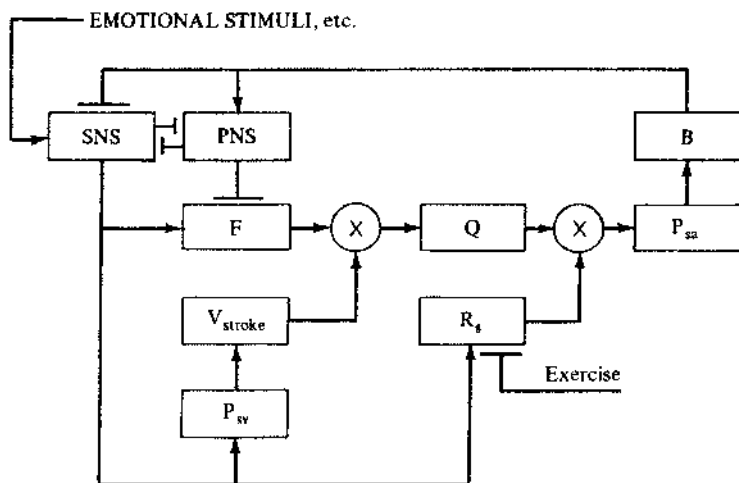


Figure 1.7. The baroreceptor loop. This diagram depicts the qualitative relationships among the many factors that regulate systemic arterial blood pressure. Arrowheads indicate positive influences, whereas bars indicate negative influences. The symbol $\times$ indicates multiplication. SNS = activity of the sympathetic nervous system, which increases the heart rate (F), the systemic venous pressure (P_{sv}), and the systemic resistance (R_s); PNS = activity of the parasympathetic nervous system, which slows the heart; B = activity of the baroreceptors, which stimulates the parasympathetic nervous system and inhibits the sympathetic nervous system. (Note that these two arms of the autonomic nervous system also inhibit each other.) V_{stroke} is the stroke volume, Q is the cardiac output, and P_{sa} is the arterial blood pressure. Note that any path through this diagram from P_{sa} back to itself involves an odd number of inhibitory influences. Thus, the whole system is a negative feedback loop (actually, several negative feedback loops working in parallel) that tends to stabilize the arterial blood pressure against changes such as emotional stimuli or exercise, which are shown here as inputs to the system.

impulses to the brain stem at a rate that increases with increasing arterial pressure (see Chapter 6).

2. The *parasympathetic nervous system* (PNS) is excited by activity of the baroreceptors. Its effect is to slow the heart rate (F).
3. The *sympathetic nervous system* (SNS) is inhibited by activity of the baroreceptors. It has several effects on the circulation, including:
 - (a) increased heart rate;
 - (b) increased venous pressure, and so increased stroke volume;
 - (c) increased systemic resistance.

The loop is closed through the mechanics of the circulation, which implies that $Q = FV_{\text{stroke}}$ and that $P_{\text{sa}} = QR_s$.

Tracing any closed loop from P_{sa} back to P_{sa} in Figure 1.7, we find an odd number of inhibitory influences. This means that any changes in P_{sa} lead to *compensatory* changes through the baroreceptor loop.

We will not present a detailed model of the baroreceptor loop here (see Chapter 6, for some further discussion). Instead, we model its overall function by assuming that the baroreceptor loop adjusts the heart rate F until the systemic arterial pressure achieves a target value P^* . Note that this model ignores the effects of the sympathetic nervous system on venous pressure and on systemic resistance. These effects are less important than the effect on heart rate in the normal operation of the circulation.

Thus, we have a new unknown, F , that was previously a parameter and a new parameter, $P_{\text{sa}} = P^*$, that was previously an unknown. Also, because F is no longer a parameter, we have to rewrite (1.5.1) and (1.5.2) in the form

$$Q_R = FC_R P_{\text{sv}}, \quad (1.8.1)$$

$$Q_L = FC_L P_{\text{pv}}, \quad (1.8.2)$$

where C_R and C_L are the *diastolic compliance* of the right and left hearts (see Section 1.4), so that $C_R P_{\text{sv}}$ is the right stroke volume and $C_L P_{\text{pv}}$ is the left stroke volume.

This gives us a model of the controlled circulation in which the equations are (1.8.1) and (1.8.2) together with (1.5.3) through (1.5.9) and the steady-state relation $Q_R = Q_p = Q_s = Q_L$. The unknowns are the same as before except that now F replaces P_{sa} .

Instead of solving these equations directly, we make some approximations. First, we neglect P_{sv} compared to P_{sa} in the equation of the systemic resistance. This represents a 2% error, and it gives us the equation

$$QR_s = P^*. \quad (1.8.3)$$

Next, we neglect the pulmonary volumes in comparison with the systemic volumes in the equation of the total blood volume. This represents a 10%

error, and it gives us

$$V_{sa} + V_{sv} = V_0, \quad (1.8.4)$$

which can be rewritten as

$$C_{sa}P^* + C_{sv}P_{sv} = V_0. \quad (1.8.5)$$

We can now determine Q directly from (1.8.3) and P_{sv} directly from (1.8.5):

$$Q = \frac{P^*}{R_s}, \quad (1.8.6)$$

$$P_{sv} = \frac{V_0 - C_{sa}P^*}{C_{sv}}. \quad (1.8.7)$$

Substituting these results in the equation of the right heart (1.8.1), we can solve for the heart rate

$$F = \frac{P^*C_{sv}}{R_sC_R(V_0 - C_{sa}P^*)}. \quad (1.8.8)$$

These results summarize the performance of the controlled circulation. We have achieved what we set out to do: Since $P_{sa} = P^*$, which is constant, $\sigma_{P_{sa}R_s} = 0$. Again, since P^* is constant, $\sigma_{QR_s} = -1$.

Thus, our model of the controlled circulation responds to changes in R_s with (inversely) proportional changes in cardiac output while the arterial pressure is maintained. In the model as in life, the mechanism responsible for the increased cardiac output is an increase in heart rate, since the venous pressure and stroke volume ($V_{stroke} = C_R P_{sv}$) are independent of R_s in the model.

In the uncontrolled circulation the cardiac output depends on all of the parameters of the model; in the controlled circulation it depends only on P^* and R_s . This isolation of cardiac output from extraneous influences is just as important as the heightened sensitivity to R_s . We give one example. In the uncontrolled circulation, we had $\sigma_{QV_0} = 1$, which means that the cardiac output is proportional to the blood volume (see equation 1.5.24). In the controlled circulation, $\sigma_{QV_0} = 0$, which means that the cardiac output is protected against blood loss, for example. From (1.8.8), we see that the mechanism of adaptation to blood loss is an increase in heart rate that compensates for the decrease in stroke volume. We also see that the model breaks down when $V_0 = C_{sa}P^*$, which corresponds to complete depletion of the systemic venous blood.

We have shown that the response of the controlled circulation to stress is very different from that of the uncontrolled circulation. It is remarkable that such dramatic changes in behavior emerge when the only change in the mathematical model is to make one parameter into an unknown and one unknown into a parameter.

1.9 Autoregulation

Up to this point, we have treated the systemic resistance as a parameter. In this section we shall consider the local control of systemic resistance. Central control of systemic resistance was mentioned, but not modeled, in the previous section (see also Chapter 6).

There are two phenomena that come under the term autoregulation:

1. When the pressure-flow relation of a tissue is measured, it often turns out that there is a range of pressures in which the flow is relatively insensitive to the pressure difference.
2. At constant pressure difference, the flow through many tissues depends on the rate of O_2 consumption of the tissue.

In the normal function of the circulation, the second phenomenon is more important than the first, since the pressure difference is relatively constant, as we have just seen in the previous section. In pathological conditions, the first phenomenon may be important for regulating blood flow in the face of fluctuating pressures.

In this section we outline a simple model that accounts for both phenomena through a single mechanism. The model that we shall describe is a simplified version of a model proposed by Huntsman, Attinger, and Noordergraaf.

The key hypothesis is that the resistance of a tissue is regulated by the venous O_2 concentration of the tissue. In general, *concentration* means the amount per unit volume, and its units depend on how the amount is measured. In the case of a gas, it is convenient to measure the amount of gas in terms of the volume that the gas would occupy under some specified conditions of temperature and pressure. In physiology, the most natural conditions are atmospheric pressure and body temperature. When this is done, the concentration becomes dimensionless (volume/volume.)

For example, the concentration of O_2 in blood, denoted by $[O_2]$, is the number of liters of O_2 that are carried in one liter of blood. Oxygen is carried in blood bound to hemoglobin, and when all of the O_2 -carrying sites in the hemoglobin molecules are occupied, the concentration of O_2 in blood is $\frac{1}{5}$. Coincidentally, this is the same as the concentration of O_2 in the atmosphere itself. Under normal conditions, the hemoglobin becomes saturated as it passes through the lungs, so that the arterial concentration $[O_2]_a = \frac{1}{5}$. The O_2 concentration of arterial blood is constant for all of the tissues of the body, but it may vary under conditions of high altitude or anemia. In the former case, the hemoglobin may fail to be saturated in the lung. In the latter case, the concentration of hemoglobin in blood is lower than normal. In both cases, $[O_2]_a$ is reduced, but this reduction is felt by all tissues of the body.

The venous O_2 concentration, $[O_2]_v$, is different in the different tissues of the body. Let M (metabolic rate) stand for the rate of O_2 consumption of a tissue (M has units of liters/minute). Also, let Q be the blood flow to the tissue in question. The rate at which O_2 is delivered to the tissue in the arterial blood is $Q[O_2]_a$, the units of which are (liters of blood/minute) $\times$ (liters of O_2 /liter of blood). Similarly, the rate at which O_2 leaves the tissue in its venous blood is $Q[O_2]_v$. If the tissue is in a steady state, the difference must be accounted for by the metabolic rate of the tissue. This gives the equation

$$Q[O_2]_a - Q[O_2]_v = M, \quad (1.9.1)$$

which is called *Fick's principle*. Thus,

$$[O_2]_v = [O_2]_a - M/Q. \quad (1.9.2)$$

This formula shows that $[O_2]_v$ may serve as an index of the adequacy of the blood supply in relation to the metabolic rate of the tissues. When the blood supply is just barely sufficient to sustain the metabolic rate, then

$$Q = Q^* = M/[O_2]_a \quad (\text{this defines } Q^*), \quad (1.9.3)$$

and we get $[O_2]_v = 0$. As Q is raised above Q^* , $[O_2]_v$ rises, and finally, $[O_2]_v \rightarrow [O_2]_a$ as $Q \rightarrow \infty$. This shows why it might be reasonable to use $[O_2]_v$ to regulate the resistance of a tissue to blood flow.

A problem with this hypothesis is that resistance is regulated on the arterial side of the tissue, not on the venous side. The venous O_2 concentration is determined by the *tissue* O_2 concentration, however, and the arterioles run through the tissue and may therefore be influenced by the tissue O_2 concentration.

Suppose, for example, that

$$R = R_0[O_2]_v, \quad (1.9.4)$$

where

$$R = P/Q \quad (1.9.5)$$

is the resistance of the tissue to blood flow, P is the arteriovenous pressure difference ($P = P_{sa} - P_{sv}$), and R_0 is the constant of proportionality that relates the resistance of the tissue to the venous O_2 concentration.

Equation (1.9.4) simply asserts that tissue resistance to blood flow is proportional to venous O_2 concentration. This is the simplest of a class of models in which tissue resistance is regulated by venous O_2 concentration.

To study the consequences of this simple hypothesis, we substitute (1.9.2) and (1.9.5) into (1.9.4) to obtain the pressure-flow relation of a tissue in

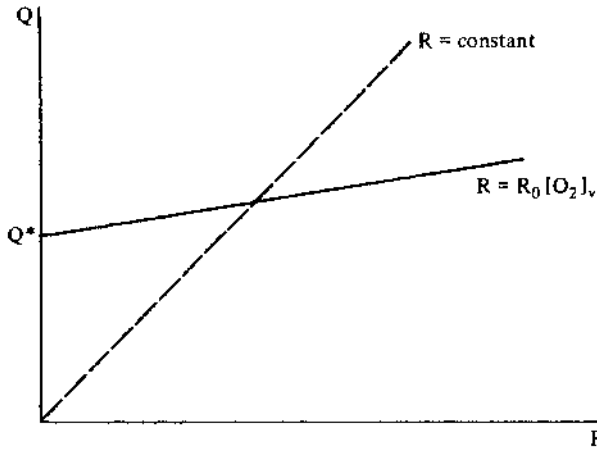


Figure 1.8. Pressure-flow relation of a tissue with and without autoregulation. $P = P_{sa} - P_{sv}$ = pressure difference driving flow through the tissue, and Q is the resulting flow. The dashed line shows the unregulated relationship characterized by a constant resistance R , and the solid line shows the autoregulated response, in which the resistance is proportional to the venous oxygen concentration $[O_2]_v$, with R_0 as the constant of proportionality. Note that the unregulated relationship includes the origin ($Q = 0$ when $P = 0$), but that in the autoregulated case there is still nonzero flow ($Q = Q^*$) even when ($P = 0$).

which resistance is regulated in this way. The result is

$$Q = \frac{M}{[O_2]_a} + \frac{P}{R_0 [O_2]_a} \quad (1.9.6)$$

$$= Q^* + \frac{P}{R_0 [O_2]_a}, \quad (1.9.7)$$

where $Q^* = M/[O_2]_a$. This result is plotted in Figure 1.8.

The behavior of the model tissue is summarized by the following statements, which the reader should be able to verify:

1. The sensitivity of flow to pressure (σ_{QP}) is less when $R = R_0 [O_2]_v$ than when $R = \text{constant}$.
2. The tissue always receives at least the minimum flow Q^* required to sustain its metabolic rate. (Think about how this works. When $P \rightarrow 0$, why doesn't $Q \rightarrow 0$ in the model? What happens to R and $[O_2]_v$?)
3. At constant P , if M changes, then $\Delta Q = \Delta Q^* = \Delta M/[O_2]_a$. This means that the change in blood flow is just what is needed to support the extra O_2 consumption.

4. At constant P , if M increases, then R automatically decreases. (Plot R as a function of M with P , $[O_2]_a$, and R_0 constant.)
5. If $[O_2]_a$ changes (with P and R_0 constant), then Q automatically adjusts in such a way that $Q[O_2]_a = \text{constant}$. Therefore, the rate of O_2 supply to all of the tissues is the same as it was before the change in $[O_2]_a$.

In summary, the simple device of setting $R = R_0[O_2]_v$ (instead of $R = \text{constant}$) makes the blood supply to a tissue less sensitive to pressure changes and more responsive to the needs of the tissue.

1.10 Changes in the Circulation Occurring at Birth

The circulation forms a simple loop after birth. Before birth, however, the configuration of the circulation is complicated by additional connections. One of these is a vessel called the *ductus arteriosus*, which connects the pulmonary and systemic arteries near the heart. Another is an opening in the wall that separates the right and left atria. This opening, called the *foramen ovale*, is guarded by a flap of tissue that acts as a valve to ensure that blood flow through the foramen always goes from right to left.

The function of these extra connections is to shunt blood away from the lungs, which are collapsed before birth and which therefore present high resistance to blood flow.

In this section we present a simple model of the *fetal circulation* (the circulation before birth), and we shall use this model to explain the sequence of changes (initiated by the first breath) that close the shunts and establish the single-loop configuration of the circulation that persists into adult life.

The model is shown in Figure 1.9. The shunt flows are the ductus flow Q_d and the foramen flow Q_f . If these are both zero, then the model takes on the configuration of a simple loop in which blood flows through the right heart; the pulmonary arteries, tissues, and veins; the left heart; and the systemic arteries, tissues, and veins.

Note that the flow through the ductus arteriosus is not always in the direction indicated by the arrow in Figure 1.9. The arrow points in the direction of the flow that we have chosen to call positive, which is also the normal direction of flow during fetal life. After birth but before the closure of the ductus arteriosus, however, the flow through the ductus is in the opposite direction. The reasons for this sudden reversal at birth will be explained below. The two situations are easily accommodated by a single system of equations if we consider the reverse flow through the ductus as being negative. Thus, $Q_d > 0$ means that the flow is in the direction defined by the arrow in Figure 1.9, whereas $Q_d < 0$ means that flow is in the opposite direction.

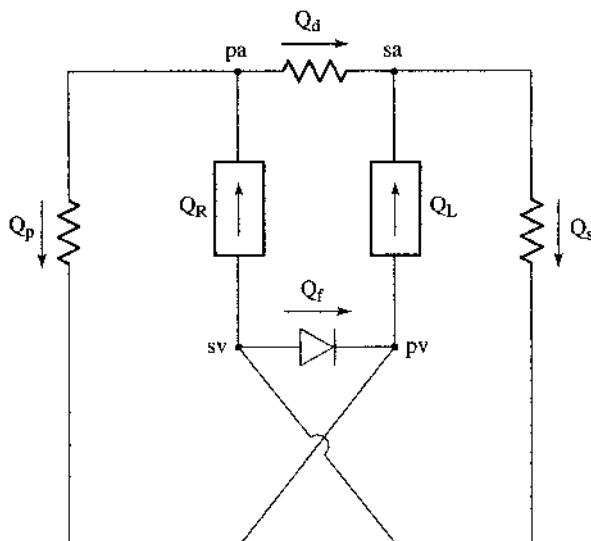


Figure 1.9. The fetal circulation. There are six distinct flows to consider, namely Q_L = left heart output, Q_R = right heart output, Q_p = pulmonary flow (which is small, through the collapsed lungs), Q_s = systemic flow (which is large, since it includes the flow to the placenta), Q_d = flow through the ductus arteriosus, Q_f = flow through the foramen ovale. All of these flows except Q_d are always in the direction of the arrows, but in the case of Q_d its arrow merely indicates the direction of flow that we choose to call positive, from the pulmonary artery (pa) to the systemic arteries (sa). This is, in fact, the direction of flow during fetal life ($Q_d > 0$), but after birth the ductus flow normally reverses ($Q_d < 0$) until the ductus actually closes and Q_d becomes zero. The electrical symbol for a diode is used for the foramen ovale to indicate its role as a valve, which allows flow to pass freely (zero resistance) from the right to the left atrium, i.e., in effect, from the systemic veins (sv) to the pulmonary veins (pv), but which prevents flow in the reverse direction. The valvular nature of the foramen ovale becomes important immediately after birth. If the ductus arteriosus and foramen ovale are removed ($Q_d = Q_f = 0$), then we are left with the much simpler nonfetal circulation in the form of a simple closed loop (although drawn here in a somewhat convoluted way).

The equations we use are those of the uncontrolled circulation:

$$Q_R = K_R P_{sv}, \quad (1.10.1)$$

$$Q_L = K_L P_{pv}, \quad (1.10.2)$$

$$Q_p = (P_{pa} - P_{pv})/R_p, \quad (1.10.3)$$

$$Q_s = (P_{sa} - P_{sv})/R_s, \quad (1.10.4)$$

which must be supplemented by the equations for the ductus and the foramen. The ductus is modeled as being a simple resistance

$$Q_d = \frac{P_{pa} - P_{sa}}{R_d}. \quad (1.10.5)$$

The foramen is modeled as an ideal valve. This means that the valve may be *open*, in which case

$$Q_f \geq 0 \quad \text{and} \quad P_{sv} = P_{pv}, \quad (1.10.6)$$

or *closed*, in which case

$$Q_f = 0 \quad \text{and} \quad P_{sv} \leq P_{pv}. \quad (1.10.7)$$

Unfortunately, the natural terminology for a fluid valve is precisely the opposite of the natural terminology for an electrical switch, where "closed" means conducting and "open" means nonconducting. We shall use the natural fluid terminology here, with apologies for the confusion that this may cause to the reader who is accustomed to the electrical terminology.

How do we decide whether the foramen is open or closed? The answer is to try either alternative and see whether or not it is self-consistent. For example, we could start by assuming the foramen to be open and use the equation $P_{sv} = P_{pv}$. This will give enough equations to determine Q_f , and we can check whether or not $Q_f \geq 0$. If not, our original guess was incorrect. (The reader may want to worry about what to do if both assumptions or neither turns out to be self-consistent. Such concern is unnecessary because of the following theorem, which the reader may enjoy trying to prove: For any set of parameter values, at least one of the conditions (1.10.6) or (1.10.7) is self-consistent, and if both are self-consistent, the parameters must be such that $P_{sv} = P_{pv}$ and $Q_f = 0$. In the latter case, it makes no difference whether we say that the foramen is open or closed.)

We now have equations for each element of the model, but we need the steady-state relationships among the different flows. In the nonfetal circulation, these relationships were very simple ($Q_R = Q_L = Q_p = Q_s$), and there was really only one independent flow, the cardiac output. Here, the situation is much more complicated.

At each junction in Figure 1.9 we have the relationship that the total flow *in* equals the total flow *out*. This gives the following four equations:

$$pa : Q_R = Q_p + Q_d, \quad (1.10.8)$$

$$sa : Q_d + Q_L = Q_s, \quad (1.10.9)$$

$$sv : Q_s = Q_R + Q_f, \quad (1.10.10)$$

$$pv : Q_p + Q_f = Q_L. \quad (1.10.11)$$

Note that each of the six flows appears exactly once on the left-hand side and exactly once on the right-hand side of equations (1.10.8) through

(1.10.11). (Why?) Thus, the sum of these four equations gives the identity

$$Q_R + Q_d + Q_L + Q_s + Q_p + Q_f = Q_R + Q_d + Q_L + Q_s + Q_p + Q_f.$$

It follows that any one of these equations can be derived from the other three. (Try it and see.) In short, we have three independent relations among six flows in the model, so we can express all of the flows in terms of any three of them. If we (arbitrarily) choose Q_R , Q_d , Q_L as these three flows, then we have as a consequence of (1.10.8) through (1.10.11) that

$$Q_s = Q_L + Q_d, \quad (1.10.12)$$

$$Q_p = Q_R - Q_d, \quad (1.10.13)$$

$$Q_f = Q_L - Q_R + Q_d. \quad (1.10.14)$$

The last equation shows the importance of having both an arterial and an atrial (venous) shunt. If we want $Q_R = Q_L$, then we must have $Q_f = Q_d$. Conversely, if Q_d or Q_f is zero, then an imbalance between the two sides of the heart will be required to shunt blood away from the lungs. Presumably, it is important to maintain at least a rough equality of output between the two sides of the heart during fetal life in order that the heart should develop properly and be prepared for the postnatal situation in which the configuration of the circulation is a single closed loop and in which both sides of the heart must therefore pump the same amount of blood per unit time. As we have just shown, the need to shunt blood away from the high-resistance fetal lungs can be reconciled with the need to balance the outputs of the two sides of the heart only if *two* shunts are provided, one on the venous side (the foramen ovale) and one on the arterial side (the ductus arteriosus).

We are now ready to solve the equations of the fetal circulation. First, we remark that the fetal circulation is characterized by a high pulmonary resistance (because the lungs are collapsed) and a low systemic resistance (because of blood flow to the placenta through the umbilical cord). The result is that $R_p > R_s$, a relationship that reverses after birth. A second remark is that the two sides of the heart are more nearly equal in their compliance properties before birth than afterwards. We model this by setting

$$K_R = K_L = K \quad (\text{this defines } K). \quad (1.10.15)$$

Third, the ductus is wide open before birth, and we assume that $R_d = 0$. This gives the equation

$$P_{sa} = P_{pa} = P_a \quad (\text{this defines } P_a). \quad (1.10.16)$$

Finally, we make the provisional assumption that the foramen ovale is open before birth. This gives the equation

$$P_{sv} = P_{pv} = P_v \quad (\text{this defines } P_v). \quad (1.10.17)$$

Later, we will have to solve for Q_f and check that $Q_f \geq 0$.

The analysis of the model is now simple. Since there is a common venous pressure and the two sides of the heart have the same pump coefficients ($K_R = K_L$), we have

$$KP_v = Q_R = Q_L = Q \quad (\text{this defines } Q). \quad (1.10.18)$$

Since there is also a common arterial pressure, we have

$$R_s Q_s = R_p Q_p = P_a - P_v. \quad (1.10.19)$$

Thus, the ratio of the pulmonary to the systemic flow is controlled (inversely) by the ratio of resistances

$$\frac{Q_p}{Q_s} = \frac{R_s}{R_p}. \quad (1.10.20)$$

Using (1.10.12), (1.10.13), and (1.10.18), we get

$$\frac{Q - Q_d}{Q + Q_d} = \frac{R_s}{R_p}, \quad (1.10.21)$$

which can be solved for Q_d/Q :

$$\frac{Q_d}{Q} = \frac{R_p - R_s}{R_p + R_s}. \quad (1.10.22)$$

The foramen flow is determined by (1.10.14), which reduces to $Q_f = Q_d$, since $Q_R = Q_L$. Thus, we also have

$$\frac{Q_f}{Q} = \frac{R_p - R_s}{R_p + R_s}. \quad (1.10.23)$$

Finally, from equations (1.10.12) and (1.10.13), we see that

$$\frac{Q_s}{Q} = 1 - \frac{Q_d}{Q} = 1 + \frac{R_p - R_s}{R_p + R_s} = 2 \frac{R_p}{R_p + R_s}, \quad (1.10.24)$$

$$\frac{Q_p}{Q} = 1 - \frac{Q_d}{Q} = 1 - \frac{R_p - R_s}{R_p + R_s} = 2 \frac{R_s}{R_p + R_s}. \quad (1.10.25)$$

We have now determined all of the ratios of flows in the model in terms of the parameters R_p and R_s . Note that $Q_f > 0$ as long as $R_p > R_s$, so we have found a self-consistent solution with the foramen ovale *open*.

Perhaps the best way to appreciate the effectiveness of the model in shunting blood flow away from the high-resistance lungs is to consider the extreme case $R_p = \infty$. None of our formulae yield infinite values in this case. We simply get the limiting solution $Q_f = Q_d = Q$, $Q_s = 2Q$, and $Q_p = 0$.

At birth, the lungs expand with the first breath, and R_p falls suddenly. The umbilical cord constricts (either naturally or because it is clamped), and R_s rises suddenly. The net effect is that $R_p < R_s$. This makes $Q_f < 0$ in (1.10.23). This means that the solution found above is no longer self-consistent. Therefore, the flap of tissue guarding the foramen ovale suddenly slams shut.

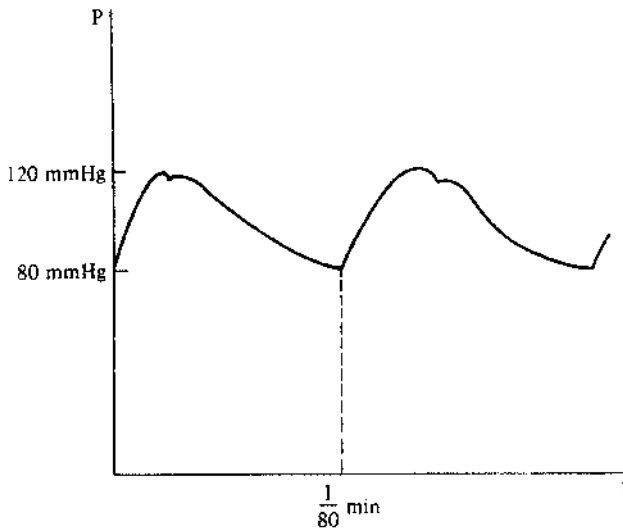


Figure 1.10. Systemic arterial blood pressure (P) as a function of time (t). The normal maximum (systolic) and minimum (diastolic) pressures are indicated on the pressure axis, and a typical period of the heartbeat is indicated on the time axis.

A self-consistent closed foramen solution is found in the Exercises at the end of the chapter. This describes the circulation after birth when $R_p < R_s$. One property of this solution is $Q_d < 0$, which means that the flow in the ductus has reversed compared to the fetal case. After birth the flow through the ductus is a *left-to-right shunt*, and it carries fully oxygenated blood from the aorta back into the pulmonary arterial tree. The high O_2 content of this blood stimulates the closure of the ductus, completing the transition to a single-loop circulation.

1.11 Dynamics of the Arterial Pulse

In the foregoing sections we have treated the circulation as though all of the pressures, flows, and volumes are constant in time. This is not correct. Actually, the heart ejects blood into the arteries in discrete bursts. During these contractions of the heart, the blood pressure rises rapidly, and it falls again between contractions as blood runs out of the arteries through the tissues. The result of this process is the *arterial pulse*, which can be felt wherever an artery is convenient to press (e.g., at the wrist) and which can be used to count the heart rate. The waveform of the arterial pulse is sketched in Figure 1.10.

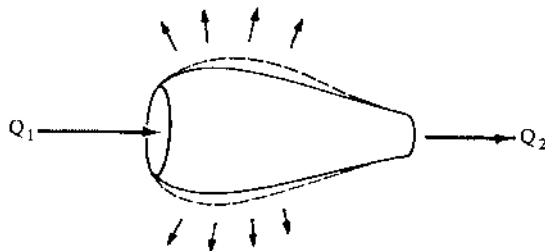


Figure 1.11. Volume conservation in pulsatile flow. The rate of change of volume of a vessel is equal to its inflow (Q_1) minus its outflow (Q_2).

When blood pressure is measured with an air cuff on the upper arm, the actual quantities determined by this measurement are the maximum (systolic) and minimum (diastolic) pressures achieved by the arterial pulse. A blood pressure of 120/80 means that the systolic arterial blood pressure is 120 mmHg and the diastolic arterial blood pressure is 80 mmHg. The difference between these values (in this case 40 mmHg) is called the *pulse pressure*.

The names *systolic* and *diastolic* refer to the phases of the cardiac cycle (see Section 1.4). Under normal conditions, the systolic pressure in the systemic arteries is essentially the same as the systolic left ventricular pressure, since the aortic valve is open during systole. The diastolic pressure in the ventricle is much lower than that in the arteries. This is possible because the aortic valve is closed during diastole.

In this section we shall describe the simplest model that can account for the qualitative form of the arterial pulse. This model will be used to show how the systolic and diastolic pressures depend on the parameters of the heart and circulation. It will also be used to justify the simpler steady flow models that were studied in the foregoing sections of this chapter.

We begin by considering a compliance vessel that is not in the steady state (see Figure 1.11). Thus, the inflow $Q_1(t)$ is not equal to the outflow $Q_2(t)$ at every instant. When they are not equal, the volume of the vessel changes. In fact, if $V(t)$ denotes the volume of the vessel at time t , we have a differential equation:

$$\dot{V} = Q_1 - Q_2. \quad (1.11.1)$$

(Here and below we write $\dot{V}$ for dV/dt .) This says that the rate of change of the vessel's volume is the difference between the flow in and the flow out. When $V = \text{constant}$, $Q_1 = Q_2$, which is the steady-state relation that we have used up to now.

This differential equation describes how the volume changes, but it can be converted into one relating the pressure in the vessel to the flows in and

out by using the equation of a compliance vessel: Either

$$V(t) = CP(t) \quad \text{or} \quad V(t) = CP(t) + V_d$$

(see Section 1.3). In either case, we have $\dot{V} = C \dot{P}$, so

$$C \dot{P} = Q_1 - Q_2. \quad (1.11.2)$$

This is the equation that governs pressure changes in a compliance vessel in the case of unsteady flow.

Next, we use this equation to study the systemic arterial tree. Now, $P = P_{sa}$, the systemic arterial pressure; $C = C_{sa}$, the systemic arterial compliance, $Q_1 = Q_L$, the output of the left heart, and $Q_2 = Q_s$, the blood flow through the systemic tissues. For Q_s , we have the equation $Q_s = (P_{sa} - P_{sv})/R_s$, which we approximate by

$$Q_s = \frac{P_{sa}}{R_s} \quad (1.11.3)$$

since $P_{sv} \ll P_{sa}$. Thus, (1.11.2) becomes

$$C_{sa} \dot{P}_{sa} = Q_L(t) - \frac{P_{sa}}{R_s} \quad (1.11.4)$$

During diastole, when the aortic valve is closed, $Q_L = 0$. In that case, the solution of (1.11.4) is

$$P_{sa}(t) = P_{sa}(0) \exp\left(-\frac{t}{R_s C_{sa}}\right). \quad (1.11.5)$$

The constant $P_{sa}(0)$ remains to be determined.

To find it, we consider ventricular systole. We make the simplifying assumption that the entire stroke volume ΔV_0 is ejected from the heart instantaneously. Then we cannot use equation (1.11.4) for systole, at least not in an elementary way. We can, however, figure out what happens during systole by considering the change in arterial pressure produced in the arteries by a sudden change in volume of magnitude ΔV_0 : From the compliance equation $V = CP + V_d$, it is clear that

$$\Delta V_0 = C_{sa} \Delta P_{sa}. \quad (1.11.6)$$

Now suppose that the heartbeat is a *periodic* phenomenon. That is, suppose that everything repeats exactly from one beat to the next. Let the duration of each heartbeat be T , so that the heart rate is $1/T$. Then the diastolic arterial pressure is $P_{sa}(T)$ and the systolic arterial pressure is $P_{sa}(0)$ (see Figure 1.12). Thus, the jump in pressure caused by cardiac ejection is given by the formula

$$\Delta P_{sa} = P_{sa}(0) - P_{sa}(T). \quad (1.11.7)$$

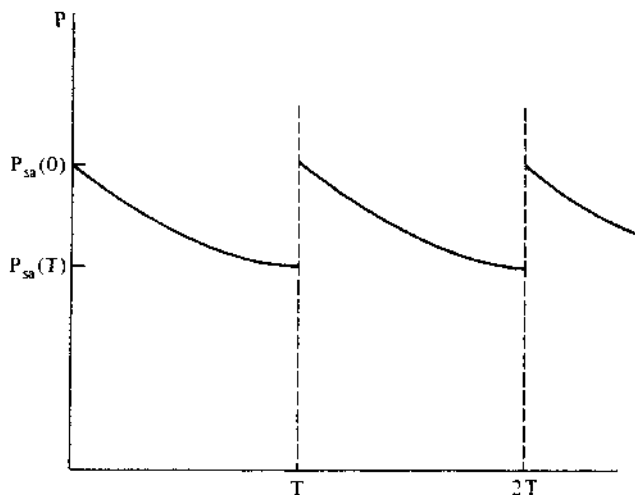


Figure 1.12. Idealized arterial pulse, under the assumption that ejection of the entire stroke volume from the left ventricle occurs instantaneously. T = period of the heartbeat, $P_{sa}(0)$ = systolic pressure, $P_{sa}(T)$ = diastolic pressure.

Now let $t = T$ in (1.11.5) and substitute (1.11.7) into (1.11.6) to obtain a pair of equations for $P_{sa}(0)$ and $P_{sa}(T)$. The result is

$$P_{sa}(T) = \Theta P_{sa}(0), \quad (1.11.8)$$

$$\frac{\Delta V_0}{C_{sa}} = P_{sa}(0) - P_{sa}(T), \quad (1.11.9)$$

where

$$\Theta = \exp\left(-\frac{T}{(R_s C_{sa})}\right). \quad (1.11.10)$$

Note that $0 < \Theta < 1$. Solving these equations gives

$$P_{sa}(0) = \frac{\Delta V_0}{C_{sa}(1 - \Theta)}, \quad (1.11.11)$$

$$P_{sa}(T) = \frac{\Delta V_0 \Theta}{C_{sa}(1 - \Theta)}, \quad (1.11.12)$$

which are formulae for the systolic and diastolic pressures in terms of the stroke volume, the arterial compliance, the systemic resistance, and the heart rate. Subtracting these two equations, we recover (1.11.9), which is the formula for the pulse pressure.

What about the mean arterial pressure? A useful definition of the *mean* of a periodic function, say $f(t)$, is

$$\langle f \rangle = \frac{1}{T} \int_0^T f(t) dt,$$

where the period of f is T and where integration is over any period, e.g., over the interval $(0, T)$. With this notation, we define the mean arterial pressure to be

$$\langle P_{sa} \rangle = \frac{1}{T} \int_0^T P_{sa}(t) dt. \quad (1.11.13)$$

(This is approximately the average of N samples of the function $P_{sa}(t)$ taken at equally spaced times that span the interval $[0, T]$. The approximation becomes exact as $N \rightarrow \infty$.) We leave it as an exercise to check that

$$\langle P_{sa} \rangle = \frac{R_s \Delta V_0}{T}. \quad (1.11.14)$$

Since $\Delta V_0/T$ is the cardiac output, this equation can be interpreted as $\langle P_{sa} \rangle = Q R_s$, which is the equation that holds in the steady state case (if we neglect P_{sv} as we are doing here). This explains how a steady-state model still has significance for a pulsatile circulation: The quantities appearing in the steady-state model are the time averages of the corresponding pulsatile quantities.¹

We have determined the form of the arterial pulse in the periodic case, where everything repeats exactly from one beat to the next. This is not quite correct even in the normal circulation, where the heart rate and stroke volumes change slightly in response to the phases of breathing. The assumption of periodicity is even less appropriate for individuals with abnormal rhythms of the heart where successive heartbeats may be considerably different from each other, both in their durations and in their stroke volumes. As an extreme example of an aperiodic arterial pulse, consider what happens if the heart has just been started following a period of cardiac arrest. The arterial pressure is initially very low, and it has to build up toward its equilibrium values over the first few beats. The rest of this section studies these transient situations.

If the stroke volume and timing vary from beat to beat, we need notation to tell us what happens on each beat. Let $j = 1, 2, \dots$ be an index counting beats of the heart. Let t_j be the time that the j th beat occurs, and let ΔV_j be the corresponding stroke volume. Since the pressure $P_{sa}(t)$ jumps at the times t_j (see Figure 1.13), we need notation to distinguish the arterial pressures just before and just after cardiac ejection. Let

$P_{sa}(t_j^-)$ = arterial pressure just before ejection (diastolic);

¹This statement is only approximate. In a pulsatile version of the whole circulation model considered above, the pressures that determine the cardiac outputs would be the end-diastolic venous pressures, not the mean pressures, since it is the end-diastolic pressures that determine the volumes of the ventricular chambers just prior to ejection. More generally, any nonlinearity that might be introduced to make the model more realistic would further degrade the correspondence between the steady-flow results and the mean values of the pulsatile results.

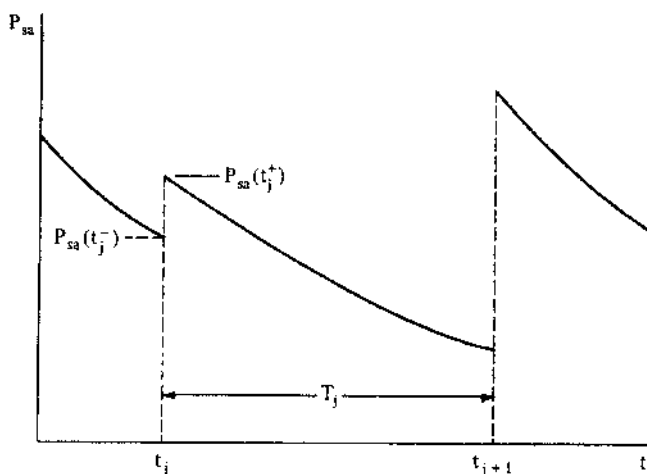


Figure 1.13. Idealized arterial pulse in the presence of an irregular heart rhythm. t_j = time of the j th ventricular systole (idealized here as instantaneous), $P_{sa}(t_j^-)$ = diastolic pressure immediately before the j th ventricular systole, $P_{sa}(t_j^+)$ = systolic pressure immediately after the j th ventricular systole.

$P_{sa}(t_j^+)$ = arterial pressure just after ejection (systolic).

Between beats of the heart, we have, as before, the differential equation

$$C_{sa} \dot{P}_{sa} = -P_{sa}/R_s,$$

but now it is more convenient to write the solutions in the form

$$P_{sa}(t) = P_{sa}(t_j^+) \exp(-(t - t_j)/(R_s C_{sa})) \quad (1.11.15)$$

for $t_j < t < t_{j+1}$. Setting $t = t_{j+1}^-$ gives

$$P_{sa}(t_{j+1}^-) = P_{sa}(t_j^+) \Theta_j, \quad (1.11.16)$$

where

$$\Theta_j = \exp(-T_j/(R_s C_{sa})) \quad (1.11.17)$$

and

$$T_j = t_{j+1} - t_j. \quad (1.11.18)$$

Equation (1.11.16) gives the diastolic pressure just before beat $j + 1$ in terms of the systolic pressure just after beat j .

The equation for the jump in arterial pressure on beat j now takes the form

$$P_{sa}(t_j^+) = P_{sa}(t_j^-) + \Delta V_j / C_{sa}. \quad (1.11.19)$$

Now suppose that we are given any sequence of times t_j and stroke volumes ΔV_j together with the constant parameters C_{sa} and R_s . If we are told the diastolic pressure just before the first beat, we can use (1.11.19) to find the systolic pressure just after that beat. Then we can use (1.11.16) to find the diastolic pressure just before the next beat. Repeating this process we can predict the entire sequence of diastolic and systolic pressures, however irregular it might be.

The equations that we have just developed for the aperiodic situation should include the periodic arterial pulse as a special case. Suppose the heartbeat is regular, so that $t_{j+1} - t_j = T$ and $\Delta V_j = \Delta V_0$ for all j . Then Θ_j reduces to Θ , and equations (1.11.16) and (1.11.19) become

$$P_{sa}(t_{j+1}^-) = P_{sa}(t_j^+) \Theta, \quad (1.11.20)$$

$$P_{sa}(t_j^+) = P_{sa}(t_j^-) + \Delta V_0 / C_{sa}. \quad (1.11.21)$$

Now we can look for a solution of these equations in which $P_{sa}(t_j^+)$ and $P_{sa}(t_j^-)$ are independent of j . We express this by means of the notation

$$P_{sa}(t_j^+) = P_{sa}^+ \text{ (systolic pressure),} \quad (1.11.22)$$

$$P_{sa}(t_j^-) = P_{sa}^- \text{ (diastolic pressure).} \quad (1.11.23)$$

Thus, P_{sa}^+ and P_{sa}^- satisfy

$$P_{sa}^- = P_{sa}^+ \Theta \quad (1.11.24)$$

$$P_{sa}^+ = P_{sa}^- + \Delta V_0 / C_{sa}. \quad (1.11.25)$$

These are the same equations as (1.11.8) and (1.11.9) for the periodic systolic and diastolic pressures. This confirms that our theory of irregular arterial pulses contains the periodic pulse as a special case. (There are other solutions of (1.11.20) and (1.11.21) that do not correspond to a periodic pressure pulse even though the heartbeat and stroke volume are regular. Can you show that these solutions approach the periodic pulse as time increases?)

1.12 Computer Simulation of Pulsatile Blood Flow

In this section we show how computers can be used to solve the equations of time dependent pressure, flow, and volume in the circulatory system. We start with a simple case, the systemic arterial pulse, and work up to a model of the circulation as a whole. Already in the case of the arterial pulse, we shall see that the use of computers gives us the freedom to make our mathematical models more complicated and more realistic than before.

The equations of the systemic arterial pulse that we shall use in this section are as follows:

$$\dot{V}_{sa} = Q_{Ao} - Q_s, \quad (1.12.1)$$

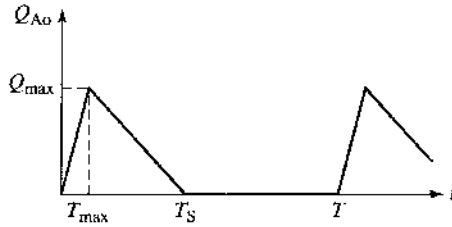


Figure 1.14. Outflow from the left ventricle through the aortic valve into the systemic arterial tree is here denoted $Q_{Ao}(t)$. It is a periodic function whose triangular waveform is determined by the parameters Q_{max} = peak flow, T_{max} = time of peak flow, T_S = duration of systole, and T = duration of heartbeat. Note that the stroke volume is given by the integral of $Q_{Ao}(t)$ over one heartbeat, which is equal to $Q_{max}T_S/2$, and that the cardiac output is the stroke volume times the heart rate, which is therefore equal to $Q_{max}T_S/(2T)$.

$$V_{sa} = (V_{sa})_d + C_{sa}P_{sa}, \quad (1.12.2)$$

$$Q_s = P_{sa}/R_s. \quad (1.12.3)$$

In these three equations, the three unknowns (all of which are functions of time) are V_{sa} , the systemic arterial volume; P_{sa} , the systemic arterial pressure; and Q_s , the flow through the systemic resistance. Note that Q_s can also be described as the outflow from the systemic arteries. The systemic arterial pulse is driven by the inflow to the systemic arteries, which is the flow through the aortic valve, denoted here by Q_{Ao} . This is a function of time, which for purposes of modeling just the systemic arterial pulse we shall treat as a *given* function of time. Its specific form will be specified later. The parameters appearing in these equations are $(V_{sa})_d$, the volume of the systemic arteries when $P_{sa} = 0$; C_{sa} , the compliance of the systemic arteries; and R_s , the systemic resistance. These three parameters are all (given) constants. (But $(V_{sa})_d$ drops out of the equations when we express everything in terms of the pressure, below.)

Equation (1.12.1) expresses conservation of volume for the systemic arteries: The rate of change of volume is the inflow minus the outflow. Equation (1.12.2) is the compliance relationship between systemic arterial pressure and volume. Equation (1.12.3) describes the flow through the systemic resistance in terms of the systemic arterial pressure. In writing (1.12.3), we have made the approximation of neglecting the systemic venous pressure in comparison to the systemic arterial pressure.

By substituting equations (1.12.2) and (1.12.3) into equation (1.12.1), we reduce the above system to a single equation for the systemic arterial pressure:

$$C_{sa} \dot{P}_{sa} = Q_{Ao} - P_{sa}/R_s. \quad (1.12.4)$$

Except for slight differences in notation, the theory that we have outlined here is the same as in Section 1.11, but with this important difference: Here we allow Q_{Ao} to be any (given) function of time. The specific function that we shall use is shown in Figure 1.14 and may be described mathematically as follows. First, Q_{Ao} is a periodic function of time:

$$Q_{Ao}(t + T) = Q_{Ao}(t), \quad (1.12.5)$$

where T is the period of the heartbeat ($1/T$ is the heart rate). Because Q_{Ao} is periodic, we can define it completely by specifying it in any one period, say $(0, T)$. This is done as follows:

$$Q_{Ao}(t) = \begin{cases} Q_{\max} t/T_{\max}, & 0 \leq t \leq T_{\max} \\ Q_{\max}(T_S - t)/(T_S - T_{\max}), & T_{\max} \leq t \leq T_S \\ 0, & T_S \leq t \leq T \end{cases} \quad (1.12.6)$$

where $Q_{\max}$ is the maximum flow, $T_{\max}$ is the time at which maximum flow occurs (relative to the beginning of systole), and T_S is the duration of systole. Note that the stroke volume (integral of Q_{Ao} over one cycle) is given by $Q_{\max}T_S/2$ and that the cardiac output is therefore given by $Q_{\max}T_S/(2T)$. Reasonable choices of the parameters that determine Q_{Ao} are $T = 0.0125$ minutes, $T_S = 0.0050$ minutes, $T_{\max} = 0.0020$ minutes, $Q_{\max} = 28$ liters/minute. Note that this gives a stroke volume of 0.070 liters and a cardiac output of 5.6 liters/minute.

The next step is to introduce a numerical method for the (approximate) solution of equation (1.12.4). Such a method can be found by replacing the time derivative in that equation by a difference quotient. There are several ways to do this, but we choose a "backward" method, for reasons that will be discussed later (see the discussion following equation 1.12.33). When we do this, equation (1.12.4) becomes

$$C_{sa}(P_{sa}(t) - P_{sa}(t - \Delta t))/\Delta t = Q_{Ao}(t) - P_{sa}(t)/R_s. \quad (1.12.7)$$

The way that we shall use this equation is to solve it for $P_{sa}(t)$ in terms of $P_{sa}(t - \Delta t)$ and other known quantities, including $Q_{Ao}(t)$. The solution is:

$$P_{sa}(t) = \frac{P_{sa}(t - \Delta t) + \Delta t Q_{Ao}(t)/C_{sa}}{1 + \Delta t/(R_s C_{sa})}. \quad (1.12.8)$$

Now if we have a starting value $P_{sa}(0)$, we can use (1.12.8) repeatedly to find $P_{sa}(\Delta t)$, $P_{sa}(2\Delta t)$, $P_{sa}(3\Delta t)$, and so on, until we have spanned whatever interval of time may be of interest.

Two questions may be bothering the reader at this point: How should I choose the starting value $P_{sa}(0)$, and how should I choose the time step Δt ?

The answer to the first question is that you can choose any value at all for $P_{sa}(0)$. No matter what pressure you start at, the arterial pulse will eventually settle down into the same periodic waveform. When you get your computer program working, try different starting values and compare

the results. The arterial pulse will gradually “forget” what the starting value was. So you don’t have to worry about this, except that you have to specify some starting value, and you have to run your program long enough for the pulse waveform to settle down. Now you might say in response to this, “Why don’t I just start at the diastolic pressure, since I know what that is (80 mmHg)?” The trouble with this is that you know the diastolic pressure only under normal conditions. A computer program such as the one we are creating here wouldn’t be of much use if it could only handle one case. What you want to do is to change parameters and see what happens. As soon as you change parameters, you don’t know the diastolic pressure anymore, so you will just have to let the program run until the pressure waveform settles into its periodic pattern.

The answer to the second question is that Δt has to be “small enough.” Recall that we made an approximation in replacing $\dot{P}_{sa}$ by $(P_{sa}(t) - P_{sa}(t - \Delta t))/\Delta t$. This approximation gets better as Δt gets smaller. To tell whether Δt is small enough, you should rerun your program with Δt replaced by $\Delta t/2$. (Note that with Δt half as big you will have to take twice as many steps to span the same amount of time.) If the results are essentially the same (to within whatever tolerance you think is reasonable), then Δt is indeed small enough. If not, then you have to keep making Δt smaller. To get into the right ballpark in the first place, you should choose Δt small compared to any of the times that define the problem. In our case, a reasonable choice might be $\Delta t = 0.01T$. With this choice, it takes 100 steps to span a cardiac cycle.

The numerical method that we have just outlined is implemented by the following Matlab program. It involves one script and two functions. We shall describe the functions first, so it will be clear what the script is doing when it calls these functions. The functions are `QAo_now(t)`, which computes the flow through the aortic valve at time t according to equations (1.12.5-1.12.6), and `Psa_new(Psa_old,QAo)`, which computes the “new” pressure $P_{sa}(t)$ in terms of the “old” pressure $P_{sa}(t - \Delta t)$ and the given flow $Q_{Ao}(t)$ through the aortic valve. These functions read as follows:

```
function Q=QAo_now(t)
%filename: QAo_now.m
global T TS TMAX QMAX;
tc=rem(t,T); % tc=time elapsed since
%the beginning of the current cycle
%rem(t,T) is the remainder when t is divided by T
if(tc<TS)
    %SYSTOLE:
    if(tc<TMAX)
        %BEFORE TIME OF MAXIMUM FLOW:
        Q=QMAX*tc/TMAX;
    else
```

```

    %AFTER TIME OF PEAK FLOW:
    Q=QMAX*(TS-tc)/(TS-TMAX);
end
else
    %DIASTOLE:
    Q=0;
end

function Psa=Psa_new(Psa_old,QAo)
%filename: Psa_new.m
global Rs Csa dt;
Psa=(Psa_old+dt*QAo/Csa)/(1+dt/(Rs*Csa));

```

With these two functions in hand, we can write a script that we shall call `sa` for “systemic arteries.” It loops over time and computes $P_{sa}(t)$. It will call another script `in_sa` that does initialization. These scripts read as follows:

```

%filename: sa.m
clear all % clear all variables
clf % and figures
global T TS TMAX QMAX;
global Rs Csa dt;
in_sa %initialization
for klok=1:klokmax
    t=klok*dt;
    QAo=QAo_now(t);
    Psa=Psa_new(Psa,QAo); %new Psa overwrites old
    %Store values in arrays for future plotting:
    t_plot(klok)=t;
    QAo_plot(klok)=QAo;
    Psa_plot(klok)=Psa;
end
%Now plot results in one figure
%with QAo(t) in upper frame
%and Psa(t) in lower frame
subplot(2,1,1), plot(t_plot,QAo_plot)
subplot(2,1,2), plot(t_plot,Psa_plot)

%filename: in_sa.m (initialization for the script sa)
T =0.0125 %Duration of heartbeat (minutes)
TS=0.0050 %Duration of systole (minutes)
TMAX=0.0020 %Time at which flow is max (minutes)
QMAX=28.0 %Max flow through aortic valve (liters/minute)
Rs=17.86 %Systemic resistance (mmHg/(liter/minute))
Csa=0.00175 %Systemic arterial compliance (liters/(mmHg))
%This value of Csa is approximate and will need adjustment

```

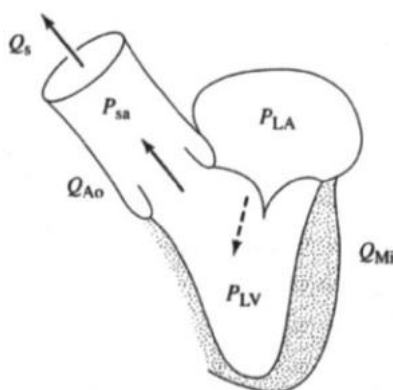


Figure 1.15. Model of the left heart and systemic arteries. Pressures and flows are P_{LA} = left atrial pressure, P_{LV} = left ventricular pressure, P_{sa} = systemic arterial pressure, Q_{Mi} = flow through the mitral valve (indicated here by a dotted arrow because the mitral valve is closed at the moment shown), Q_{Ao} = flow through the aortic valve, and Q_s = outflow from the systemic arterial tree, i.e., aggregate flow through all of the tissues of the body.

```
%to make the blood pressure be 120/80.
dt=0.01*T    %Time step duration (minutes)
%This choice implies 100 timesteps per cardiac cycle.
klokmax=15*T/dt %Total number of timesteps
%This choice implies simulation of 15 cardiac cycles.
Psa=0        %Initial value of Psa (mmHg)
%Any initial value is OK here; try some others.
%Initialize arrays to store data for plotting:
    t_plot=zeros(1,klokmax);
    QAo_plot=zeros(1,klokmax);
    Psa_plot=zeros(1,klokmax);
```

The program for the systemic arterial pulse is now complete. It is the collection of four files `sa.m`, `in_sa.m`, `QAo_now.m`, and `Psa_new.m`. To invoke it from within a Matlab session launched from the same directory where these files reside, just type “`sa`” at the Matlab prompt “`>>`”. Physiological applications of this computer program are outlined in Section (1.13).

The next step in building up towards a computer model of the whole circulation is to attach the left heart to the systemic arteries. This model is shown in Figure 1.15. The model of the left heart that we shall use was already outlined in Section 1.4; it treats the left ventricle as a compliance vessel the compliance of which is not constant but instead is some given function of time. This given function $C_{LV}(t)$ will replace $Q_{Ao}(t)$ as the periodic time function that drives the cardiac cycle, and $Q_{Ao}(t)$ will become one of the unknowns of the model.

An important complication here in comparison with the model of the systemic arteries alone is that we now have to consider the valves of the heart. The left ventricle has an inflow valve called the *mitral valve*, and an outflow valve called the *aortic valve*. When these valves are healthy, they present a very low resistance to forward flow, but they completely block backflow. Valve diseases may alter either or both of these properties, i.e., the resistance to forward flow may be increased ("stenosis"), or the valve may become leaky and allow backflow ("regurgitation" or "incompetence"). Valve malfunctions such as these may be caused by congenital malformations of the valves, by the aftermath of certain diseases like rheumatic fever, or by bacterial infection of the valves (which can happen as a result of intravenous drug abuse). The computer program that we develop here (and also the next one, which models the whole circulation) can be used to study some of the circulatory consequences of heart valve malfunction.

We shall use the following system of equations to model the left heart and systemic arteries. (Those describing the systemic arteries are the same as above but will be repeated here so that we have the whole model in one place.)

First, we have the equations that give the rate of change of volume of each chamber as the inflow minus the outflow:

$$\dot{V}_{LV} = Q_{Mi} - Q_{Ao}, \quad (1.12.9)$$

$$\dot{V}_{sa} = Q_{Ao} - Q_s. \quad (1.12.10)$$

Next, we have the compliance relation for each chamber:

$$V_{LV} = (V_{LV})_d + C_{LV}P_{LV}, \quad (1.12.11)$$

$$V_{sa} = (V_{sa})_d + C_{sa}P_{sa}. \quad (1.12.12)$$

Although these two equations are of the same form, we should keep in mind the important difference that C_{LV} is a function of time, whereas C_{sa} is constant.

The next group of equations specify the flows in terms of pressure differences. In the case of the valve flows, they involve "switching variables" S_{Mi} and S_{Ao} , which take on the value 1 when the valve in question is open, allowing forward flow to proceed, and 0 when that valve is closed, preventing backflow.

$$Q_{Mi} = S_{Mi}(P_{LA} - P_{LV})/R_{Mi}, \quad (1.12.13)$$

$$Q_{Ao} = S_{Ao}(P_{LV} - P_{sa})/R_{Ao}, \quad (1.12.14)$$

$$Q_s = P_{sa}/R_s. \quad (1.12.15)$$

Finally, we have the equations that determine the valve states. Whether a valve is open or closed depends on the direction in which the flow is "trying" to go, i.e., on the sign of the pressure difference across the valve.

(We assume here that the valve always has some resistance, so there is some pressure drop, albeit a small one, even in forward flow.) Thus we have

$$S_{Mi} = \begin{cases} 0, & P_{LA} < P_{LV}, \\ 1, & P_{LA} > P_{LV}, \end{cases} \quad (1.12.16)$$

$$S_{Ao} = \begin{cases} 0, & P_{LV} < P_{sa}, \\ 1, & P_{LV} > P_{sa}. \end{cases} \quad (1.12.17)$$

What if the pressures upstream and downstream of the valve are exactly equal? Then it doesn't matter whether we say that the valve is open or closed, since the flow through it will be zero in any case, see equations (1.12.13)–(1.12.14).

Our model of the left heart and systemic arteries is defined by the nine equations (1.12.9)–(1.12.17). The nine unknowns in these equations are the left ventricular and systemic arterial volumes V_{LV} and V_{sa} ; the corresponding pressures P_{LV} and P_{sa} ; the mitral, aortic, and systemic flows Q_{Mi} , Q_{Ao} , and Q_s ; and the switching variables of the mitral and aortic valves S_{Mi} and S_{Ao} . All of these are unknown functions of time. There is one given function of time that drives everything else, the left ventricular compliance $C_{LV}(t)$. It will be specified below. The given constant parameters are the left atrial pressure P_{LA} ; the mitral and aortic valve resistances R_{Mi} and R_{Ao} ; the systemic resistance R_s ; and the systemic arterial compliance C_{sa} . Of these, R_s and C_{sa} have been assigned values above. A reasonable choice for P_{LA} is 5 mmHg. For a healthy valve, the valve resistance is very small, and the exact value of the valve resistance is therefore not very important. A valve resistance that is small enough to have essentially no impact on the rest of the circulation is 0.01 mmHg/(liter/minute), so this is a reasonable value to use for R_{Mi} or R_{Ao} in modeling healthy valves.

We now specify the left ventricular compliance $C_{LV}(t)$. A qualitative sketch of how this function should look was given in Figure 1.4, except that here we shall take time zero to be at the end of diastole, when the left ventricular compliance has its *maximum* value. The function that we shall use is periodic with period T ,

$$C_{LV}(t + T) = C_{LV}(t), \quad (1.12.18)$$

and it is defined on one period as follows:

$$C_{LV}(t) = \begin{cases} C_{LVD} \left(\frac{C_{LVS}}{C_{LVD}} \right)^{\frac{1 - \exp(-(t/\tau_S))}{1 - \exp(-T_S/\tau_S)}}, & 0 \leq t \leq T_S, \\ C_{LVS} \left(\frac{C_{LVD}}{C_{LVS}} \right)^{\frac{1 - \exp(-(t - T_S)/\tau_D)}{1 - \exp(-(T - T_S)/\tau_D)}}, & T_S \leq t \leq T. \end{cases} \quad (1.12.19)$$

The form of this function has been chosen so that during systole $C_{LV}(t)$ makes a transition from its maximum value C_{LVD} to its minimum value C_{LVS} , and during diastole it makes a transition back from C_{LVS} to C_{LVD} . The speeds of these transitions are determined by the time constants τ_S

and τ_D . Smaller values of these time constants yield faster transitions. Reasonable values for the parameters that appear in the function $C_{LV}(t)$ are as follows: $C_{LVD} = 0.0146$ liters/mmHg; $C_{LVS} = 0.00003$ liters/mmHg; $\tau_S = 0.0025$ minutes; $\tau_D = 0.0075$ minutes. (The remaining parameters, T and T_S , were defined above.)

We now write a Matlab function that can be called to evaluate $C_{LV}(t)$. We shall make it general enough so that later, when we are doing the whole circulation, we can call it for either side of the heart, with the appropriate arguments:

```
function CV=CV_now(t, CVS, CVD)
%filename: CV_now.m
global T TS tauS tauD;
tc=rem(t,T); %tc=time in the current cycle,
               %measured from start of systole.
if(tc<TS)
    e=(1-exp(-tc/tauS))/(1-exp(-TS/tauS));
    CV=CVD*(CVS/CVD)^e;
else
    e=(1-exp(-(tc-TS)/tauD))/(1-exp(-(T-TS)/tauD));
    CV=CVS*(CVD/CVS)^e;
end
```

When this function is being used on the left side of the heart, CVS will be set equal to CLVS, and CVD will be set equal to CLVD, and the output CV will be the left ventricular compliance CLV. The timing parameters that we have made global are essentially the same on both sides of the heart.

The next step is to introduce a numerical method for solving the equations of the left side of the heart and the systemic arteries. Before doing so, however, we simplify these equations by expressing everything in terms of pressures. This is done by combining equations (1.12.9)–(1.12.15) to obtain

$$\frac{d(C_{LV}P_{LV})}{dt} = \frac{S_{Mi}(P_{LA} - P_{LV})}{R_{Mi}} - \frac{S_{Ao}(P_{LV} - P_{sa})}{R_{Ao}}, \quad (1.12.20)$$

$$C_{sa}\dot{P}_{sa} = \frac{S_{Ao}(P_{LV} - P_{sa})}{R_{Ao}} - \frac{P_{sa}}{R_s}. \quad (1.12.21)$$

Note that the arterial compliance C_{sa} has been factored out of the time derivative term in equation (1.12.21), which is legitimate since C_{sa} is constant. In equation (1.12.21), however, C_{LV} is time dependent and must therefore remain inside the time derivative. (We could, of course, use the product rule to expand the time derivative in equation (1.12.21), but there is no reason to do so.) Our model is now reduced to equations (1.12.16)–(1.12.21).

As before, we derive a numerical method by approximating the time derivatives by backward differences:

$$\begin{aligned}
 & \frac{C_{LV}(t)P_{LV}(t) - C_{LV}(t - \Delta t)P_{LV}(t - \Delta t)}{\Delta t} \\
 &= \frac{S_{Mi}(t)}{R_{Mi}} (P_{LA} - P_{LV}(t)) - \frac{S_{Ao}(t)}{R_{Ao}} (P_{LV}(t) - P_{sa}(t)), \quad (1.12.22) \\
 & \frac{C_{sa}P_{sa}(t) - P_{sa}(t - \Delta t)}{\Delta t} \\
 &= \frac{S_{Ao}(t)}{R_{Ao}} (P_{LV}(t) - P_{sa}(t)) - \frac{P_{sa}}{R_s}(t). \quad (1.12.23)
 \end{aligned}$$

If the states of the valves, $S_{Mi}(t)$ and $S_{Ao}(t)$, were known, this would be a 2×2 linear system in the two unknowns $P_{LV}(t)$ and $P_{sa}(t)$. In fact, it is a nonlinear system, since $S_{Mi}(t)$ and $S_{Ao}(t)$ depend on $P_{LV}(t)$ and $P_{sa}(t)$; see equations (1.12.16)–(1.12.17). Nevertheless, it will be useful to solve for $P_{LV}(t)$ and $P_{sa}(t)$ in terms of $S_{Mi}(t)$ and $S_{Ao}(t)$. To do this, we first put equations (1.12.22)–(1.12.23) into standard form:

$$C_{11}(t)P_{LV}(t) + C_{12}(t)P_{sa}(t) = B_1(t), \quad (1.12.24)$$

$$C_{21}(t)P_{LV}(t) + C_{22}(t)P_{sa}(t) = B_2(t), \quad (1.12.25)$$

where

$$C_{11}(t) = C_{LV}(t) + \Delta t \left(\frac{S_{Mi}(t)}{R_{Mi}} + \frac{S_{Ao}(t)}{R_{Ao}} \right), \quad (1.12.26)$$

$$C_{12}(t) = C_{21}(t) = -\Delta t \frac{S_{Ao}(t)}{R_{Ao}}, \quad (1.12.27)$$

$$C_{22}(t) = C_{sa} + \Delta t \left(\frac{S_{Ao}(t)}{R_{Ao}} + \frac{1}{R_s} \right), \quad (1.12.28)$$

$$B_1(t) = C_{LV}(t - \Delta t)P_{LV}(t - \Delta t) + \Delta t \frac{S_{Mi}(t)}{R_{Mi}} P_{LA}, \quad (1.12.29)$$

$$B_2(t) = C_{sa}P_{sa}(t - \Delta t). \quad (1.12.30)$$

The solution, which can be found for example by determinants, is as follows:

$$P_{LV}(t) = \frac{B_1(t)C_{22}(t) - B_2(t)C_{12}(t)}{D(t)}, \quad (1.12.31)$$

$$P_{sa}(t) = \frac{B_2(t)C_{11}(t) - B_1(t)C_{21}(t)}{D(t)}, \quad (1.12.32)$$

where

$$D(t) = C_{11}(t)C_{22}(t) - C_{12}(t)C_{21}(t). \quad (1.12.33)$$

Exercise for the reader: Show that $D(t) > 0$, so there will never be any trouble about dividing by zero in equations (1.12.31)–(1.12.32).

We can now give a brief discussion of the reason why backward differences are preferable to forward differences in this particular problem.

Recall that the valve resistances R_{Mi} and R_{Ao} are very small in healthy valves. This suggests that we consider the limits $R_{Mi} \rightarrow 0$ and $R_{Ao} \rightarrow 0$, to see whether our numerical scheme is well behaved no matter how small the valve resistances may be. These limits should be taken with all other parameters fixed, including Δt . There are four cases that must be considered separately, corresponding to the four possible valve configurations: $(S_{Mi}(t), S_{Ao}(t)) = (0, 0), (0, 1), (1, 0), (1, 1)$. When either valve is closed, its resistance drops out, thus simplifying the process of taking the limit. In case $(0, 0)$, $P_{LV}(t)$ and $P_{sa}(t)$ are independent of the valve resistances, so they clearly have finite limits as the valve resistances approach zero. In case $(0, 1)$, the limiting values of $P_{LV}(t)$ and $P_{sa}(t)$ can be found by multiplying the numerator and denominator in equations (1.12.31)–(1.12.32) by R_{Ao} and then setting $R_{Ao} = 0$. In case $(1, 0)$, do the same but with R_{Mi} instead of R_{Ao} . And finally, in case $(1, 1)$, multiply numerator and denominator by the product $R_{Mi}R_{Ao}$ and then set both valve resistances equal to zero. The point here is that in all four cases, we get finite limits as the valve resistances approach zero, and this means that we can set the valve resistances as small as we like. If we had used forward differences, instead of equations (1.12.22)–(1.12.23), we would have

$$\begin{aligned} & \frac{C_{LV}(t + \Delta t)P_{LV}(t + \Delta t) - C_{LV}(t)P_{LV}(t)}{\Delta t} \\ &= \frac{S_{Mi}(t)}{R_{Mi}} (P_{LA} - P_{LV}(t)) - \frac{S_{Ao}(t)}{R_{Ao}} (P_{LV}(t) - P_{sa}(t)), \end{aligned} \quad (1.12.34)$$

and

$$C_{sa} \frac{P_{sa}(t + \Delta t) - P_{sa}(t)}{\Delta t} = \frac{S_{Ao}(t)}{R_{Ao}} (P_{LV}(t) - P_{sa}(t)) - \frac{1}{R_s} P_{sa}(t). \quad (1.12.35)$$

Here we don't have to bother with solving a 2×2 system, since each of the unknowns $P_{LV}(t + \Delta t)$ and $P_{sa}(t + \Delta t)$ appears only once in its own equation. This is an advantage, at least in terms of simplicity, but if we do solve for these unknowns and then try taking the limit as the valve resistances approach zero, we shall see immediately that there are no finite limits. What this means in practice is that if we use the forward scheme, our computed solution will "blow up" if we try to make the valve resistances too small. How small we can make them will depend on the choice of Δt , or to put it another way, the smaller we make the valve resistances, the smaller we shall have to make Δt to get a sensible computed solution. The backward scheme has no such restriction. To get a better feel for what is going on here, try programming both schemes and compare their behaviors as the valve resistances are reduced.

Returning now to the scheme based on backward differences, we write a Matlab function that solves for $P_{LV}(t)$ and $P_{sa}(t)$ given the following data: the values of these pressures at the previous time step, $P_{LV}(t - \Delta t)$ and $P_{sa}(t - \Delta t)$; the left ventricular compliance at both the previous and

the current time step, $C_{LV}(t - \Delta t)$ and $C_{LV}(t)$; the state of the valves at the current time, $S_{Mi}(t)$ and $S_{Ao}(t)$; and, of course, all of the constant parameters (which will be listed as global variables in the Matlab function). One point that requires clarification is how we can treat the valve states as known, even though they depend on the very pressures that we are trying to determine. This will be revealed later. For now, just keep in mind that the function we are creating has the limited purpose of solving for the pressures given the state of the valves. From the point of view of this function, it is as though the valves had been locked in position, and we wanted to determine the resulting pressures. The function for solving the problem reads as follows:

```
function [PLV,Psa]=PLV_Psa_new(PLV_old,Psa_old,CLV_old, ...
                                CLV,SMi,SAo)

%filename PLV_Psa_new.m
global Csa Rs Rmi RAo dt CHECK PLA;
C11=CLV+dt*((SMi/Rmi)+(SAo/RAo));
C12=-dt*(SAo/RAo);
C22=Csa+dt*((SAo/RAo)+(1/Rs));
B1=CLV_old*PLV_old+dt*(SMi/Rmi)*PLA;
B2=Csa*Psa_old;
D=C11*C22-C12^2;
PLV=(B1*C22-B2*C12)/D;
Psa=(B2*C11-B1*C12)/D;
if (CHECK)
    LHS1=(CLV*PLV-CLV_old*PLV_old)/dt;
    RHS1=(SMi/Rmi)*(PLA-PLV)-(SAo/RAo)*(PLV-Psa);
    CH1=RHS1-LHS1
    LHS2=Csa*(Psa-Psa_old)/dt;
    RHS2=(SAo/RAo)*(PLV-Psa)-(1/Rs)*Psa;
    CH2=RHS2-LHS2
end
```

Note that this routine includes the global parameter CHECK, which should be set during initialization to the value 0 (false) or 1 (true). When CHECK has the value 1, the function PLV_Psa_new checks itself by substituting the computed solution back into the equations that it is trying to solve. Note that these equations are written for checking purposes in as close to their original form as possible (but with all terms on one side, so that the result of the check should be CH1=CH2=0). Because of this, we are checking not only the computer program but also any algebraic manipulations that were made in deriving the computer program, e.g., putting the equations in standard form. Because the lines in which CH1 and CH2 are defined have no “;” at the end, Matlab will put the values of CH1 and CH2 on the screen. If all is well, they will be zero to within roundoff error. To prevent seeing this output every time the function is called, just set CHECK=0.

One could, of course, handle the results of the check in other ways, e.g., by setting a tolerance and having the program report the results of the check only if the tolerance is exceeded, but we leave such refinements to the reader.

Now we come to the part of the program in which we have to determine the state of the valves. There is something circular going on here, since it would seem that the pressures can be determined only when the state of the valves is known (as in `PLV_Psa_new`, above), but we also know that the valve states are determined by the pressures, see equations (1.12.16)–(1.12.17)! What we are seeking is a *self-consistent* choice of the valve states and the pressures. One way to find such a choice is by trial and error: Start with any state of the valves, and solve for the pressures. Then reset the valves according to the pressures, solve for the pressures again, and so on, until the valve states stop changing. This idea is implemented in the following Matlab script:

```
%filename: set_SMi_SAO.m
%script to find
%self-consistent valve states and pressures:
done=0; %not done yet!
while(~done) %keep trying if not done (see below)
    SMi_noted=SMi; %note the value of SMi
    SAo_noted=SAo; %note the value of SAo
    % set pressures based on valve states:
    [PLV,Psa]=PLV_Psa_new(PLV_old,Psa_old,CLV_old,CLV,SMi,SAo);
    %and then set valve states based on pressures:
    SMi=(PLA>PLV); %evaluates to 1 if PLA>PLV, 0 otherwise
    SAo=(PLV>Psa); %evaluates to 1 if PLV>Psa, 0 otherwise
    %we're done if both valve states are unchanged:
    done=(SMi==SMi_noted)&(SAo==SAo_noted);
end
```

To avoid confusion, note that the above `while` loop is *not* a loop over time. It happens within a single time step. Therefore, the values of `PLV_old`, `Psa_old`, `CLV_old`, and `CLV` all remain constant during the execution of this loop. Once the loop has finished executing, `PLV`, `Psa`, `SMi`, and `SAo` are the self-consistent pressures and valve states at the end of the time step.

Note that the initial guesses for `SMi` and `SAo` are not set within the above script. Instead, they are whatever values these variables happen to have when the script is called. In practice, these will be the valve states at the end of the previous time step. On most time steps, the valve states will not change. Therefore, only one pass through the `while` loop will be required on most time steps. This makes the above procedure very efficient.

At this point, the mathematically alert reader may have some questions: How do we know that there is any self-consistent combination of valve states and pressures? If there is such a combination, how do we know that there

is only one of them? And even if there is exactly one such combination, how do we know that the above procedure will find it? (Maybe it will loop forever without encountering the right state of the valves.) We leave these questions as a challenge to the reader. (We know of a proof that resolves the first two but not the third.) Another challenge is to write a script that does not presume the answers to these questions, but instead finds out in each case exactly how many solutions there are. This can be done by considering all possible valve configurations, of which there are $2^2 = 4$ in this case.

With the above tools in place, we are now ready to write the main program and the initialization program for simulating the left heart and the systemic arteries:

```
%filename: LV_sa.m
clear all %clear all variables
clf %and figures
global T TS tauS tauD;
global Csa Rs Rmi RAO dt CHECK PLA;
in_LV_sa %initialize
for klok=1:klokmax
    t=klok*dt;
    PLV_old=PLV;
    Psa_old=Psa;
    CLV_old=CLV;
    CLV=CV_now(t,CLVS,CLVD);
    %find self-consistent
    %valve states and pressures:
    set_SMi_SAO
    %store in arrays for future plotting:
    t_plot(klok)=t;
    CLV_plot(klok)=CLV;
    PLV_plot(klok)=PLV;
    Psa_plot(klok)=Psa;
    VLV_plot(klok)=CLV*PLV+VLVD;
    Vsa_plot(klok)=Csa*Psa+Vsad;
    QMi_plot(klok)=SMi*(PLA-PLV)/Rmi;
    QAO_plot(klok)=SAO*(PLV-Psa)/RAO;
    Qs_plot(klok)=Psa/Rs;
    SMi_plot(klok)=SMi;
    SAO_plot(klok)=SAO;
end
%plot results:
figure(1)
subplot(3,1,1), plot(t_plot,CLV_plot)
subplot(3,1,2), plot(t_plot,PLV_plot,t_plot,Psa_plot)
```

```

subplot(3,1,3), plot(t_plot,QMi_plot,t_plot,QAo_plot, ...
                    t_plot,Qs_plot)
%left ventricular pressure-volume loop
figure(2)
plot(VLV_plot,PLV_plot)
%systemic arterial pressure-volume 'loop'
figure(3)
plot(Vsa_plot,Psa_plot)

%filename: in_LV_sa.m (initialization for LV_sa)
T =0.0125    %Duration of heartbeat (minutes)
TS=0.0050    %Duration of systole (minutes)
tauS=0.0025  %CLV time constant during systole (minutes)
tauD=0.0075  %CLV time constant during diastole (minutes)
Rs=17.86     %Systemic resistance (mmHg/(liter/minute))
%Valve resistances here are not supposed to be realistic,
%just small enough to be negligible:
RMi=0.01     %mitral valve resistance (mmHg/(liter/minute))
RAo=0.01     %aortic valve resistance (mmHg/(liter/minute))
%The following value of Csa is approximate;
%needs adjustment to make blood pressure 120/80:
Csa=0.00175  %Systemic arterial compliance (liters/mmHg)
CLVS=0.00003 %Min (systolic) value of CLV (liters/mmHg)
CLVD=0.0146  %Max (diastolic) value of CLV (liters/mmHg)
Vsad=0.825   %Systemic arterial volume when Psa=0 (liters)
VLVD=0.027   %Left ventricular volume when PLV=0 (liters)
PLA=5        %Left atrial pressure (mmHg)
dt=0.01*T    %Time step duration (minutes)
%This choice implies 100 timesteps per cardiac cycle.
klokmax=15*T/dt %Total number of timesteps
%This choice implies simulation of 15 cardiac cycles.
PLV=5        %Initial value of PLV (mmHg)
Psa=80       %Initial value of Psa (mmHg)
%set initial valve states:
SMi=(PLA>PLV); %evaluates to 1 if PLA>PLV, 0 otherwise
SAo=(PLV>Psa); %evaluates to 1 if PLV>Psa, 0 otherwise
CLV=CV_now(0,CLVS,CLVD) %Initial value of CLV (liters/mmHg)
%Initialize arrays used to store data for plotting:
%Although the program will work without doing this,
%it will run MUCH faster if Matlab knows in advance
%how much space is needed for these arrays.
t_plot=zeros(1,klokmax);
CLV_plot=zeros(1,klokmax);
PLV_plot=zeros(1,klokmax);
Psa_plot=zeros(1,klokmax);

```

```

VLV_plot=zeros(1,klokmax);
Vsa_plot=zeros(1,klokmax);
QMi_plot=zeros(1,klokmax);
QAO_plot=zeros(1,klokmax);
Qs_plot=zeros(1,klokmax);
SMi_plot=zeros(1,klokmax);
SAO_plot=zeros(1,klokmax);
%For self-checking in the function PLV_Psa_new, set CHECK=1
%To skip the self-checking, set CHECK=0
CHECK=1

```

We are now ready to tackle the problem of simulating the whole circulation. This will be done from a very general point of view, so that the program we develop will be easily adaptable to a variety of circumstances. One may want to model configurations of the circulation that differ from the standard one, e.g., the fetal circulation, or the circulation of a patient with a congenital heart disease. Even in the standard case, one may want to modify the model by changing the amount of detail that is included. The atria, for example, might be lumped in with the veins, or they might be represented in the model as separate chambers in their own right. A part of the circulation that is normally considered as a single chamber, such as the systemic arteries, might have to be partitioned to model a condition like coarctation of the aorta that introduces significant resistance where none is normally present.

Let us therefore model the generic situation of an arbitrary collection of N compliance vessels (possibly with time dependent compliances, to simulate a heart chamber) connected by resistance vessels equipped with valves. In fact, to be completely general, let us assume that every pair of compliance vessels is connected by two such resistance vessels, with the valves pointing in opposite directions; see Figure 1.16. If any of these connections is not actually wanted, we can make the corresponding resistance infinite. If a connection without a valve is wanted, we can get that effect by making the two resistances equal, so that the resistance experienced by flow going in either direction is the same, just as if there were no valve. When both resistances are finite but unequal, that arrangement models a leaky valve, with the leak being worse the more nearly equal the two resistances are. In summary, by assigning the resistances in different ways, we can model any kind of connection that we want, including no connection at all, within this general framework.

The equations of the general network that we have just described are as follows. First we have the equation of conservation of volume:

$$\dot{V}_i = \sum_{j=1}^N (Q_{ji} - Q_{ij}), \quad i = 1, \dots, N, \quad (1.12.36)$$

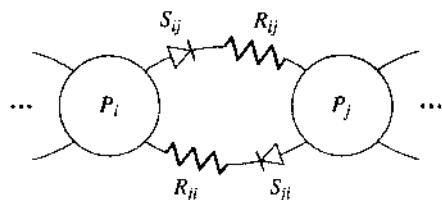


Figure 1.16. Part of a general model of the whole circulation. A pair of compliance chambers is shown, each chamber labeled by its pressure P_i or P_j . Every such pair in the model is connected by a pair of resistance vessels equipped with valves, the two valves of the pair pointing in opposite directions. $S_{ij} = 0$ or 1, denoting the state of the valve that allows flow from compliance chamber i into compliance chamber j . R_{ij} = resistance of the vessel in series with this valve.

where V_i is the volume of compliance chamber i , and where Q_{ij} is the flow out of chamber i into chamber j .

Next, we have the compliance relation for each chamber:

$$V_i = (V_d)_i + C_i P_i, \quad i = 1, \dots, N, \quad (1.12.37)$$

where C_i is the compliance of chamber i , P_i is the pressure in that chamber, and $(V_d)_i$ is the "dead volume" of chamber i , i.e., its volume when $P_i = 0$. Note that C_i will be constant for arteries and veins, but time dependent for chambers of the heart.

Finally, we have the pressure-flow relationship for each of the valve-equipped resistances that connect any pair of chambers in our model:

$$Q_{ij} = \frac{S_{ij}}{R_{ij}} (P_i - P_j) = S_{ij} G_{ij} (P_i - P_j), \quad i, j = 1, \dots, N, \quad (1.12.38)$$

where

$$S_{ij} = \begin{cases} 1, & P_i > P_j, \\ 0, & P_i \leq P_j, \end{cases} \quad (1.12.39)$$

and where we have introduced the notation $G_{ij} = 1/R_{ij}$. The main reason for the use of "conductance" (G) instead of resistance (R) is to cover the case $R = \infty$, which becomes $G = 0$. Note that S_{ij} and S_{ji} cannot both be 1 at the same time, and they can both be zero at the same time only in the borderline case that $P_i = P_j$, in which case there cannot be any flow, since there is no pressure difference to drive it.

Following our (by now) standard practice, we combine equations (1.12.36)–(1.12.38) to obtain

$$\begin{aligned}\frac{d}{dt}(C_i P_i) &= \sum_{j=1}^N (S_{ji} G_{ji} (P_j - P_i) - S_{ij} G_{ij} (P_i - P_j)) \\ &= \sum_{j=1}^N (S_{ij} G_{ij} + S_{ji} G_{ji}) (P_j - P_i),\end{aligned}\quad (1.12.40)$$

which holds for $i = 1, \dots, N$. This is a system of N differential equations in the N unknowns $P_1, \dots, P_N$. (The valve states S_{ij} are also unknown but they may be regarded as functions of the pressures; see equation (1.12.39).

A numerical method for these equations is obtained, as we have done previously, by replacing the time derivative by a backward difference quotient:

$$\begin{aligned}&\frac{C_i(t)P_i(t) - C_i(t - \Delta t)P_i(t - \Delta t)}{\Delta t} \\ &= \sum_{j=1}^N (S_{ij}(t)G_{ij} + S_{ji}(t)G_{ji}) (P_j(t) - P_i(t))\end{aligned}\quad (1.12.41)$$

If we temporarily regard the valve states $S_{ij}(t)$ as known quantities, this is a linear system in the unknowns $P_i(t)$. It can be put in standard form as follows:

$$\sum_{j=1}^N A_{ij}(t)P_j(t) = C_i(t - \Delta t)P_i(t - \Delta t), \quad i = 1, \dots, N, \quad (1.12.42)$$

where

$$A_{ij}(t) = -\Delta t (S_{ij}(t)G_{ij} + S_{ji}(t)G_{ji}), \quad i \neq j, \quad (1.12.43)$$

$$A_{ii}(t) = C_i(t) - \sum_{j:j \neq i} A_{ij}(t), \quad (1.12.44)$$

The Matlab program that sets up and solves this linear system takes advantage of the matrix–vector features of Matlab. It reads as follows:

```
function P=P_new(P_old,C_old,C,S)
%filename: P_new.m
global G dt CHECK N;
A=-dt*((S.*G)+(S.*G)');
A=diag(C-(sum(A)'))+A;
P=A\'(C_old.*P_old);
if(CHECK)
    for i=1:N
        CH(i)=-(C(i)*P(i)-C_old(i)*P_old(i))/dt;
    for j=1:N
```

```

CH(i)=CH(i)+S(j,i)*G(j,i)*(P(j)-P(i));
CH(i)=CH(i)-S(i,j)*G(i,j)*(P(i)-P(j));
end
end
CH      %Write out the values of CH,
%which should be zero to within roundoff.
end

```

The notation here is very compact, and some comments on what is going on may be helpful (See also Appendix A). First, note that P , P_{old} , C_{old} , and C are column vectors of length N . On the other hand, G , S , and A are $N \times N$ matrices. The line $A = -dt * ((S.*G) + (S.*G)')$; defines the off-diagonal part of A , corresponding to equation (1.12.43). The notation $.*$ denotes the element-by-element multiplication of two matrices or vectors. (This is *not* matrix multiplication as defined in linear algebra, nor is it a "dot product," despite the notation.) The notation $'$ denotes the operation of taking the transpose of a matrix, i.e., forming a new matrix whose columns are the rows of the original matrix. The next line, $A = \text{diag}(C - (\text{sum}(A))') + A$;, adds in the diagonal elements of A ; see equation (1.12.44). This line makes use of two built-in Matlab functions, sum and diag . When A is a matrix with more than one row, $\text{sum}(A)$ is a row vector whose j th element is the sum of the j th column of A . In our case, it doesn't matter whether we sum the rows or the columns, since A is symmetric, but we do need to convert $\text{sum}(A)$ to a column vector so that it can be subtracted from C . The diag function takes a (row or column) vector as input, and generates as output a square matrix whose diagonal elements are those of the input vector and whose off-diagonal elements are zero. (Alternatively, diag can take a square matrix as input and form a vector from the diagonal elements of the input matrix, but that is not how we are using it here.) Finally, in the line $P = A \backslash (C_{old}.*P_{old})$;, the notation $A \backslash$ denotes multiplication on the left by the inverse of the matrix A . That is, $x = A \backslash b$ is the solution of $Ax = b$.

Note that the function P_{new} is self-checking if the global variable $CHECK$ has been set equal to 1 (true), but not if it has been set equal to 0 (false). The checking program uses the equations in their original form, so that it will check the algebra that was done to put them into standard form as well as the solution process itself.

The function P_{new} finds the pressures given the state of the valves, and it is easy to set the state of the valves if we are given the pressures. We now write a script that finds self-consistent valve states and pressures by setting each according to the other until the valve states stop changing:

```

%filename: set_valves.m
%script to find self-consistent
%valve states and pressures:
done=0; % not done yet!
while(~done) %if not done, keep trying

```

```

S_noted=S; %note valve states
%set pressures based on valve states:
P=P_new(P_old,C_old,C,S);
%then set valve states based on pressures:
P_matrix=P*ones(1,N);
S=((P_matrix) > (P_matrix'));
%done if all valve states are unchanged:
done=all(all(S==S_noted));
end

```

Note the similarity of this to the routine `set_SMi_SAO` that is used in the case of the left heart and systemic arteries. The way in which the valves are set here is a bit tricky, though, and requires some explanation. First, we take the column vector of pressures P and convert it into a square matrix P_matrix by multiplying it on the right by a row vector of length N containing all 1's, denoted in Matlab by `ones(1,N)`. The result is that the i th row of P_matrix contains the number $P(i)$ repeated N times. Similarly, the j th column of P_matrix' , the transpose of P_matrix , contains the element $P(j)$ repeated N times. Thus the line

```
S=((P_matrix) > (P_matrix'));
```

has the same effect as executing $S(i,j)=(P(i)>P(j))$ for all (i,j) . That is, it puts a 1 in the (i,j) location of S whenever $P(i)>P(j)$, and it puts a 0 in that location otherwise.

The built-in Matlab function `all` operates as follows: Given a matrix as input, `all` performs a logical AND operation on each column and reports the results as a row vector. Given a vector (in our case, the result of the application of `all` to a matrix), `all` applies a logical AND to the elements of the vector. Thus the expression `all(all(S==S_noted))` evaluates to 1 (true) if *every* element of S agrees with the corresponding element of S_noted , and it evaluates to 0 (false) if *any* element of S differs from the corresponding element of S_noted .

The main program for the whole circulation simulation is as follows:

```

%filename: circ.m
clear all % clear all variables
clf % and figures
global T TS tauS tauD;
global G dt CHECK N;
in_circ %initialize
for klok=1:klokmax
    t=klok*dt;
    P_old=P;
    C_old=C;
    %find current values of left and right
    %ventricular compliance and store each
    %of them in the appropriate slot in the array C:

```

```

C(iLV)=CV_now(t,CLVS,CLVD);
C(iRV)=CV_now(t,CRVS,CRVD);
%find self-consistent valve states and pressures:
set_valves
%store variables in arrays for future plotting:
t_plot(klok)=t;
C_plot(:,klok)=C;
P_plot(:,klok)=P;
V_plot(:,klok)=Vd+C.*P;
Pdiff=P(iU)-P(iD); %pressure differences
                    %for flows of interest:
Q_plot(:,klok)=(Gf.*(Pdiff>0)+Gr.*(Pdiff<0)).*Pdiff;
%(the net flow is computed in each case)
end
%plot results:
circ_out

```

As mentioned above, this is a very general program with many potential applications. The following initialization script sets up a standard case for the normal adult circulation:

```

%filename: in_circ.m (initialization for circ)
T =0.0125 %Duration of heartbeat (minutes)
TS=0.0050 %Duration of systole (minutes)
tauS=0.0025 %CLV time constant during systole (minutes)
tauD=0.0075 %CLV time constant during diastole (minutes)
Rs=17.5 %Systemic resistance (mmHg/(liter/minute))
Rp= 1.79 %Pulmonary resistance (mmHg/(liter/minute))
%Unrealistic valve resistances,
%Chosen small enough to be negligible.
RMi=0.01 %mitral valve resistance (mmHg/(liter/minute))
RAo=0.01 %aortic valve resistance (mmHg/(liter/minute))
RTr=0.01 %tricuspid valve resistance (mmHg/(liter/minute))
RPu=0.01 %pulmonic valve resistance (mmHg/(liter/minute))
%The following values of Csa and Cpa are approximate.
%They will need adjustment to make the systemic
%blood pressure be roughly 120/80 mmHg
%and to make the pulmonary
%blood pressure be roughly 25/8 mmHg.
Csa=0.00175 %Systemic arterial compliance (liters/mmHg)
Cpa=0.00412 %Pulmonary arterial compliance (liters/mmHg)
Csv=1.75 %Systemic venous compliance (liters/mmHg)
Cpv=0.08 %Pulmonary venous compliance (liters/mmHg)
CLVS=0.00003 %Min (systolic) value of CLV (liters/mmHg)
CLVD=0.0146 %Max (diastolic) value of CLV (liters/mmHg)
CRVS=0.0002 %Min (systolic) value of CRV (liters/mmHg)

```

```

CRVD=0.0365 %Max (diastolic) value of CRV (liters/mmHg)
Vsad=0.825 %Systemic arterial volume at P=0 (liters)
Vpad=0.0382 %Pulmonary arterial volume at P=0 (liters)
Vsvd=0 %Systemic venous volume at P=0 (liters)
Vpvd=0 %Pulmonary venous volume at P=0 (liters)
VLVd=0.027 %Left ventricular volume at P=0 (liters)
VRVd=0.027 %Right ventricular volume at P=0 (liters)
dt=0.01*T %Time step duration (minutes)
%This choice implies 100 timesteps per cardiac cycle.
klokmax=15*T/dt %Total number of timesteps
%This choice implies simulation of 15 cardiac cycles.
%Assign an index to each compliance vessel
%of the model circulation:
iLV=1
isa=2
isv=3
iRV=4
ipa=5
ipv=6
N=6
%Enter parameters and initial values
%into correct slots in arrays.
%Note that the code that follows is independent
%of the specific numbering scheme chosen above.
%Compliance vector:
C=zeros(N,1);
%This makes C a column vector of length N.
C(iLV)=CV_now(0,CLVS,CLVD); %initial value
C(isa)=Csa;
C(isv)=Csv;
C(iRV)=CV_now(0,CRVS,CRVD); %initial value
C(ipa)=Cpa;
C(ipv)=Cpv;
C %This writes the result on the screen.
%Pressure vector (initial values) at end of diastole:
P=zeros(N,1);
%This makes P a column vector of length N.
P(iLV)= 5;
P(isa)=80;
P(isv)= 2;
P(iRV)= 2;
P(ipa)= 8;
P(ipv)= 5;
P %This writes the result on the screen.
%Vector of dead volumes (volume at zero pressure);

```

```

%Note: Vd is only needed for output purposes.
%It drops out of the equations we solve for P,
%but we need it if we want to output (e.g., plot)
%the volume of any compliance vessel.
Vd=zeros(N,1);
%This makes Vd a column vector of length N.
Vd(iLV)=VLVd;
Vd(isa)=Vsad;
Vd(isv)=Vsvd;
Vd(iRV)=VRVd;
Vd(ipa)=Vpad;
Vd(ipv)=Vpvd;
Vd
%This writes the results on the screen.
%Conductance matrix:
G=zeros(N,N);
%This makes G an NxN matrix filled with zeros.
%Any element of G that is not explicitly
%made nonzero remains zero,
%thus modeling an infinite resistance connection,
%that is, no connection at all.
G(iLV,isa)=1/RAo; %But G(isa,iLV)=0 (no leak)
G(isa,isv)=1/Rs; %no valve
G(isv,isa)=1/Rs; %no valve
G(isv,iRV)=1/RTr; %But G(iRV,isv)=0; (no leak)
G(iRV,ipa)=1/RPu; %But G(ipa,iRV)=0; (no leak)
G(ipa,ipv)=1/Rp; %no valve
G(ipv,ipa)=1/Rp; %no valve
G(ipv,iLV)=1/RMi; %But G(iLV,ipv)=0; (no leak)
G %This writes the result on the screen.
%Matrix of initial valve states:
S=zeros(N,N)
%This makes S an NxN matrix filled with zeros
%(and writes it on the screen).
%Start with all valves closed.
%Valves will adjust to pressures
%during first time step.
%Initialize arrays to store data for plotting:
t_plot=zeros(1,klokmax);
C_plot=zeros(N,klokmax);
P_plot=zeros(N,klokmax);
%Other variables that we might want to plot
%can be found from these.
%For self-checking in P_new, set CHECK=1.
%To skip self-checking set CHECK=0.

```

```

%(should be much faster with CHECK=0)
CHECK=1
%Initialize flow computation (for output purposes only)
%assign an index to each flow of interest:
jAo=1;
js =2;
jTr=3;
jPu=4;
jp =5;
jMi=6;
Nflows=6;
%note index of upstream and downstream chamber
%for each flow:
iU=zeros(Nflows,1);
iD=zeros(Nflows,1);
iU(jAo)=iLV;
iD(jAo)=isa;
iU(js )=isa;
iD(js )=isv;
iU(jTr)=isv;
iD(jTr)=iRV;
iU(jPu)=iRV;
iD(jPu)=ipa;
iU(jp )=ipa;
iD(jp )=ipv;
iU(jMi)=ipv;
iD(jMi)=iLV;
%extract the conductances from the matrix G:
Gf=zeros(Nflows,1);
Gr=zeros(Nflows,1);
for j=1:Nflows
    Gf(j)=G(iU(j),iD(j)); %forward conductance
    Gr(j)=G(iD(j),iU(j)); %reverse conductance
end
%create arrays to store current pressure differences
%and history over time of the net flows:
Pdiff=zeros(Nflows,1);
Q_plot=zeros(Nflows,klokmax);

We conclude this section with an example plotting program circ_out:

%filename: circ_out.m
%script to plot results of computer simulation
%of the entire circulation.
%left ventricular compliance, pressures, and flows:
figure(1)

```



```

subplot(3,1,1),plot(t_plot,C_plot(iLV,:))
subplot(3,1,2),plot(t_plot,P_plot([ipv,iLV,isa],:))
subplot(3,1,3),plot(t_plot,Q_plot([jMi,jAo],:))
%right ventricular compliance, pressures, and flows:
figure(2)
subplot(3,1,1),plot(t_plot,C_plot(iRV,:))
subplot(3,1,2),plot(t_plot,P_plot([isv,iRV,ipa],:))
subplot(3,1,3),plot(t_plot,Q_plot([jTr,jPu],:))
%systemic and pulmonary flows:
figure(3)
plot(t_plot,Q_plot([js,jp],:))
figure(4)%pressure-volume loops for both ventricles
subplot(2,1,1),plot(V_plot(iLV,:),P_plot(iLV,:))
subplot(2,1,2),plot(V_plot(iRV,:),P_plot(iRV,:))

```

1.13 Suggestions for Computing Projects Concerning the Circulation

The purpose of this section is to suggest some projects that use the computer programs introduced above. Following the outline of the previous section, we begin with the model of the systemic arteries only, and work up to the model of the circulation as a whole.

For the model of the systemic arteries only, the first thing to do is adjust the systemic arterial compliance by trial and error in order to achieve a blood pressure of 120/80 mmHg. The adjusted compliance should be considered as the normal compliance in all subsequent studies. Next, check that the time step is small enough by reducing it by a factor of two and comparing results. (Note that one has to run through twice as many steps at the smaller time step to cover the same amount of time.) Try different initial pressures (some above and some below 80 mmHg) to check that the pressure settles down to the same periodic steady state, independent of the initial pressure, and to see how long it takes to get close enough to the periodic steady state that you don't notice any changes from one heartbeat to the next.

In the following paragraphs, all of the parameter changes mentioned are supposed to be made starting from a standard set of "normal" parameters. The student has to be careful to organize the work in this way, and *not* to let the changes made in one computer experiment stay around and incorrectly define the reference state for the next experiment. One good way to avoid this pitfall is to run a standard initializing script that sets up normal values, and then run a separate script that makes the required parameter changes for the experiment at hand. This is much better than editing the initializing script, for at least two reasons: It documents all of the changes that define

a particular experiment in one place, and it avoids the task of undoing all edits to get back to the standard state.

With the preliminary studies out of the way, try modeling the aging process by reducing the systemic arterial compliance ("hardening of the arteries"), e.g., by a factor of 2. What happens to the systolic, diastolic, mean, and pulse pressures? Can you explain these results?

Next, consider normal exercise. During exercise, systemic resistance falls because of autoregulation in the working muscles. As an example, consider a reduction in systemic resistance by a factor of 2. The response of the body is to double the cardiac output (so that the systemic arterial pressure will not fall), and this is achieved primarily by an increase in heart rate, not stroke volume. We can double the heart rate in our model by halving all three of the times T (the duration of the cardiac cycle), TS (the duration of systole), and $TMAX$ (the time of peak aortic flow, relative to the onset of systole). To keep the stroke volume at its normal value in the face of these changes in timing, we must double $QMAX$. (The above prescription is, of course, simplified. In reality, when heart rate changes, the corresponding changes in the duration of systole are less pronounced than those in the duration of the whole cardiac cycle.)

As a variation on the exercise theme, consider a patient whose heart rate cannot respond to exercise. In such a patient, the needed increase in cardiac output is accomplished by increasing the stroke volume. Thus, the response to reduction in systemic resistance by a factor of 2 will be simply a doubling of $QMAX$, with no changes in T , TS , or $TMAX$. (It is interesting that the same change, doubling $QMAX$, that was needed in the normal exercise case to maintain stroke volume is needed here to increase the stroke volume!) Try this and compare the pressure and flow waveforms to those of normal exercise.

Yet another variation on the exercise theme comes about when heart rate increases without any need for increased cardiac output, as in anxiety. To simulate this, leave the systemic resistance at its normal value, halve the timing parameters T , TS , and $TMAX$ as in normal exercise, but leave $QMAX$ at its normal value so that the stroke volume falls just enough to keep cardiac output normal despite the increased heart rate. How does the pulse of an anxious person compare with the pulse of an exercising person when both have the same heart rate?

Finally, it is of interest to see what happens to the arterial pulse when the normal periodic pattern of heartbeats is interrupted. Suppose cardiac arrest occurs for some period of time, and then the heart is restarted. What happens to arterial pressure during the arrest period, and how does it recover afterwards? To simulate this, we have to modify the function $QAo_now(t)$ so that its output is zero between two specified times t_arrest and $t_restart$, and normal otherwise. (Of course, this ignores many complexities, since it assumes that the heart immediately resumes its normal rate, rhythm, and stroke volume, which is unlikely to be the case.) We leave the programming details to the reader.

Let us now turn our consideration to the program that simulates the left heart coupled to the systemic arteries. Of course, one can essentially repeat the foregoing studies with this model (although some modifications have to be made, since different parameters are now under our control). But it is more interesting to consider the new opportunities opened up by the new model, and these concern the function of the valves.

Recall that the left heart has an inflow ("mitral") and an outflow ("aortic") valve. A heart valve can malfunction in either of two distinct ways (or in some combination of the two). These are "stenosis," in which the valve is narrowed and presents a high resistance to forward flow, and "incompetence" or "insufficiency," in which the valve is leaky and fails to prevent backflow. In our model of the left heart and systemic arteries, stenosis is easier to model than insufficiency, since we already have the necessary parameters in the model, R_{Mi} (resistance of the mitral valve) and R_{Ao} (resistance of the aortic valve). Increasing either of these resistances simulates stenosis of the corresponding valve. Try this in each case, over a range of different valve resistances, and observe the effects on stroke volume, left ventricular blood pressures, and systemic arterial blood pressure. Compare the effects of mitral stenosis to those of aortic stenosis. Note, however, that these are the uncompensated effects of these conditions. (See below for suggestions on how to study physiological compensations for valve disease.)

To study mitral and/or aortic insufficiency in our model of the left heart and systemic arteries, we have to modify the model by providing a parallel path for blood flow in the reverse direction through the valve. For example, the flow through the mitral valve, which was previously given by $S_{Mi}(P_{LA} - P_{LV})/R_{Mi}$, will now be given by $(S_{Mi}/R_{MiF} + (1 - S_{Mi})/R_{MiB})(P_{LA} - P_{LV})$, where R_{MiF} is the resistance of the mitral valve to *forward* flow (previously called R_{Mi}), and R_{MiB} is the *backflow* resistance of the mitral valve. For a healthy mitral valve R_{MiF} is very small, and R_{MiB} is very large (essentially infinite). Mitral stenosis is modeled by increasing R_{MiF} , and mitral insufficiency is modeled by decreasing R_{MiB} . (Presumably, though, we can never have $R_{MiF} > R_{MiB}$, since that would mean that the valve was installed backwards!). The corresponding modifications in the formula for aortic flow need to be made in order to model aortic insufficiency. Once these changes are in place, one can do similar studies of insufficiency to those outlined above for stenosis.

So far, we have considered only the uncompensated effects of valvular malfunction. In fact, however, the circulation makes various automatic adjustments which can be viewed as compensations for the effects of valve disease. Simulations that include some of these compensations will be discussed in the following paragraphs.

In mitral stenosis, left atrial pressure rises to maintain stroke volume on the left side of the heart. To simulate this, first do an uncompensated simulation of mitral stenosis, and observe the fall in stroke volume. Then see how much the left atrial pressure (a parameter of the model) has to

be increased in order to bring the stroke volume back to normal. Plot the compensated left atrial pressure as a function of the resistance of the mitral valve. Note that left atrial pressure is essentially the same as pulmonary venous pressure, and that once pulmonary venous pressure exceeds about 28 mmHg, fluids are pressed out into the air spaces of the lung (pulmonary edema). This puts a ceiling on the severity of mitral stenosis that can be tolerated. It would be of great interest to repeat the above study under exercise conditions (reduced systemic resistance, increased heart rate), since patients with mitral stenosis may be asymptomatic at rest but may experience pulmonary edema upon attempting to exercise.

In aortic stenosis, the primary compensation is an increase in thickness of the left ventricular wall, in response to the elevated left ventricular pressure. This shows up in the model as a reduction in compliance. Both the systolic and diastolic compliances will be reduced. The reduction in systolic compliance is helpful to the left ventricle during ejection, but the reduction in diastolic compliance impedes filling, and this causes a secondary compensation, which is a rise in left atrial pressure. To model this whole story, first postulate a definite relationship between ventricular compliance and ventricular pressure. For example, let both the systolic and diastolic left ventricular compliances be *inversely* proportional to the maximum (systolic) left ventricular pressure, the constants of proportionality being determined by the normal values. It will require an iterative process to find the right compliances for any given level of aortic stenosis, since the compliance depends on the pressure, which depends on the compliance. This iterative process involves running the program, observing the systolic left ventricular pressure, setting the diastolic compliances accordingly, re-running the program, etc. Then, for any given level of aortic stenosis, use trial and error to find the left atrial pressure that makes the stroke volume normal. Plot the compensated left atrial pressure, left ventricular pressures (systolic and diastolic), systemic arterial pressures (systolic and diastolic), and left ventricular compliances (systolic and diastolic), all as functions of the resistance of the aortic valve.

Finally, we consider studies that can be done with the whole circulation model. Of course, these include essentially all of the studies described above for the two partial circulation models that we have considered, except that certain changes have to be made to take into account the differences with regard to which quantities are subject to our direct control. For example, in the whole circulation model PLA is an unknown, and we are not free to vary it at will. On the other hand, at least part of the compensatory change in PLA that we have described in the case of mitral or aortic stenosis will just happen by itself in the whole circulation model, as a consequence of a shift in volume from the systemic to the pulmonary circulation. It would be interesting to see how effective this compensatory mechanism is. The rest of the change in PLA can be produced as needed by changing the blood volume, a parameter that is subject to our control in the whole circulation model.

In the body, such compensatory changes in blood volume are produced by feedback mechanisms involving the kidney.

Let us focus our discussion, here, however, on some applications that require the whole circulation model. One class of applications of this type concerns the fetal circulation and congenital heart disease. Such simulations exploit the ability of the model to handle an arbitrary pattern of connections.

To simulate the fetal circulation, it is necessary only to lay out the whole circulation model in the manner shown in Figure 1.9, and then to choose appropriate fetal parameters, some of which at least can be inferred from data on fetal pressures and flows such as those available in Rudolph's book, cited in Section 1.14. Other parameters may have to be guessed or adjusted by trial and error. It would then be very interesting to watch the changes in the circulation that occur at birth. These would be induced by suddenly decreasing the pulmonary resistance (first breath) and suddenly increasing the systemic resistance (constricting or clamping of the umbilical cord). Note the resulting closure of the foramen ovale and the reversal of mean flow in the ductus arteriosus.

Some congenital heart diseases that can be simulated with the whole circulation model include ventricular septal defect (VSD), which is a connection between the two ventricles; atrial septal defect (ASD), which is a connection (not usually the foramen ovale, but a separate connection) between the two atria; and patent ductus arteriosus (PDA), which is the persistence of the fetal connection between the pulmonary artery and the aorta. All of these connections are unvalved; i.e., their resistances are the same in both directions. In severe cases, these resistances are very low. All three conditions result in postnatal left-to-right shunts, with high pulmonary flow, a dangerous condition for the development of the lung. One might consider banding the pulmonary artery (increasing the pulmonary resistance) as a way of reducing or eliminating the left-to-right shunt and thereby equalizing the pulmonary and systemic flows. What are the consequences of this strategy in each of the three conditions? (As a practical matter, banding of the pulmonary artery would not be used in PDA, since it would be as easy to close the defect itself. In the other two cases, though, the defect lies within the heart, so an extracardiac procedure such as banding the pulmonary artery might be useful as a temporary measure. Your mission is to use computer simulation to see what the circulatory consequences of this procedure might be.)

Of course, there are many other possible applications of the whole circulation model. One way you could use it is to test some of the principles discussed in this chapter. For example, how well does the steady-state model used earlier in this chapter predict the mean pressures, flows, and volumes of the pulsatile whole circulation model? Can you put in autoregulation and/or impose neural control via a simulated baroreceptor loop? Have fun thinking up some applications of your own!

1.14 Annotated References

Many of the concepts used in this chapter were pioneered by A.C. Guyton. See, for example:

Guyton A.C.: *Circulatory Physiology: Cardiac Output and Its Regulation*. Saunders, Philadelphia, PA, 1963.

The idea that each cardiac ventricle can be modeled as a time-varying compliance and hence that the stroke volume can be determined from a pressure-volume loop (Figure 1.5) comes from the work of K. Sagawa and his colleagues:

Sagawa, K., Suga, H., and Nakayama, K.: Instantaneous pressure-volume ratio of the left ventricle versus instantaneous force-length relation of papillary muscle. In: *Cardiovascular System Dynamics* (Baan, J., Noordergraaf, A., and Raines, J., eds.), M.I.T. Press, Cambridge, MA, 1978, 99-105.

Neural control of the circulation is discussed in the following references:

Karloff, et al.: Adaptation of the left ventricle to sudden changes in heart rate in patients with artificial pacemakers. *Cardiovascular Research* 7: 322, 1973.

Korner, P.I.: Integrative neural control of the circulation, *Physiological Reviews* 51, 312-367, 1971.

Rowell, L.B.: Human cardiovascular adjustments to exercise and thermal stress, *Physiological Reviews* 54, 75-159, 1974.

Topham, W.S. and Warner, H.R.: The control of cardiac output during exercise. In: *Physical Bases of Circulatory Transport: Regulation and Exchange* (Reeve and Guyton, eds.), Saunders, Philadelphia, PA, 1967.

Our emphasis on oxygen as the key factor in autoregulation (Section 1.9) can be traced back to the work of Guyton:

Guyton, A.C., Ross, J.M., Carrier, O., Jr., and Walker, J.R. : Evidence for tissue oxygen demand as the major factor causing autoregulation, *Circulation Research* 14, 60, 1964.

The particular model of autoregulation that we use is simplified from:

Huntsman, L.L., Attinger, E.O., and Noordergraaf, A.: Metabolic autoregulation of blood flow in skeletal muscle: A model. In: *Cardiovascular System Dynamics* (Baan, J., Noordergraaf, A., and Raines, J., eds.), M.I.T. Press, Cambridge, MA, 1978, 400-414.

For a comprehensive discussion of the fetal circulation and the changes in the circulation occurring at birth, see the first few chapters of:

Rudolph, A.M.: *Congenital Diseases of the Heart*, Clinical-Physiological Considerations, 2nd Edition. Futura Publishing Company, Inc., 2001.

Rudolph's book is written in such a way that the mathematically inclined reader will find many opportunities for the construction of medically relevant mathematical models.

A computer program similar to `circ.m` is applied to congenital heart disease in the following references:

Peskin, C.S. and Tu, C.: Hemodynamics in congenital heart disease. *Comput. Biol. Med.* 16:331-359, 1986.

Tu, C. and Peskin, C.S.: Hemodynamics in transposition of the great arteries with compensation to ventricular septal defect. *Comput. Biol. Med.* 19: 95-128, 1989.

Finally, for a slightly more advanced look at some of the material presented in this chapter, see:

Peskin, C.S.: Control of the heart and circulation, In: *Mathematical Aspects of Physiology* (Hoppensteadt, F.C., ed.), *Lectures in Applied Mathematics* 19, American Mathematical Society, Providence, RI, 1981, 1-38.

Exercises

1.1. Circulation Times²

Given that the total blood volume is 5 liters and that the cardiac output is 5.6 liters/minute, calculate the time it takes for a "typical" red blood cell to go once around the circulation. This is called the *circulation time*.

First, consider the simplest possible model in which the circulation is a single-loop and all cells take the same amount of time to traverse the loop.

Next, consider a more realistic model in which the circulation consists of several parallel loops with different circulation times. Let the volumes occupied by these loops be $V_1, \dots, V_n$, let their blood flows be $Q_1, \dots, Q_n$, and their circulation times be $T_1, \dots, T_n$, respectively. Note that

$$V_0 = V_1 + \dots + V_n$$

and

$$Q = Q_1 + \dots + Q_n.$$

Make up a reasonable definition of average circulation time and show that the result is the same as in the single loop case.

²Warning: Exercise 1.1 is not related to the rest of the chapter, and many students find it confusing. It should probably be skipped.

Table 1.4. The Effect of Parameter Changes on the Uncontrolled Circulation

	Normal	50% Reduction In					units
		K_R	K_L	R_s	R_p	V_0	
Q	5.6						liters/min
P_{sa}	100						mmHg
P_{sv}	2						mmHg
P_{pa}	15						mmHg
P_{pv}	5						mmHg
V_{sa}	1.0						liters
V_{sv}	3.5						liters
V_{pa}	0.1						liters
V_{pv}	0.4						liters

1.2. Table 1.2

Write out the formulae used to construct Table 1.2, and check that the numerical values in the table are correct.

1.3. Parameter Changes

Starting from the parameter values given in Table 1.2, find the effect of varying the parameters *one at a time*. Fill in Table 1.4. (In each case, the named parameter is reduced to half its normal value while all other parameters are held at their normal values.)

1.4. Sensitivity to Parameter Changes

Transform Table 1.4 into a table of sensitivities. Note that all of the sensitivities fall in the range $-1 \leq \sigma \leq 1$.

1.5. Discussion of the Effects of Parameter Changes

Discuss the results of Exercises 1.3 and 1.4. In particular, compare the effects of changing K_R with those of changing K_L , and compare the effects of changing R_s with those of changing R_p . Note the shifts in blood volume that accompany the changes in parameters and explain these shifts as best you can.

1.6. Comparison of the Controlled and Uncontrolled Circulation

Make a graph that shows how cardiac output and systemic arterial pressure vary as functions of systemic resistance in the *uncontrolled* circulation. Consider the interval $0 < R_s < P_{sa}^{\text{normal}}$. Construct a similar graph for the *controlled* circulation. (Use the same scales for the controlled and the uncontrolled cases.) Discuss the difference between the two cases.

1.7. Uncontrolled Circulation: More Realistic Compliance Equation

This exercise and the next two develop some improvements in the model of the uncontrolled circulation. First, in each compliance vessel, replace the relation $V = CP$ by the more realistic formula $V = CP + V_d$. Show that the model remains the same as before, but that the volumes appearing in it must now be

interpreted as excess volumes. Thus, our procedure for estimating C is no longer correct for the revised model. See Exercise 1.17.

1.8. Uncontrolled Circulation: Nonzero P_{thorax}

In our model of the uncontrolled circulation, we also neglected P_{thorax} . Fix this by changing the equations of the right and left hearts and the pulmonary compliance to the following:

$$\begin{aligned} Q_R &= (FC_R)(P_{sv} - P_{\text{thorax}}) = K_R(P_{sv} - P_{\text{thorax}}) \\ Q_L &= (FC_L)(P_{pv} - P_{\text{thorax}}) = K_L(P_{pv} - P_{\text{thorax}}) \\ V_{pa} &= C_{pa}(P_{pa} - P_{\text{thorax}}) \\ V_{pv} &= C_{pv}(P_{pv} - P_{\text{thorax}}). \end{aligned}$$

Here P_{thorax} is a (negative) parameter and the volumes are understood to mean *excess* volumes as in the foregoing exercise. Solve the equations of the improved model and plot Q and P_{sv} as functions of P_{thorax} for $P_{\text{thorax}} \leq 0$.

1.9. Uncontrolled Circulation: Localized Partial Venous Collapse

When P_{thorax} is sufficiently negative, P_{sv} becomes negative in the preceding exercise. This has the effect that the systemic veins experience partial collapse at the point where they enter the thorax. The partial collapse prevents P_{sv} from falling below zero, but it puts a (variable) resistance R_c between the systemic veins and the right atrium. In this situation we have two additional unknowns: R_c = resistance of the partially collapsed veins, and P_{ra} = pressure in the right atrium. The additional equations are

$$\begin{aligned} P_{sv} &= 0, \\ Q_s R_c &= P_{sv} - P_{ra} = 0 - P_{ra}. \end{aligned}$$

Moreover, the equation of the right heart is replaced by

$$Q_R = (FC_R)(P_{ra} - P_{\text{thorax}}) = K_R(P_{ra} - P_{\text{thorax}}),$$

in which P_{ra} replaces P_{sv} . (Note that the excess systemic venous volume is now zero; see Exercise 1.7. Thus, C_{sv} no longer plays any role in the model.)

Solve the equations of this revised model and show that Q is now independent of P_{thorax} and that R_c increases as P_{thorax} becomes more negative. Note that Q is also independent of K_R . (Warning: This solution applies only when P_{thorax} is sufficiently negative. At the transition value of P_{thorax} , $R_c = 0$.) Correct the graph of Exercise 1.8 by taking into account the results of this problem.

1.10. Relationship between Sensitivities

Using the formula $P_{sa} = Q R_s$, derive the relationship

$$(-\sigma_{Q R_s}) + \sigma_{P_{sa} R_s} = 1.$$

1.11. Controlled Circulation

In the model of the controlled circulation in Section 1.8 we assumed that the heart rate F is adjusted to achieve a certain target arterial pressure $P_{sa} = P^*$. There are patients in whom the heart rate cannot respond in this way. Then F is a fixed parameter. In these patients, however, the baroreceptor loop is still

important because of its effects on the veins (see Figure 1.7). These effects were neglected in the model presented in Section 1.8, but they become very important when the heart rate is fixed.

Develop a model in which F is a parameter but C_{av} is adjusted to achieve $P_{sa} = P^*$. Compare this model to the model of Section 1.8. What are the similarities and differences of the two models with regard to their responses to changes in parameters? Consider changes in R_a and V_0 and the responses of the cardiac output, the stroke volume, the heart rate, and the systemic arterial pressure.

Some patients have *pacemakers* in which the heart rate is an adjustable parameter that the patient can control by turning a knob with an external magnet. If such a patient increases his heart rate while he is *resting*, the cardiac output does not go up. Instead, the stroke volume falls. Does your model predict this? See what happens to cardiac output and stroke volume in your model when F changes while all of the other parameters are held constant.

1.12. Venous Oxygen as an Index of Blood Flow in Relation to Oxygen Consumption

Using equation (1.9.2), plot $[O_2]_v / [O_2]_a$ as a function of $Q[O_2]_a/M$.

1.13. Autoregulation

Demonstrate the validity of the five characteristics of autoregulation that are listed at the end of Section 1.9.

1.14. Fetal Circulation

In our discussion of the fetal circulation, we made no use of the compliance equations, so we could not solve the problem completely. We determined only the ratio of flows. Use the appropriate compliance equations and solve for Q , P_a , P_v , and the four volumes in the fetal case.

1.15. The State of the Circulation Immediately After Birth

In the equations of the fetal circulation, assume that $R_d = 0$, $K_R = K_L = K$ and $R_p < R_a$. This is roughly the situation just *after* birth with the ductus still open, the two sides of the heart still roughly equal, but with the lungs expanded and the blood flow to the placenta shut off. Show that the closed-foramen solution is self-consistent in this case. Find a formula for Q_d and show that $Q_d < 0$.

1.16. Closure of the Ductus Arteriosus

Consider the situation described in Exercise 1.15, but with $R_d \neq 0$ and with $K_R \neq K_L$. Solve for Q_d , Q_R , Q_L , P_{sa} , and P_{pa} as functions of the parameters. Fix all of the parameters except R_d at their normal adult values and plot these flows and pressures as functions of R_d . Plot all of the flows on one graph and the two pressures on another. (This shows roughly the changes that occur in the circulation as the ductus closes ($R_d \rightarrow \infty$), except, of course, that the other parameters may be changing and they are certainly not at their adult values.)

1.17. Estimate of Arterial Compliance from the Arterial Pulse

The dynamics of the arterial pulse give us an alternative way to estimate C_{sa} . We have $\Delta V_0 = C_{sa} \Delta P_{sa}$, where $\Delta V_0 = 0.070$ liters is the stroke volume and where $\Delta P_{sa} = 40$ mmHg is the pulse pressure. Calculate C_{sa} from these data.

Note that the result does not agree with the value of C_{sa} in Table 1.2, which was calculated from the formula $V_{sa} = C_{sa}P_{sa}$ using the normal resting (mean) values $\langle V_{sa} \rangle = 1$ liter and $\langle P_{sa} \rangle = 100$ mmHg.

A possible explanation of this discrepancy is that the equation $V_{sa} = C_{sa}P_{sa}$ is not a good approximation. Suppose we use $V_{sa} = C_{sa}P_{sa} + (V_d)_{sa}$, where $(V_d)_{sa}$ is constant. Then $\Delta V_{sa} = C_{sa}\Delta P_{sa}$, so the arterial compliance can be determined from the stroke volume and pulse pressure, as above. Once C_{sa} is known, we can use the mean arterial pressure and volume to determine $(V_d)_{sa}$ from the equation

$$(V_d)_{sa} = \langle V_{sa} \rangle - C_{sa}\langle P_{sa} \rangle.$$

Evaluate $(V_d)_{sa}$ in this way and make a graph showing the systemic arterial pressure as a function of volume. Mark the points corresponding to systolic and diastolic pressures on this plot when the blood pressure is 120/80.

1.18. Mean Arterial Pressure

Derive the formula for the mean arterial pressure (1.11.14) in two different ways:

1. by evaluating the integral (1.11.13) using the fact that an exponential function is proportional to its own derivative
2. by evaluating the average of N equally spaced samples of $P_{sa}(t)$ and then taking the limit as $N \rightarrow \infty$. When evaluating the average, it is helpful to note that the samples of $P_{sa}(t)$ form a (finite) geometric sequence, since $P_{sa}(t)$ is exponential.

1.19. Calculation Concerning the Periodic Arterial Pulse

Using $R_s = 17.5$ mmHg/(liter/min), $C_{sa} = 0.00175$ liters/mmHg (see Exercise 1.17), $\Delta V_0 = 0.070$ liters, and $T = (1/80)$ min, evaluate $\Theta = \exp(-T/(R_s C_{sa}))$ and then use Θ to evaluate the systolic pressure, the diastolic pressure, the pulse pressure, and the mean arterial pressure in the periodic case.

1.20. Pulse Pressure Versus Mean Arterial Pressure

Think of two different limiting cases in which the pulse pressure approaches zero while the mean arterial pressure and the cardiac output remain constant.

1.21. Circulation During Exercise

During exercise in the model of the controlled circulation, R_s falls but T falls proportionally (heart rate goes up) and stroke volume is constant. Also, C_{sa} is constant. What happens to the systolic, diastolic, and pulse pressures in this situation?

1.22. Circulation During Exercise with the Heart Rate Fixed

Suppose that a patient with a fixed heart rate performs exercise (so that R_s is reduced) and suppose she manages to raise her stroke volume enough so that her mean arterial pressure remains constant. What happens to the systolic, diastolic, and pulse pressures in this case?

1.23. Cardiac Arrest

Suppose that a patient has suffered cardiac arrest for a long enough time that his arterial blood pressure is essentially zero. (Estimate how long this will take

using the exponential decay of the arterial pressure in the absence of heartbeats. To be specific, what is the "half-life" of the arterial pressure as it decays in the absence of heartbeats?) Then his heart suddenly recovers. Make the unrealistic assumption that T , ΔV_0 , R_s , and C_{sa} immediately assume their normal values as given in Exercise 1.19. Calculate the arterial systolic and diastolic pressures on the first 10 beats. Plot the results in graph form.

1.24. Arrhythmia

Consider a patient who has an *arrhythmia* in which the heartbeats alternate so that

$$\begin{aligned} \Delta V_j &= \Delta V_1, & j \text{ odd,} \\ &= \Delta V_2, & j \text{ even,} \\ T_j &= T_1, & j \text{ odd,} \\ &= T_2, & j \text{ even,} \end{aligned}$$

(recall that T_j is the time interval *following* beat j). The arrhythmia is regular, so that her arterial pulse also repeats every other beat. Thus,

$$\begin{aligned} P(t_j^+) &= P_1^+, & j \text{ odd,} \\ &= P_2^+, & j \text{ even,} \end{aligned}$$

and

$$\begin{aligned} P(t_j^-) &= P_1^-, & j \text{ odd,} \\ &= P_2^-, & j \text{ even.} \end{aligned}$$

Solve for P_1^+ , P_2^+ , P_1^- , and P_2^- as functions of the parameters ΔV_1 , ΔV_2 , T_1 , T_2 , R_s , and C_{sa} .

Check your solution for symmetry (it should be invariant when the subscripts 1 and 2 are interchanged). Also, check it by considering the special case where $\Delta V_1 = \Delta V_2$ and $T_1 = T_2$. What should happen to your solution then?

Plot the form of the pulse when T_1 and T_2 are such that $\Theta_1 = \frac{2}{3}$, and $\Theta_2 = \frac{1}{2}$ while $\Delta V_1 = 0.1$ liters and $\Delta V_2 = 0.05$ liters. (Note that the larger stroke volume comes after the larger filling period, since $T_2 > T_1$ and T_2 comes before ΔV_1 .)

1.25. Twins

Consider *hypothetical* twins who have the same values of R_s and C_{sa} and who both have abnormal rhythms that produce *identical* sequences of heart beats. This is, the times t_j and the stroke volumes ΔV_j are the same for both individuals although these sequences of times and stroke volumes are not at all regular.

At some time in the past, though, one of these individuals had an experience that shifted his blood pressure with respect to that of his twin. Write down the equations for each individual's blood pressure. Subtract these equations and show that their difference approaches zero exponentially with time.

2

Gas Exchange in the Lungs

The lungs contain about 3×10^8 *alveoli* (little sacs) in which air and blood are brought close together so that gas exchange can occur. The principal gases exchanged are O_2 , which is picked up by the blood, and CO_2 , which leaves the blood stream and enters the air spaces of the lung. These gases need to cross the thin alveolar capillary membrane; this crossing occurs by diffusion. In normal circumstances, the alveolar-capillary membrane presents so slight a barrier to diffusion that the blood in the alveolar capillaries achieves *equilibrium* with the alveolar air before leaving the capillaries.

The blood leaving the right heart is subdivided many times by the branching pulmonary arterial tree before it reaches the alveoli. After passing through the alveoli, it is collected by the pulmonary veins. The alveolar capillaries are therefore connected *in parallel* in the sense that the pulmonary blood flow is the sum of the blood flows of the individual alveoli.

In a similar way, air entering the trachea is subdivided many times by the bronchial tree before it reaches the alveoli, and the total alveolar ventilation is the sum of the individual alveolar ventilations. Unlike the blood, however, the air leaves the alveoli on expiration by way of the same bronchial tree through which it entered on inspiration.

In this chapter we discuss the laws that govern the transport of gases in the lung,¹ and we consider the conditions that lead to optimal transport of gases.

2.1 The Ideal Gas Law and the Solubility of Gases

The *ideal gas law* is approximately satisfied for many gases. This is stated as

$$PV = nkT, \quad (2.1.1)$$

where

- P = pressure of the gas,
- V = volume occupied by the gas,
- n = number of molecules of the gas,
- k = Boltzmann's constant,
- T = absolute temperature.

Note that k is a universal constant that does not depend on the identity of the gas.² In a given volume at a given temperature, this law asserts that the pressure is proportional to the number of molecules. The physical reason for this is that the pressure is actually the force per unit area exerted by the molecules on the walls of the container as they collide with the walls and bounce back into the interior of the gas. If there are twice as many molecules, the pressure is twice as great.

In (2.1.1) the identity of the gas makes no difference. It does not matter, for example, whether the molecules are heavy or light. At a given temperature, the light molecules move faster and the force that they exert on the wall turns out to be the same as that of the heavy molecules. Only the *number* of molecules enters into the formula. This suggests that if we have a mixture of several ideal gases in a chamber, and if n_j is the number of molecules of gas j , then the pressure is given by the formula

$$PV = kT \sum_j n_j. \quad (2.1.2)$$

¹Throughout this chapter, the phrase "the lung" is used to refer to the pulmonary tissue as a whole, which in fact is made up of two separate lungs.

²In the practical application of the ideal gas law, it is inconvenient that n , the number of molecules, is so large and that k , Boltzmann's constant, is so small. This is generally circumvented by multiplying and dividing by Avogadro's number $N_A \approx 6 \times 10^{23}$, which is the number of atoms in one gram of hydrogen. Thus, we get $PV = (n/N_A)(N_A k)T$. The constant $N_A k$ is called the "gas constant" and is usually denoted by R . The quantity n/N_A is called the number of "moles" and is usually denoted by n , but we shall use m to avoid confusion. Then the ideal gas law takes the form $PV = mRT$, in which the numbers that typically appear are less extreme than before. We shall use the form $PV = nkT$, however, because of its simpler physical interpretation: Molecules are more fundamental than moles.

Let the *partial pressure* of the j th gas P_j be defined by

$$P_j V = n_j k T. \quad (2.1.3)$$

Then summing over j and comparing the result with (2.1.2), we see that

$$P = \sum_j P_j, \quad (2.1.4)$$

so the total pressure is the sum of the partial pressures.

It is sometimes useful to relate the partial pressure of a gas to its concentration (i.e., the number of molecules per unit volume). The required formula is easily obtained by dividing both sides of (2.1.3) by V and letting $c_j = n_j/V$. The result is

$$P_j = c_j k T. \quad (2.1.5)$$

The concept of partial pressures can be extended to gases in solution. Consider the situation where a liquid and a gas are in contact across the surface of the liquid. Molecules of the gas can enter the liquid and wander for a time among its molecules; they are then said to be *dissolved* in the liquid, forming a *solution*. Molecules in solution can also leave the liquid at its surface and become part of the gas again.

After the system has been standing for a sufficiently long time, the rate at which gas molecules enter the solution becomes equal to the rate at which they leave. When this is true for all of the gases present, the system is said to be at *equilibrium*. If the gas molecules move independently of each other, it is not hard to see that the concentration of the gas in solution is proportional (at equilibrium) to its partial pressure in the gas. Solutions that obey this law are called *simple solutions*. For such solutions, we have

$$c_j = \sigma_j P_j, \quad (2.1.6)$$

where c is the concentration, σ is the *solubility*, and P is the partial pressure. As before, the subscript j labels the gas. Of course, equation (2.1.6) can be solved to yield

$$P_j = c_j / \sigma_j. \quad (2.1.7)$$

Given a simple solution in which the concentration of gas j is c_j , equation (2.1.7) tells us the partial pressure of gas j that is needed to maintain this concentration if the solution is allowed to equilibrate with a gaseous phase. We summarize this by saying that the partial pressure of gas j in the *solution* is P_j .

These concepts can be extended to gases that do not form simple solutions. (An important example is O_2 in blood, see Section 2.6.) First, the linear relation (2.1.6) is replaced by a nonlinear relation

$$c_j = C_j(P_j) \quad (2.1.8)$$

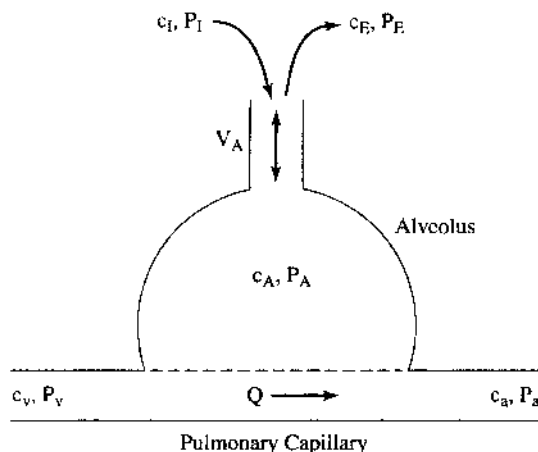


Figure 2.1. Gas exchange in one alveolus. c = concentration, P = partial pressure, Q = blood flow, V_A = alveolar ventilation, I = inspired, E = expired, A = alveolar, v = venous, a = arterial. Note that the designations “venous” and “arterial” refer to the chemical character of the blood, not to the anatomy of the vessels involved. Thus, “venous” refers to the bluish oxygen-depleted systemic venous blood (which flows through the pulmonary arteries), and “arterial” refers to the bright red oxygen-enriched blood that is destined to become systemic arterial blood (but leaves the lungs through the pulmonary veins).

in which C_j is an increasing function. Because C_j is increasing, it has an inverse function C_j^{-1} such that

$$P_j = C_j^{-1}(c_j). \quad (2.1.9)$$

Note that C_j^{-1} is also an increasing function. Equation (2.1.9) tells us the partial pressure that we need in a gaseous phase in contact with the liquid to maintain the concentration c_j . We call this the partial pressure of the gas in the liquid even when no gaseous phase is physically present.

Throughout most of this chapter we shall consider gases that form simple solutions in blood. These include certain anesthetics that are cleared from the body by the lungs. Then in Section 2.6 we consider the transport of oxygen. Because oxygen is carried in blood by hemoglobin, its behavior is different from that of a simple solution.

2.2 The Equations of Gas Transport in One Alveolus

We consider the transport of one particular gas in the alveolus of Figure 2.1.

Let

- V_A = alveolar ventilation,
- Q = blood flow,
- c = concentration of the gas of interest,
- P = partial pressure of the gas of interest,
- σ = solubility of the gas of interest,
- k = Boltzmann constant,
- T = absolute temperature.

On the variables c and P , the subscripts I, A, E, v, a have the following meanings:

- I = inspired air,
- A = alveolar air,
- E = expired air,
- v = venous blood,
- a = arterial blood.

Several of the terms used in the foregoing require discussion.

The *alveolar ventilation* is the volume of fresh air delivered to the alveolus per unit time. This is equal to the product of the volume of fresh air brought in on each breath and the number of breaths per unit time. In this definition it is important that we specify *fresh* air, since the first part of the air that reaches the alveolus on each breath is simply the air that was expelled on the previous breath and remained in the bronchial tree between breaths. This *dead space* volume is not included in the alveolar ventilation. Strictly speaking, we should distinguish between the alveolar ventilation of inspiration and the alveolar ventilation of expiration. These are not quite equal, because the consumption of O_2 by the body is not exactly equal to the production of CO_2 . This distinction will be ignored here, and we therefore need only one quantity, V_A , the alveolar ventilation.

By *venous blood* we mean the blood entering the alveolus, despite the fact that this blood flows in vessels that are part of the pulmonary arterial tree. The reason for this designation is that the chemical composition of this blood is identical to that in the systemic veins. In particular, the venous blood is relatively depleted of O_2 and relatively rich in CO_2 . By arterial blood we mean the blood leaving the alveolus, which is relatively rich in O_2 and relatively free of CO_2 and which is destined to become the systemic arterial blood. Again, this designation ignores the fact that this arterial blood flows in the pulmonary veins. Thus, we name the blood according to its chemical composition instead of the type of vessels through which it travels.

To derive the equations of gas transport in one alveolus, we make the following assumptions:

1. Steady state. The number of molecules of the gas of interest entering the alveolus per unit time is equal to the number of molecules of that gas leaving per unit time. Note that molecules may enter by the air or blood and they may also leave by the air or blood. This can be summarized by an equation that describes the conservation of molecules:

$$V_A c_I + Q c_v = V_A c_E + Q c_a. \quad (2.2.1)$$

2. Expired air is a sample of alveolar air. Thus,

$$c_E = c_A. \quad (2.2.2)$$

3. Ideal gas law. The gas of interest behaves like an ideal gas in the alveolar air:

$$P_A = kT c_A. \quad (2.2.3)$$

4. Simple solution. The gas of interest forms a simple solution in blood. In particular, in the arterial blood

$$\sigma P_a = c_a. \quad (2.2.4)$$

5. Equilibrium. As the blood passes through the alveolus, it achieves equilibrium with alveolar air. Thus, the partial pressure of each gas in the blood leaving the alveolus is the same as in the alveolar air:

$$P_n = P_A. \quad (2.2.5)$$

The first two equations can be combined to yield

$$V_A (c_I - c_A) = Q (c_a - c_v). \quad (2.2.6)$$

The quantity on the left side of (2.2.6) represents the number of molecules of gas per unit time that are given up by the air; the quantity on the right represents the number of molecules of the gas picked up per unit time by the blood. In the steady state, these must be equal. Either side of (2.2.6) is therefore an expression for the net transport of the gas of interest in the alveolus. This immediately tells us something important (and perhaps surprising): For net transport of gas to occur, it is necessary that alveolar air have a different composition than inspired air ($c_I \neq c_A$). For example, the partial pressure of O_2 in the inspired air is about 160 mmHg whereas in the alveolus it is about 100 mmHg. (The concentrations are proportional to partial pressures.) The partial pressure of CO_2 in the inspired air is essentially zero, whereas in the alveolus it is about 40 mmHg.

Note that the sign of $c_I - c_A$ is positive for O_2 and negative for CO_2 . This reflects the fact that the direction of gas transport is inward for O_2 and outward for CO_2 .

Equations (2.2.3) through (2.2.5) can be combined to yield

$$c_a = \sigma k T c_A. \quad (2.2.7)$$

We now look on (2.2.6) and (2.2.7) as a pair of simultaneous equations in the two unknowns c_a and c_A . The solution is

$$\begin{aligned} c_A &= \frac{V_A c_I + Q c_v}{V_A + Q \sigma k T} \\ &= \frac{r c_I + c_v}{r + \sigma k T}, \end{aligned} \quad (2.2.8)$$

$$c_a = \sigma k T \frac{r c_I + c_v}{r + \sigma k T}, \quad (2.2.9)$$

where

$$r = V_A/Q. \quad (2.2.10)$$

It is an important fact that the compositions of the alveolar air and the arterial blood are determined not by the ventilation and blood flow separately, but only by the *ventilation-perfusion ratio* r . As $r \rightarrow \infty$, the composition of the alveolar air approaches that of the inspired air, since

$$\begin{aligned} c_A &= \frac{r c_I + c_v}{r + \sigma k T} \\ &= \frac{c_I + c_v/r}{1 + \sigma k T/r} \\ &\rightarrow c_I \text{ as } r \rightarrow \infty, \end{aligned} \quad (2.2.11)$$

and the composition of arterial blood approaches equilibrium with inspired air. As $r \rightarrow 0$, $c_A \rightarrow c_v/\sigma k T$, so the composition of arterial blood $c_a = \sigma k T c_A$ approaches that of the venous blood. In this limit, the composition of alveolar air approaches equilibrium with the venous blood. In fact, the ventilation-perfusion ratio is not too different from unity, and the alveolar and arterial partial pressures are between the corresponding inspired and venous partial pressures.

We now derive a formula for the net transport of the gas of interest. As mentioned above, this is given by the right- or by the left-hand side of (2.2.6). Using the right-hand side and substituting the formula for c_a , (2.2.9), we get

$$\begin{aligned} f &= Q(c_a - c_v) \\ &= Q r \frac{\sigma k T c_I - c_v}{r + \sigma k T} \\ &= Q r \sigma \frac{P_I - P_v}{r + \sigma k T}, \end{aligned} \quad (2.2.12)$$

where we have used $P_I = k T c_I$ and $\sigma P_v = c_v$. The reader should verify that the same result is obtained if we start from $f = V_A(c_I - c_A)$ and use the formula for c_A , (2.2.8).

The formula (2.2.12) for the flux of gas has the following simple interpretation. The quantity $Q\sigma(P_i - P_v)$ is the flux of gas that would occur if the venous blood were allowed to equilibrate directly with inspired air. To get the actual flux, we multiply this by the fraction $r/[r + \sigma kT]$, which involves the ventilation-perfusion ratio r .

2.3 Gas Transport in the Lung

We now take account of the fact that the lung consists of some 3×10^8 alveoli connected in parallel. Several of the physical quantities that appear in the previous section have the same value in all of the different alveoli of the lung. Because the inflowing venous blood is simply partitioned among the various alveoli, its composition is constant throughout the lung. For the same reason, the composition of the inspired air is the same for all of the different alveoli. The solubility of any gas in blood is the same in different parts of the lung. The Boltzmann constant certainly does not vary, and we make the assumption that the temperature is uniform throughout the lung.

The ventilation V_A and the perfusion Q can certainly differ in different alveoli. Because of the effects of gravity, there is a systematic gradient of perfusion in the lung: The lower parts have a substantially larger blood flow than the upper parts of the lung. This is caused by the effects of hydrostatic pressure on the freely distensible pulmonary vessels. The ventilation is also greater in the lower parts of the lung. To understand this, recall that alveolar ventilation depends on the *difference* between the maximum and minimum volume of the alveolus during the breathing cycle. The maximum volumes achieved during inspiration are fairly uniform throughout the lung, but the minimum volumes achieved during expiration are smaller in the lower part of the lung because the weight of the lung tissue participates in the compression of the alveoli. Accordingly, the alveolar ventilation, like the blood flow, is larger in the lower parts of the lung. These effects do not exactly balance, however, so the ratio of ventilation to perfusion is different in different alveoli.

More extreme variations in the ratio of ventilation to perfusion occur in pulmonary disease. Either the air supply or the blood supply to an alveolus can be completely obliterated, resulting in a ventilation-perfusion ratio of zero or infinity.

Thus, we cannot regard the ventilation-perfusion ratio as constant. From equations (2.2.8) and (2.2.9) it is clear that variations in r , the ratio of ventilation to perfusion, will result in variations in the composition of the alveolar air and of the blood leaving the alveolus. From (2.2.12) it is clear that the flux of gas is also different in alveoli with different values of r .

To analyze the situation in which the ventilation-perfusion ratio may vary, we number the alveoli using an index i , $i = 1, \dots, 3 \times 10^8$. Quantities that do not vary from one alveolus to the next are written without the index i . Accordingly, we have

$$(c_A)_i = \frac{r_i c_I + c_v}{r_i + \sigma k T}, \quad (2.3.1)$$

$$(c_a)_i = \sigma k T \frac{r_i c_I + c_v}{r_i + \sigma k T}, \quad (2.3.2)$$

where

$$r_i = (V_A)_i / Q_i, \quad (2.3.3)$$

and the flux of gas in the i th alveolus is given by

$$f_i = Q_i r_i \sigma \frac{P_I - P_v}{r_i + \sigma k T}. \quad (2.3.4)$$

By summing (2.3.4) over all the alveoli, we derive an equation for the transport of the gas of interest in the lung as a whole. The formula is

$$f = \sum_i f_i = \sigma (P_I - P_v) \sum_i \frac{Q_i r_i}{r_i + \sigma k T}. \quad (2.3.5)$$

Multiplying and dividing by the total pulmonary blood flow, we find that f can be written in the form

$$f = \sigma (P_I - P_v) Q_0 E, \quad (2.3.6)$$

where

$$E = \frac{1}{Q_0} \sum_i \frac{Q_i r_i}{r_i + \sigma k T}, \quad (2.3.7)$$

$$Q_0 = \sum_i Q_i. \quad (2.3.8)$$

As in the case of a single alveolus, we remark that $\sigma (P_I - P_v) Q_0$ is the transport that would occur if venous blood were brought directly into contact with room air. The quantity E is a fraction ($0 < E < 1$) that can be thought of as the *efficiency* of gas transport.

2.4 Optimal Gas Transport

Suppose we are given the total alveolar ventilation

$$(V_A)_0 = \sum_i (V_A)_i \quad (2.4.1)$$

and the total pulmonary blood flow

$$Q_0 = \sum_i Q_i. \quad (2.4.2)$$

How should this airflow and blood flow be distributed to maximize the magnitude of f for given P_1 and P_v ? Alternatively, we could ask how the airflow and blood flow should be distributed to minimize the magnitude of $P_1 - P_v$ for given f ? The first problem is appropriate when the rate of transport is determined by the lung, as in the case of anesthetic gases that are cleared from the body by the lungs. The second problem is appropriate for gases like O_2 and CO_2 , for which the rate of transport is fixed by the tissues. For such a gas, the venous partial pressure automatically adjusts to whatever value is needed to maintain adequate flux across the lung. In any case, it is clear from (2.3.6) that we can solve both problems simultaneously by maximizing the efficiency

$$E = \frac{1}{Q_0} \sum_i \frac{Q_i r_i}{r_i + \sigma k T}. \quad (2.4.3)$$

We now show that

$$E \leq E_{\max}, \quad (2.4.4)$$

where

$$\begin{aligned} E_{\max} &= \frac{1}{Q_0} \sum_i \frac{Q_i r_0}{r_0 + \sigma k T}, \\ &= \frac{r_0}{r_0 + \sigma k T} \end{aligned} \quad (2.4.5)$$

and

$$r_0 = (V_A)_0 / Q_0. \quad (2.4.6)$$

Note that r_0 is the ventilation-perfusion ratio for the lung as a whole and that $E_{\max}$ is the efficiency of gas transport when the ventilation-perfusion ratio of each alveolus matches that of the lung as a whole. Our goal is to prove that the efficiency of gas transport cannot be greater than $E_{\max}$. (The efficiency $E_{\max}$ is actually achieved by setting $r_i = r_0$ for all i .)

To prove this, we use the fact that the function

$$g(r) = r / [r + \sigma k T] \quad (2.4.7)$$

lies below all of its tangent lines (see Figure 2.2). This can be shown by taking the second derivative (curvature) of g , which is

$$g''(r) = -\frac{2\sigma k T}{[r + \sigma k T]^3} < 0. \quad (2.4.8)$$

It follows that

$$g(r_i) \leq g(r_0) + (r_i - r_0)g'(r_0). \quad (2.4.9)$$

(This inequality just expresses the fact that the function lies below the line that is tangent to g at r_0 .) Now we use (2.4.9) to derive a bound on E as

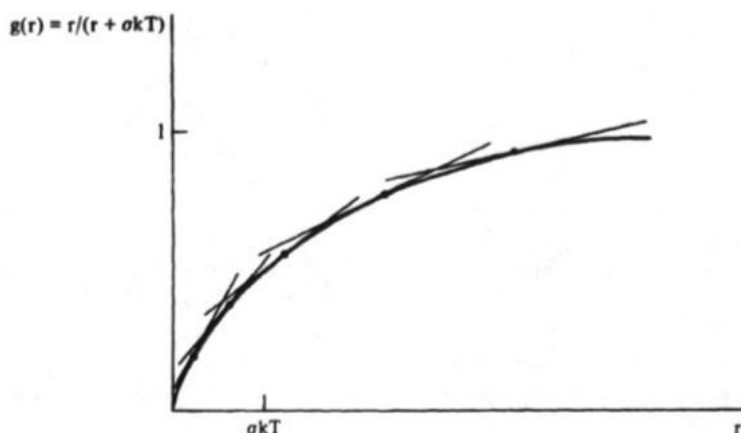


Figure 2.2. The function $g(r)$ lies below all of its tangent lines.

follows:

$$Q_0 E = \sum_i Q_i g(r_i) \leq \sum_i Q_i g(r_0) + \sum_i Q_i (r_i - r_0) g'(r_0). \quad (2.4.10)$$

Using the constraints and the definitions of r_i and r_0 , the reader can show that the last term on the right is zero (see Exercise 2.2). Thus,

$$Q_0 E \leq \sum_i Q_i g(r_0) = Q_0 E_{\max}, \quad (2.4.11)$$

so

$$E \leq E_{\max} \quad (2.4.12)$$

as promised.

We have shown that optimal gas transport is achieved when the *ventilation-perfusion ratio* is constant throughout the lung. Note that the separate distributions of airflow and blood flow are not important, as long as these distributions are *matched* in the sense that their ratio is constant.

2.5 Mean Alveolar and Arterial Partial Pressures

We have just seen that inhomogeneity of the ventilation-perfusion ratio degrades the efficiency of gas transport. Another manifestation of the same phenomenon is the lack of equilibrium between alveolar air and arterial blood that occurs as a result of nonuniform ventilation-perfusion ratio. The "alveolar air" mentioned in the foregoing is the air that is expelled from the whole population of pulmonary alveoli during exhalation. Note that this is a mixture of air samples from the different alveoli; accordingly, we refer to

it as the *mixed alveolar air*. Each alveolus contributes to this mixture in proportion to its own alveolar ventilation. Similarly, the (systemic) arterial blood is a mixture obtained by combining the streams of blood flowing through the pulmonary capillaries of the various alveoli. The contribution of each alveolus to the *mixed (systemic) arterial blood* is proportional to the blood flow (perfusion) of the alveolus in question.

The following argument seems to prove that mixed alveolar air and mixed arterial blood are in equilibrium. In each alveolus, the expired air and the blood leaving the alveolus are in equilibrium. Accordingly, the mixed alveolar air must be in equilibrium with the mixed arterial blood. This argument is incorrect, and the mean alveolar partial pressure is not necessarily equal to the mean arterial blood pressure, as we now show.

The *mean arterial concentration* of the gas of interest, $\langle c_a \rangle$, is calculated as follows:

$$\langle c_a \rangle = \frac{1}{Q_0} \sum_i Q_i (c_a)_i. \quad (2.5.1)$$

The numerator of this fraction is the number of molecules of the gas of interest entering the arterial blood from the alveolar capillaries per unit of time. The mean arterial partial pressure of the gas is computed from $\langle c_a \rangle$ as follows:

$$\langle P_a \rangle = \frac{\langle c_a \rangle}{\sigma} = \frac{\sum_i Q_i (c_a)_i}{Q_0 \sigma}, \quad (2.5.2)$$

where σ is the solubility of the gas in blood.

Since expired air is a sample of alveolar air, we define the mean alveolar concentration in terms of the expired air:

$$\langle c_A \rangle = \frac{\sum_i (V_A)_i (c_A)_i}{(V_A)_0}. \quad (2.5.3)$$

The mean alveolar partial pressure is then given by

$$\langle P_A \rangle = kT \langle c_A \rangle = kT \frac{\sum_i (V_A)_i (c_A)_i}{(V_A)_0}. \quad (2.5.4)$$

Using equations (2.3.1) through (2.3.3) and (2.4.6), we get

$$\langle P_A \rangle = \frac{kT}{Q_0} \sum_i Q_i \frac{(r_i/r_0)(r_i c_l + c_v)}{r_i + \sigma kT}, \quad (2.5.5)$$

$$\langle P_a \rangle = \frac{kT}{Q_0} \sum_i Q_i \frac{r_i c_l + c_v}{r_i + \sigma kT}. \quad (2.5.6)$$

Subtracting these equations gives

$$\langle P_A \rangle - \langle P_a \rangle = \frac{kT}{r_0 Q_0} \sum_i Q_i \frac{(r_i - r_0)(r_i c_l + c_v)}{r_i + \sigma kT}. \quad (2.5.7)$$

But

$$\begin{aligned} \frac{r_i c_I + c_v}{r_i + \sigma k T} &= \frac{r_0 c_I + c_v}{r_0 + \sigma k T} \\ &+ (r_i - r_0) \frac{\sigma k T c_I - c_v}{(r_i + \sigma k T)(r_0 + \sigma k T)}. \end{aligned} \quad (2.5.8)$$

Substituting (2.5.8) in (2.5.7) and using the fact that

$$\sum_i Q_i (r_i - r_0) = 0,$$

(see Exercise 2.2), we get

$$\langle P_A \rangle - \langle P_a \rangle = \frac{\sigma k T (P_I - P_v)}{(r_0 + \sigma k T)(Q_0 r_0)} \sum_i \frac{(r_i - r_0)^2 Q_i}{r_i + \sigma k T}. \quad (2.5.9)$$

Note that

$$\sum_i Q_i \frac{(r_i - r_0)^2}{r_i + \sigma k T} \geq 0, \quad (2.5.10)$$

with equality only when $r_i = r_0$ for all i . The condition $r_i = r_0$ for all i means that the ventilation-perfusion ratio is constant throughout the lung.

With these calculations we have shown that mixed alveolar air and mixed arterial blood are in equilibrium ($\langle P_A \rangle = \langle P_a \rangle$) only if the ventilation-perfusion ratio is uniform throughout the lung. Otherwise, $\langle P_A \rangle - \langle P_a \rangle$ has the same sign as $P_I - P_v$. Since the latter quantity determines the direction of gas transport (see equation 2.3.5), we conclude that for any gas transported, into the body, $\langle P_A \rangle \geq \langle P_a \rangle$, with equality only for a homogeneous lung and that for any gas transported out of the body, $\langle P_A \rangle \leq \langle P_a \rangle$, with equality only for a homogeneous lung.

Thus, a nonuniform distribution of ventilation in relation to perfusion creates an effective barrier between air and blood, even when there is no such barrier in each individual alveolus. The practical consequence of such nonuniformity in the ventilation-perfusion ratio is that the body must increase the total ventilation, the cardiac output, or both in order to achieve normal levels of gas transport at normal partial pressures.

2.6 Transport of O₂

As mentioned in Section 2.1, O₂ does not form a simple solution in blood. The relation between partial pressure and concentration of O₂ in blood is plotted in Figure 2.3 in two ways: concentration as a function of partial pressure and partial pressure as a function of concentration. In this section we use the latter relation,

$$P = C^{-1}(c) = H(c), \quad (2.6.1)$$

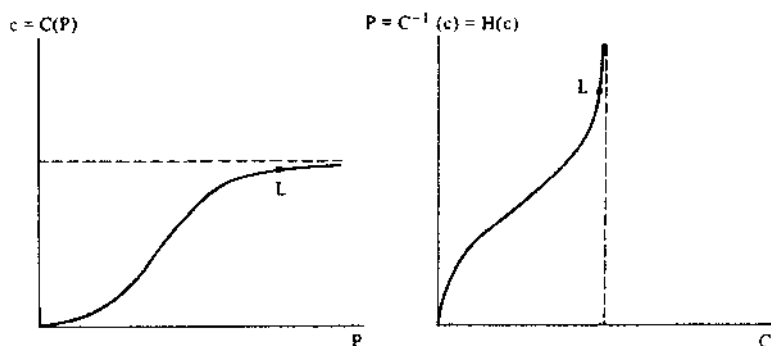


Figure 2.3. Relationship between the concentration (c) and partial pressure (P) of oxygen in blood. This relationship is plotted in two ways: $c = C(P)$ on the left and $P = H(c)$ on the right. The functions C and H are each the inverse of the other. Although each of these functions has both positive and negative curvature, the curvature of H is positive ($d^2H/dc^2 > 0$) in the neighborhood of the point marked L , which is where the lung actually operates.

and we note that the function $H = C^{-1}$ has the following properties:

$$H(0) = 0, \quad (2.6.2)$$

$$H'(c) > 0, \quad (2.6.3)$$

where H' denotes the derivative of H with respect to c , and also, in the range of partial pressures that are relevant for the lung,

$$H''(c) > 0. \quad (2.6.4)$$

In Section 2.2 the equations of gas transport in an alveolus were reduced to a pair of simultaneous equations for c_a and c_A . The corresponding equations for the case of O_2 are

$$r(c_I - c_A) = c_a - c_v, \quad (2.6.5)$$

$$kTc_A = H(c_a). \quad (2.6.6)$$

Substituting (2.6.6) into (2.6.5), we get a nonlinear equation for c_a :

$$r \left(c_I - \frac{H(c_a)}{kT} \right) = c_a - c_v, \quad (2.6.7)$$

or

$$rc_I + c_v = c_a + \frac{rH(c_a)}{kT}. \quad (2.6.8)$$

The expression on the left in (2.6.8) is a positive constant (for fixed r). As a function of c_a , the expression on the right is monotone increasing. It has the value zero at $c_a = 0$, and it increases without bound as $c_a \rightarrow \infty$. It follows that (2.6.8) has a unique solution c_a for each r . The solution

depends on r , so we express it as a function $c_a(r)$. It is important to note that this function is the same in all of the different alveoli of the lung. (The value of r , of course, can be different in different alveoli, and so the value of c_a can also be different, but the function as a whole is the same for all alveoli of the lung.)

Although we cannot find the function $c_a(r)$ explicitly, we can find some of its properties. For instance, putting $r = 0$ in (2.6.8), we find that

$$c_a(0) = c_v. \quad (2.6.9)$$

Putting $r = \infty$, we obtain

$$kTc_I = H(c_a(\infty)), \quad (2.6.10)$$

which asserts that arterial blood is in equilibrium with inspired air at $r = \infty$.

Differentiating (2.6.7) twice with respect to r , we get

$$c'_a = c_I - \frac{H(c_a) + rH'(c_a)c'_a}{kT} \quad (2.6.11)$$

$$c''_a = -\frac{2H'(c_a)c'_a + r[H'(c_a)c''_a + H''(c_a)(c'_a)^2]}{kT} \quad (2.6.12)$$

Solving for c'_a and c''_a and using (2.6.7) gives

$$\begin{aligned} c'_a &= \frac{c_I - (1/kT)H(c_a)}{1 + (r/kT)H'(c_a)} \\ &= \frac{c_a - c_v}{r(1 + (r/kT)H'(c_a))}, \end{aligned} \quad (2.6.13)$$

$$c''_a = -\frac{(2/kT)H'(c_a)c'_a + (r/kT)H''(c_a)(c'_a)^2}{1 + (r/kT)H'(c_a)}. \quad (2.6.14)$$

Since O₂ is transported into the body, $c_a > c_v$. It follows from the properties of H listed above (6.6.2–6.6.3) that

$$c'_a > 0, \quad (2.6.15)$$

$$c''_a < 0. \quad (2.6.16)$$

The function $c_a(r)$ is sketched in Figure 2.4.

The meaning of these results is as follows. The inequality $c'_a > 0$ means that the arterial oxygen concentration increases with increasing values of the ventilation-perfusion ratio. The inequality $c''_a < 0$, on the other hand, means that the rate of increase diminishes as the ventilation-perfusion ratio is raised.

We are now ready to consider O₂ transport for the lung as a whole. The flow of O₂ is given by

$$f = \sum_i Q_i(c_a(r_i) - c_v). \quad (2.6.17)$$

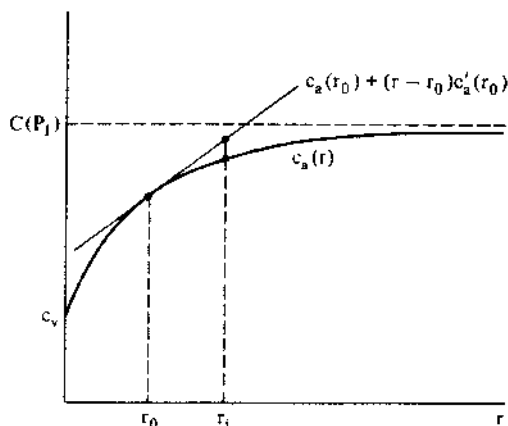


Figure 2.4. The function $c_a(r)$ lies below all of its tangent lines. In particular, it lies below the line that is tangent to it at $r = r_0$, which is shown in the figure and labeled by the linear function of r that defines that particular tangent line. r = ventilation-perfusion ratio, r_0 = ventilation-perfusion ratio for the lung as a whole, r_i = ventilation-perfusion ratio for the i -th alveolus, $c_a(r)$ = concentration of oxygen in the blood leaving an alveolus as a function of the ventilation-perfusion ratio of that alveolus. Note the extreme values $c_a(0) = c_v$ and $c_a(\infty) = C(P_1)$. The latter is the concentration of oxygen in blood that has equilibrated directly with room air.

We shall prove that

$$f \leq f_0 = \sum_i Q_i (c_a(r_0) - c_v) \quad (2.6.18)$$

where as before, $r_i = (V_A)_i / Q_i$ and

$$r_0 = (V_A)_0 / Q_0 = \sum_i (V_A)_i / \sum_i Q_i. \quad (2.6.19)$$

The proof is the same as for gases that form simple solutions. We have already shown that $c_a'' < 0$. It follows that c_a lies below its tangent lines. In particular,

$$c_a(r_i) \leq c_a(r_0) + (r_i - r_0)c'_a(r_0). \quad (2.6.20)$$

Multiplying by Q_i and summing, we get

$$\sum_i Q_i c_a(r_i) \leq \sum_i Q_i c_a(r_0) + c'_a(r_0) \sum_i Q_i (r_i - r_0). \quad (2.6.21)$$

The last term is zero (see Exercise 2.2), so we can subtract $c_v Q_0$ from both sides to get

$$f \leq f_0 \quad (2.6.22)$$

as required.

The inequality $f \leq f_0$ can be given two different interpretations depending on what we regard as given. If we are given c_1 and c_v , then the flux of O₂ is determined by the lung, and $f \leq f_0$ is a restriction on how big this flux can be.

Suppose, instead, that we are given c_1 and f . This is the physiological situation since f is really determined by the tissues and since the venous O₂ concentration adjusts itself accordingly. Then the inequality $f \leq f_0$ can be reinterpreted as a restriction on c_v . To see what form this restriction takes, we have to make explicit the dependence of c_a on c_v . Thus, we replace the function $c_a(r)$ by $c_a(r, c_v)$. This function is defined as the solution of (2.6.8), from which it follows that c_a increases as c_v increases with r fixed. From (2.6.7), though, we see that $c_a - c_v$ decreases as c_v (and so c_a) increases with r fixed. Let

$$\Delta_0(c_v) = c_a(r_0, c_v) - c_v. \quad (2.6.23)$$

We have just shown that Δ_0 is a decreasing function. Our inequality $f \leq f_0$ can be rewritten in terms of Δ_0 as follows:

$$f \leq Q_0 \Delta_0(c_v). \quad (2.6.24)$$

Let c_v^0 be defined by the equation

$$f = Q_0 \Delta_0(c_v^0). \quad (2.6.25)$$

Thus, c_v^0 is the venous oxygen concentration (for given f) when the ventilation-perfusion ratio is uniform throughout the lung. Combining (2.6.24) and (2.6.25), we see that

$$\Delta_0(c_v^0) \leq \Delta_0(c_v). \quad (2.6.26)$$

Since Δ_0 is decreasing, this implies that

$$c_v \leq c_v^0. \quad (2.6.27)$$

That is, for a given rate of O₂ consumption by the body, the venous O₂ concentration is maximized when the ventilation-perfusion ratio is uniform throughout the lung. This is of importance because the systemic veins are in equilibrium with the tissues of the body. Thus, a high c_v means that oxygen is plentiful and that the performance of the various tissues is not limited by the oxygen supply.

2.7 Computer Solution of the Equations for O₂ Transport in the Lung

The equations for oxygen transport in the lung are complicated by the nonlinear relationship between the partial pressure of oxygen in blood

and the concentration of oxygen in blood. Because of this, we have been limited up to now in this chapter to qualitative theory. With the help of computers, though, we can solve these nonlinear equations and calculate the partial pressures and oxygen concentrations in arterial and venous blood, as well as in alveolar air, under a variety of physiological and pathophysiological situations. The aim of this section is to develop a Matlab program for that purpose. The student can then use this program to explore what happens under conditions of exercise, altitude, anemia, or polycythemia, ventilation-perfusion mismatch, or various combinations of these conditions.

We begin by summarizing the equations that have to be solved. They are essentially the same as the equations introduced above, with the following differences. One change concerns the units that will be used. In order to make the numbers more convenient, we shall measure concentration in moles/liter instead of molecules/liter. One mole contains Avogadro's number (6.02×10^{23}) of molecules. The only change that this makes in our equations is that Boltzmann's constant k is replaced by the gas constant R , which is Avogadro's number times k . In these units, the ideal gas law reads $P = RTc$. Another change in our equations is that we shall introduce a specific formula for the nonlinear relationship between partial pressure and concentration of oxygen in blood. Finally, we shall introduce a parameter n that is the total number of lung compartments that we consider. In principle, each alveolus is a separate compartment, but any collection of alveoli with the same ventilation-perfusion ratio can be lumped together to form a larger compartment with the same equations as for a single alveolus (see Exercise 2.3). This is fortunate from a computational point of view, since our computer program would run very slowly if we had to set $n = 3 \times 10^8$. In the conceptual development of the program, though, it is probably best to think of the compartments as alveoli, and of n as the number of alveoli. That is how we shall describe the program as we develop it.

There are two types of variables and equations to consider, those that pertain to a single alveolus (or compartment), and those that pertain to the organism as a whole. We use the subscript $i = 1, \dots, n$ to designate a particular alveolus. Those variables that lack a subscript i pertain to the organism as a whole. An important example of such a whole-organism variable is the concentration of oxygen in the venous blood, which is denoted by c_v . This plays a pivotal role, as we shall see. With these conventions, our equations read

$$(V_A)_i(c_i - (c_A)_i) = Q_i((c_a)_i - c_v), \quad (2.7.1)$$

$$(P_A)_i = RT(c_A)_i, \quad (2.7.2)$$

$$(P_a)_i = H((c_a)_i), \quad (2.7.3)$$

$$(P_A)_i = (P_a)_i. \quad (2.7.4)$$

These equations hold for $i = 1, \dots, n$. The function H , which appears in equation (2.7.3), is defined as follows:

$$H(c) = P_* \left(\frac{c}{c^* - c} \right)^{1/3}. \quad (2.7.5)$$

This particular form of the function $H(c)$ is chosen because it gives a reasonably good fit to experimental data, not for any theoretical reason. The number 3 in the exponent $\frac{1}{3}$ is known as a "Hill coefficient." In the equation that defines $H(c)$, $P_* = 25$ mmHg is the partial pressure of oxygen at which hemoglobin is half-saturated with oxygen. On the average, then, two of the four oxygen-binding sites on each hemoglobin molecule are occupied when the partial pressure of oxygen is equal to P_* . The constant c^* is the theoretical concentration of oxygen in blood bound to hemoglobin at infinite partial pressure of oxygen, i.e., when all four sites on every hemoglobin molecule are occupied. Thus c^* is equal to four times the concentration of hemoglobin in blood. It happens that the normal value of c^* is roughly equal to the concentration of oxygen in the inspired air, c_I , at sea level. Of course, this equality can be broken either by changing c^* (anemia or polycythemia), or by changing c_I (changing altitude, or breathing air enriched with oxygen), so there is no universal relationship between c_I and c^* . Finally, it should be mentioned that equation (2.7.5) takes into account only oxygen bound to hemoglobin. It ignores dissolved oxygen, which may become important under special circumstances, such as breathing 100% O₂, especially in a hyperbaric chamber.

In addition to the foregoing individual-alveolus equations, we have one very important whole-body equation, namely

$$M = \sum_{i=1}^n (V_A)_i (c_I - (c_A)_i) = \sum_{i=1}^n Q_i ((c_a)_i - c_v). \quad (2.7.6)$$

Although this looks like two equations, they are clearly the same, on account of equation (2.7.1). The content of equation (2.7.6) is that the rate at which oxygen is consumed by the body has to be equal (in the steady state) to the net rate at which oxygen is taken up by the lungs. That net rate of oxygen uptake can either be expressed as the rate at which oxygen is given up by the air, or as the rate at which oxygen is taken up by the blood. This explains the two different ways of writing equation (2.7.6).

Our task, then, is to solve the $4n + 1$ equations (2.7.1)–(2.7.4) and (2.7.6) in the $4n + 1$ unknowns $(c_a)_i$, $(P_a)_i$, $(c_A)_i$, $(P_A)_i$, $i = 1, \dots, n$, and c_v . The given parameters are the overall rate of oxygen consumption M , the concentration of oxygen in the inspired air, c_I ; the individual alveolar ventilations $(V_A)_i$ and the individual blood flows Q_i , the gas constant R and the absolute body temperature T (but note that these always occur together in the combination RT), and the constants that characterize the relationship

between the partial pressure of oxygen and the concentration of oxygen in blood, namely P_* and c^* .

It is worth emphasizing that we regard M as known, and c_v as unknown. The reason for this is that, within limits, the rate of oxygen consumption is set by the tissues, and the lungs have to adjust accordingly. If the rate of oxygen consumption by the tissues increases, then the venous oxygen concentration will start to fall. This by itself, for purely physical reasons and even in the absence of any physiological control mechanism being called into play, will tend to increase the rate of oxygen uptake by the blood as it passes through the lungs. The progressive decline in venous oxygen concentration will continue until the rate of oxygen uptake by the lungs catches up with the rate of oxygen consumption by the tissues. In solving the system of equations stated above, we shall be finding the value of c_v at which this balance is struck. On the other hand, c_v cannot become negative. That is why the tissues are free to adjust M only "within limits." Our Matlab program will have to check for situations in which the value of M that has been specified is impossibly large, given the other parameters.

Our strategy in writing a computer program to solve the problem stated above is to begin by pretending that we know c_v . This uncouples the different alveoli, so we have n separate problems, one for each i , each with four equations and four unknowns. The four equations (2.7.1)–(2.7.4) can be combined to yield one equation in the single unknown $(c_a)_i$:

$$\phi((c_a)_i, c_v, r_i) = 0, \quad (2.7.7)$$

where

$$r_i = \frac{(V_A)_i}{Q_i}, \quad (2.7.8)$$

and where the function $\phi(c_a, c_v, r)$ is defined as follows:

$$\phi(c_a, c_v, r) = c_a - c_v + r \left(\frac{H(c_a)}{RT} - c_l \right). \quad (2.7.9)$$

Given c_v and r , we need a method that will solve the nonlinear equation $\phi(c_a, c_v, r) = 0$ for the unknown variable c_a . The first step is to define an interval that is known to contain the solution. This interval will be given by $[0, c_a^{\max}]$, where $c_a^{\max}$ is defined as the solution of

$$H(c_a^{\max}) = RTc_l. \quad (2.7.10)$$

This equation can be solved explicitly. The formula for $c_a^{\max}$ is

$$c_a^{\max} = c^* \frac{(RTc_l)^3}{P_*^3 + (RTc_l)^3}. \quad (2.7.11)$$

The physical meaning of $c_a^{\max}$ is that it is the oxygen concentration in blood that is allowed to equilibrate directly with the inspired air, without the intervention of the lungs. Intuitively, this should be an upper bound on

the arterial oxygen concentration (and therefore it should also be an upper bound on the venous oxygen concentration).

Now let us consider the two ends of the interval $[0, c_a^{\max}]$ proposed above. When $c_a = 0$, we have $\phi(0, c_v, r) = -c_v - rc_1 < 0$. When $c_a = c_a^{\max}$, on the other hand, we have $\phi(c_a^{\max}, c_v, r) = c_a^{\max} - c_v > 0$. Thus, ϕ has different signs at the two ends of the interval in question. Moreover, $\phi(c_a, c_v, r)$ is a continuous function of c_a on the interval $[0, c_a^{\max}]$, and it is also an *increasing* function of c_a on this interval. It follows that there is exactly one solution of $\phi(c_a, c_v, r) = 0$ with c_a in the interval $[0, c_a^{\max}]$, and that the solution can be found by the method of bisection, which we explain next.

Bisection is a method for finding an approximate solution to an equation of the form $F(x) = 0$, where F is a continuous, increasing function of x . (The term “increasing” means that $F(x_2) > F(x_1)$ whenever $x_2 > x_1$.) The starting point for bisection is an interval $[x_{\text{lower}}, x_{\text{upper}}]$ with the property that $F(x_{\text{lower}}) \leq 0 \leq F(x_{\text{upper}})$. It follows that there is a solution x somewhere in $[x_{\text{lower}}, x_{\text{upper}}]$, and we want to “zero in” on that solution. The trick is to cut the interval in half, and then decide which half to keep and which to throw away. If we make sure that the half interval we keep still contains the solution (see below), then we are back in the starting situation, but with an interval that has been reduced in length by a factor of 2. Clearly, we can then repeat this process as often as we like. Each time we repeat it, the uncertainty in the solution (i.e., the length of the interval that is known to contain the solution) is cut in half. Knowing this, we can decide in advance how many steps to take, based on the amount of uncertainty we are willing to tolerate in the computed solution. For example, if we take 20 steps, we reduce the length of the interval of uncertainty by a factor of 2^{20} , which is about a million.

How do we decide which half interval to keep? That’s easy. Just evaluate the function F at the point where the cut was made. If the result is negative, keep the upper half interval, i.e., move x_{lower} up to the position of the cut. If the result is positive, keep the lower half interval, i.e., move x_{upper} down to the position of the cut. If the result is zero, keep either half, since the solution is then right on the cut and belongs to both half-intervals. (The intervals we consider are closed; i.e., they include their boundary points.) One might think that we should test for this special case ($F(x) = 0$) and stop if we encounter it, since the solution has then been found, but it is more efficient in general not to waste time testing for this unlikely event to happen.

The following Matlab program implements the method of bisection in our case. In particular, it takes advantage of the vector operations in Matlab to perform bisection for all of the alveoli at once:

```
function ca=carterial(cv,r)
%filename: carterial.m
global camax n maxcount;
```

```

%initialize:
ca_lower=zeros(n,1);           %lower end of interval
ca_upper= ones(n,1)*camax;     %upper end of interval
ca=(ca_lower+ca_upper)/2;      %midpoint of interval
%start main loop:
for count=1:maxcount
    %check sign of function at midpoint of interval:
    pos=(phi(ca,cv,r)>0);
    neg=~pos;
    %if (phi>0) ca_upper=ca; else ca_lower=ca:
    ca_upper=ca_upper.*neg+ca.*pos;
    ca_lower=ca_lower.*pos+ca.*neg;
    %ca=midpoint of new interval:
    ca=(ca_lower+ca_upper)/2;
end

```

In our use of this program, *cv* is just a number, but *r* and *ca* are column vectors of length *n*. Also, the internal variables *ca_lower*, *ca_upper*, *ca*, *pos*, and *neg* are column vectors of length *n*. In particular, *pos* and *neg* are logical vectors; their components can be only 1 or 0. The line *pos*=(*phi*(*ca*,*cv*,*r*)>0); sets each component of *pos* equal to 1 if and only if the output vector of the function *phi* has a positive component at the corresponding location. The next line makes the vector *neg* complementary to *pos*; i.e., *neg* contains a 1 wherever *pos* contains a 0, and vice versa. (A more precise name for *neg* would be *not_pos*, since it is also 1 where *phi* returns 0.) Once the vectors *pos* and *neg* have been defined, we can use them to move *ca_upper* or *ca_lower* as appropriate without any if statements. In each case, the upper or lower limit of the interval is either left alone or moved to the midpoint of the interval. In reading these lines, recall that *.** is Matlab notation for the element-by-element multiplication of vectors (or matrices). Finally the new midpoint of the interval is defined, in preparation for the next bisection step.

The Matlab functions that the above program needs in order to run are as follows:

```

function c=phi(ca,cv,r)
%filename:phi.m
global RT cI;
c=ca-cv+r.*(H(ca)/RT-cI);

function P=H(ca)
%filename: H.m
global Pstar cstar;
ratio=ca/cstar;
P=Pstar*(ratio./(1-ratio)).^(1/3);

```

We have now gone as far as we can pretending that c_v is known, and we have to face the task of solving for c_v . This will also be done by bisection. First we need to formulate the equation that determines c_v . This equation is

$$M_{\text{diff}}(c_v, r) = 0, \quad (2.7.12)$$

where

$$M_{\text{diff}}(c_v, r) = M - \sum_{i=1}^n Q_i ((c_{\text{arterial}}(c_v, r_i) - c_v). \quad (2.7.13)$$

Here, M_{diff} is the difference between the rate of oxygen consumption by the tissues and the rate of oxygen uptake by the lungs. In the steady state this must be zero. The symbol r denotes the vector whose components are the individual r_i . Equations (2.7.12)–(2.7.13) are essentially the same as equation (2.7.6), but note that we have here put that equation in the form $M_{\text{diff}}(c_v, r) = 0$. In order to do this, we have had to make use of the function $c_{\text{arterial}}(c_v, r_i)$, for which we have no explicit formula. Instead $c_{\text{arterial}}(c_v, r_i)$ is implicitly defined as the solution of equation (2.7.7). That is,

$$\phi(c_{\text{arterial}}(c_v, r_i), c_v, r_i) = 0. \quad (2.7.14)$$

Note that the function c_{arterial} is the same for all alveoli, even though its value may differ from one alveolus to another because of different values of the ventilation-perfusion ratio r_i . Recall that a Matlab function `carterial(cv, r)` has already been introduced above. It evaluates $c_{\text{arterial}}(c_v, r_i)$ for all of the different r_i at once.

The function M_{diff} defined by equation (2.7.13) is an increasing function of c_v . This is not obvious, since the two appearances of c_v in M_{diff} drive M_{diff} in opposite directions, and the question is which one wins. This has already been discussed in the previous section, however, where we have shown that $(c_a - c_v)$ is a decreasing function of c_v in each alveolus. It follows from this that M_{diff} is an increasing function of c_v . (One can also establish the same result by differentiating M_{diff} with respect to c_v , with all of the r_i held fixed. In carrying this out, one needs the derivative of c_{arterial} with respect to c_v , and this can be found by implicit differentiation of equation (2.7.14), taking into account the specific formula for ϕ given by equation (2.7.9). From this it is easy to see that $dc_{\text{arterial}}/dc_v < 1$, and that is what is needed to show that $dM_{\text{diff}}/dc_v > 0$.)

In order to apply the method of bisection to the equation $M_{\text{diff}} = 0$, we need to start with an interval of values of c_v such that $M_{\text{diff}} < 0$ at the lower end of the interval, and $M_{\text{diff}} > 0$ at the upper end. It turns out that the same interval as we previously used in solving for the $(c_a)_i$, namely $[0, c_a^{\text{max}}]$, can be used for this purpose, with an interesting complication that we shall discuss in a moment. First, consider the upper end of this interval. Recall that c_a^{max} is the concentration of oxygen in blood that has been allowed to

equilibrate directly with the inspired air, without the intervention of the lungs. Clearly, if the venous blood already has this oxygen concentration, there is nothing more for the lungs to do, and the arterial blood leaving all of the alveoli will also have the oxygen concentration $c_a^{\max}$. To see this mathematically, recall the definition of $c_a^{\max}$, equation (2.7.10). Substituting this into the formula for $\phi(c_a, c_v, r)$, equation (2.7.9), we see that $\phi = 0$ when $c_a = c_v = c_a^{\max}$. Thus, $M_{\text{diff}}(c_a^{\max}, r) = M > 0$, so $c_a^{\max}$ works as an upper limit of the interval that we need to start the bisection process. Next, consider the proposed lower limit $c_v = 0$. Here we *cannot* guarantee that M_{diff} will be negative. Once all of the other parameters have been fixed, one can always choose M large enough to make $M_{\text{diff}} > 0$. Such M values correspond, however, to rates of oxygen consumption that the lungs cannot support. Thus, we should always test the sign of $M_{\text{diff}}(0, r)$. If it is positive, we should stop the program with an error message, and if it is negative, we may proceed with the bisection method to solve for c_v . This procedure is implemented in the following Matlab script:

```
%filename: cvsolve.m
%This script solves the equation
%Mdiff(cv,r)=0
%by the method of bisection.
if(Mdiff(0,r)>0)
    %stop program and issue error message:
    error('M is too large')
end
%initialize:
cv_lower=0;           %lower limit of interval
cv_upper=camax;       %upper limit of interval
cv=(cv_lower+cv_upper)/2; %midpoint of interval
%begin main loop:
for count=1:maxcount
    %check sign of Mdiff at midpoint of interval
    pos=(Mdiff(cv,r)>0);
    neg=~pos;
    %if(pos) cv_upper=cv; else cv_lower=cv;
    cv_upper=cv_upper*neg+cv*pos
    cv_lower=cv_lower*pos+cv*neg
    %set cv=midpoint of current interval:
    cv=(cv_lower+cv_upper)/2;
end
```

The logic here is the same as in our previous bisection routine, except that here we are dealing with variables whose values are individual numbers rather than vectors. (There is one vector that appears in `cvsolve`, namely `r`, but it is merely passed along as an argument to `Mdiff`.) The function `Mdiff(cv,r)` is as follows:

```
function Md=Mdiff(cv,r)
%filename: Mdiff.m
global M Q;
Md=M-sum(i=1:n)(Q(i)*(carterial(cv,r(i))-cv)):
Md=M-Q'*(carterial(cv,r)-cv)
```

This completes the oxygen transport program, except for details such as initialization, output, and checking. These will be discussed below. Note how the parts of the program that we have already written work together. A call to `cvsolve` initiates the process of solving for `cv` by the method of bisection. At *each* of the `maxcount` iterations of this process, it is necessary to call the function `Mdiff`, which in turn calls `carterial`. Now `carterial` itself employs the method of bisection (in vector form, with vectors of length `n`) to solve for the oxygen concentrations in the arterial blood leaving each of the separate alveoli. This, too, requires `maxcount` iterations (each of which requires a call to `phi`, which calls `H`). Thus we have bisection nested within bisection, and the overall running time of the program will be proportional to `maxcount`². (It will also be proportional to `n`, on a serial computer. The possibility of doing all alveoli in parallel could eliminate this factor of `n` on a parallel computer.)

The following program calculates output variables, checks that all of the original equations are satisfied, and plots results. It assumes that `cv` has already been found by the script `cvsolve`:

```
%filename: outchecklung.m
%starting from the value of cv
%determined by cvsolve,
%solve for all other unknowns:
%concentrations and partial pressures
%in the individual alveoli:
%
%vector of oxygen concentrations in
%arterial blood leaving each alveolus:
ca=carterial(cv,r);
%
%vector of oxygen partial pressures
%in arterial blood leaving each alveolus:
Pa=H(ca);
%
%vector of oxygen partial pressures
%in alveolar air:
PA=Pa;
%
%vector of oxygen concentrations
%in alveolar air:
cA=PA/RT;
```

```

%
%concentrations and partial pressures
%for the whole organism:
%(write out these results)
%oxygen concentration in (systemic) venous blood:
cv
%
%mean arterial oxygen concentration:
cabar=Q' *ca/sum(Q)
%
%mean alveolar oxygen concentration:
cAbar=VA'*cA/sum(VA)
%
%oxygen concentration is the inspired air:
cI
%
%oxygen partial pressure in venous blood:
Pv=H(cv)
%
%mean arterial oxygen partial pressure:
Pabar=H(cabar)
%
%mean alveolar oxygen partial pressure:
PAbar=RT*cAbar
%
%oxygen partial pressure in the inspired air:
PI=RT*cI
%
%check that partial pressures are in expected order
if((Pv<Pabar)&(Pabar<=PAbar)&(PAbar<PI))
    'partial pressures ordered as expected'
else
    'WARNING: partial pressures NOT ordered as expected'
    Pv
    Pabar
    PAbar
    PI
end
%
%check that all equations are satisfied
%output a measure of relative residual in each case
%first consider individual-alveolus equations:
fba= Q.*(ca-cv);
faa=VA.*(cI-cA);
check1=max(abs(fba-faa))/max(abs(fba))

```

```

check2=max(abs(PA-Pa))/max(abs(PA))
check3=max(abs(PA-RT*cA))/max(abs(PA))
ca_check=cstar*((Pa/Pstar).^3)./(1+(Pa/Pstar).^3);
check4=max(abs(ca-ca_check))/max(abs(ca))
%
%now check that total rate of oxygen transport
%matches rate of consumption
fb=Q'*(ca-cv); %sum(Q.*(ca-cv))
fa=VA'*(cI-cA); %sum(VA.*(cI-cA))
check5=abs(M-fb)/abs(M)
check6=abs(M-fa)/abs(M)
%
%plot various results
%against the ventilation-perfusion ratio, r:
%(plot individual points instead of lines
%since r values are not in order)
%define vector with all unit entries:
u=ones(n,1);
cblood=[cv*u cabar*u ca];
cair= [cI*u cAbar*u cA];
Pressures=[Pv*u Pabar*u PAbar*u PI*u Pa PA];
figure(2)
subplot(2,1,1),plot(r,cblood, '.')
subplot(2,1,2),plot(r,cair, '.')
figure(3)
plot(r,Pressures, '.')

```

Initialization is accomplished by the following script, `setup_lung`. It distributes ventilations and perfusions at random, in a particular way that may not be realistic but makes it possible to study the influence of heterogeneity on lung performance. In particular, there is a "heterogeneity parameter," `beta`, which should be set in the interval $[0, 1]$. When `beta=0`, the ventilation-perfusion ratio is constant, and when `beta=1`, there is no correlation at all between ventilation and perfusion. (The setup that we do here does not allow for the possibility of a *negative* correlation between ventilation and perfusion, which would be even worse than no correlation at all.) A scatter plot of ventilation vs. perfusion is generated by `setup_lung` in order to show the results of the random initialization.

```

% filename: setup_lung.m
%heterogeneity parameter (0<=beta<=1):
%beta=0 for homogenous lung
%beta=1 for no ventilation/perfusion correlation
beta=0.5
%
%number of iterations used in bisection:

```

```

maxcount=20
%
%number of 'alveoli':
n=100
%
%reference oxygen concentration (moles/liter):
cref=0.2/(22.4*(310/273))
%cref=concentration of oxygen
%in air at sea level at body temperature
%
%oxygen concentration in the inspired air:
cI=cref
%
%blood oxygen concentration
%at full hemoglobin saturation:
cstar=cref
%cstar=4*(concentration of hemoglobin
%in blood expressed in moles/liter)
%
%rate of oxygen consumption (moles/minute):
M=0.25*cref*5.6
%
%oxygen partial pressure
%at which hemoglobin is half-saturated:
Pstar=25
%
%gas constant*absolute temperature
%(mmHg*liters/mole):
RT=760*22.4*(310/273)
%
%oxygen partial pressure
%in the inspired air (mmHg):
PI=RT*cI
%
%oxygen concentration
%in blood exposed directly to inspired air:
camax=cstar*(PI/Pstar)^3/(1+(PI/Pstar)^3)
%camax is an upper bound
%on oxygen concentration in blood
%
%expected value of total alveolar ventilation:
VAtotal=5.0      %(liters/minute)
%
%expected value of total perfusion:
Qttotal=5.6      %(liters/minute)

```



```

%
%expected alveolar ventilation per alveolus:
VAbar=VAtotal/n
%
%expected perfusion per alveolus:
Qbar=Qttotal/n
%
%random initialization
%of ventilation and perfusion:
%create two independent vectors
%of exponential random variables
%with mean 1:
a1=-log(rand(n,1));
a2=-log(rand(n,1));
av=(a1+a2)/2;
%components of av have mean 1
%and distribution like  $t \cdot \exp(-t)$ .
VA=VAbar*(a1*beta+av*(1-beta));
Q = Qbar*(a2*beta+av*(1-beta));
%when beta=0, VA and Q are in a fixed proportion
%when beta=1, VA and Q are independent
r=VA./Q;
figure(1)
plot(Q,VA,'.')
%
%find actual values of
%VAtotal, Qttotal, VAbar, and Qbar:
VAtotal=sum(VA)
Qttotal =sum(Q)
VAbar=VAtotal/n
Qbar= Qttotal/n

```

Now that we have all of the tools, the main program for oxygen transport by the lung can be very short:

```

%filename: lung.m (main program)
clear all
clf
global Pstar cstar n maxcount M Q camax RT cI;
setup_lung
cvsolve
outchecklung

```

2.8 Computing Projects Concerning Oxygen Transport by the Lung

Note that each run of the program `lung.m` as written above will handle only one specific set of parameter values. The projects described in this section, however, all involve varying some parameter and plotting results as functions of that parameter. To automate this, you will need to make some changes. First, `lung.m` will need to involve a loop over the desired values of the parameter in question. Second, since the parameter of interest is then set in `lung.m`, it should not also be set in `setup_lung`. Finally, `lung.m` will have to save results for future plotting, and also do the plotting that summarizes how the behavior of the lung changes as the parameter in question is varied.

The programs introduced in the foregoing section make it possible to study the consequences of changing the extent to which airflow and bloodflow are coordinated throughout the lung. This is done by varying the parameter `beta`, which controls the setup of the simulated lung. When `beta=0`, airflow and bloodflow are in a fixed proportion, i.e., the ratio of ventilation to perfusion is constant throughout the lung. At the opposite extreme, when `beta=1`, then the airflow and bloodflow are chosen independently, so some alveoli may get high ventilation with very little perfusion, and vice versa.

Try varying `beta` with all other parameters constant. Plot the inspired partial pressure of oxygen, the mean alveolar partial pressure of oxygen, the mean arterial partial pressure of oxygen, and the venous partial pressure of oxygen, all on the same graph as functions of `beta`. Explain the results in terms of the theory developed in this chapter.

In the foregoing study the overall rate of oxygen consumption was assumed constant. But we can also ask, what is the maximum sustainable rate of oxygen consumption for a given set of parameters, and how does it depend on the extent to which ventilation is matched to perfusion? To study this, use trial and error to find the maximum sustainable rate of oxygen consumption for each of several values of `beta`, and plot this rate as function of `beta`. (Recall that the program will stop with an error message if the assumed rate of oxygen consumption is too high.) Keep all parameters other than `beta` and `M` constant throughout this study.

Next, consider the effects of altitude, with and without adaptation. (These studies should be done with `beta` fixed, but it would be interesting to repeat the whole study for several different values of `beta` to see what the effects of ventilation-perfusion mismatch are on oxygen transport at altitude.) The primary change at altitude is, of course, a reduction in the inspired partial pressure of oxygen, and also, therefore, in the corresponding concentration of oxygen in the inspired air. You can use the inspired concentration or partial pressure of oxygen as your principal independent

parameter for this study, but it will make the results more interesting if you look up how this translates to actual altitude in feet or meters. An important long-term adaptation to altitude is the production of more red blood cells, thus increasing the concentration of hemoglobin in blood. In terms of our computer program, cI is the inspired concentration of oxygen, $cstar$ is 4 times the concentration of hemoglobin, i.e., the maximum concentration of oxygen that the blood can carry, and c_{ref} is the concentration of oxygen in air at sea level. At sea level, then, we have $cI = c_{ref}$ by definition, and it happens to be the case that $cstar = c_{ref}$, too. As we go up in altitude, cI becomes smaller than c_{ref} , and if we stay at altitude long enough, $cstar$ becomes larger than c_{ref} .

This suggests the following protocol. First vary cI with all other parameters constant to see the effects of acute exposure to altitude. Plot the mean alveolar, mean arterial, and venous partial pressures and concentrations of oxygen as functions of the altitude. At what altitude does it become impossible to sustain the normal resting rate of oxygen consumption (without breathing harder or increasing the cardiac output; see below for discussion of how to include these changes as well)? These results are for a person adapted to sea level.

Now repeat the above study with $cstar = 1.5 * c_{ref}$ instead of $cstar = c_{ref}$. This corresponds to someone who has been living at high enough altitude for so long that his/her red blood cells make up 60% of the blood volume instead of the usual 40%. At what altitude does it become impossible even for this altitude-adapted person to sustain the normal resting rate of oxygen consumption (without breathing harder or increasing cardiac output)? Compare the results obtained for this altitude-adapted person to those obtained above for a person who lacks such adaptation to altitude.

Don't forget to do all of these altitude studies for several different choices of β to see the effect of ventilation-perfusion mismatch on one's ability to cope with altitude!

Another condition involving a change in the hemoglobin concentration in blood is anemia. To simulate anemia in our model, simply reduce $cstar$ while keeping all other parameters constant. Plot the inspired, mean alveolar, mean arterial, and venous partial pressures and concentrations of oxygen as functions of $cstar$ as that parameter is reduced below c_{ref} while all other parameters are held at their normal resting (sea-level) values. What is the minimum value of $cstar$ at which the normal resting rate of oxygen consumption can be maintained (without breathing harder or increasing cardiac output). How do these results depend on the ventilation-perfusion mismatch parameter β ? (Repeat the study for several different β to find out.)

What if a person with anemia attempts to go to high altitude? How does the outcome depend on the ventilation-perfusion mismatch parameter β ?

Another important aspect of gas exchange in the lung is the response to exercise. Both cardiac output and total alveolar ventilation increase during exercise. We can make a simplified model of this by assuming that both quantities are inversely proportional to the venous oxygen concentration. (In the case of cardiac output, this is essentially the model of autoregulation discussed in the foregoing chapter; see Section 1.9.) To implement this, we must first find the normal resting value of the venous oxygen concentration, which we shall call `cvzero`. This can be done by running the lung program exactly as written except that the parameter `beta` should be set equal to zero, to express the fact that the normal lung has a constant ventilation-perfusion ratio. (Of course, in reality, it can't be perfectly constant, so we are idealizing somewhat here.) The value of `cv` that the program computes in this case is `cvzero`.

Once we know `cvzero`, we can call `setup_lung` to initialize the arrays `VA` and `Q` just as before, but when we use either of these arrays we should always multiply every element of each of them by the factor $(cvzero/cv)$. Note that this has no effect on the ventilation-perfusion ratio! Thus, there are only a few places in the program where we have to make any change. The most important of these is in `Mdiff`, where the line `Md=M-Q'*(carterial(cv,r)-cv)` needs to be replaced by `Md=M-(Q'*(carterial(cv,r)-cv))*(cvzero/cv)`, so that all flows are effectively multiplied by the factor $(cvzero/cv)$. Of course, the function `Mdiff` needs to know the value of `cvzero`. This can be handled by making `cvzero` global and including it on the global statement in `Mdiff` as well as on the global statement in the main program. Also, don't forget to initialize `cvzero`, presumably in `setup_lung`. There are also a few places in `outchecklung` where similar changes to the one made in `Mdiff` must also be made. We'll leave it to the reader to find those.

At this point, the alert reader may ask, "Doesn't the bisection routine that solves for `cv` assume that `Mdiff` is an increasing function of `cv`? Now that we have modified this function, is that still the case?" The answer is yes, but we leave the proof to the reader.

With the above changes in place, we have a lung simulation that realistically adjusts to increases in the rate of oxygen consumption by breathing harder and increasing cardiac output. To study this, plot cardiac output and total alveolar ventilation as functions of the rate of oxygen consumption. Do this for several different values of the ventilation-perfusion mismatch parameter `beta`. What are the implications of such mismatch for the difficulty of performing exercise at any given level of intensity (as measured by the rate of oxygen consumption)?

Also, we now have a lung simulation that can adjust to some of the other stresses we have considered (altitude, anemia) by breathing harder and increasing cardiac output. Therefore, it would be interesting to repeat the altitude and anemia studies suggested above with this more realistic lung model. Note that now it will never happen that a particular metabolic

rate cannot be achieved. Instead, in the model, it will be achieved by a sufficiently drastic increase in cardiac output and alveolar ventilation. Thus, if you want to report your results in terms of limits beyond which a person cannot go, you will have to do so by imposing a ceiling on the cardiac output or on the alveolar ventilation. That would be realistic, of course, since there are limits to what the heart and lungs can do.

Finally, you might want to use the modified lung simulation program to study some interactions, such as exercise at altitude for lungs with different ventilation-perfusion mismatch parameters.

2.9 Annotated References

This chapter is strongly influenced by the following book, which emphasizes the importance of the ratio of ventilation to perfusion:

West, J.B.: *Ventilation/Blood Flow and Gas Exchange*, 5th Edition, Blackwell, Oxford, UK, 1990.

The following paper also addresses the specific issue of optimal gas transport, which is the dominant theme of this chapter:

Evans, J.W., Wagner, and West, J.B.: Conditions for reduction of pulmonary gas transfer by ventilation-perfusion inequality. *Journal of Applied Physiology* **36**, 535-567, 1974.

Exercises

2.1. Alternate derivation of the formula for the net gas transport

Verify that (2.2.12) can also be derived starting from the formula

$$f = (V_A)(c_I - c_A).$$

2.2. Derivation of an important identity

Show that

$$\sum_i Q_i(r_i - r_0) = 0,$$

where $r_i = (V_A)_i/Q_i$ and $r_0 = \sum_i (V_A)_i / \sum_i Q_i$.

2.3. Reduction of Alveolar Population to a Single Effective Alveolus

Consider a collection of alveoli that all have the same ratio of ventilation to perfusion. Show that the whole collection is functionally equivalent to a single alveolus.

2.4. An Extreme Mismatch of Ventilation and Perfusion

Consider a lung with two parts. In each part, the ventilation-perfusion ratio is constant. Suppose that the airflow and blood flow are as follows:

$$\begin{aligned}(V_A)_1 &= 5.0 \text{ liters/min}, & Q_1 &= 0.0 \text{ liters/min}, \\ (V_A)_2 &= 0.0 \text{ liters/min}, & Q_2 &= 5.0 \text{ liters/min}.\end{aligned}$$

Calculate f , $\langle P_A \rangle$, and $\langle P_a \rangle$ in terms of P_1 , P_v , σ , and kT . Explain your results in words.

2.5. Optimal Gas Transport in a Two-Compartment Lung

Consider a lung with two parts where in each part the ventilation-perfusion ratio is constant. Assume that $P_1 = 0$, $\sigma kT = 1$, $Q_1 = 2.0$ liters/min, $Q_2 = 3.0$ liters/min, and $(V_A)_v = (V_A)_1 + (V_A)_2 = 5.0$ liters/min.

1. Find a formula for E as a function of $(V_A)_1$. Note that $(V_A)_2 = (V_A)_v - (V_A)_1$.
2. Determine the value of $(V_A)_1$ that maximizes E by setting $dE/d(V_A)_1 = 0$. Interpret your result in terms of ventilation/perfusion ratios.
3. Evaluate $E_{\max}$, the value of E when $(V_A)_1$ has the optimal value determined above.
4. Plot E , $\langle P_a \rangle/P_v$, $\langle P_A \rangle/P_v$ as functions of $(V_A)_1$ over the interval $0 < (V_A)_1 < 5.0$. The mean partial pressures should both be plotted on the same graph. Explain how these graphs illustrate the theory of this chapter.

2.6. Arterial O_2 Saturation as a Function of r

A formula that fits reasonably well the measured relationship between the concentration of O_2 in blood and the partial pressure of O_2 in blood is as follows:

$$c = C(P) = c^* P^3 / [P_*^3 + P^3],$$

where c^* and P_* are constants.

1. Solve this equation to obtain $P = H(c)$.
2. Divide equation (2.6.8) by c^* to obtain

$$r(c_1/c^*) + (c_v/c^*) = (c_a/c^*) + (rH(c_a)/kTc^*).$$

Assume that $c_1 = c^*$, $c_v = 0.4c^*$, $kTc^* = 150$ mmHg, and $P_* = 25$ mmHg. Solve for c_a/c^* when $r = 0, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, \infty$, and plot your results as a graph of c_a/c^* as a function of r . The solution has to be obtained by some approximation method. For example, fix r and plot the right-hand side as a function of c_a . The solution is the value of c_a where the right-hand side crosses the (constant) left-hand side. Whatever solution method you use, it has to be carried out separately for each of the values of r that are listed above.

3

Control of Cell Volume and Electrical Properties of Cell Membranes

Cells contain proteins, nucleic acids, and other macromolecules that often carry many negative charges per molecule. This electrical charge is balanced by positive ions (especially potassium, denoted by K^+) that are dissolved in the intracellular water. These ions tend to draw water into the cells by osmosis, and the cells would swell and eventually burst if this osmotic effect were not offset by other factors.

The thick cell wall of plant cells can withstand considerable hydrostatic pressure, and an equilibrium is achieved in which the elevated pressure in the interior of the cell balances the osmotic effect.

Animal cells lack a cell wall, and their thin membranes cannot withstand any measurable pressure difference. Therefore, a more subtle mechanism is required to control cell volume. As we shall see, a by-product of this mechanism is that the membranes become electrically charged. In a spectacular example of evolutionary spin-off, the electrical potential difference across the cell membrane is used by nerve, muscle, and certain endocrine cells as the basis of a signaling mechanism.

3.1 Osmotic Pressure and the Work of Concentration

In Figure 3.1a, two chambers filled with water are separated by a rigid membrane. The pores make it possible for water to flow from one compartment to the other. Each compartment is equipped with a piston that can

be used to apply pressure to that compartment and push water through the pores.

The volume rate of flow through the membrane is linearly related to the applied pressure difference as shown in Figure 3.1b. Thus, the flow Q is given by the simple equation

$$P_1 - P_2 = RQ, \quad (3.1.1)$$

where R is the *flow resistance* of the membrane.

Now suppose that sugar is added to the compartment on the left, as in Figure 3.1c, and that the sugar molecules are too large to get through the pores in the membrane. The pressure-flow relation is now shifted from the origin, as shown in Figure 3.1d, so that the flow Q is now given by

$$P_1 - P_2 - kTc = RQ, \quad (3.1.2)$$

where c is the concentration of sugar, measured in molecules per unit volume.

According to equation (3.1.2), a pressure difference given by $P_1 - P_2 = kTc$ must be applied merely to prevent the water from flowing from right to left into the sugar solution. If no such pressure difference is applied (i.e., $P_1 = P_2$), then the flow Q is given by $Q = -kTc/R$, as though a pressure difference equal to $-kTc$ had been applied in the absence of the sugar solution. The pressure kTc is called the *osmotic pressure*. Note that it tends to draw water into the compartment containing the sugar solution. If we write $c = n/V_1$, where n is the number of sugar molecules and V_1 is the volume in which they are distributed, then we see that

$$P_{\text{osmotic}} = kTc = nkT/V_1, \quad (3.1.3)$$

so that the osmotic pressure is given by a formula that is identical to the ideal gas law!

What does an ideal gas have in common with a sugar solution? A gas is *ideal* if the interactions between its molecules can be neglected. The internal energy of an ideal gas is entirely kinetic (i.e., there is no energy of interaction between molecules). Similarly, in a (dilute) solution, the solute molecules wander around independently of each other. They can interact with the solvent, but only rarely (relatively speaking) do they encounter each other.

We now use these ideas to derive the formula (3.1.2) from energetic arguments. The assumptions are:

1. When the volume rate of flow through the membrane is Q , the rate at which work is *performed* by the pistons is given by

$$\dot{W}_{\text{piston}} = (P_1 - P_2)Q \quad (3.1.4)$$

(See Exercise 3.1).

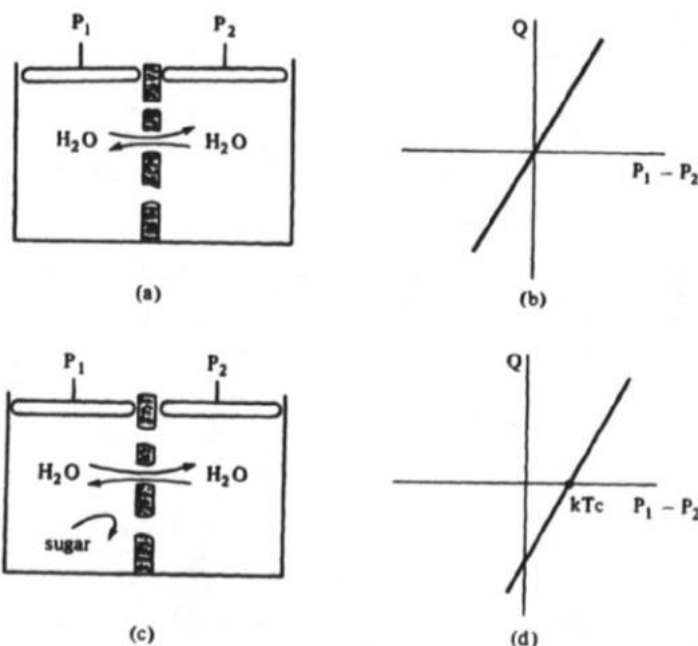


Figure 3.1. Osmotic pressure and flow. Panels (a) and (c) show a semipermeable membrane containing pores that allow water flow but are too small to admit sugar molecules. Pistons are provided to control the pressures P_1 and P_2 on the two sides of the membrane. The flow Q of water through the membrane is a function of the pressure difference $P_1 - P_2$ across the membrane, as plotted in panels (b) and (d). Positive Q denotes flow from compartment 1 to 2, and negative Q denotes flow in the opposite direction. With pure water on both sides of the membrane as in (a), the pressure-flow relation goes through the origin as in (b), but when sugar solution is used instead of pure water on one side of the membrane as in (c), the pressure-flow relation is offset, as in (d). The pressure difference required to prevent the flow of water in this situation is called the "osmotic pressure." It is equal to kTc , where k = Boltzmann's constant, T = absolute temperature, and c is the concentration of the sugar solution that was used (on one side of the membrane only). If there is no pressure difference across the membrane, then the water flows as though to dilute the sugar solution. Note the negative value of Q when $P_1 = P_2$ in (d). The magnitude of this unopposed osmotic flow is kTc/R , where R is the hydraulic resistance of the membrane.

2. The rate at which work is *required* to drive water through the pores in the membrane at the volume rate of flow Q is

$$\dot{W}_{\text{membrane}} = RQ^2. \quad (3.1.5)$$

3. The rate at which work is *required* to concentrate the sugar solution is the same as if the solute molecules were an ideal gas consisting of n molecules distributed in the volume V_1 . In that case, we would have $PV_1 = nkT$, or simply $P = kTc$, since $c = n/V_1$. The work of concentration would be given by

$$\dot{W}_{\text{concentration}} = -P\dot{V}_1 = -kTc \dot{V}_1. \quad (3.1.6)$$

We assume that this formula holds for a sugar solution as well as for an ideal gas. Since the flow through the membrane Q decreases the volume of solution on side 1 at the rate $\dot{V}_1 = -Q$, we have

$$\dot{W}_{\text{concentration}} = kTc Q. \quad (3.1.7)$$

4. Finally, by conservation of energy

$$\dot{W}_{\text{piston}} = \dot{W}_{\text{membrane}} + \dot{W}_{\text{concentration}}. \quad (3.1.8)$$

Substituting in (7.1.8) and dividing by Q , we obtain

$$(P_1 - P_2) = RQ + kTc, \quad (3.1.9)$$

which is the same as (3.1.2). Note that (3.1.1) is a special case that holds when $c = 0$.

What happens to (3.1.2) when sugar is present on both sides of the membrane? The reader can probably guess that the flow of water is then given by

$$(P_1 - P_2) - kT(c_1 - c_2) = RQ, \quad (3.1.10)$$

where c_1 and c_2 are the two concentrations (see Exercise 3.2). Thus, the concentration *difference* between the two solutions determines their *relative osmotic pressure*.

Finally, what happens when a mixture of solutes is present? It should be clear from the foregoing derivation that the crucial determinant of osmotic pressure is the *number* of independently moving solute particles per unit volume. Thus, in a mixture of solutes, the *partial osmotic pressures* add up, just as the partial pressures add up to determine the total pressure of a mixture of gases. An important example of this is the case of ionic solutions. If NaCl is dissolved in water, the Na^+ and Cl^- ions move independently, and the osmotic pressure of the solution is

$$kT[\text{Na}^+] + kT[\text{Cl}^-],$$

where $[]$ denotes concentration.

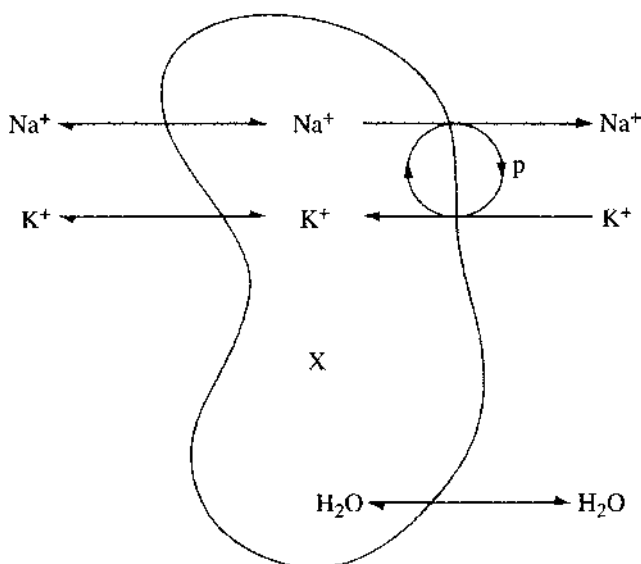


Figure 3.2. A simplified model for control of cell volume with electrical effects ignored. Sodium ions (Na^+), potassium ions (K^+), and water (H_2O) diffuse across the cell membrane. Moreover, sodium ions are pumped out of the cell in a 1:1 exchange with potassium ions, at the pump rate p . Finally, there are X macromolecules trapped within the cell. The osmotic effect of these trapped macromolecules is to draw water into the cell. This must somehow be compensated by the pump, but how can this help if each sodium ion pumped out is replaced by one potassium ion pumped in?

3.2 A Simple Model of Cell Volume Control

In this section we describe a simplified model of the mechanism that animal cells use to achieve osmotic balance and control their volume. The simplification is that we ignore all electrical effects. This will be remedied in later sections.

Consider the cell shown in Figure 3.2. In its interior there are some molecules that cannot get through the cell membrane. The number of these molecules is designated by X , so their concentration is X/V , where V is the volume of the cell. The cell also contains Na^+ and K^+ ions. These ions can diffuse through the cell membrane, which is also permeable to water.

In addition, the cell membrane is equipped with a “pump” that pumps internal Na^+ ions out of the cell in exchange for external K^+ ions. This active process is needed to maintain a constant cell volume, as we shall see.

We make the assumption that the pressure inside the cell equals the pressure outside it. That is, we assume that the cell membrane cannot withstand any mechanical pressure difference. In this respect, the cell membrane differs from the rigid membrane considered earlier.

The equations for the net flux of Na^+ , K^+ , and H_2O in the outward direction are

$$f_{\text{Na}} = \alpha_{\text{Na}}([\text{Na}^+]_i - [\text{Na}^+]_o) + p, \quad (3.2.1)$$

$$f_{\text{K}} = \alpha_{\text{K}}([\text{K}^+]_i - [\text{K}^+]_o) - p, \quad (3.2.2)$$

$$R_{(\text{H}_2\text{O})}Q = -kT([\text{Na}^+]_i + [\text{K}^+]_i + \frac{X}{V} - [\text{Na}^+]_o - [\text{K}^+]_o). \quad (3.2.3)$$

In these equations f stands for flux (ions per unit time), and p is the pump rate. The quantities α_{Na} and α_{K} are the (passive) permeabilities of the cell to Na^+ and K^+ , and $R_{(\text{H}_2\text{O})}$ is the resistance of the cell membrane to the flow of water. The subscripts i and o designate the inside and the outside of the cell, respectively. Equation (3.2.3) is the osmotic pressure law derived in the previous section, except that here we have $P_i = P_o$, as discussed above.

Now suppose that the cell is in a *steady state*, which means that the net flux of all quantities is zero. Then (3.2.1) through (3.2.3) become

$$[\text{Na}^+]_i - [\text{Na}^+]_o = -\frac{p}{\alpha_{\text{Na}}}, \quad (3.2.4)$$

$$[\text{K}^+]_i - [\text{K}^+]_o = +\frac{p}{\alpha_{\text{K}}}, \quad (3.2.5)$$

$$[\text{Na}^+]_i + [\text{K}^+]_i + \frac{X}{V} = [\text{Na}^+]_o + [\text{K}^+]_o. \quad (3.2.6)$$

Equations (3.2.4) and (3.2.5) give the concentration differences produced by the pump, whereas (3.2.6) simply asserts that the total concentration of solute must be the same inside and outside the cell. Otherwise, water would flow. Substituting (3.2.4) and (3.2.5) in (3.2.6), we get

$$\frac{X}{V} = p\left(\frac{1}{\alpha_{\text{Na}}} - \frac{1}{\alpha_{\text{K}}}\right) = p\frac{\alpha_{\text{K}} - \alpha_{\text{Na}}}{\alpha_{\text{K}}\alpha_{\text{Na}}}. \quad (3.2.7)$$

That is,

$$V = \frac{X}{p} \frac{\alpha_{\text{K}}\alpha_{\text{Na}}}{\alpha_{\text{K}} - \alpha_{\text{Na}}}. \quad (3.2.8)$$

This result has some interesting implications. First, we get a finite positive¹ cell volume V only if $\alpha_{\text{K}} > \alpha_{\text{Na}}$. That is, the cell membrane has to be more permeable to K^+ than to Na^+ . This asymmetry is necessary because

¹The negative value of V that is obtained when $\alpha_{\text{K}} < \alpha_{\text{Na}}$ has no physical significance; it merely tells us that there is no steady-state volume in this situation. If we studied the time-dependent equations for the case $\alpha_{\text{K}} < \alpha_{\text{Na}}$, we would find that V increases indefinitely. In practice, the cell would swell until it burst.

the pump pushes Na^+ out and K^+ in. Diffusion through the membrane partially offsets the work of the pump. Therefore, when $\alpha_K > \alpha_{\text{Na}}$, the net effect of the pump is to reduce the concentration of ions in the cell. This offsets the molecules that are trapped in the cell and makes it possible for the cell to achieve osmotic balance and a steady state.

The second interesting point is the dependence of V on p . Equation (3.2.8) shows that the cell volume is inversely proportional to the pump rate. As $p \rightarrow 0$, $V \rightarrow \infty$. This shows the necessity of the pump for holding the cell volume down. Dead cells swell.

Finally, we note that V is proportional to X . Thus, the volume of our model cell is automatically adjusted to the number of protein and other macromolecules that are trapped within the cell's membrane. As the cell produces more molecules, say through growth, its volume automatically increases proportionally.

3.3 The Movement of Ions Across Cell Membranes

A battery is shown in Figure 3.3 to be driving an ionic current across a membrane. The purpose of the battery is to specify the voltage difference across the membrane (also known as the membrane potential) so that we can study the relationship between the membrane potential and the ionic current. Later, we shall see how the membrane potential is determined in the physiological situation, where there is, of course, no battery applied to the system from the outside as in Figure 3.3. The ionic current in Figure 3.3 is carried by a particular ionic species designated by A^{z+} . This notation means that each ion carries z elementary (positive) charges. That is, its charge is zq , where q is the charge on a proton. We are interested here in Na^+ and K^+ , for which $z = +1$, and in Cl^- , for which $z = -1$.

The battery establishes a potential difference

$$v = v_1 - v_2,$$

and we want to derive the relationship between this potential difference and the current I_A flowing through the membrane.

The relationship shown in Figure 3.3b can be derived from energetic arguments similar to the ones used in Section 3.1. The assumptions that we need are:

1. The work *performed* by the battery per unit time is

$$\dot{W}_{\text{battery}} = I_A(v_1 - v_2) \equiv I_A v. \quad (3.3.1)$$

This follows from the definitions of current and potential, since current is charge per unit time and potential (voltage) is work per unit charge.

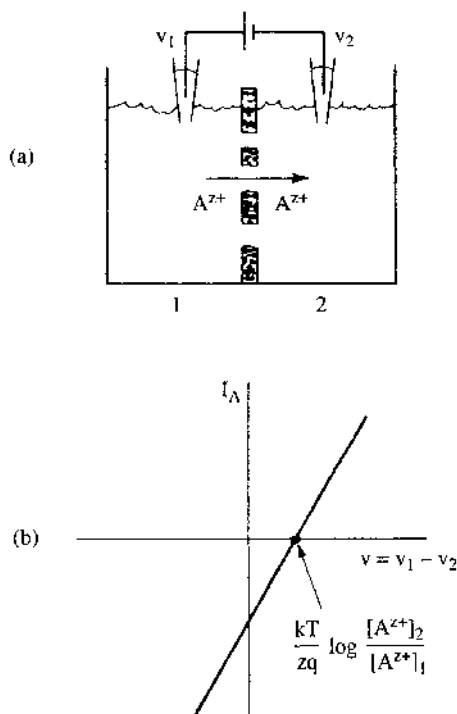


Figure 3.3. Ionic current across a membrane. In panel (a), a generic ion A^{z+} carries z positive charges. A membrane containing channels through which this ion can flow separates two solutions of the ion, in which the concentrations ($[A^{z+}]_1$ and $[A^{z+}]_2$) may be different. A voltage difference $v = v_1 - v_2$ is applied by a battery. The current carried by the ion A^{z+} is denoted I_A and is plotted in panel (b). The result is a straight line that does not go through the origin (unless $[A^{z+}]_1 = [A^{z+}]_2$). The arrow indicates the equilibrium potential for the ion A^{z+} , i.e., the voltage difference that is just sufficient to prevent any net current from being carried by this species of ion.

2. The rate at which work is *required* to drive current through the membrane is

$$\dot{W}_{\text{membrane}} = R_A I_A^2. \quad (3.3.2)$$

The reader should be warned that this assumption is correct only in particularly simple situations. The actual form of $\dot{W}_{\text{membrane}}$ depends on the details of the mechanism that transports ions across the membrane. We make the assumption (3.3.2) because it leads to the simplest possible current-voltage relation (as shown in Figure 3.3b) and because this simple relationship is actually observed in some biologically important membranes, such as the squid axons in which the fundamental properties of nerve membranes were discovered.

3. The rate at which work is *required* to change the concentration of the ions from $[A^{z+}]_1$ to $[A^{z+}]_2$ as the ions are driven across the membrane is the same as if the ions were the molecules of an ideal gas. This amount of work is $kT \log([A^{z+}]_2/[A^{z+}]_1)$ per ion transported across the membrane. Since the number of ions transported across the membrane per unit time is $I_A/(zq)$, we have

$$\dot{W}_{\text{concentration}} = I_A \frac{kT}{zq} \log \left(\frac{[A^{z+}]_2}{[A^{z+}]_1} \right). \quad (3.3.3)$$

By the conservation of energy law, we have that

$$\dot{W}_{\text{battery}} = \dot{W}_{\text{membrane}} + \dot{W}_{\text{concentration}}. \quad (3.3.4)$$

Substituting into this and dividing by I_A , we obtain

$$v = R_A I_A + \frac{kT}{zq} \log \left(\frac{[A^{z+}]_2}{[A^{z+}]_1} \right). \quad (3.3.5)$$

Dividing by R_A and rearranging terms gives

$$I_A = g_A \left(v - \frac{kT}{zq} \log \left(\frac{[A^{z+}]_2}{[A^{z+}]_1} \right) \right), \quad (3.3.6)$$

where $g_A = 1/R_A$ is called the *membrane conductance*. This is the current-voltage relationship plotted in Figure 3.3b. The quantity

$$E_A = \frac{kT}{zq} \log \left(\frac{[A^{z+}]_2}{[A^{z+}]_1} \right) \quad (3.3.7)$$

is called the *equilibrium potential* of the ion in question. When $v = E_A$, the current is zero. It is worth mentioning here that this expression for the equilibrium potential holds much more generally than the linear current-voltage relation derived here. If assumption (2) were changed, then the current-voltage relation would become nonlinear, but it would still have the same equilibrium potential.

Finally, we apply the results of this section to a situation where several ions are present simultaneously. In that case, we assume that the different

ions cross the membrane through different channels; this has been demonstrated experimentally in many cases. Each ion has its own equilibrium potential determined by its concentration ratio, but all of the ions feel the same membrane potential v . The total current is simply the sum of the individual ionic currents, and each ionic current is given by an expression similar to (3.3.6).

3.4 The Interaction of Electrical and Osmotic Effects

We now consider the model cell of Figure 3.4, which is an extension of the model that we considered in Section 3.2. Because we now include electrical effects, it is necessary to take into account the Cl^- ions, even though these are not actively transported across the membrane. Also, we now assume that each of the x molecules that are trapped inside the cell carries z negative charges. We therefore designate these molecules by the chemical symbol X^{z-} .

It turns out that the electrical effect of these trapped charges produces an indirect osmotic effect because of the positive ions that are needed to balance the trapped negative charge. This indirect effect is much more important than the direct osmotic effect of X^{z-} when z is large.

We begin by stating the equations of the model cell. Then we discuss the various assumptions contained in these equations. First, we have expressions for the three ionic currents (positive is outward):

$$I_{\text{Na}} = g_{\text{Na}}(v - (kT/q) \log([Na^+]_o/[Na^+]_i)) - pq, \quad (3.4.1)$$

$$I_{\text{K}} = g_{\text{K}}(v - (kT/q) \log([K^+]_o/[K^+]_i)) - pq, \quad (3.4.2)$$

$$I_{\text{Cl}} = g_{\text{Cl}}(v + (kT/q) \log([Cl^-]_o/[Cl^-]_i)). \quad (3.4.3)$$

Next, we have an equation for the membrane potential $v = v_i - v_o$ in terms of the excess charge in the cell:

$$Cv = q(V[Na^+]_i + V[K^+]_i - V[Cl^-]_i - xz), \quad (3.4.4)$$

where C is the capacitance of the cell membrane, which has the units of charge divided by voltage.

Finally, we have the equation for the osmotic flux of water (again, positive is outward)

$$R_{H_2O}Q = -kT([Na^+]_i + [K^+]_i + [Cl^-]_i + \frac{x}{V} - [Na^+]_o - [K^+]_o - [Cl^-]_o). \quad (3.4.5)$$

The first three equations are the current-voltage relations of the previous section with extra terms added for the currents produced by the pump. If p is the pump rate, say in terms of ions exchanged per unit time, then pq is the Na^+ current produced by the pump and $-pq$ is the corresponding

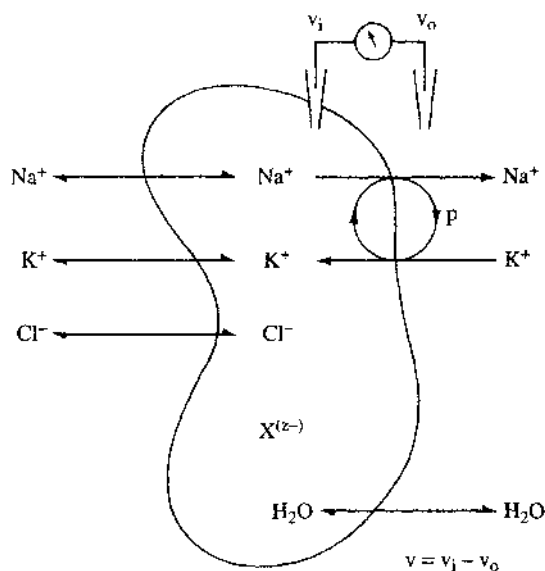


Figure 3.4. Control of cell volume with electrical effects included. At top, a voltmeter measures the potential difference $v = v_i - v_o$ across the cell membrane, where v_i is the inside potential and v_o is the outside potential. Sodium ions (Na^+), potassium ions (K^+), chloride ions (Cl^-), and water molecules (H_2O) all move passively through channels in the cell membrane. In addition to these passive fluxes, sodium ions are pumped out of the cell in exchange for potassium ions at the pump rate p . Macromolecules trapped within the cell carry, on average, z negative charges, and are therefore denoted by X^{z-} .

K^+ current. Note that the net pump current is zero. We have assumed for simplicity that the pump ratio is 1:1. That is, one Na^+ ion is pumped out of the cell for every K^+ ion that flows inwards via the pump. Equations (3.4.1) and (3.4.2) could easily be changed to accommodate other pump ratios. In the case of a 3:2 pump, for example, equation (3.4.1) would contain the term $3pq$, and equation (3.4.2) would contain the term $-2pq$ (instead of pq and $-pq$, respectively).

Since Cl^- is negatively charged, the current I_{Cl} is in the opposite direction to the physical flux of chloride. This explains the change in sign in (3.4.3). It is noteworthy that this change of sign applies only to the equilibrium potential and not to the membrane potential term. This is because the qualitative effect of membrane potential on ionic current is the same for both positive and negative ions. To understand this, recall that a flow of positive charges in one direction and a flow of negative charges in the other are considered as electrical currents flowing in the same direction, that of the positive charges. Thus, a membrane potential that generates positive outward current by pushing positive charges out of the cell also generates positive outward current by pulling negative charges into the cell.

Equation (3.4.4) is based on the assumption that the cell acts like a capacitor, so that the membrane potential is proportional to the excess charge. It is important to remark that the excess charge is *very* small compared to the total positive or negative charge. The excellent approximation of setting the excess charge equal to zero is called the condition of *electroneutrality*. It can be obtained mathematically by letting $C \rightarrow 0$ in (3.4.4); we shall do this later.

Equation (3.4.5) is the expression for the osmotic flux of water, provided that we make the assumption, as in Section 3.2, that the membrane cannot withstand any hydrostatic pressure difference, so that the internal and external pressures must be equal.

As in Section 3.2, we seek a *steady state* in which the currents I_{Na} , I_K , and I_{Cl} and the water flux Q are all equal to zero. In addition, we make the following simplifications:

1. Let $C = 0$. As remarked above, this yields the condition of electroneutrality, an excellent approximation.
2. Let $xz = N$, so that N is the total number of negative charges that are trapped within the cell. Now consider the limit as $z \rightarrow \infty$ and $x \rightarrow 0$ with N fixed. This expresses the fact that the cell contains a relatively small number of large molecules with a large number of negative charges per molecule. The effect of this limit is that the direct osmotic effect of X^{z-} drops out of (3.4.5), but the electrical effect of the negative charge remains in (3.4.4).

When these simplifications have been made, our equations can be rewritten as follows:

$$0 = g_{\text{Na}}(v - (kT/q) \log([Na^+]_o/[Na^+]_i)) + pq, \quad (3.4.6)$$

$$0 = g_{\text{K}}(v - (kT/q) \log([K^+]_o/[K^+]_i)) - pq, \quad (3.4.7)$$

$$0 = g_{\text{Cl}}(v + (kT/q) \log([Cl^-]_o/[Cl^-]_i)), \quad (3.4.8)$$

$$0 = [Na^+]_i + [K^+]_i - [Cl^-]_i - N/V, \quad (3.4.9)$$

$$0 = [Na^+]_i + [K^+]_i + [Cl^-]_i - [Na^+]_o - [K^+]_o - [Cl^-]_o. \quad (3.4.10)$$

We regard these as five simultaneous equations in the five unknowns $[Na^+]_i$, $[K^+]_i$, $[Cl^-]_i$, v , and V . The external concentrations are taken to be known. It is important to notice, however, that the given external concentrations must satisfy the equation

$$[Na^+]_o + [K^+]_o = [Cl^-]_o \quad (3.4.11)$$

so that the external solution is electrically neutral.

To solve the system (3.4.6) through (3.4.10), we proceed as follows. First, we use (3.4.6) through (3.4.8) to express the internal concentrations in terms of the membrane potential v :

$$[Na^+]_i/[Na^+]_o = \exp(-(qv/kT) - (pq^2/g_{\text{Na}}kT)), \quad (3.4.12)$$

$$[K^+]_i/[K^+]_o = \exp(-(qv/kT) + (pq^2/g_{\text{K}}kT)). \quad (3.4.13)$$

$$[Cl^-]_i/[Cl^-]_o = \exp(qv/kT). \quad (3.4.14)$$

To simplify the notation, let

$$\gamma = \exp(qv/kT), \quad (3.4.15)$$

$$\beta_{\text{Na}} = \exp(-pq^2/g_{\text{Na}}kT), \quad (3.4.16)$$

$$\beta_{\text{K}} = \exp(pq^2/g_{\text{K}}kT), \quad (3.4.17)$$

so that

$$[Na^+]_i = \gamma^{-1}\beta_{\text{Na}}[Na^+]_o, \quad (3.4.18)$$

$$[K^+]_i = \gamma^{-1}\beta_{\text{K}}[K^+]_o, \quad (3.4.19)$$

$$[Cl^-]_i = \gamma[Cl^-]_o. \quad (3.4.20)$$

These expressions can be substituted into (3.4.9) and (3.4.10) to obtain a pair of equations for γ and V . Note that γ determines v . After dividing through these equations by $2[Cl^-]_o$ and using (3.4.11), we get

$$0 = \beta/(2\gamma) - \gamma/2 - b, \quad (3.4.21)$$

$$0 = \beta/(2\gamma) + \gamma/2 - 1, \quad (3.4.22)$$

where

$$b = \frac{N}{2[Cl^-]_o V} \quad (3.4.23)$$

and

$$\beta = (\beta_{\text{Na}}[\text{Na}^+]_o + \beta_{\text{K}}[\text{K}^+]_o)/([\text{Na}^+]_o + [\text{K}^+]_o). \quad (3.4.24)$$

We shall use (3.4.21), which is the equation of electroneutrality, to express γ in terms of b . This means that we are finding the potential in terms of the cell volume, which is temporarily regarded as given. Later, we use the equation of osmotic balance (3.4.22) to determine the volume.

Equation (3.4.21) can be written as the quadratic

$$\gamma^2/2 + b\gamma - \beta/2 = 0. \quad (3.4.25)$$

The positive solution of (3.4.25) is

$$\gamma = -b + \sqrt{b^2 + \beta}. \quad (3.4.26)$$

We are forced to take the positive solution here because $\gamma = \exp(qv/kT)$, which is positive for all v . Note that

$$1/\gamma = \frac{b + \sqrt{b^2 + \beta}}{\beta}. \quad (3.4.27)$$

Finally, we can determine b , and hence the cell volume V , by substituting (3.4.26) and (3.4.27) in the equation of osmotic balance (3.4.22). The result is

$$\sqrt{b^2 + \beta} = 1, \quad (3.4.28)$$

$$b = \sqrt{1 - \beta}, \quad (3.4.29)$$

$$V = \frac{N}{2[\text{Cl}^-]_o \sqrt{1 - \beta}}. \quad (3.4.30)$$

Thus, we only get real, finite volumes if $\beta < 1$. When $\beta \geq 1$, there is no steady volume possible, and the cell swells until it bursts.

We next investigate the dependence of β upon the pump rate p .

We have

$$\beta(p) = \frac{[\text{Na}^+]_o \exp(-pq^2/g_{\text{Na}}kT) + [\text{K}^+]_o \exp(pq^2/g_{\text{K}}kT)}{[\text{Na}^+]_o + [\text{K}^+]_o}. \quad (3.4.31)$$

Clearly, $\beta(0) = 1$ and $\beta(p) \rightarrow +\infty$ as $p \rightarrow +\infty$.

Differentiating this expression, we find that

$$\frac{d\beta}{dp} = a(-[\text{Na}^+]_o \beta_{\text{Na}}/g_{\text{Na}} + [\text{K}^+]_o \beta_{\text{K}}/g_{\text{K}}), \quad (3.4.32)$$

so

$$\frac{d\beta}{dp}(0) = a(-[\text{Na}^+]_o/g_{\text{Na}} + [\text{K}^+]_o/g_{\text{K}}), \quad (3.4.33)$$

where a is independent of p . We now impose the condition

$$[\text{Na}^+]_o/g_{\text{Na}} > [\text{K}^+]_o/g_{\text{K}}, \quad (3.4.34)$$

so that $d\beta/dp < 0$ at $p = 0$. With this inequality satisfied, there is a range of pump rates that lead to finite-volume, steady-state solutions, since $\beta(p) < 1$ for small enough values of p . If (3.4.34) is not satisfied, it is easy to see that $\beta > 1$ for all $p > 0$, and no such solutions exist.

When (3.4.34) holds, it is interesting to note that β has a minimum at a value of p found by setting $d\beta/dp = 0$. The result is

$$p_{\text{opt}} = (kT/q^2)(g_{\text{Na}}g_{\text{K}}/(g_{\text{Na}} + g_{\text{K}})) \log([Na^+]_o g_{\text{K}}/g_{\text{Na}}[K^+]_o). \quad (3.4.35)$$

From (3.4.30), we see that this is also the value of p that minimizes the cell volume V . It seems reasonable to speculate that real cells operate their pumps at a rate that is near p_{opt} because at this pump rate, the cell volume is insensitive to small changes in the pump rate. The quantitative relationship between cell volume and pump rate is further investigated in Exercise 3.5.

At a fixed pump rate, we can see from (3.4.30) and (3.4.31) that the cell volume depends on the number of trapped negative charges and also on the ionic concentrations in the external medium. During cell growth, as more macromolecules carrying trapped negative charges are produced, the cell volume increases proportionately. The dependence of cell volume on external ionic concentrations can be described as follows: First, note that β can be rewritten in terms of the ratio $[Na^+]_o/[K^+]_o$. When this ratio is fixed, the cell volume is inversely proportional to the total external ion concentration $2[Cl^-]_o = [Cl^-]_o + [Na^+]_o + [K^+]_o$. Thus, cells swell in a more dilute external environment.

Moreover, if the ratio $[Na^+]_o/[K^+]_o$ is reduced, then β increases and the cell volume also increases. In fact, as we have remarked above, if the ratio $[Na^+]_o/[K^+]_o$ is so low that (3.4.34) is violated, then the cell volume becomes infinite; i.e., the cell bursts. Thus, the cell volume is sensitive both to the total external ion concentration and also to the composition of the external medium.

Finally, with the cell volume known, we can determine the steady state, or resting, membrane potential v . To do this, we substitute (3.4.29) in (3.4.26). The result is

$$\gamma = -\sqrt{1 - \beta} + 1, \quad (3.4.36)$$

so that

$$v = (kT/q) \log \gamma = (kT/q) \log(1 - \sqrt{1 - \beta}). \quad (3.4.37)$$

Since $\gamma < 1$, $v < 0$. Thus, the negative membrane potential of animal cells is a by-product of the mechanism that regulates cell volume. In the next section we shall see how the membrane potential can be transiently modified by changes in the passive conductances g_{Na} and g_{K} . This is the basis of the signaling mechanism that is used for communication, most notably by nerve and muscle cells, but also by certain endocrine cells.

3.5 The Hodgkin-Huxley Equations for the Nerve Action Potential

The basic mechanism for control of cell volume that we have outlined in this chapter operates in all animal cells. But some of these cells, most notably neurons, have developed the ability to make brief changes in the membrane potential by adjusting the conductances g_K and g_{Na} . The resulting changes in membrane potential typically take the form of a stereotyped, pulse-shaped waveform, which is known as the "action potential." The level of activity of a neuron is measured in terms of the number of action potentials that it generates per unit time. The electrical activity of neurons is responsible for sensation, action, and thought, so it is no exaggeration to say that we are here concerned with a fundamental and important physiological mechanism.

Changes in the membrane conductances g_K and g_{Na} lead to departures from the steady state, so that I_{Na} , I_K and I_{Cl} are no longer equal to zero. Because the changes in question are rapid, taking place on a time scale of milliseconds, the capacitance of the cell membrane cannot be neglected, as it was when we were considering the slower changes associated with cell volume regulation. Under these circumstances of rapid conductance change, the fundamental equation governing the potential difference v across the cell membrane can be derived as follows. First, differentiate equation (3.4.4) with respect to time to obtain

$$C\dot{v} = \frac{d}{dt}(Vq[Na^+]_i) + \frac{d}{dt}(Vq[K^+]_i) + \frac{d}{dt}(V(-q)[Cl^-]_i). \quad (3.5.1)$$

Note that $(Vq[Na^+]_i)$ is the amount of charge inside the cell that is contributed by the Na^+ ions. Therefore, the rate of change of this quantity is $-I_{Na}$ (since our sign convention is that outward currents are considered positive). Similarly, the rate of change of $(Vq[K^+]_i)$ is $-I_K$, and the rate of change of $(V(-q)[Cl^-]_i)$ is $-I_{Cl}$. Therefore, taking into account the formulae for I_{Na} , I_K , and I_{Cl} (equations 3.4.1-3.4.3), we have

$$C\dot{v} + g_{Na}(v - E_{Na}) + g_K(v - E_K) + g_{Cl}(v - E_{Cl}) = 0, \quad (3.5.2)$$

where

$$E_{Na} = \frac{kT}{q} \log \left(\frac{[Na^+]_o}{[Na^+]_i} \right), \quad (3.5.3)$$

$$E_K = \frac{kT}{q} \log \left(\frac{[K^+]_o}{[K^+]_i} \right), \quad (3.5.4)$$

$$E_{Cl} = \frac{kT}{-q} \log \left(\frac{[Cl^-]_o}{[Cl^-]_i} \right). \quad (3.5.5)$$

Note that the pump currents have canceled out. This is because we have made the simplifying assumption of a 1:1 Na^+ - K^+ exchange. With other ratios, the pump will make a contribution to equation (3.5.1).

Now we come to an important simplifying assumption: that the cell is so large, and the fluxes of ions are so small, that the internal concentrations of the various ions change only slightly during an action potential. This is a point that is often missed by the student, who imagines large swings of ion concentrations occurring during an action potential. In fact, the changes in concentration are so small that even a cell in which the Na^+ - K^+ exchange pump mechanism has been poisoned can generate hundreds of action potentials before the concentrations "run down." Mathematically, we express this by treating the quantities E_{Na} , E_{K} , and E_{Cl} as given constants. Their values will be specified later. For now, we just note the qualitative relationship

$$E_{\text{K}} < E_{\text{Cl}} < 0 < E_{\text{Na}}. \quad (3.5.6)$$

For purposes of discussion, it is helpful to rewrite equation (3.5.2) in the following form:

$$C\dot{v} + g(v - E) = 0, \quad (3.5.7)$$

where

$$g = g_{\text{Na}} + g_{\text{K}} + g_{\text{Cl}} \quad (3.5.8)$$

is the total membrane conductance, and where

$$E = \frac{g_{\text{Na}}E_{\text{Na}} + g_{\text{K}}E_{\text{K}} + g_{\text{Cl}}E_{\text{Cl}}}{g_{\text{Na}} + g_{\text{K}} + g_{\text{Cl}}} \quad (3.5.9)$$

is a weighted average of the equilibrium potentials E_{Na} , E_{K} , and E_{Cl} , the weight being given in each case by the corresponding conductance. Given these definitions of g and E , it is easy to check that equations (3.5.7) and (3.5.2) are equivalent.

According to equation (3.5.7), the membrane potential v is always approaching the instantaneous value of E . Whenever $v < E$, we have $\dot{v} > 0$; and when $v > E$, we have $\dot{v} < 0$. It is therefore significant that the cell can adjust E simply by changing the membrane conductances. In the cell at rest, v and E are both close to E_{Cl} . Early in an action potential, Na^+ channels open (how this happens will be discussed below), dramatically increasing g_{Na} . This makes E become close to E_{Na} , and v follows, actually reversing sign in the process. Later on in the action potential, the Na^+ channels close and K^+ channels open. This brings E close to E_{K} , and the membrane potential v again follows E , thus becoming more negative than the resting potential of the cell. Finally, all of the conductances come back to normal, and E and v return to their resting values, close to E_{Cl} .

We now turn to the important question of how the Na^+ and K^+ conductances actually change. This was discovered by Hodgkin and Huxley,

through experiments on the squid giant axon. (The axon is a fiber that projects out from the cell body of a neuron and carries the signals generated by that neuron to other, possibly distant, locations.) These experimenters employed a "voltage clamp," which is an electronic circuit that uses feedback to force the membrane potential to follow a command signal specified by the experimenter. The current that is needed to keep the voltage tracking the command signal is recorded. Hodgkin and Huxley used command signals in the form of voltage steps, so that in between jumps, the voltage would be constant. The significance of this is that when v is constant, $C\dot{v} = 0$. Thus, except during the jumps in voltage, all of the current generated by the voltage clamp circuit would actually flow through the membrane channels; none of it would be involved in charging or discharging the membrane itself. A constant voltage is also useful for another reason. As we shall see, the properties of the membrane channels turn out to be voltage dependent. These properties are therefore easiest to investigate under constant voltage conditions. It was part of the genius of Hodgkin and Huxley to realize that voltage was the right variable to control.

Another experimental simplification made by Hodgkin and Huxley was to thread a silver wire down the length of the nerve axon that they were investigating. (Only in a *giant* axon would this be possible!) Since silver is an excellent conductor of electricity, this has the effect of eliminating any voltage differences that might otherwise develop between one location and another along the length of the axon. Mathematically speaking, this "space clamp" has the effect of ensuring that the membrane potential v is a function only of the time t , and not also of the position x along the axon. A major triumph of the theory introduced by Hodgkin and Huxley is that it was able to predict the behavior of the nerve action potential in an axon without a space clamp, despite the fact that the theory was derived from experiments performed with the space clamp in place. Specifically, Hodgkin and Huxley were able to predict the speed with which the action potential propagates along the (non-space-clamped) axon. Propagation, of course, cannot even be discussed without introducing the spatial variable x . Here, however, we shall restrict consideration to the space-clamped case. For this reason, the version of the Hodgkin-Huxley equations that we consider will be a system of *ordinary* differential equations, with time as the sole independent variable. This version is adequate for modeling essentially all phenomena associated with the nerve action potential *except* for the propagation of the action potential as a traveling wave. For that, the Hodgkin-Huxley *partial* differential equations are needed, and these are beyond the scope of this book.

Having ensured by means of voltage clamp and space clamp that all of the measured current would flow through membrane channels, Hodgkin and Huxley then went a step further and used pharmacological intervention to block one channel type or another, and thereby obtain separately the different ionic currents. By studying how these currents varied over

time in response to various voltage steps, they arrived at a mathematical description of the individual currents. We shall present this description in a deductive manner, as though it were derived from first principles, which we shall state. But of course, there was no way to know what those principles would turn out to be until the measurements had been made, and even now, no one is sure what is happening in detail within the channels that control the different ionic currents. Thus, the reader should take the following description with a grain of salt. It is a theory of membrane channels that is consistent with measurements made on the squid giant axon, but its details are not to be taken too seriously or applied too universally, since there are many different types of membrane channels in different types of neurons with quite different properties, and since even in the case of the squid giant axon, there may be many different models of the membrane channels that would be consistent with the experimental data.

The great achievement of Hodgkin and Huxley was to characterize the behavior of the Na^+ and K^+ channels of the squid giant axon. We shall begin with the K^+ channel, which is simpler. Its behavior can be derived from the following postulates:

1. Each K^+ channel has four gates. Each gate can be OPEN or CLOSED. The K^+ channel as a whole is OPEN if and only if all four of its gates are OPEN.
2. All four gates within a K^+ channel are identical. (And since we assume that there is only one type of K^+ channel, all of the different K^+ channels are identical, too.)
3. The different gates within a K^+ channel operate independently of one another. (And of course, the different gates in different K^+ channels also operate independently of one another.)
4. The rate constant (probability per unit time) for opening or closing a gate of a K^+ channel is a specified function of voltage for the opening rate and another specified function of voltage for the closing rate.

Note that the term "OPEN" in the foregoing refers to the conducting state of a gate or a channel. This is opposite to the standard electrical terminology (which comes from switches that use air as an insulator), but when talking about channels and gates it seems irresistible to say that they are "OPEN" when they allow ionic current to flow.

Let us now reduce the foregoing postulates to a mathematical theory. Consider the whole population (which we assume is large) of K^+ channels, and the population of gates within those channels, which we shall call K^+ gates. Let $n(t)$ be the fraction of K^+ gates that are in the OPEN state at time t . Alternatively, one can say that $n(t)$ is the probability that a given K^+ gate is OPEN at time t . Then the dynamics of $n(t)$ are governed by

the following equation:

$$\frac{dn}{dt} = \alpha_n(v)(1 - n) - \beta_n(v)n. \quad (3.5.10)$$

Here $\alpha_n(v)$ is the opening rate constant for the K^+ gates, which is a function of voltage. Similarly, $\beta_n(v)$ is the closing rate constant for the K^+ gates, which is also a function of voltage. These functions will be specified later. For now we just mention that $\alpha_n(v)$ is an increasing function and that $\beta_n(v)$ is a decreasing function. Thus, increasing the membrane potential (inside relative to outside) encourages K^+ gates to open and discourages them from closing, thus increasing the K^+ conductance. (We are speaking here of an algebraic increase, i.e., a change that makes the inside potential less negative or more positive relative to the outside potential. Since the rest potential is negative, an increase in the membrane potential starting from rest is, in fact, a decrease in the magnitude of the potential, and for this reason such an increase is often called "depolarization.")

At any particular v , $\alpha_n(v)$ and $\beta_n(v)$ are just numbers, with units of reciprocal time, i.e., rate. The opening rate constant $\alpha_n(v)$ is multiplied by $(1 - n)$ in equation (3.5.10) because $(1 - n)$ is the fraction of CLOSED K^+ gates, and a gate has to be CLOSED in order to open. For example, if $n = 1$ at some instant, then all of the gates are already OPEN, so the rate of opening has to be zero at that particular instant. Similarly, the closing rate constant $\beta_n(v)$ is multiplied by n in equation (3.5.10) because n is the fraction of OPEN K^+ gates, and a gate has to be OPEN in order to close.

To complete the theory of the K^+ channel, we need to specify the K^+ conductance, g_K , in terms of the fraction n of open K^+ gates. This is where the assumed independence of the K^+ gates comes into the picture. Since there are four *independent* gates in each channel, and since n is the probability that each of them is OPEN, then the probability that all four are OPEN (and hence that the channel is OPEN) is equal to n^4 . Thus g_K should be proportional to n^4 . Hodgkin and Huxley used the notation $\bar{g}_K$ for the constant of proportionality. Thus

$$g_K = \bar{g}_K n^4. \quad (3.5.11)$$

Note that $\bar{g}_K$ is the K^+ conductance in the hypothetical situation of all K^+ gates in all K^+ channels being open. The value of the constant $\bar{g}_K$ will be specified later.

We now turn to the Hodgkin-Huxley theory of the Na^+ channel. The behavior of the Na^+ channel is qualitatively different from that of the K^+ channel in the following respect. When an upward step of voltage is applied, the K^+ conductance rises monotonically and settles down to a new, elevated, level that is maintained as long as the elevated voltage is maintained. In response to the same kind of voltage step, however, the Na^+ conductance rises only briefly, and then falls back to a low level, even when the elevated voltage level is maintained. This phenomenon is known

as Na^+ inactivation. To describe it, Hodgkin and Huxley had to assume that there are two types of gates within each Na^+ channel: fast "m-gates" that are encouraged to open by increasing membrane potential, and slower "h-gates" that have the opposite response to membrane potential. Thus, in the voltage step described above, the transient increase in Na^+ conductance happens when the m-gates have opened but the h-gates have not yet closed. Inactivation is a consequence of the eventual closure of the slower h-gates.

More precisely, the Hodgkin-Huxley theory of the Na^+ channel can be derived from the following postulates:

1. Each Na^+ channel has four gates. Each gate can be OPEN or CLOSED. The Na^+ channel as a whole is OPEN if and only if all four of its gates are OPEN.
2. The four gates, however, are not identical. Three of them are of a type that we shall call an "m-gate," and one is of a type that we shall call an "h-gate." All m-gates are identical to one another (whether they live within the same Na^+ channel or not), and all h-gates are identical to one another, but the m-gates and the h-gates have different properties.
3. The different gates within a Na^+ channel operate independently of one another, regardless of their type. (And of course, the different gates in different Na^+ channels also operate independently of one another.)
4. The rate constant (probability per unit time) for opening or closing a gate of a Na^+ channel is a specified function of voltage for the opening rate and another specified function of voltage for the closing rate. The opening and closing rates of the m-gates are different from those of the h-gates. In fact, they are qualitatively different: The m-gates are encouraged to open by increasing voltage, whereas the h-gates are encouraged to close by increasing voltage.

As we did in the case of the K^+ channel, we now translate these postulates into a mathematical theory. Consider a large population of Na^+ channels. Let $m(t)$ be the fraction of open m-gates at time t , and let $h(t)$ be the fraction of open h-gates at that time. Then $m(t)$ and $h(t)$ satisfy differential equations of the same form as equation (3.5.10):

$$\dot{m} = \alpha_m(v)(1 - m) - \beta_m(v)m, \quad (3.5.12)$$

$$\dot{h} = \alpha_h(v)(1 - h) - \beta_h(v)h. \quad (3.5.13)$$

The functions $\alpha_m(v)$, $\beta_m(v)$, $\alpha_h(v)$, and $\beta_h(v)$ that give the opening and closing rate constants for the m-gates and the h-gates will be specified later. For now, we note that $\alpha_m(v)$ and $\beta_h(v)$ are increasing functions of v , whereas $\beta_m(v)$ and $\alpha_h(v)$ are decreasing functions of v . Thus, increasing

voltage encourages opening and discourages closure of the m-gates, but has just the opposite effect on the h-gates.

Another qualitative point worth making about the different gates that we have considered is that the m-gates respond to voltage changes about $10 \times$ faster than the h-gates. The gates of the K^+ channel (which we may now call "n-gates") are comparable in speed to the slower gates of the Na^+ channel, i.e., to the h-gates.

To complete the theory of the Na^+ channel, we have to specify the Na^+ conductance, g_{Na} , as a function of m and h , which are the fractions of open m-gates and h-gates, respectively. Here again we are helped by the probability interpretation of m and h and by the assumption that the different gates of the Na^+ channel operate independently. Since there are three m-gates and one h-gate in each channel, with probability m that any particular m-gate is OPEN and probability h that any particular h-gate is OPEN, and since the gates open and close independently, the probability that all four gates are OPEN, and hence that the channel as a whole is OPEN, is equal to m^3h . The Na^+ conductance is proportional to this, and hence is given by

$$g_{Na} = \bar{g}_{Na} m^3 h, \quad (3.5.14)$$

where the constant $\bar{g}_{Na}$ is the hypothetical Na^+ conductance when all of the gates in all of the Na^+ channels are simultaneously open. The value of $\bar{g}_{Na}$ will be specified later.

In addition to the Na^+ and K^+ channels, Hodgkin and Huxley also found a third type of channel in the membrane of the squid giant axon. They called this a "leakage" channel. Unlike the Na^+ and K^+ channels, the leakage channel population has a constant conductance

$$g_L = \bar{g}_L. \quad (3.5.15)$$

The leakage channel corresponds roughly to the Cl^- channel that was considered earlier in this chapter, but it is not specific to Cl^- ions. When a channel allows more than one ion type to flow through it, the current-voltage relation of that channel may still be of the form $I = g(v - E)$, but there is no longer a simple formula for E in terms of the ion concentrations on the two sides of the membrane. Under these circumstances, E is called the "reversal potential," a more general term that includes "equilibrium potential" as a special case. This change in terminology conveys an important physical distinction. In a channel that is specific for a single ion, when the voltage across the membrane is equal to the equilibrium potential, the situation is one of true thermodynamic equilibrium. No heat is being generated, for example, and the ion concentrations are not running down. In a channel that admits more than one species of ion, though, even when the voltage is at the reversal potential and there is no net current, there are nevertheless opposing currents of the different ions that do generate heat and that will eventually tend to equalize the concentrations on the two sides

of the membrane. This is *not* a case of thermodynamic equilibrium; hence the change in terminology. The reversal potential of the leakage channel, which we shall call E_L , turns out to be somewhat less negative than the resting potential of the squid giant axon. This would not be possible for a pure Cl^- channel. Since Cl^- is not pumped, its equilibrium potential has to coincide with the rest potential. In the equations that follow, the leakage channel, denoted by the subscript L, will replace the chloride channel considered earlier.

We may now summarize the Hodgkin-Huxley equations for the nerve action potential in the space-clamped case (no propagation). Those equations are as follows:

$$C \frac{dv}{dt} + g(v - E) = i_0(t), \quad (3.5.16)$$

where $i_0(t)$ is an applied current (per unit area of membrane) as a function of time. It may represent current applied through an electrode by an investigator, or synaptic current flowing naturally into a neuron because channels have been opened by a neurotransmitter. In either case, we regard $i_0(t)$ as a known function of time.

Recall that

$$g = g_{\text{Na}} + g_{\text{K}} + g_{\text{L}} \quad (3.5.17)$$

is the total membrane conductance (per unit area), and that

$$E = \frac{g_{\text{Na}}E_{\text{Na}} + g_{\text{K}}E_{\text{K}} + g_{\text{L}}E_{\text{L}}}{g_{\text{Na}} + g_{\text{K}} + g_{\text{L}}} \quad (3.5.18)$$

is the weighted average of the reversal potentials weighted by the conductances. The reversal potentials E_{Na} , E_{K} , and E_{L} are constants. Their values will be given below. The conductances g_{Na} , g_{K} , and g_{L} are given by

$$g_{\text{Na}} = \bar{g}_{\text{Na}} m^3 h, \quad (3.5.19)$$

$$g_{\text{K}} = \bar{g}_{\text{K}} n^4, \quad (3.5.20)$$

$$g_{\text{L}} = \bar{g}_{\text{L}}, \quad (3.5.21)$$

where $\bar{g}_{\text{Na}}$, $\bar{g}_{\text{K}}$, and $\bar{g}_{\text{L}}$ are constants whose values will be given below.

The gating variables m , h , and n obey the following differential equations (all of the same form):

$$\dot{m} = \alpha_m(v)(1 - m) - \beta_m(v)m, \quad (3.5.22)$$

$$\dot{h} = \alpha_h(v)(1 - h) - \beta_h(v)h, \quad (3.5.23)$$

$$\dot{n} = \alpha_n(v)(1 - n) - \beta_n(v)n. \quad (3.5.24)$$

The six functions of v that appear as coefficients in these equations represent the opening and closing rate constants for the different types of gates described earlier. These rates were measured by Hodgkin and Huxley, who then fit formulae to the data they obtained. This curve-fitting technique

yields the following expressions for the opening and closing rate constants as functions of voltage. These formulae are written in a specific system of units in which v is in millivolts (10^{-3} volts), and the α and β rate constants are all in reciprocal milliseconds ($1/(10^{-3}\text{seconds})$ or $10^3/\text{second}$).

$$\alpha_m(v) = 1.0 \frac{(v + 45)/10}{1 - \exp(-(v + 45)/10)}, \quad (3.5.25)$$

$$\beta_m(v) = 4.0 \exp(-(v + 70)/18), \quad (3.5.26)$$

$$\alpha_h(v) = 0.07 \exp(-(v + 70)/20), \quad (3.5.27)$$

$$\beta_h(v) = 1.0 \frac{1}{1 + \exp(-(v + 40)/10)}, \quad (3.5.28)$$

$$\alpha_n(v) = 0.1 \frac{(v + 60)/10}{1 - \exp(-(v + 60)/10)}, \quad (3.5.29)$$

$$\beta_n(v) = 0.125 \exp\left(-\frac{v + 70}{80}\right). \quad (3.5.30)$$

Note that some of the above expressions evaluate to 0/0 at particular values of voltage. In those special cases, L'Hospital's rule may be used to find the missing value.

Finally, the constants appearing in the Hodgkin-Huxley equations have the following values:

$$C = 1.0 \frac{\text{microamperes} \times \text{milliseconds}}{\text{centimeter}^2}, \quad (3.5.31)$$

$$\bar{g}_{Na} = 120 \frac{\text{microamperes/millivolt}}{\text{centimeter}^2}, \quad (3.5.32)$$

$$\bar{g}_K = 36 \frac{\text{microamperes/millivolt}}{\text{centimeter}^2}, \quad (3.5.33)$$

$$\bar{g}_L = 0.3 \frac{\text{microamperes/millivolt}}{\text{centimeter}^2}, \quad (3.5.34)$$

$$E_{Na} = 45 \text{ millivolts}, \quad (3.5.35)$$

$$E_K = -82 \text{ millivolts}, \quad (3.5.36)$$

$$E_L = -59 \text{ millivolts}. \quad (3.5.37)$$

3.6 Computer Simulation of the Nerve Action Potential

In this section we describe a numerical method and Matlab program for solving the (space-clamped) Hodgkin-Huxley equations that were introduced in the previous section. The numerical method is derived by replacing the derivatives that appear in the Hodgkin Huxley equations by the corresponding difference quotients. To save writing, we shall exploit the fact

that m , h , and n all obey equations of the same form by using s as a generic variable that can stand for m , h , or n . Also, we write the equations in the order in which they will be used in the computational scheme:

$$\frac{s(t) - s(t - \Delta t)}{\Delta t} = \alpha_s (v(t - \Delta t)) (1 - s(t)) - \beta_s (v(t - \Delta t)) s(t), \quad (3.6.1)$$

where $s = m$, h , or n ;

$$g_{Na}(t) = \bar{g}_{Na} (m(t))^3 h(t), \quad (3.6.2)$$

$$g_K(t) = \bar{g}_K (n(t))^4, \quad (3.6.3)$$

$$g(t) = g_{Na}(t) + g_K(t) + \bar{g}_L, \quad (3.6.4)$$

$$E(t) = \frac{g_{Na}(t)E_{Na} + g_K(t)E_K + \bar{g}_L E_L}{g_{Na}(t) + g_K(t) + \bar{g}_L}, \quad (3.6.5)$$

$$C \frac{v(t) - v(t - \Delta t)}{\Delta t} + g(t) (v(t) - E(t)) = i_0(t). \quad (3.6.6)$$

These equations are used to compute $s(t)$ (i.e., $m(t)$, $h(t)$, and $n(t)$) and $v(t)$, given $s(t - \Delta t)$ and $v(t - \Delta t)$. In equation (3.6.1), on the right-hand side, we use $v(t - \Delta t)$ instead of $v(t)$ because the former is known but the latter is yet to be computed. On the other hand, in that same expression we use $s(t)$ instead of $s(t - \Delta t)$, even though $s(t)$ is unknown, because we can easily solve for it and because the resulting scheme is more stable (see below) than if $s(t - \Delta t)$ were used. The solution of equation (3.6.1) for $s(t)$ is

$$s(t) = \frac{s(t - \Delta t) + \Delta t \alpha_s v(t - \Delta t)}{1 + \Delta t (\alpha_s v(t - \Delta t) + \beta_s v(t - \Delta t))}, \quad (3.6.7)$$

where $s = m$, h , or n . It follows that $s(t)$ lies between $s(t - \Delta t)$ and

$$\frac{\alpha_s v(t - \Delta t)}{\alpha_s v(t - \Delta t) + \beta_s v(t - \Delta t)},$$

which in turn lies between 0 and 1. Thus, if s starts out in the interval $(0, 1)$ it remains within that interval forever, no matter how large a time step Δt may be chosen. This is as it should be, since s (i.e., m , h , or n) represents the fraction of open gates of a particular type.

Now suppose, in contrast to what was done above, that we had chosen to use $s(t - \Delta t)$ instead of $s(t)$ on the right-hand side of equation (3.6.1). Then the formula for $s(t)$ would have been

$$s(t) = (1 - (\alpha_s + \beta_s)\Delta t) s(t - \Delta t) + \alpha_s \Delta t, \quad (3.6.8)$$

in which we have omitted the arguments of α_s and β_s for simplicity. Note that this can also be written

$$s(t) = (1 - (\alpha_s + \beta_s)\Delta t) s(t - \Delta t) + \frac{\alpha_s}{\alpha_s + \beta_s} (\alpha_s + \beta_s)\Delta t, \quad (3.6.9)$$

which has the form of a weighted average of $s(t - \Delta t)$ and $\alpha_s/(\alpha_s + \beta_s)$, and therefore lies between these two values, provided that $(\alpha_s + \beta_s)\Delta t < 1$. But if this inequality is violated, one of the “weights” becomes negative, the interpretation of a weighted average breaks down, and one cannot conclude anymore that the result lies in the expected interval. Note that there was no such restriction on Δt when we used $s(t)$ on the right-hand side of equation (3.6.1), which is why we have done so.

Similarly, the solution of equation (3.6.6) for $v(t)$ is as follows:

$$v(t) = \frac{v(t - \Delta t) + \frac{\Delta t}{C} (g(t)E(t) + i_0(t))}{1 + \frac{\Delta t}{C} g(t)}. \quad (3.6.10)$$

It follows that $v(t)$ lies between $v(t - \Delta t)$ and $E(t) + i_0(t)/g(t)$. Consider the special case of $i_0(t) = 0$ (no applied current). Then $v(t)$ lies between $v(t - \Delta t)$ and $E(t)$. Moreover, we know that $E_K < E(t) < E_{Na}$ (recall that $E(t)$ is a weighted average of E_K , E_L , and E_{Na} with positive weights; and also that $E_K < E_L < E_{Na}$). Thus, in the case of no applied current, if v starts out in the interval (E_K, E_{Na}) , it remains in that interval forever, regardless of the choice of Δt . Just as in the case of $s(t)$, discussed above, we would not have this stability result for all choices of Δt if we had used $v(t - \Delta t)$ instead of $v(t)$ in the term $g(t)(v(t) - E(t))$ in equation (3.6.6).

Let us now turn to the task of writing a Matlab program that implements the foregoing scheme.

First consider the task of updating m , h , or n . We can write one function `snew` that will serve this purpose in all three cases:

```
function s=snew(s_old,alpha,beta,dt)
%filename: snew.m
global check;
s=(s_old+dt*alpha)/(1+dt*(alpha+beta));
if(check)
    chsnew=(s-s_old)/dt-(alpha*(1-s)-beta*s)
end
```

Note that this function is self-checking, provided that the global variable `check` is set equal to 1 (true). In that case the value of `chsnew` will be printed out each time the function `snew` is called. If all is well, it should be within roundoff error of 0.

The function `snew` can be used to update m , h , or n , depending on how it is called. To update m , for example, we use the following line of code:

```
m=snew(m,alphan(v),betam(v),dt)
```

In this particular way of calling `snew`, the old value of m is overwritten by the new value. We are assuming here the existence of Matlab functions `alphan(v)` and `betam(v)`. These will be specified later. This particular line of code makes rather extensive use of the notion of a function, and it is worth thinking through exactly what happens when it is executed. First,

the old value of `m` is sent to the function `snew`, where it is used as the value of `s_old`. (Matlab uses "call by value," so the function actually gets a copy of the number that is stored in the variable `m`; it assigns that copy to `s_old`.) Next, the value of `v` is sent to the functions `alpham` and `betam`. The results come back and are sent on to `snew`, where they are called `alpha` and `beta`. Finally, the value of `dt` is sent to `snew`, which happens also to call it `dt`. When `snew` has all of its arguments, it runs, and its output, which within `snew` is known as `s`, is assigned to the variable `m`, thus overwriting the old value of `m`, which is forgotten.

The functions `alpham(v)`, `betam(v)`, etc., that give the opening and closing rate constants for the various gates were specified in equations (3.5.25)–(3.5.30). The Matlab implementation of these functions is as follows:

```
function a=alpham(v)
%filename: alpham.m
theta=(v+45)/10;
if(theta==0)    %check for case that gives 0/0
    a=1.0; %in that case use L'Hospital's rule
else
    a=1.0*theta/(1-exp(-theta));
end

function b=betam(v)
%filename: betam.m
theta = (v+70)/18;
b=4.0*exp(-theta);

function a=alphah(v)
%filename: alphah.m
theta = (v+70)/20;
a=0.07*exp(-theta);

function b=betah(v)
%filename: betah.m
theta = (v+40)/10;
b=1.0/(1+exp(-theta));

function a=alphan(v)
%filename: alphan.m
theta=(v+60)/10;
if(theta==0)    %check for case that gives 0/0
    a=0.1; %in that case use L'Hospital's rule
else
    a=0.1*theta/(1-exp(-theta));
end

function b=betan(v)
```

```
%filename: betan.m
theta = (v+70)/80;
b=0.125*exp(-theta);
```

It is now time to write the main program for solving the space-clamped Hodgkin-Huxley equations and computing the nerve action potential:

```
%filename HH.m
%numerical solution of the space-clamped Hodgkin-Huxley equations
clear all
clf
global check;
global t1p t2p ip; %parameters for the function izero(t)
in_HH
in_mhmv
for klok=1:klokmax
    t=klok*dt; %note time
    m=snew(m,alphan(v),betam(v),dt); %update m
    h=snew(h,alphah(v),betah(v),dt); %update h
    n=snew(n,alphan(v),betan(v),dt); %update n
    gNa=gNabar*(m^3)*h; %sodium conductance
    gK =gKbar*(n^4); %potassium conductance
    g=gNa+gK+gLbar; %total conductance
    gE=gNa*ENa+gK*EK+gLbar*EL; %gE=g*E
    %save old value of v for checking purposes:
    v_old=v;
    %update v:
    v=(v+(dt/C)*(gE+izero(t)))/(1+(dt/C)*g);
    if(check)
        E=gE/g;
        chv=C*(v-v_old)/dt+g*(v-E)-izero(t)
    end
    %store results for future plotting:
    mhmv_plot(:,klok)=[m h n]';
    v_plot(klok)=v;
    t_plot(klok)=t;
end
subplot(2,1,1),plot(t_plot,v_plot)
subplot(2,1,2),plot(t_plot,mhmv_plot)
```

The function `izero(t)`, which appears in the line that updates `v`, specifies the current applied to the neuron as a function of time. It could be anything, depending what experiment is being simulated. To give a specific example that might be useful, consider a square pulse of current beginning at time `t1p` and ending at time `t2p`, with current equal to `ip` between those two times (and zero, otherwise). This would be coded as follows:

```

function i=izero(t)
%filename: izero.m
global t1p t2p ip;
if((t1p<t)&(t<t2p))
    i=ip;
else
    i=0;
end

```

Other functions `izero(t)` can, of course, be substituted for the above example, including a function that simply returns 0 for all t , to simulate a neuron with no applied current. Note that the list of global parameters may have to be modified depending on what parameters the function `izero(t)` requires.

The initial values of m , h , n , and v are set in the script `in_mhmv`, given below, which requires quite a bit of explanation even though it is only a few lines long. This script makes use of two parameters set by the user, `vhold` and `vstart`. We assume that the interesting part of the experiment starts at $t=0$, but that for all time prior to $t=0$, the voltage was fixed at the “holding voltage” `vhold`. The simplest example of this would be if the neuron were isolated and at rest prior to $t=0$, in which case `vhold` would be equal to the rest potential (`vhold=-70`). Another possibility, though, is that the experimenter has intervened and has used a voltage clamp circuit to hold the membrane potential at some level other than rest for a long time prior to $t=0$. At $t=0$, we assume that a brief shock of current may be applied to the nerve membrane, sufficient to change its voltage suddenly from `vhold` to `vstart`. (In the special case that no such current shock is applied, `vstart=vhold`.) Because the shock is of brief duration, m , h , and n do not have time to change. (The applied current affects the voltage directly but affects m , h , and n only indirectly, through its effect on the voltage. Since the time derivatives of m , h , and n remain finite even during a step in voltage, these gating variables cannot themselves change suddenly.) Therefore, our strategy is to set m , h , and n equal to the values to which they would settle down during the long period of time prior to $t=0$ during which $v=vhold$, and then to set $v=vstart$.

From equation (3.6.1), we can see that if v is constant for a long time, and if s settles down (i.e., stops changing, so that $s(t) = s(t - \Delta t)$), then the value to which s settles down must be such that the right-hand side of equation (3.6.1) is zero. Therefore, solving for s , we get

$$s = \frac{\alpha_s(v)}{\alpha_s(v) + \beta_s(v)}. \quad (3.6.11)$$

Recalling that s is shorthand for m , h , or n , we are now in a position to write the script `in_mhmv`:

```

%filename: in_mhmv.m

```

```

%this script simulates holding the
%voltage constant for a long time
%prior to t=0 at the value v=vhold,
%and then applying a current shock
%at t=0 to step the voltage
%to a new level, v=vstart.
%The gating variables m,h,n
%cannot change suddenly and therefore
%remain constant during the current shock.
%
%simulate time prior to t=0:
v=vhold
%set m,h,n equal to their steady values
%under constant-v conditions:
m=alphan(v)/(alphan(v)+betan(v))
h=alphah(v)/(alphah(v)+betah(v))
n=alphan(v)/(alphan(v)+betan(v))
%
%now let voltage jump to its value
%just after t=0, without making
%any further change in m,h,n:
v=vstart

```

The only thing left to do is to write the script `in_HH`, which initializes all parameters:

```

%filename: in_HH.m
%Units:
%voltage is in millivolts (mV)
%current is in microamperes (muA)
%length is in centimeters (cm)
%time is in milliseconds (ms)
%note that microfarads =
%microamperes*milliseconds/millivolt
%= muA*ms/mV
%
%initialize membrane parameters:
%membrane capacitance per unit area:
C=1.0      % (muF/cm^2)
%max possible Na+ conductance per unit area:
gNabar=120 % ((muA/mV)/cm^2)
%max possible K+ conductance per unit area:
gKbar=36   % ((muA/mV)/cm^2)
%leakage conductance per unit area:
gLbar=0.3  % ((muA/mV)/cm^2)
%Na+ equilibrium potential:

```

```

ENa = 45    %(mV)
%K+ equilibrium potential:
EK = -82    %(mV)
%leakage channel reversal potential:
EL = -59    %(mV)
%
%initialize time step and experiment duration:
dt=0.1      %time step duration (ms)
tmax=50     %duration of experiment (ms)
%total number of time steps in the experiment:
klokmax=ceil(tmax/dt)
%
%initialize arrays that hold data for plotting:
mhn_plot=zeros(3,klokmax);
v_plot=zeros(1,klokmax);
t_plot=zeros(1,klokmax);
%
%initialize parameters that define the experiment:
%the neuron is at rest (v= -70 mV) prior to t=0;
%at t=0 a current shock is applied after which v= -55 mV;
%then a subsequent 15  $\mu$ A current pulse of 1 ms duration
%is applied beginning at t=10 ms.
%voltage prior to t=0:
vhold= -70 %(mV)
%voltage just after t=0:
vstart= -55 %(mV)
%(change in v is result of current shock applied at t=0)
%
%initialize parameters of subsequent current pulse:
t1p=10      %starting time (ms)
t2p=11      %stopping time (ms)
ip=15       %current applied ( $\mu$ A)
%
%initialize checking parameter
check=1     %set check=1 to enable self-checking
           %set check=0 to disable self-checking

```

3.7 Suggestions for Computing Projects Concerning the Nerve Impulse

The first thing to do is to run the program HH as written, and see what happens. It is set up to describe a neuron that has been at rest prior to $t = 0$. At $t = 0$ a current shock is delivered that is sufficient to displace the

membrane potential from -70 millivolts to -55 millivolts. This is enough to generate an action potential. Then, 10 milliseconds later, a current of 15 microamperes is turned on for a duration of 1 millisecond (via the function `izero`). This will again displace the membrane potential upward by about 15 millivolts, but will that be enough to generate a second action potential? Try it and see. (This issue will be explored more systematically below.) Once you have the program running, check whether the time step ($dt=0.1$ ms) is small enough. Try a smaller time step, and see whether the results at corresponding times are essentially the same.

An important property of the nerve action potential is its "all-or-none" character. There is a *threshold* for excitation. To find it, do essentially the same experiment as above, but turn off the second current pulse by setting `izero=0` at all time steps. Now keeping `vhold=-70`, perform a series of computer experiments in which `vstart` is varied in, say, 3 millivolt increments, starting from -70 millivolts, and increasing (depolarization) in the direction of 0 millivolts membrane potential. It would be best to plot the results for different values of `vstart` on the same axes, so that they are all to the same scale. If you plot them on different axes, Matlab will choose the scale separately in each case (unless you specify the scale yourself, which can certainly be done using `axis`), and then it will be harder to see what is going on. Once you see where the threshold is, you might want to take smaller steps in the neighborhood of that critical value of `vstart` in order to determine the threshold more precisely and study the near-threshold behavior.

Considering the size of the stimulus as being `vstart-vhold`, and the size of the response as being the maximum deviation of the membrane potential from rest, use your results to answer these questions: Below threshold, how does the size of the response depend on the size of the stimulus? Above threshold, how does the size of the response depend on the size of the stimulus?

So far, we have only considered depolarizing current shocks applied at $t = 0$. What about hyperpolarizing shocks? Repeat the above experiment, but with `vstart` decreasing starting from rest (-70 millivolts) in increments of (for example) -3 millivolts. (This is called "hyperpolarization" because the magnitude of the membrane potential is increasing.) Eventually, you will get action potentials this way, too! Can you explain why? It will help to look at the behavior of the variables `m`, `h`, and `n`.

In all of the neuron computer experiments discussed up to now, the "preparation" of the neuron has been to leave it at rest for all time prior to $t = 0$. Suppose instead that the neuron is "clamped" at some different potential for a long time prior to $t = 0$, and then suppose that the voltage clamp is simply removed at $t = 0$ with no current shock being applied. In that case, we can set `vhold` equal to any voltage, but then we must set `vstart=vhold`. Try a range of values of `vhold` that are both below and above rest, say from about -100 millivolts to about -40 millivolts, in 5 millivolt intervals. How do these results (in which `vhold=vstart`) differ

from those found above (in which $v_{hold} = -70$) when you use the same value of v_{start} ? Can you explain the differences that you see? (Compare the starting values of m , h , and n .)

Let us now consider the effect of one action potential on the threshold for obtaining a subsequent action potential. To study this, restore the function `izero` to the way it was written above, set $v_{hold} = -70$ and $v_{start} = -55$ to generate the first action potential, and then use the parameters $t1p$, $t2p$, and ip to control the timing and strength of a second stimulus. Keep $t2p - t1p = 1$, so that the duration of the second stimulus is always 1 millisecond, and increase $t1p$ in (for example) 2 millisecond increments starting at 2 milliseconds, and continuing until the results become essentially independent of the time since the last action potential. For each choice of $t1p$, use the parameter ip to control the strength of the second stimulus and vary that parameter to determine the strength that is just enough to generate a second action potential. Plot this threshold value of ip as a function of $t1p$, i.e., as a function of the time interval between the two stimuli. (For some $t1p$, it may be impossible to generate a second action potential. The period of time in which this is the case is called the "absolute refractory period." It is followed by the "relative refractory period," in which a second action potential can be generated but only by a stronger stimulus than in the case of a neuron at rest.)

If you want to be very thorough about the foregoing study, you might repeat it with a different duration for the second stimulus, e.g., $t2p - t1p = 0.5$ for a half-millisecond current pulse. How does the threshold value of the current strength ip depend on the duration of the stimulus? (For short enough stimulus durations there should be a simple relationship. See whether you can find it. For longer stimulus durations the situation is more complicated.)

Up to now, we have considered only stimuli that are localized in time. Neurons can also respond to steady or to slowly varying stimuli by generating a long sequence of action potentials. Sensory neurons typically work in this way. The sensory signal is detected by a specialized transducer (rod or cone in the eye, hair cell in the inner ear, Pacinian corpuscle in the skin, . . .), which produces a "generator current" that is some function of the strength of the stimulus, and this current, flowing into a sensory neuron, causes that neuron to emit action potentials at a rate that depends on the magnitude of the generator current. Thus, stimulus strength is encoded as the rate at which action potentials are generated (and not, of course, by the size of any individual action potential, since all action potentials produced by any given neuron are essentially the same).

To study this process, use the function `izero` to generate a constant current. One way to do this is to set $t1p$ equal to zero and $t2p$ equal to the time at which the experiment is supposed to end. Then ip is the magnitude of the constant current that is applied to the neuron. With ip in a certain range, you should find that the neuron generates periodic action potentials.

Plot the frequency of these action potentials as a function of the applied current, i_p .

As the foregoing study should show, the squid giant axon (from which the specific instance of the Hodgkin-Huxley equations we are using was derived) has a rather limited "dynamic range," i.e., a limited range of frequencies over which it is capable of repetitive firing. (What is the ratio of the highest to the lowest frequency that you were able to obtain?) But slight modifications of the Hodgkin-Huxley equations can produce drastically better behavior in this respect. Consider, for example, the "Traub neuron," which also obeys the Hodgkin-Huxley equations, but with different parameters and different rate functions, as follows:

```
%max possible Na+ conductance per unit area:
gNabar=100 %((muA/mV)/cm^2)
%max possible K+ conductance per unit area:
gKbar=80 %((muA/mV)/cm^2)
%leakage conductance per unit area:
gLbar=0.2 %((muA/mV)/cm^2)

%Na+ equilibrium potential:
ENa = 50 %(mV)
%K+ equilibrium potential:
EK = -100 %(mV)
%leakage channel reversal potential:
EL = -67 %(mV)

function a=alpham(v)
%filename: alpham.m
theta=(v+54)/4;
if(theta==0) %check for case that gives 0/0
    a=1.28; %in that case use L'Hospital's rule
else
    a=1.28*theta/(1-exp(-theta));
end

function b=betam(v)
%filename: betam.m
theta = (v+27)/5;
if (theta==0) %check for case that gives 0/0
    b=1.4; %in that case, use L'Hospital's rule
else
    b=1.4*theta/(exp(theta)-1);
end

function a=alphah(v)
```



```

%filename: alphah.m
theta = (v+50)/18;
a=0.128*exp(-theta);

function b=betah(v)
%filename: betah.m
theta = (v+27)/5;
b=4.0/(1+exp(-theta));

function a=alphan(v)
%filename: alphan.m
theta=(v+52)/5;
if(theta==0) %check for case that gives 0/0
    a=0.16; %in that case use L'Hospital's rule
else
    a=0.16*theta/(1-exp(-theta));
end

function b=betan(v)
%filename: betan.m
theta = (v+57)/40;
b=0.5*exp(-theta);

```

Try these modifications and study the relationship between applied current and firing rate for the Traub neuron. In particular, note how the firing rate varies as a function of applied current at the lower end of the range of currents that results in repetitive firing, and compare this behavior with that of the squid giant axon. Which type of neuron would you rather have encoding a sensory signal like pressure on your skin, assuming that it is desirable to know the magnitude of the pressure, and not just that something is pressing there?

3.8 Annotated References

This chapter introduces the subject of osmotic pressure and osmotic flow across membranes. For a comprehensive treatment of this subject, see the following monograph:

Finkelstein, A.: *Water Movement Through Lipid Bilayers, Pores, and Plasma Membranes: Theory and Reality* (Distinguished Lecture Series of the Society of General Physiologists, Volume 4), Wiley, New York, 1987.

The type of model used here for the control of cell volume was pioneered by Tosteson and Hoffman:

Tosteson, D.C. and Hoffman, J.F.: Regulation of cell volume by active cation transport in high and low potassium sheep red cells. *Journal of General Physiology* **44**: 169–194, 1960.

Tosteson, D.C.: Regulation of cell volume by sodium and potassium transport. In: *The Cellular Functions of Membrane Transport* (Hoffman, J.F., ed.) Society of General Physiologists Symposium #10, Prentice Hall, New York, NY, 1964.

This chapter provides an introduction to the membrane physiology that is needed to understand the electrical signaling mechanisms of nerve and muscle. This is a vast subject, in which the most important single work is the following:

Hodgkin, A.L. and Huxley, A.F.: A quantitative description of membrane current and its application to conduction and excitation in nerve. *Journal of Physiology* **117**, 500–544, 1952.

Section 3.5 is based on this paper, except that we use different conventions in the definition of the membrane potential. These differences will now be briefly explained, to avoid confusion in comparing the original Hodgkin Huxley paper with Section 3.5: Hodgkin and Huxley work with the deviation of membrane potential from rest, and they consider hyperpolarization of the membrane as positive. We work with the actual potential difference across the cell membrane, with the sign convention that membrane potential equals inside potential minus outside potential. For us, therefore, the rest potential is -70 mV, whereas Hodgkin and Huxley consider the rest potential to be 0 mV (by definition). A hyperpolarized membrane potential that we would call -90 mV would be described by Hodgkin and Huxley as $+20$ mV, since that is the deviation from rest in the hyperpolarizing direction.

For more information about the Traub neuron mentioned in the suggested computing projects, see

Traub, R.D., and Miles, R.: Pyramidal cell-to-inhibitory cell spike transduction explicable by active dendritic conductances in inhibitory cell. *J. Comput. Neurosci.* **2**: 291–298, 1995.

Exercises

3.1. Work in terms of Pressure and Volume

In mechanics, work = force $\times$ distance. If a force f is applied through a distance ds , the amount of work performed is $dW = f ds$.

Consider a piston with cross-sectional area A that applies a pressure (force/area) P to a fluid as it moves through a distance dx . Show that the work done is $dW = P dV$, where dV is the volume displaced by the piston.

3.2. Equation for Osmotic Flow When There Is Solute on Both Sides of the Membrane

Derive equation (3.1.10).

3.3. Osmotic Pressure

Suppose that 180 g of sugar are dissolved in enough water to make up 10^3 cm^3 of sugar solution. Calculate the osmotic pressure of this solution at $T = 300^\circ\text{K}$, given the following data:

1. the weight of a sugar molecule is $3.0 \times 10^{-22} \text{ g}$.
2. Boltzmann's constant is $k = 1.4 \times 10^{-16} \text{ dyne-cm/}^\circ\text{K}$.

How high could this amount of pressure raise a column of water whose cross-sectional area is 1 cm^2 ? To figure this out, you need to know that the force of gravity on 1 cm^3 of water is 980 dynes. What if the cross-sectional area of the water column is 10^6 cm^2 ?

3.4. Work of Compression in terms of Concentration

Suppose that n molecules of an ideal gas are compressed by a piston from volume V_1 , where the concentration is $c_1 = n/V_1$, to a volume V_2 , where the concentration is $c_2 = n/V_2$. While the compression is occurring, the temperature is held constant at the value T . Show that the work performed by the piston is $nkT \log(c_2/c_1)$. (Hint: Use $dW = -P dV$ and $P = nkT/V$. Integrate from V_1 to V_2 , and express the result in terms of concentration.)

3.5. Effect of Pump Rate on Cell Volume and Membrane Potential

In this exercise, use the following data:

$$\begin{aligned} g_{\text{Na}} &= 3.3 \mu\text{mho/cm}^2, \\ g_{\text{K}} &= 240 \mu\text{mho/cm}^2, \\ [\text{Na}^+]_o &= 145 \mu\text{M/cm}^3, \\ [\text{K}^+]_o &= 4 \mu\text{M/cm}^3, \\ q &= 1.6 \times 10^{-19} \text{ coulombs}, \\ kT/q &= 25 \times 10^{-3} \text{ volts}, \end{aligned}$$

where $\mu\text{mho} = 10^{-6} \text{ (coulomb/sec)/volt}$, a mho is the basic unit of conductance (the reciprocal of resistance, measured in ohms), the coulomb is a standard unit of charge, and $1 \mu\text{M} = 10^{-6} \times (6 \times 10^{23})$ ions.

1. Check that inequality (3.4.34) is satisfied.
2. Plot β , V , and v against p over the range of p values for which $\beta < 1$. Plot V in units of $N/2[\text{Cl}^-]_o$ and v in units of kT/q .
3. Evaluate p_{opt} from equation (3.4.35) and check that V and v have minima at $p = p_{\text{opt}}$.
4. Assuming that $p = p_{\text{opt}}$, evaluate $[\text{Na}^+]_i$, $[\text{K}^+]_i$, and $[\text{Cl}^-]_i$. Also, evaluate v .

5. Still assuming that $p = p_{\text{opt}}$, evaluate E_{Na} , E_{K} , and E_{Cl} in equations (3.5.3) through (3.5.5). Note that $E_{\text{Cl}} = v$. Why is this? Similarly, this problem is solved without specifying a value for g_{Cl} . How is this possible?

3.6. Sketch of the Nerve Impulse

Using equation (3.5.9) and the values of E_{Na} , E_{K} , and E_{Cl} calculated above, and making the approximation that $v = E$, evaluate v under the following conditions (i.e., complete the following table):

	I	II	III	
t	0	0.25	1.0	ms (10^{-3} s)
g_{Na}	3.3	30,000	3.3	$\mu\text{mho}/\text{cm}^2$
g_{K}	240	3000	3000	$\mu\text{mho}/\text{cm}^2$
g_{Cl}	300	300	300	$\mu\text{mho}/\text{cm}^2$
v	...	...	...	

Sketch the graph of the behavior of v as a function of time if the conductances go through a sequence of values $I \rightarrow II \rightarrow III \rightarrow I$. The return to state I takes about 10 ms. The signal that you have constructed is a nerve impulse.

4

The Renal Countercurrent Mechanism

The function of the kidney is to regulate the composition of the blood *plasma*. The plasma is the part of the blood that remains when the cells have been removed. We shall consider only the regulation of the Na^+ content of the blood plasma, but the reader should keep in mind that the kidney is actually regulating the concentrations of many other substances at the same time.

In an organism with a variable diet, the kidney can achieve a constant Na^+ concentration in blood plasma only if it is able to produce urine in which Na^+ is either more or less concentrated than in plasma according to the needs of the moment. If a salty meal has just been consumed, for example, the kidney needs to excrete urine in which the Na^+ concentration is greater than that of the blood plasma. If, on the other hand, a large quantity of fresh water has just been consumed, then the kidney needs to excrete urine in which the salt concentration is less than the plasma's.

In this chapter we describe the simplest mathematical theory that can account for this dual capability in a reasonably realistic way.

4.1 The Nephron

The basic unit of the renal function is the *nephron*, shown in Figure 4.1. The kidney is made up of about 10^6 nephrons, which are connected in parallel with each other, both with respect to their blood supply and with

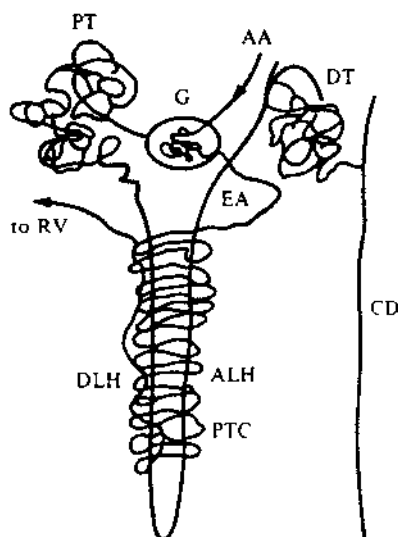


Figure 4.1. Plan of a nephron, the fundamental unit of kidney function. Blood enters through the afferent arteriole (AA), which leads to a tuft of glomerular capillaries within the glomerulus (G). This is the site of filtration, where tubular fluid first forms. Blood leaving the glomerular capillaries flows through the efferent arteriole (EA) to the peritubular capillaries (PTC), which surround the loop of Henle. These capillaries, in a process of reverse filtration, pick up much but not quite all of the water and solutes that were initially filtered at the glomerulus. From the peritubular capillaries, blood is collected by the renal vein (RV). Note the unusual arrangement of two capillary beds in connected in series. Tubular fluid, formed in the glomerulus (G), flows through the proximal convoluted tubule (PT), the descending limb of the loop of Henle (DLH), the ascending limb of the loop of Henle (ALH), the distal convoluted tubule (DT), and finally through the collecting duct (CD), which may be shared with a few other nephrons. Fluid leaving the collecting duct has the composition of urine.

respect to the production of urine. We shall describe the function of the nephron in terms of its different parts.

The transformation from blood plasma to urine begins at the *glomerulus* (G). Blood enters the glomerulus through an *afferent arteriole* (AA) that breaks up into a network of glomerular capillaries. The walls of these capillaries are permeable to water and small molecules. These substances are said to be *filtered* out of the capillaries.

The protein molecules in blood plasma are too large to be filtered, and their osmotic pressure opposes the process of filtration. Moreover, the small ions that are needed to balance the fixed charges on the proteins also contribute to the effective osmotic pressure of the proteins. (Protein molecules are often charged because several of the amino acids from which proteins

are assembled contain chemical groups on their side chains that are ionized at physiological pH values.) This combined osmotic effect opposing the process of filtration is called *oncotic pressure*. It must be overcome by the blood pressure for filtration to proceed.

When fluid leaves the glomerulus, it enters the *proximal convoluted tubule* (PT). The proximal tubular fluid has essentially the composition of blood plasma without the proteins. In the proximal tubule, Na^+ is actively reabsorbed from the tubular fluid, but the walls are permeable to water, which follows the Na^+ for osmotic reasons. Thus, the Na^+ concentration in the proximal tubule remains that of blood plasma, but the volume of fluid flowing out the end of the proximal tubule per unit time is less than the volume of fluid filtered.

Changes in Na^+ concentration begin in the *loop of Henle*. The active part of the loop, i.e., the place where metabolic energy is used, is the *ascending limb* (ALH), where Na^+ is pumped out of the tubules against a concentration difference into the interstitial fluid. The ascending limb is impermeable to water, so water is trapped in the tubule and cannot follow the Na^+ . The consequences of this Na^+ flux out of the ascending limb are not obvious, and they can best be predicted with the help of a mathematical model: The result of the Na^+ flux is that the Na^+ concentration in the interstitial fluid (outside the tubules) increases dramatically with depth, so that the concentration is highest at the bend of Henle's loop where the descending limb joins the ascending limb.

Thus, fluid in the *descending limb* of Henle's loop (DLH) flows through a medium in which the Na^+ concentration increases progressively. The walls of the descending limb are permeable to Na^+ and water, so Na^+ flows in and water flows out, concentrating the tubular fluid as it descends. In fact, the walls of the descending tubule are so permeable that the internal and external Na^+ concentrations are nearly equal.

Overall, the operation of Henle's loop results in the removal of Na^+ and water from the tubular fluid. The excess Na^+ and water are picked up by the *peritubular capillaries* (PTC) and returned to the *renal vein* (RV). These capillaries are remarkable in that they are connected in series with the glomerular capillaries via the *efferent arteriole* (EA), a structure that is peculiar to the kidney. Because the filtration process at the glomerulus holds back the plasma proteins, blood entering the peritubular capillaries has a higher concentration of plasma proteins than usual. Presumably, the resulting oncotic pressure helps draw the excess water and Na^+ into the peritubular capillaries.

Because of the operation of Henle's loop, the fluid outside the tubules of the loop is more concentrated than the blood plasma, while the tubular fluid leaving the loop is more dilute. This is the basis of the kidney's dual capability to excrete urine that is either more concentrated or more dilute than blood plasma.

The choice between these two modes of operation is made by the pituitary gland, which secretes *antidiuretic hormone* (ADH). When ADH is absent, the walls of the *distal convoluted tubule* (DT) and of the *collecting duct* (CD) are impermeable to water, so the volume of urine is essentially that of the fluid that leaves the top of the ascending limb of Henle's loop. In this situation, which is sometimes called a state of *diuresis*, the kidney excretes a large volume of dilute urine. In some individuals, production of ADH is reduced or absent, and a state of diuresis persists. This is the disease *diabetes insipidus*.

When ADH is present, however, the walls of the distal convoluted tubule and collecting ducts become permeable to water. In the distal convoluted tubule enough water is withdrawn (osmotically) to bring the concentration of solutes back to the level of the blood plasma. Then, as the tubular fluid descends in the collecting duct, enough water is withdrawn to equilibrate with the local interstitial concentration at each level. Thus, the urine ends up with a solute concentration equal to that at the bottom of Henle's loop, a concentration much higher than that of the blood plasma. Moreover, because this high concentration is achieved by withdrawal of water, the volume of urine excreted per unit time is much lower than before. This state, achieved by the action of ADH on the distal nephron, is called a state of *antidiuresis*.

Note that the operation of the concentrating engine of Henle's loop is the same whether the kidney is concentrating the urine or diluting it.

In the foregoing description, we omitted any specific discussion of Na^+ flux in the *distal* nephron. In fact, Na^+ is actively reabsorbed in the distal nephron in exchange for other solutes, and this exchange is important in the control of total body Na^+ content. The (adrenocortical) hormonal mechanisms that regulate this exchange are outside the scope of this chapter, however. We model the distal nephron as though its walls were impermeable to Na^+ ions.

4.2 Dynamics of Na^+ and H_2O Transport along the Renal Tubules

First, consider water transport in the tubule shown in Figure 4.2. The volume rate of flow along the tubule past x is designated by $Q(x)$. We assume that the flow is *steady*, which means that it is independent of time. The volume rate of flow through the walls per unit length near x is designated by $f_{\text{H}_2\text{O}}(x)$. Our sign convention is that positive Q stands for flow in the direction of increasing x and that positive f stands for outward flux through walls.

To get a differential equation that relates Q and f , consider the segment of tubule that lies between $x = a$ and $x = b$. Since the flow is steady, the

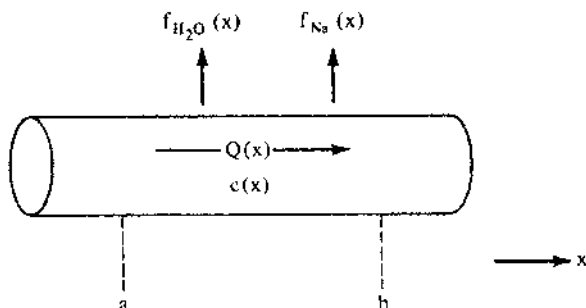


Figure 4.2. Diagram for deriving the equations of sodium ion and water transport along and through the walls of a renal tubule. x = distance along the tubule, $c(x)$ = concentration of sodium ion inside the tubule at position x , $Q(x)$ = flow of water (volume per unit time) along the tubule at x in the direction of increasing x , $f_{\text{H}_2\text{O}}(x)$ = outward transport of water (volume per unit time per unit length of tubule) across the walls of the tubule, $f_{\text{Na}}(x)$ = outward transport of sodium ions (number of ions per unit time per unit length of tubule) across the walls of the tubule. a and b are arbitrary positions used in the derivation of the equations.

volume of this segment cannot change, and the flow into the segment must equal the sum of the flow out of the segment and the flux of water out through the walls of the tube between $x = a$ and $x = b$. In symbols, we have:

$$Q(a) = Q(b) + \int_a^b f_{\text{H}_2\text{O}}(x) dx. \quad (4.2.1)$$

Thinking of b as a variable and differentiating with respect to b (with a fixed), we get

$$0 = \frac{dQ}{dx}(b) + f_{\text{H}_2\text{O}}(b). \quad (4.2.2)$$

Since b is arbitrary, this can be written as

$$0 = \frac{dQ}{dx} + f_{\text{H}_2\text{O}}(x), \quad (4.2.3)$$

which is the required relationship. If the walls are impermeable to water, $f_{\text{H}_2\text{O}} = 0$ and (4.2.3) says that $Q = \text{constant}$.

Next, consider Na^+ transport in the same tubule. Let Na^+ concentration in the tubule be $c(x)$. Then the amount of Na^+ per unit time transported along the tubule by the flow past the point x is $Q(x)c(x)$. Let the amount of Na^+ per unit time per unit length transported outward through the walls near x be $f_{\text{Na}}(x)$. Then an argument identical to the one given above shows that

$$0 = \frac{d}{dx}(cQ) + f_{\text{Na}}(x). \quad (4.2.4)$$

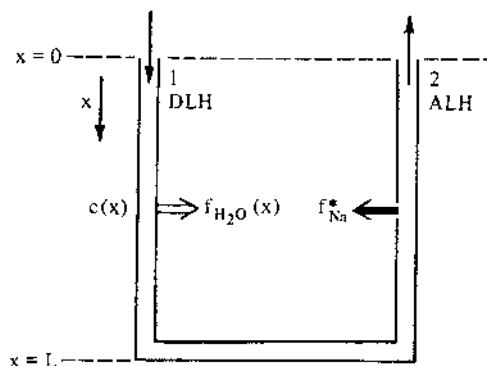


Figure 4.3. Schematic of the loop of Henle. The descending limb (DLH, also called tube 1) is freely permeable to water but impermeable to sodium ions. The ascending limb (ALH, also called tube 2) is impermeable to water, but sodium ions are actively pumped out of it at a fixed rate f_{Na}^* . x = distance, with $x = 0$ at the top of the loop and $x = L$ at the bottom. The concentration of sodium ion in the interstitium, i.e., outside the tubules, is denoted by $c(x)$. Internal concentrations and flow rates in the different tubules are denoted by $c_1(x)$, $c_2(x)$, $Q_1(x)$, and $Q_2(x)$.

In a tubule whose walls are impermeable to Na^+ , $f_{Na} = 0$, and (4.2.4) says that $cQ = \text{constant}$.

Equations (4.2.3) and (4.2.4) are general, and they apply to all of the tubules of the kidney. The tubules differ according to the properties of their walls as reflected in the fluxes f_{H_2O} and f_{Na} .

4.3 The Loop of Henle

We shall construct a model of the nephron beginning with the most interesting part of the system, the loop of Henle (see Figure 4.3). The descending limb of the loop is designated tube 1 and the ascending limb tube 2. The Na^+ concentrations and water flows in the tubules are written $c_i(x)$ and $Q_i(x)$, respectively, where $i = 1, 2$. The external Na^+ concentration is denoted by $c(x)$. By our sign conventions, the flow is positive in the descending limb and negative in the ascending limb, since x increases downward.

The physiological assumptions of our model of Henle's loop are as follows:

1. We assume that the walls of the descending limb are permeable to water but not to Na^+ . This is a simplification: The walls are also permeable to Na^+ , but this is not an essential feature of the operation of Henle's loop, and we leave it out. Moreover, we assume that the water permeability is so large that the water flux makes the internal

and external Na^+ concentrations equal. This gives the equations

$$0 = \frac{dQ_1}{dx} + f_{\text{H}_2\text{O}}^{(1)}(x), \quad (4.3.1)$$

$$0 = \frac{d}{dx}(Q_1 c_1), \quad (4.3.2)$$

$$c_1(x) = c(x). \quad (4.3.3)$$

2. We assume that Na^+ is pumped out of the ascending limb at a fixed rate f_{Na}^* per unit length. We also assume that the ascending limb is impermeable to water. This gives

$$0 = dQ_2/dx, \quad (4.3.4)$$

$$0 = (d/dx)(Q_2 c_2) + f_{\text{Na}}^*. \quad (4.3.5)$$

3. At the turn of Henle's loop ($x = L$) we assume that all of the salt and water leaving the descending limb enter the ascending limb. This gives the boundary conditions

$$c_1(L) = c_2(L), \quad (4.3.6)$$

$$Q_1(L) = -Q_2(L). \quad (4.3.7)$$

4. Finally, we need an assumption about the *peritubular capillaries*, which pick up the Na^+ that is actively pumped out of the ascending limb and the water that passively flows out of the descending limb. First, we assume that the capillaries pick up this Na^+ and water locally. That is, we do not allow for any longitudinal flow outside of the tubules and capillaries. Since we are considering a steady state model, it follows that the peritubular capillaries pick up water at the rate $f_{\text{H}_2\text{O}}^{(1)}(x)$ and Na^+ at the rate f_{Na}^* per unit length. The detailed mechanism of water and Na^+ transport across the capillary walls can give us additional equations, however. Here, we assume that the interstitial fluid is picked up by a process of filtration analogous to the process that occurs in the glomerular capillaries but running here in the opposite direction. The driving force for this reverse filtration is provided by the oncotic pressure of the plasma proteins. As remarked above, these proteins are present in higher concentrations in the peritubular capillaries than in the rest of the circulation because they were held back by the filtration process at the glomerulus. In reverse filtration at the peritubular capillaries, we assume that the Na^+ is carried passively by the water at its local tissue concentration. This implies a relationship between the flux of Na^+ and the flow of water:

$$f_{\text{Na}}^* = c(x)f_{\text{H}_2\text{O}}^{(1)}(x). \quad (4.3.8)$$

It is a remarkable feature of the model that we do not need to specify any more details about the peritubular capillaries than the considerations that led to (4.3.8). We now have enough equations to determine $c(x)$ without

reference to any further properties of the peritubular capillaries. This would not have been true if different assumptions about Na^+ and water transport across the capillary walls had been made.

We begin the study of Henle's loop by deriving a differential equation for the interstitial concentration $c(x)$. Note that (4.3.1) and (4.3.2) can be rewritten using (4.3.3) and (4.3.8) as follows:

$$0 = (dQ_1/dx) + f_{\text{Na}}^*/c(x), \quad (4.3.9)$$

$$0 = (d/dx)(Q_1 c). \quad (4.3.10)$$

Now, (4.3.10) implies that $Q_1 c = \text{constant}$. We use this to get an expression for $Q_1(x)$ in terms of $c(x)$:

$$Q_1(x) = Q_1(0)c(0)/c(x). \quad (4.3.11)$$

Note that $c(0)$ is the Na^+ concentration $c(x)$ in the fluid entering the loop of Henle from the proximal tubule. This is the same as the Na^+ concentration in blood plasma, and we can therefore regard $c(0)$ as given. Similarly, $Q_1(0)$ is the volume rate of flow entering the loop. This flow is *less* than the filtration rate by the total amount of fluid reabsorbed in the proximal tubule per unit time. We shall see later how $Q_1(0)$ is determined.

Another consequence of (4.3.10) is that

$$c \frac{dQ_1}{dx} = -Q_1 \frac{dc}{dx} = -Q_1(0)c(0) \frac{dc/dx}{c(x)}. \quad (4.3.12)$$

It follows that (4.3.9) can be rewritten in the form

$$\frac{dc}{dx} = \frac{f_{\text{Na}}^* c}{(Q_1(0)c(0))}, \quad (4.3.13)$$

which implies that

$$c(x) = c(0) \exp\left(\frac{f_{\text{Na}}^* x}{Q_1(0)c(0)}\right) \quad (4.3.14)$$

and, in particular, that

$$c(L) = c(0) \exp\left(\frac{f_{\text{Na}}^* L}{Q_1(0)c(0)}\right). \quad (4.3.15)$$

Note that $f_{\text{Na}}^* L$ is the total rate at which Na^+ is actively pumped out through the walls of the ascending limb of Henle's loop, while $Q_1(0)c(0)$ is the rate at which Na^+ enters the loop from the proximal tubule. Thus, the ratio of these fluxes

$$\alpha = \frac{f_{\text{Na}}^* L}{Q_1(0)c(0)} < 1 \quad (4.3.16)$$

determines the maximum concentrating ability of the model nephron for Na^+ through the equation

$$c(L) = c(0) \exp(\alpha). \quad (4.3.17)$$

We are now ready to consider the ascending limb of Henle's loop. Since this tubule is impermeable to water (equation 4.3.4) we have

$$Q_2(x) = Q_2(L) = -Q_1(L) = -Q_1(0)c(0)/c(L) = -Q_1(0)\exp(-\alpha). \quad (4.3.18)$$

Since Q_2 is constant, equation (4.3.5) reduces to

$$\frac{dc}{dx} = \frac{f_{Na}^*}{-Q_2} = \frac{f_{Na}^* \exp(\alpha)}{Q_1(0)}. \quad (4.3.19)$$

It follows that

$$c_2(x) = c_2(L) + (x - L)f_{Na}^* \exp(\alpha)/Q_1(0). \quad (4.3.20)$$

But, $c_2(L) = c(L) = c(0)\exp(\alpha)$. Thus,

$$c_2(x) = c(0)\exp(\alpha) + (x - L)f_{Na}^* \exp(\alpha)/Q_1(0). \quad (4.3.21)$$

In particular,

$$\begin{aligned} c_2(0) &= c(0)\exp(\alpha) + (0 - L)f_{Na}^* \exp(\alpha)/Q_1(0) \\ &= c(0)\exp(\alpha)(1 - \alpha). \end{aligned} \quad (4.3.22)$$

It is now easy to check (see Exercise 4.2) that $\exp(\alpha)(1 - \alpha) < 1$ when $\alpha \neq 0$. Thus, $c_2(0) < c(0)$, and the fluid leaving the top of the ascending limb is more dilute than blood plasma.

We have succeeded in expressing the outputs of Henle's loop in terms of its inputs. In the next section we shall find a good physiological reason for reversing this procedure. That is, we will find that the nephron actually adjusts $Q_1(0)$ to achieve a specified Na^+ concentration $c_2(0)$ at the top of the ascending limb. Thus, we should regard $c_2(0)$ as being a parameter, and we should express $Q_1(0)$ in terms of it.

4.4 The Juxtaglomerular Apparatus and the Renin-Angiotensin System

Near the top of the ascending limb of Henle's loop there is a specialized cluster of cells called the *juxtaglomerular apparatus*. These cells monitor the tubular fluid and secrete a hormone, *renin*, into the afferent arteriole just before it enters the glomerulus. Renin is converted in the blood to *angiotensin*, a potent vasoconstrictor, i.e., a substance that stimulates the constriction of blood vessels. Although the details are not certain, it is a plausible hypothesis that the cells of the juxtaglomerular apparatus monitor specifically the Na^+ concentration at the top of the ascending limb, and that they secrete enough renin to make the glomerular filtration, and perhaps reabsorption from the proximal tubule, proceed at whatever rate is needed to achieve a target level Na^+ concentration at that site.

Evidence that such a feedback mechanism is in operation comes from experiments of Thurauf, who was able to shut down flow in the proximal nephron by perfusing the distal nephron with salt solutions of higher than normal concentration.

We shall model this feedback mechanism in the simplest possible way: We assume that the inflow $Q_1(0)$ to the loop of Henle takes on whatever value is needed to satisfy the equation

$$c_2(0) = c^*, \quad (4.4.1)$$

where $c^* < c(0)$ is the target concentration sought by the juxtaglomerular apparatus. Thus, we do not model the details of the renin-angiotensin system. We simply assume that it is working, and we study its effects on the performance of the nephron.

Substituting (4.3.22) in (4.4.1), we get

$$a = \exp(\alpha)(1 - \alpha), \quad (4.4.2)$$

where

$$a = c^*/c(0) < 1. \quad (4.4.3)$$

Here a is regarded as known ($0 < a < 1$), so (4.4.2) is an equation for α and hence for $Q_1(0)$.

The graphical solution of (4.4.2) is discussed in Exercise 4.3, where it is shown that (4.4.2) has a unique positive solution for each a such that $0 < a < 1$ and that the solution satisfies $0 < \alpha < 1$. Armed with this knowledge, we can see directly from (4.4.2) that as $a \rightarrow 0$, $\alpha \rightarrow 1$. Thus, for small values of a ,

$$\alpha \approx 1, \quad (4.4.4)$$

$$1 - \alpha \approx a/\exp(1) = a/e. \quad (4.4.5)$$

from the definition of α , (4.3.16), we see that $\alpha \approx 1$ means that the Na^+ flux entering the loop of Henle has been adjusted to the pumping capacity of the ascending limb.

We can now rewrite the results of the previous section with c^* as a parameter under the assumption that a is small. We have

$$Q_1(0) = f_{\text{Na}}^* L/c(0), \quad (4.4.6)$$

$$-Q_2 = f_{\text{Na}}^* L/(e c(0)), \quad (4.4.7)$$

$$c(L) = e c(0), \quad (4.4.8)$$

$$c_2(0) = c^*. \quad (4.4.9)$$

These simple results summarize the behavior of the model of Henle's loop as controlled by the juxtaglomerular apparatus. Equation (4.4.6) states that the inflow to Henle's loop is adjusted so that the amount of Na^+ entering the loop per unit time ($Q_1(0)c(0)$) is equal to the amount that can be pumped out of the ascending limb per unit time ($f_{\text{Na}}^* L$). Equation (4.4.7)

Table 4.1. Outputs of the Model Nephron

	Diluting Mode (ADH Absent)	Concentrating Mode (ADH Present)
Na^+ concentration in urine	c^*	$e\,c(0)$
Rate of H_2O excretion	$f_{\text{Na}}^* L / (ec(0))$	$f_{\text{Na}}^* L c^* / (ec(0))^2$
Rate of Na^+ excretion	$f_{\text{Na}}^* L\,c^* / (ec(0))$	$f_{\text{Na}}^* L c^* / (ec(0))$

states that the flow ($-Q_2$) in the ascending limb of the loop is smaller by a factor of $e \approx 2.7$ than the flow entering the loop ($Q_1(0)$). The difference, of course, was reabsorbed from the descending limb and carried away by the peritubular capillaries along with the Na^+ pumped out of the ascending limb. Equation (4.4.8) states that the Na^+ concentration outside the tubules (and also in the descending limb of Henle's loop) increases by a factor of e as we follow the descending limb from its origin (where the Na^+ concentration is $c(0)$, the same as in blood plasma) to its turning point (where the Na^+ concentration is $c(L)$). Finally, equation (4.4.9) states that the Na^+ concentration in the fluid leaving the loop of Henle at the top of the ascending limb has the "target" Na^+ concentration c^* , which was indirectly established by the flow regulation achieved by the juxtaglomerular apparatus. Equations (4.4.6) through (4.4.9) are approximate results that are valid only when $c^* \ll c(0)$.

4.5 The Distal Tubule and Collecting Duct: Concentrating and Diluting Modes

We now come to the stage in the formation of urine where a decision has to be made whether to excrete a large volume of dilute urine or a small volume of concentrated urine. The hormone that determines which possibility will occur is antidiuretic hormone (ADH). When ADH is absent, we assume that the distal tubule and the collecting ducts are simple conduits, impermeable to both salt and water. In these circumstances, the fluid that leaves the top of the ascending limb becomes urine without further modification. Its properties are summarized in the first column of Table 4.1.

When ADH is present, the situation is more complicated. The effect of ADH is to make the distal tubule and the collecting duct permeable to water. We assume that this permeability is so great that equilibrium is achieved at every stage. In the distal convoluted tubule, then, enough water is withdrawn to make the Na^+ concentration equal to that of the blood plasma. Then, in the collecting duct, enough water is withdrawn to equilibrate with $c(x)$ at each x . We make the approximation that this water flux is negligible compared to the flux coming out of the descending limb

{see Exercise 4.6}. It follows that the concentration of Na^+ in the urine is $c(L) = ec(0)$. To figure out the volume rate of urine production, we first note that since only water is withdrawn, the Na^+ flux at the end of the collecting duct must be the same as that leaving Henle's loop. This Na^+ flux is given by

$$-Q_2 c^* = f_{\text{Na}}^* L c^* / (ec(0)). \quad (4.5.1)$$

Thus, the flow leaving the collecting duct when ADH is present must be

$$Q_3(L) = -Q_2 c^* / c(L) = f_{\text{Na}}^* L c^* / (ec(0))^2. \quad (4.5.2)$$

These results are entered in the second column of Table 4.1.

Comparison of the two modes of operation of the model nephron can be summarized as follows: First, the urine is more dilute than blood plasma when ADH is absent and more concentrated when ADH is present. Second, the volume rate of flow is much smaller when ADH is present by a factor of $c^*/ec(0)$. The rate at which Na^+ is excreted is the same in both cases, however.

Thus, the ADH mechanism cannot be used to regulate the total Na^+ content of the body. It can be used to regulate the Na^+ concentration of the blood plasma by excreting varying amounts of water in response to fluctuations in the plasma concentration of Na^+ .

4.6 Remarks on the Significance of the Juxtaglomerular Apparatus

The filtration rate in the kidney is about 100 times the rate of urine formation. Therefore, 99% of the filtrate must be reabsorbed, and a 1% error in the reabsorption rate results in a 100% error in the rate of urine formation. A nephron in which filtration is maintained, but reabsorption fails completely, excretes about 100 times as much urine as it should. It is inconceivable that such large errors would be tolerated. Clearly, the kidney needs control mechanisms that adjust the input to each nephron according to the capacity of the nephron for reabsorption of Na^+ and water.

We have seen that the juxtaglomerular apparatus can achieve this end by monitoring the Na^+ concentration at the top of the ascending limb of Henle's loop. When this has been done and when the filtration rate has been adjusted accordingly in each nephron, a heterogeneous population of nephrons can function in a coordinated fashion as easily as a homogeneous population. To see this, note that the concentrations in Table 4.1 are independent of the reabsorption capacity $f_{\text{Na}}^* L$, while the rates of excretion of both Na^+ and water are proportional to that capacity.

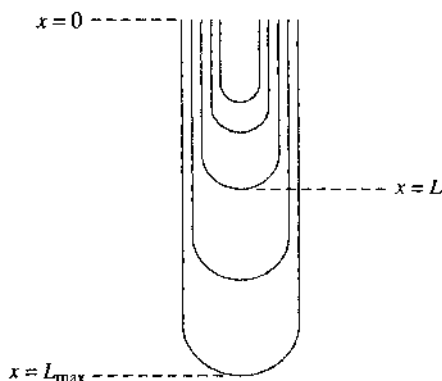


Figure 4.4. An interacting population of loops of Henle whose lengths L vary from 0 to $L_{\max}$. Note that at a given level x one sees only those loops whose lengths are at least equal to x .

Thus, each nephron contributes what it can to the production of a homogeneous urine. Coordinated performance of this kind would be impossible without the juxtaglomerular apparatus.

4.7 How Nephrons Do Better Than a Factor of e

The model nephron considered up to now in this chapter has the property that it cannot produce a urine in which the solute concentration is greater than $e \times$ the solute concentration in blood plasma, where the transcendental number e is approximately 2.7. No such limitation is observed in real kidneys, so we need to ask what feature of the foregoing model needs to be modified to account for the actual concentrating ability of the kidney. The answer, which is truly astonishing, was discovered by Harold Layton. It turns out that no modification at all in the nephron model is required. Instead, different nephrons can work together in a kind of “cascade” to produce higher solute concentrations in the urine than any one nephron could produce by itself. In order to do so, however, the nephrons must be of *different lengths*, and they must share the *same interstitial space*.

Let us consider, therefore, a population of loops of Henle, as shown in Figure 4.4. Each of these loops begins and ends at $x = 0$, and penetrates to a depth $x = L$, but L is different for the different loops in the population. We assume that L can take any value in the interval $0 \leq L \leq L_{\max}$, for some fixed $L_{\max}$, which in practice could be the “radius” of the kidney, i.e., the combined thickness of the cortex and medulla.

We shall describe the population of loops of Henle by the population density function $\rho(L)$, which has the following meaning:

$$\int_{L_1}^{L_2} \rho(L) dL = \text{number of loops with } L \text{ in the interval } (L_1, L_2). \quad (4.7.1)$$

The integral over all L , $\int_0^{L_{\max}} \rho(L) dL$, is the total number of loops in the population.

Each loop has the same properties as considered earlier in this chapter. In particular, all of the descending limbs have walls that are freely permeable to water but impermeable to Na^+ , and all of the ascending limbs have walls that are impermeable to water and pump Na^+ out at a fixed rate f_{Na}^* per unit length of tubule. We also assume that each loop comes equipped with its own juxtaglomerular apparatus that adjusts the inflow to the loop until the concentration of Na^+ at the top of the ascending limb is equal to some prescribed value c^* , independent of the length of the loop. Note that $c^* < c_0$, where c_0 is the concentration of Na^+ in blood plasma.

Since all of the loops interact with the same interstitium, the external concentration of Na^+ ion is given by a single function $c(x)$, which remains to be determined. Because the descending limbs are freely permeable to water, $c(x)$ is also the concentration of Na^+ in each of the descending limbs, regardless of the length of that limb.

The volume flow rate along the descending limbs may be different for nephrons of different lengths, even if those flow rates are measured at the same depth x . Let us therefore define $Q_1(x, L)$ = flow rate at x in the descending limb of a loop of length L . Similarly, let $(f_{\text{H}_2\text{O}})_1(x, L)$ = flux of water (volume per unit time, per unit length of tubule) at x out of a descending limb whose length is L . Of course, $Q_1(x, L)$ and $(f_{\text{H}_2\text{O}})_1(x, L)$ are defined only for $x \leq L$.

We can now write down five equations that determine the behavior of the whole population of loops of Henle. These equations (which will be explained below) are as follows:

$$c(0) = c_0, \quad (4.7.2)$$

$$\frac{\partial Q_1}{\partial x}(x, L) + (f_{\text{H}_2\text{O}})_1(x, L) = 0, \quad (4.7.3)$$

$$\frac{\partial}{\partial x} (c(x)Q_1(x, L)) = 0, \quad (4.7.4)$$

$$Q_1(L, L)c(L) = f_{\text{Na}}^*L + Q_1(L, L)c^*, \quad (4.7.5)$$

$$c(x) \int_x^{L_{\max}} (f_{\text{H}_2\text{O}})_1(x, L) \rho(L) dL = f_{\text{Na}}^* \int_x^{L_{\max}} \rho(L) dL. \quad (4.7.6)$$

The symbol $\partial/\partial x$ appearing in these equations denotes the *partial* derivative with respect to x . This is essentially the same as the familiar d/dx , also called the *ordinary* derivative, but the symbol $\partial/\partial x$ is used when there is more than one independent variable, as a reminder to hold all other vari-

ables constant when taking the derivative with respect to x . (In our case, there is one other independent variable, namely L .)

Equation (4.7.2) simply asserts that the fluid entering all of the loops of Henle has the same Na^+ concentration as blood plasma. Equation (4.7.3) expresses the conservation of volume for flows in the descending limbs in each of the loops of Henle. Equation (4.7.4) states that the transport of Na^+ along each of the descending limbs is constant (independent of x), since these descending limbs are assumed to be impermeable to Na^+ . Equation (4.7.5) expresses Na^+ balance for the ascending limbs. The lefthand side is the rate at which Na^+ enters the ascending limb of a loop of length L , and the right-hand side is the sum of the amount of Na^+ that is pumped out of the ascending limb and the amount of Na^+ that makes it all the way around the loop of Henle and leaves the ascending limb at the top.

Equation (4.7.6) requires the most explanation. The integrals that appear in it extend from x to $L_{\max}$ and therefore involve all loops of length x or greater. These are the loops that are present at level x , all others having turned before reaching this level. The integral on the righthand side is simply the *number* of loops of length at least x . This is multiplied by f_{Na}^* to get the total flux of Na^+ out of ascending limbs at level x . The integral on the left-hand side is the total flux of water out of descending limbs at level x . As this water is picked up by the peritubular capillaries, it carries with it a flux of Na^+ equal to the flux of water multiplied by $c(x)$, the local interstitial concentration of Na^+ . Thus equation (4.7.6) expresses the steady-state condition that the local rate of uptake of Na^+ by the peritubular capillaries must balance the local rate at which Na^+ is pumped out of the ascending limbs.

We now combine the above equations to get a single equation for the interstitial concentration profile $c(x)$. This is done as follows. First, solve equation (4.7.5) for $Q_1(L, L)$, the flow rate at the hairpin turn of a loop of length L . This gives

$$Q_1(L, L) = \frac{f_{\text{Na}}^* L}{c(L) - c^*}. \quad (4.7.7)$$

Next, from equation (4.7.4),

$$c(x)Q_1(x, L) = c(L)Q_1(L, L) = \frac{f_{\text{Na}}^* L}{1 - c^*/c(L)}, \quad (4.7.8)$$

$$Q_1(x, L) = \frac{1}{c(x)} \frac{f_{\text{Na}}^* L}{1 - c^*/c(L)}. \quad (4.7.9)$$

Then, from equation (4.7.3),

$$(f_{\text{H}_2\text{O}})_1(x, L) = -\frac{\partial Q_1}{\partial x}(x, L) = \frac{1}{(c(x))^2} \frac{dc}{dx}(x) \frac{f_{\text{Na}}^* L}{1 - c^*/c(L)}. \quad (4.7.10)$$

Substituting this result into equation (4.7.6), and canceling f_{Na}^* and one of the factors of $c(x)$, we get the equation

$$\frac{1}{c(x)} \frac{dc}{dx}(x) \int_x^{L_{\max}} \frac{L \rho(L) dL}{1 - c^*/c(L)} = \int_x^{L_{\max}} \rho(L) dL, \quad (4.7.11)$$

which can be put in the form

$$\frac{d}{dx} (\log c(x)) = \int_x^{L_{\max}} \rho(L) dL \left(\int_x^{L_{\max}} \frac{L \rho(L) dL}{1 - c^*/c(L)} \right)^{-1} \quad (4.7.12)$$

Finally, we can integrate this equation from $x = 0$ to $x = X$ and then apply the exponential function (which is the inverse of the natural logarithm) to both sides. Recalling the boundary condition that $c(0) = c_0$, i.e., that the concentration of Na^+ entering each loop of Henle is the same as that in the blood plasma, we obtain

$$c(X) = c_0 \exp \left(\int_0^X \frac{\int_x^{L_{\max}} \rho(L) dL}{\int_x^{L_{\max}} \frac{L \rho(L) dL}{1 - c^*/c(L)}} dx \right). \quad (4.7.13)$$

Equation (4.7.13) is an integral equation in the unknown function $c(x)$, which appears on the left-hand side evaluated at $x = X$ and on the right-hand side evaluated at $x = L$. (Both X and L are variable, and take on values between 0 and $L_{\max}$.) There is a special (or limiting) case, however, in which equation (4.7.13) reduces to a formula for $c(X)$. That is the case $c^* = 0$. Then $c(L)$ no longer appears on the right-hand side of equation (4.7.13), and we have the formula

$$c(X) = c_0 \exp \left(\int_0^X \frac{\int_x^{L_{\max}} \rho(L) dL}{\int_x^{L_{\max}} L \rho(L) dL} dx \right). \quad (4.7.14)$$

Although this result is exact only when $c^* = 0$, it is approximately correct when c^* is small, i.e., when $c^*/c_0 \ll 1$. We shall study the behavior of this approximate solution now, and later suggest a computing project to solve the integral equation (4.7.13) in order to study the effect of nonzero c^* on the interstitial concentration profile $c(x)$.

As a simple example of a nephron population, consider the case in which $\rho(L)$ is constant on the whole interval from 0 to $L_{\max}$, say $\rho(L) = \rho_0$. This means that the number of loops whose lengths fall in a given interval is just proportional to the size of that interval. Another way to say this is that the number of loops of length *at least* L declines linearly with L until that number reaches zero at $L = L_{\max}$. In this case, the integrals that appear in equation (4.7.14) are very easy to evaluate. We have

$$\int_x^{L_{\max}} \rho(L) dL = \rho_0 (L_{\max} - x), \quad (4.7.15)$$

$$\int_x^{L_{\max}} L \rho(L) dL = \frac{1}{2} \rho_0 (L_{\max}^2 - x^2). \quad (4.7.16)$$

Factoring the difference of squares, and combining these results, we get

$$c(X) = c_0 \exp \left(\int_0^X \frac{2}{L_{\max} + x} dx \right), \quad (4.7.17)$$

$$c(X) = c_0 \exp \left(2 \log \frac{L_{\max} + X}{L_{\max}} \right). \quad (4.7.18)$$

Now, $\exp(2 \log(z)) = \exp(\log(z^2)) = z^2$, and so the above result can be simplified as

$$c(X) = c_0 \left(\frac{L_{\max} + X}{L_{\max}} \right)^2. \quad (4.7.19)$$

Note in particular that $c(L_{\max}) = 4c_0$, and this exceeds the concentrating ability of an isolated nephron by the factor $4/e$ which is about 1.5, or a 50% improvement. In Exercise (4.8) you will have the opportunity to design loop-length distributions that do considerably better than this.

The theory that we have just outlined can be improved by including the effects of the collecting ducts on the concentration profile. This is in the spirit of Exercise 4.7, in which you are asked to do the same thing for the single-nephron model developed earlier in the chapter. Here, however, we give the details in the text, since they are somewhat more complicated. Our discussion here is limited to the concentrating mode (ADH present), in which the distal tubule and collecting duct are freely permeable to water but not, according to our assumptions, to Na^+ . The water that is reabsorbed from the collecting duct affects the concentration profile of the interstitium, and it is this effect that we want to take into account.

The distal convoluted tubules of several nephrons typically enter into a common collecting duct, and we shall make the assumption here that the interacting nephrons (of different loop lengths) that were considered above all share a single collecting duct. (It would not change the theory in any substantive way if the collecting ducts were separate, but it would change the notation and setup of the equations.) Note that this single collecting duct runs parallel to the tubules of the multiple loops of Henle of our model, that it shares the interstitium with them, and that it traverses the complete interstitium and therefore, in our notation, extends from $x = 0$ to $x = L_{\max}$.

As was done earlier in this chapter, we use the subscript 3 to denote the collecting duct. The inflow to the collecting duct comes from the distal convoluted tubules of the different nephrons. Recall the assumption made earlier that in the concentrating mode, enough water is withdrawn from the distal convoluted tubule to make the concentration of Na^+ in it equal to c_0 at its distal end, where it joins the collecting duct. Another assumption

made earlier is that the distal convoluted tubule is impermeable to Na^+ , so that the net rate at which Na^+ enters this tubule is equal to the net rate at which it leaves (since we assume steady-state conditions). These considerations lead to the equation

$$c^* \int_0^{L_{\max}} Q_1(L, L) \rho(L) dL = c_0 Q_3(0). \quad (4.7.20)$$

The left-hand side of this equation is the sum (integral) of the rate at which Na^+ is leaving all of the ascending limbs of all of the different loops of Henle (of various lengths L), and hence entering the distal convoluted tubules. Recall that c^* is the Na^+ concentration at the top of each of the ascending limbs; this is the result of the operation of the juxtaglomerular apparatus. Also, recall that $Q_1(L, L)$ is the flow rate of water all along the ascending limb of a loop of length L , and that $\rho(L) dL$ is the number of loops with length between L and $L + dL$. The right-hand side, of course, is the rate at which Na^+ enters the collecting duct.

The equations for flow of Na^+ and water along the collecting duct are similar to those of the descending limb of the loops of Henle, equations (4.7.3)–(4.7.4):

$$\frac{d}{dx} Q_3(x) + (f_{\text{H}_2\text{O}})_3(x) = 0, \quad (4.7.21)$$

$$\frac{d}{dx} (Q_3(x) c(x)) = 0. \quad (4.7.22)$$

Note that we use ordinary derivatives here because the functions in question depend only on the single variable x .

Finally, we have to modify equation (4.7.6) to take into account the flow of water out of the collecting duct. This water, like the water that flows out of the descending limbs of the various loops of Henle, has to be picked up by the peritubular capillaries. The revised version of equation (4.7.6) is as follows:

$$c(x) \int_x^{L_{\max}} (f_{\text{H}_2\text{O}})_1(x, L) \rho(L) dL + c(x) (f_{\text{H}_2\text{O}})_3(x) = f_{\text{Na}}^* \int_x^{L_{\max}} \rho(L) dL. \quad (4.7.23)$$

The new feature is the second term on the left-hand side, $c(x)(f_{\text{H}_2\text{O}})_3(x)$, which represents the amount of Na^+ picked up by the water that has left the collecting duct as that water flows through the interstitium on its way to being reabsorbed by the peritubular capillaries.

In summary, the equations of the improved model are equations (4.7.2)–(4.7.5) and equations (4.7.20)–(4.7.23). The improved model can be reduced to an integral equation for the unknown interstitial Na^+ concentration profile $c(x)$ in much the same way as was done above for the model that neglected the influence of the collecting duct. Much of the work has already been done. The new part is as follows:

First, we use equations (4.7.20)–(4.7.22) to express $(f_{\text{H}_2\text{O}})_3(x)$ in terms of $c(x)$. We have

$$Q_3(x)c(x) = Q_3(0)c_0 = c^* \int_0^{L_{\max}} Q_1(L, L)\rho(L) dL, \quad (4.7.24)$$

and hence

$$Q_3(x) = \frac{c^*}{c(x)} \int_0^{L_{\max}} Q_1(L, L)\rho(L) dL. \quad (4.7.25)$$

Therefore,

$$(f_{\text{H}_2\text{O}})_3(x) = -\frac{dQ_3}{dx} = \frac{c^*}{(c(x))^2} \frac{dc}{dx} \int_0^{L_{\max}} Q_1(L, L)\rho(L) dL. \quad (4.7.26)$$

Substituting into this the expression for $Q_1(L, L)$ derived above (equation (4.7.7)), and then substituting the result into equation (4.7.23), we obtain

$$\begin{aligned} \frac{1}{c(x)} \frac{dc}{dx}(x) & \left(\int_x^{L_{\max}} \frac{L\rho(L) dL}{1 - c^*/c(L)} + \int_0^{L_{\max}} \frac{(c^*/c(L)) L\rho(L) dL}{1 - c^*/c(L)} \right) \\ & = \int_x^{L_{\max}} \rho(L) dL, \end{aligned} \quad (4.7.27)$$

which should be compared to equation (4.7.11). Note that the new term is an integral from 0 to $L_{\max}$, not x to $L_{\max}$. The reason for this is that the collecting duct receives fluid from loops of all lengths. Manipulating this exactly as we did equation (4.7.11), we arrive at

$$c(X) = c_0 \exp \left(\int_0^X \frac{\int_x^{L_{\max}} \rho(L) dL}{\int_x^{L_{\max}} \frac{L\rho(L) dL}{1 - c^*/c(L)} + \int_0^{L_{\max}} \frac{(c^*/c(L)) L\rho(L) dL}{1 - c^*/c(L)}} dx \right), \quad (4.7.28)$$

which is the improved version of equation (4.7.13). (Again note the different limits on the two integrals in the denominator.) Like equation (4.7.13), this is an integral equation for the unknown function $c(x)$, which appears on the left-hand side as $c(X)$ and on the right-hand side as $c(L)$. A suggested computing project will be the numerical solution of these two integral equations, to compare their solutions. Note that in the limit $c^* \rightarrow 0$, equation (4.7.28) reduces to the same explicit formula for $c(X)$ as is obtained by taking that same limit in equation (4.7.13), namely, equation (4.7.14). This means that as $c^* \rightarrow 0$, the presence of the collecting duct has less and less influence on the concentration profile of interstitial solute. (The same phenomenon happens in a single-nephron model; see Exercise 4.7.) Think about why this is the case!

4.8 Computing Project on the Interacting Nephron Population Model

The purpose of this project is to evaluate the integral in equation (4.7.14) in the case of an arbitrary loop-length distribution $\rho(L)$, which may be given only in the form of data rather than as a mathematical expression, and then to solve the two integral equations (4.7.13) and (4.7.28), again in the case of an arbitrary loop-length distribution.

Comparison of the three results for a given $\rho(L)$ can be used to address the following questions. First, how good is the approximation $c^* = 0$? Equation (4.7.14) makes this approximation, but the other two equations do not. Next, what is the effect of the collecting duct on the concentration profile of solute in the interstitium? Equation (4.7.28) includes this influence, but equation (4.7.13) does not. A more physiological way to say this is that equation (4.7.13) applies to the diluting mode, in which ADH is absent, and the distal convoluted tubule and collecting duct are impermeable to water, whereas equation (4.7.28) applies to the concentrating mode, in which ADH is present and these tubules are freely water-permeable. How much difference does this make to the concentration profile $c(x)$ and in what direction? Think about what you expect before doing the computation! Finally, any of the three equations (but most reliably equation (4.7.28)) can be used to study the influence of the loop-length distribution on the concentrating ability of the interacting nephron population. This is done analytically in Exercise (4.8), but only for a limited class of loop-length distributions. The power of the computational approach is that it is not dependent on specific mathematical formulae but instead can handle any $\rho(L)$ at all, even one that is only given as a table of numbers.

Instead of writing out specific computer programs in this case, we just explain the main ideas, and leave the programming to the student.

The first question is how to evaluate a definite integral, something like $\int_0^{L_{\max}} f(x) dx$. We assume that $f(x)$ is known on an array of $N + 1$ equally spaced points $x = 0, \Delta x, 2\Delta x, \dots, L_{\max}$, where $\Delta x = L_{\max}/N$, for some integer N . Then a good way to approximate the integral in question is to use the *trapezoidal rule*:

$$\begin{aligned} \int_0^{L_{\max}} f(x) dx \\ \approx \left(\frac{1}{2} f(0) + f(\Delta x) + f(2\Delta x) + \dots + \frac{1}{2} f(N\Delta x) \right) \Delta x. \end{aligned} \quad (4.8.1)$$

Note that the endpoints are each weighted by $\frac{1}{2}$ relative to the interior points. (Think about why this makes sense.)

In the equations of interest we also have integrals of the form $\int_x^{L_{\max}} f(L) dL$. These are also definite integrals if we think of x as a fixed number, and so they can be evaluated by the above method for any par-

ticular x . But we want to evaluate them for all x on the same array of points that was mentioned above, namely $x = 0, \Delta x, 2\Delta x, \dots, L_{\max}$. A little thought will show that the best way to do this is *not* to treat each value of x as a separate problem, but rather to sweep down from $x = L_{\max}$ (where the integral is clearly zero, since the endpoints coincide) to $x = 0$, taking advantage of the latest result to calculate the next result in each case. (Think about what terms have to be added each time x moves down by Δx . In thinking about this, don't forget those annoying factors of $\frac{1}{2}$.)

These ideas suffice for the numerical evaluation of the integral in equation (4.7.14), so try this before proceeding further. You can check your program by running it on some cases that can be done analytically, e.g., constant $\rho(L)$, or the family of $\rho(L)$ considered in Exercise 4.8.

Now we come to the fundamentally more difficult problem of solving the integral equations (4.7.13) and (4.7.28). What makes these more difficult is that the unknown function $c(x)$ appears on both sides of the equation. In effect, either of these equations provides a definition of $c(x)$ in terms of itself. One way to solve such a system (which is not guaranteed to work, but which tells you when it is working and when it is not) is by the method of successive approximations (also known as "fixed-point iteration"). This procedure starts with an initial guess for $c(x)$, which might, for example, be obtained by setting $c^* = 0$, i.e., by doing the part of this project described above. The initial guess is then substituted into the right-hand side of equation (4.7.13) or (4.7.28), and the integrals that appear there are evaluated by the trapezoidal rule, as outlined above. The result is a new approximation to $c(x)$. This procedure is repeated, each time substituting the latest result on the right-hand side of equation (4.7.13) or (4.7.28) and obtaining the next result on the left-hand side. The hope is that the resulting sequence of results will *converge*, i.e., approach some limiting function $c(x)$, which would then be the solution to the equation. This is not guaranteed, but in practice it is easy to tell whether convergence is happening or not by monitoring the changes in the computed solution from one step to the next. If these are getting smaller in some regular manner, then the solution is converging, and the procedure may be stopped when the changes are small enough to be ignored for practical purposes. On the other hand, if the changes are getting larger instead of smaller from one step to the next, then the computation is diverging, and the solution cannot be found in this way.

Can we say anything in advance about whether the method of successive approximations will converge or not? In fact, we can. Convergence is expected when a change in any particular guess leads to a *smaller* change in the next guess, i.e., when the results of a single step are sufficiently *insensitive* to the results of the previous step. (These ideas can be made precise, but here we just want to convey the intuition.) Now, we already know qualitatively how to make such insensitivity happen: by making c^* small. In the extreme case, when $c^* = 0$, the function $c(x)$ does not even appear on the

right-hand side, and the method of successive approximations gets to the answer in one step. Therefore, we can expect convergence when c^*/c_0 is sufficiently small. This qualitative argument does not say how small, but you can find that out by trying the method for different c^*/c_0 and monitoring the results for convergence or nonconvergence. (When the method of successive approximations fails to converge, this does *not* necessarily mean that anything is wrong with the problem, just that other methods have to be used to find the solution.)

4.9 Annotated References

For students who want to learn more about the physiology of the kidney, the following two texts should be useful (the first is elementary and the second is advanced):

Sullivan, L.P., and Grantham, J.J.: *Physiology of the Kidney* (2nd edition). Lea & Febiger, Philadelphia, 1982.

Koushanpour, E. and Kriz, W.: *Renal Physiology: Principles, Structure, and Function* (2nd edition). Springer-Verlag, New York, NY, 1986.

The discussion of the juxtaglomerular apparatus in this chapter is strongly influenced by the work of Thureau:

Thureau, K. and Mason, J.: Intrarenal function of the juxtaglomerular apparatus. In: *Kidney and Urinary Tract Physiology* (Thureau, K., ed.). M.T.P. International Review of Science, Physiology Series 1, Volume 6. University Park Press, Baltimore, MD, 1974, 357–390.

The discussion in this chapter concerning the interaction of loops of Henle of different lengths is based on the following paper:

Layton, H.E.: Distribution of Henle's loops may enhance urine concentrating ability. *Biophys. J.* 49; 1033–1040, 1986.

There is a considerable body of mathematical work concerning the kidney. This literature is reviewed in the following papers:

Knepper, M.A., and Rector, F.C., Jr.: Urinary concentration and dilution. In: *The Kidney* (4th edition) Volume 1, (Brenner, B.M. and Rector, F.C., Jr., eds.) Harcourt Brace Jovanovich, Philadelphia, PA, 1991, 445–482.

Roy, D.R., Layton, H.E., and Jamison, R.L.: Countercurrent mechanism and its regulation. In: *The Kidney: Physiology and Pathophysiology* (3rd edition) (Seldin, D.W. and Giebish, G., eds.) Lippincott Williams & Wilkins, 2000.

Exercises

4.1. Equation of Na^+ Conservation in a Renal Tube

Derive equation (4.2.4).

4.2. Proof

Show that $\exp(\alpha)(1 - \alpha) < 1$ for all $\alpha \neq 0$. (Hint: Consider the sign of the derivative of this function of α when $\alpha > 0$ and when $\alpha < 0$.)

4.3. Graphical Solution of the Equation for α

Consider the equation $a = \exp(\alpha)(1 - \alpha)$ where a is given and $0 < a < 1$. Plot the right-hand side as a function of α over the interval $0 < \alpha < 1$ and indicate how such a plot could be used to find the unique positive value of α that satisfies the equation. For example, calculate α when $a = c^*/c(0) = 0.43$.

4.4. Quantitative State of the Nephron

Assume that

$$\begin{aligned} c(0) &= 140 \text{ mEq/liter,} \\ c^* &= 60 \text{ mEq/liter,} \\ f_{\text{Na}}^* L/c(0) &= 0.25/(2.0 \times 10^6) \text{ liters/minute,} \end{aligned}$$

where $1 \text{ mEq} = 10^{-3} \times 6.0 \times 10^{23} \text{ ions}$.

1. Using the approximation that $c^*/c(0) \ll 1$, calculate $Q_1(0)$, $Q_1(L)$, $Q_3(0)$, $Q_3(L)$, and $c(L)$. Do this calculation for both the concentrating and diluting modes and use your answers in each case to construct a quantitative diagram of the nephron with flows and Na^+ concentrations labeled at appropriate sites.
2. Repeat part 1, but without making the approximation that $c^*/c(0) \ll 1$. (Hint: Start from the value of α that was determined in Exercise 4.3.)

4.5. The Effect of a Change in f_{Na}^*

What happens to the nephron in Exercise 4.4 (in the concentrating mode) if f_{Na}^* is reduced by a factor of 2? Consider two cases:

1. The juxtaglomerular apparatus is working, so $c_2(0) = c^*$, which remains constant, but $Q_1(0)$ changes.
2. The juxtaglomerular apparatus is not working, so $c_2(0)$ is allowed to change, but $Q_1(0)$ remains constant.

Note: Do *not* use the approximation $c^*/c(0) \ll 1$.

Compare the two cases with regard to water output, Na^+ output, and Na^+ concentration in the urine. What happens in the two cases if $f_{\text{Na}}^* = 0$?

4.6. Comparison of Water Reabsorption from DLH and CD

Use the results of Exercise 4.4 to make a quantitative comparison of the flow of water out of the descending limb of Henle's loop with the flow of water out of the

collecting duct when ADH is present. Recall that we made the approximation that the latter flow is small in comparison with the former.

4.7. A More Complete Model

How are the equations of the single-nephron model changed when the flow of water out of the collecting duct is not neglected? (Hint: This water has to be picked up by the peritubular capillaries.)

Solve the equations of this more accurate model under the assumption that c^* is given and compare the results with those of Section 4.5. In particular, show that

1. The interstitial fluid is slightly more *dilute* when the kidney is in the concentrating mode than when it is in the diluting mode.
2. The approximation of neglecting the flux of water out of the collecting ducts gets better and better as $c^*/c(0) \rightarrow 0$.

4.8. Influence of the Loop-Length Distribution on the Solute Concentration That an Interacting Nephron Population Can Achieve

This exercise relates to the section at the end of the chapter, in which we considered a population of nephrons of different lengths interacting through their shared interstitium. Make the approximation $c^* = 0$ and therefore use equation (4.7.14), which is an explicit formula for $c(X)$. In the text we evaluated this formula for the special case $\rho(L) = \text{constant}$, independent of L , which implies a uniform distribution of loop lengths. Now consider a more general case given by $\rho(L) = a(L_{\max} - L)^p$, where a is an arbitrary constant (which should not affect your result; do you see why?), and p is a positive (but possibly fractional) power. The point of the problem is to see how the power p affects the concentrating ability of the nephron population. Therefore, evaluate $c(X)$ in terms of p , paying particular attention to $c(L_{\max})$. Make a table or graph showing how $c(L_{\max})/c_0$ depends on p . Take the limit $p \rightarrow 0$ to check that you recover the result found in the text. Hint: It will help in evaluating the integrals to make a change of variables such as $Z = L_{\max} - L$.

5

Muscle Mechanics

In this chapter we consider some macroscopic and microscopic aspects of how muscles work. The macroscopic behavior of muscle is described by a relationship between force and velocity known as the “force-velocity curve.” We shall see how this relationship is a consequence of the collective behavior of a large number of tiny crossbridges that are the microscopic agents of muscle contraction.

5.1 The Force-Velocity Curve

An important property of muscle is the relationship between the *force* P that it generates and the *velocity* V of its shortening. Observations of muscles under tension reveal a relationship between force and shortening velocity like that shown in Figure 5.1.

The force-velocity curve characterizes a muscle in a constant contractile state. Such a state can be achieved in an isolated muscle by rapid, repetitive electrical stimulation. In life, a constant contractile state is achieved by sending a steady stream of nerve impulses to signal the muscle to contract. A muscle in a constant contractile state (and within a certain range of lengths) shortens at a rate that is determined by the load (or force) at the ends of the muscle. It is this relationship that is depicted in Figure 5.1.

Two important points on the force-velocity curve are its intercepts with the coordinate axes: When $P = 0$, the muscle shortens at its maximum velocity $V_{\max}$. As the load is increased, the velocity of shortening gets

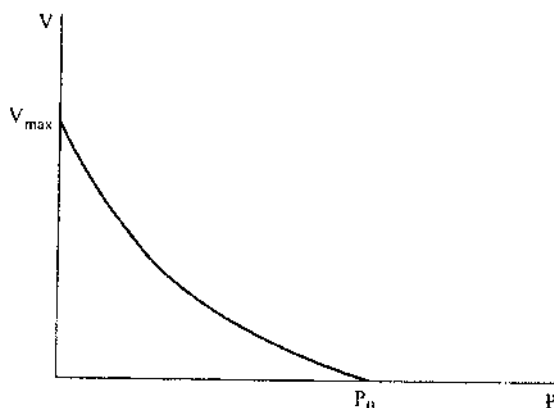


Figure 5.1. The force-velocity curve of muscle. V = velocity of shortening, P = force against which the muscle is working. $V_{\max}$ is the velocity of shortening in the absence of any load ($P = 0$), and P_0 is the isometric force, i.e., the force that the muscle develops when it cannot shorten at all ($V = 0$).

smaller, until the *isometric force*, P_0 , is reached, at which the muscle cannot shorten ($V = 0$).

A.V. Hill was first to notice that the experimental force-velocity curves are closely fitted by an equation of the form

$$V = b \frac{P_0 - P}{P + a}, \quad (5.1.1)$$

where P_0 is the isometric force, and a and b are constants that can be determined from the experimental data. To do this, we first determine P_0 as the force at which contraction does not occur. Then data points (V_n, P_n) are observed for $n = 1, \dots, N$. The relation

$$(P + a)V = b(P_0 - P)$$

is linear in the parameters a and b and it is therefore one to which the method of least squares (see Section 7.1.2) can be applied. In particular, it shows there how a and b can be chosen such that

$$\sum_n [(P_n + a)V_n - b(P_0 - P_n)]^2$$

is minimized. This gives the "best" estimate of a and b from data. Then $V_{\max}$ can be determined by setting $P = 0$ in (5.1.1) and solving for V . The result is

$$V_{\max} = bP_0/a. \quad (5.1.2)$$

The remainder of this chapter is devoted to a derivation of the form of Hill's force-velocity curve from a microscopic theory of muscles.

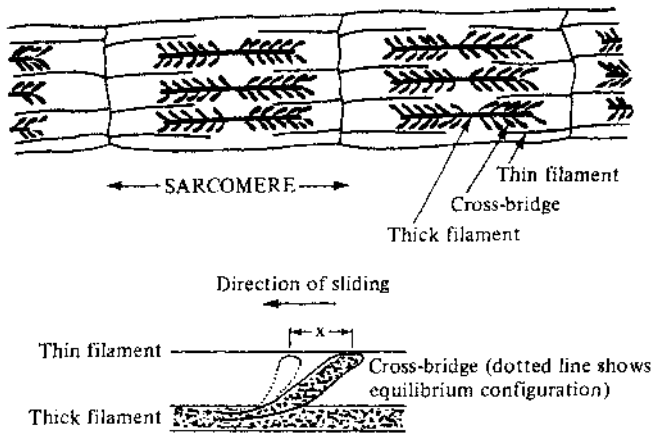


Figure 5.2. The molecular machinery of muscle contraction. Upper part of the figure shows longitudinal section of the sarcomere, the fundamental unit of muscle function. In the center of the sarcomere are the thick filaments, made of myosin. The heads of the myosin molecules protrude and form the crossbridges. These interact with the thin filaments, made of actin, which extend to the ends of the sarcomere. Note orientation of the crossbridges, which point away from the center of the sarcomere. A blowup of a crossbridge is shown in the lower part of the figure. The crossbridge has an equilibrium configuration in which it exerts no longitudinal force on the thin filament. The variable x measures longitudinal displacement from this equilibrium configuration. Note that x is undefined for a crossbridge that is not attached to the thin filament.

5.2 Crossbridge Dynamics

We now turn to a microscopic description of the contractile process in skeletal muscle. As indicated in Figure 5.2, the *sarcomere* is a basic unit of muscle function. This contains the *thick filaments* and *thin filaments* that slide past each other during the process of contraction. These filaments do not change length during contraction of the muscle, but their relative motion is brought about by the *crossbridges* between them.

The crossbridges are arm-like projections from the thick filaments that connect and disconnect with the thin filaments at rates which depend on the chemical composition of the crossbridge environment. Immediately after such a connection is established, a chemical reaction occurs that puts the crossbridge into a strained configuration. The crossbridge then pulls the thin filament toward the center of the sarcomere. A second chemical reaction leads to breakage of the crossbridge connection to the thin filament. This is referred to as the *crossbridge cycle*.

The smooth contraction of the sarcomere is brought about by the combined activity of the entire crossbridge population. The events of at-

tachment and detachment occur essentially at random and independently in different crossbridges. The chemical environment within the sarcomere governs the mean attachment and detachment rates. On the other hand, the motions of the attached crossbridges are coordinated, since these motions are determined by the sliding velocity of the filaments. The sliding velocity is the same for all of the crossbridges in a sarcomere.

As the sliding of the filament carries the crossbridge away from its initial configuration, the strain of that configuration is relieved, and the force on the thin filament is reduced. At higher velocities of shortening, this happens faster. Thus, the muscle develops smaller forces at higher velocities of shortening. This effect explains, qualitatively at least, the form of the force-velocity curve. Such considerations can be made quantitative as follows.

We assume that an attached crossbridge has an equilibrium configuration in which it exerts no force on the thin filament, and we let x denote the displacement from this equilibrium configuration measured along the thin filament. Let $p(x)$ be the force on the thin filament when the displacement of the crossbridge is equal to x . A specific formula for $p(x)$ is introduced later, but for now we note that $p(0) = 0$, since $x = 0$ is the equilibrium configuration of an attached crossbridge, and that $p(x)$ is an increasing function of x . Let n_0 be the number of crossbridges in a half sarcomere. All of these crossbridges are effectively in parallel (their forces add), and all of the half sarcomeres are effectively in series (their lengths add). Accordingly, the force on the ends of the muscle if all crossbridges were attached and had displacement x would be $n_0 p(x)$.

In general, the different crossbridges have different values of x , and we describe the crossbridge population by the *population-density function* $u(x)$ in the sense that the fraction of crossbridges with displacement x such that $x_1 < x < x_2$ is given by the integral of u over that interval:

$$\int_{x_1}^{x_2} u(x) dx \quad (5.2.1)$$

We take the point of view that the quantity x is defined only for an attached crossbridge. The total fraction of crossbridges that are attached is given by

$$U = \int_{-\infty}^{\infty} u(x) dx < 1. \quad (5.2.2)$$

The force P on the ends of the muscle is also given by an integral over the crossbridge population:

$$P = n_0 \int_{-\infty}^{\infty} p(x) u(x) dx. \quad (5.2.3)$$

To understand this formula, think of $n_0 u(x) dx$ as the number of crossbridges with displacements in an interval of length dx near x . Then

$n_0 p(x) u(x) dx$ is the force exerted by the crossbridges in this interval, and the integral over all x gives the total force.

The crossbridge *cycle* (attachment, sliding, detachment) is described in quantitative terms as follows: We assume that all crossbridges form their attachments in a certain configuration $x = A > 0$. The rate of attachment of new crossbridges is proportional to the number of crossbridges that are available for attachment at any given time. In a half sarcomere, this number is $n_0(1-U)$. Thus, we assume that the rate of formation of new crossbridges in a half sarcomere is $\alpha n_0(1-U)$, where the constant α is called the *rate constant* for attachment. Note that α has units of time^{-1} .

Attached crossbridges move with the thin filament. Thus, an attached crossbridge satisfies the equation

$$dx/dt = -v,$$

where v is the shortening velocity of the thin filament relative to the thick filament. From the anatomy of the sarcomere and the arrangement of the sarcomeres in the muscle (Figure 5.2), we conclude that $v = V/2N$, where V is the macroscopic velocity of shortening for the muscle as a whole and where N is the number of sarcomeres that are connected in series to make up the muscle.

Finally, we assume that the rate of detachment of crossbridges is independent of their configuration x . Thus, the rate that bridges break is proportional to the number of attached crossbridges. The constant of proportionality is denoted by β . The parameter β also has units of time^{-1} , and is called the rate constant for crossbridge detachment. If we restrict attention to those crossbridges that have x in some particular interval, say (x_1, x_2) , then the number of such crossbridges (per half sarcomere) is

$$n_0 \int_{x_1}^{x_2} u(x) dx,$$

and the rate at which these crossbridges break is

$$\beta n_0 \int_{x_1}^{x_2} u(x) dx.$$

The total rate of detachment is $\beta n_0 U$.

We can now derive an equation for the crossbridge population density $u(x)$. Our key assumption is that this population is in a steady state. Since crossbridges are formed at $x = A$ and are moved by sliding in the direction of decreasing x , we have $u(x) = 0$ for $x > A$.

Consider the population of crossbridges that have x in the interval $x_0 < x < A$, where x_0 is any value of x such that $x_0 < A$ (see Figure 5.3). New crossbridges enter this population by the process of attachment that occurs at the rate $\alpha n_0(1-U)$. Crossbridges leave this interval in two ways: First,

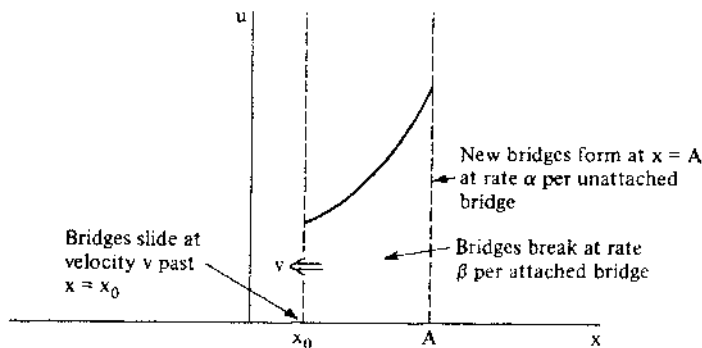


Figure 5.3. Dynamics of the crossbridge population density $u(x)$. $x = A$ is the special value of x at which new crossbridges form, and $x = x_0$ is an arbitrary value of x with $x_0 < A$.

they break at the rate

$$\beta n_0 \int_{x_0}^A u(x) dx.$$

Second, they slide across the lower boundary at $x = x_0$. The rate of this last process is $vn_0u(x_0)$, since v is the velocity of sliding and $n_0u(x_0)$ is the density (number per unit length) of crossbridges at $x = x_0$.

Because the crossbridge population is in a steady state, the three rates that we have considered must balance. This gives the equation

$$\alpha(1 - U) = \beta \int_{x_0}^A u(x) dx + v u(x_0) \quad (5.2.4)$$

where we have divided through by the common factor n_0 .

This integral equation for u can be used to derive both a differential equation that holds for $x < A$, and a boundary condition that holds at $x = A$. To get the differential equation, differentiate (5.2.4) with respect to x_0 . The result is

$$0 = -\beta u(x_0) + v (du/dx)(x_0). \quad (5.2.5)$$

Since x_0 can be any value of x , this differential equation can be written more simply as

$$v \frac{du}{dx} = \beta u. \quad (5.2.6)$$

The solutions of this equation have the form

$$u(x) = u(A) \exp(\beta(x - A)/v). \quad (5.2.7)$$

The constant $u(A)$ in (5.2.7) is not yet determined, however. Note that (5.2.7) applies only when $x < A$. On $x > A$, we have $u(x) = 0$.

To derive an equation for $u(A)$, we go back to (5.2.4) and set $x_0 = A$ (strictly speaking, we let $x_0 \rightarrow A^-$). The result is

$$\alpha(1 - U) = v u(A), \quad (5.2.8)$$

which asserts that new crossbridges are carried away from $x = A$ as fast as they are formed.

Unfortunately, (5.2.8) does not quite determine $u(A)$, since U is still unknown. Integrating (5.2.7) from $-\infty$ to A , we find that

$$U = \int_{-\infty}^A u(x) dx = v u(A) / \beta. \quad (5.2.9)$$

This gives another relationship between U and $u(A)$, so we can solve (5.2.8) and (5.2.9) as a pair of linear equations in these two unknowns. The result is

$$U = \frac{\alpha}{\alpha + \beta}, \quad (5.2.10)$$

$$u(A) = \frac{\alpha\beta}{v(\alpha + \beta)}, \quad (5.2.11)$$

so that for $x < A$,

$$u(x) = \alpha\beta \frac{\exp(\beta(x - A)/v)}{v(\alpha + \beta)} \quad (5.2.12)$$

and $u(x) = 0$ if $x > A$. This result is plotted in Figure 5.4 for two different sliding velocities v . Note that the total number of attached crossbridges $n_0 U$ is independent of v . For low v , the crossbridges are clustered near $x = A$, whereas for large v they are more spread out in the direction of negative x .

Having determined the behavior of the crossbridge population, we can now calculate the force-velocity curve. From (5.2.3), we have that

$$P = \frac{\alpha\beta}{v(\alpha + \beta)} \int_{-\infty}^A n_0 p(x) \exp(\beta(x - A)/v) dx. \quad (5.2.13)$$

To proceed further, we need to know $p(x)$, the force exerted by a single crossbridge in configuration x . The simplest hypothesis would be $p(x) = kx$, which would mean that the crossbridge acts like a linear spring. As shown in Exercise 5.1, this hypothesis leads to a linear force-velocity curve. Here, we assume that

$$p(x) = p_1(\exp(\gamma x) - 1), \quad (5.2.14)$$

which is plotted in Figure 5.5.

Substituting this expression in (5.2.13) and evaluating the integral (see Exercise 5.2), we get

$$P = \frac{\alpha n_0 p_1}{\alpha + \beta} \frac{(e^{\gamma A} - 1) - (\gamma v / \beta)}{1 + (\gamma v / \beta)}, \quad (5.2.15)$$

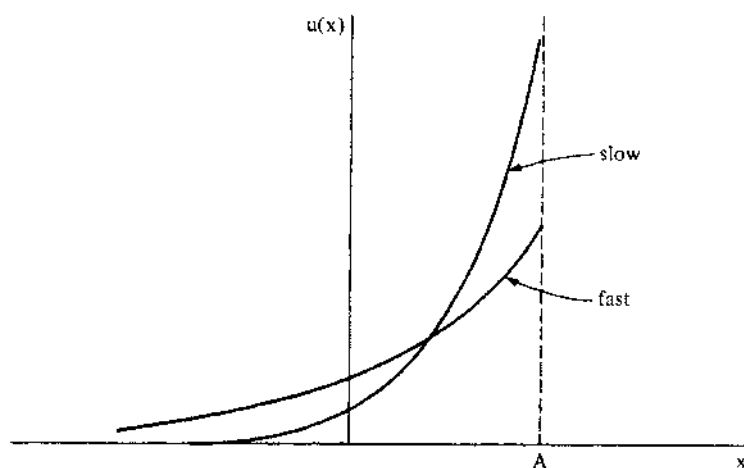


Figure 5.4. Qualitative effect of the velocity of sliding on the crossbridge population density: Faster sliding drags the crossbridges towards more negative x and so stretches out the population density function in that direction. The area under the curve (total fraction of attached crossbridges) remains unchanged.

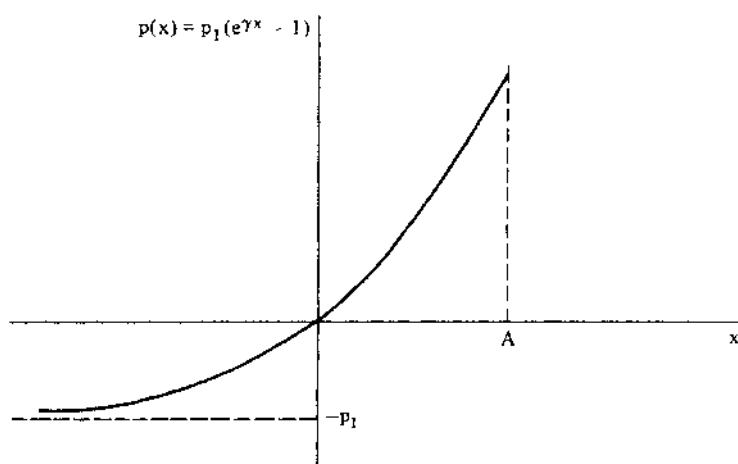


Figure 5.5. The force extension curve $p(x)$ of an individual crossbridge. Note that $p(0) = 0$ and that $p(x)$ is positive for positive x and negative for negative x . It is important that crossbridges can push (negative p) as well as pull; the balance between pulling and pushing determines V_{max} .

which is a formula for the force-velocity curve. Solving for $\gamma v/\beta$, we get

$$\frac{\gamma v}{\beta} = \frac{(\alpha n_0 p_1 / (\alpha + \beta)) (e^{\gamma A} - 1) - P}{P + \alpha n_0 p_1 / (\alpha + \beta)}. \quad (5.2.16)$$

This has the same form as the empirical force-velocity curve in equation (5.1.1). Since the velocity V at the ends of the muscle is related to v according to $V = 2Nv$, we have as before

$$V = b(P_0 - P)/(P + a), \quad (5.2.17)$$

where the constants P_0 , b , and a are now given by the formulae

$$P_0 = (\alpha n_0 p_1 / (\alpha + \beta)) (\exp(\gamma A) - 1), \quad (5.2.18)$$

$$b = 2N\beta/\gamma, \quad (5.2.19)$$

$$a = \alpha n_0 p_1 / (\alpha + \beta). \quad (5.2.20)$$

Since $V_{\max} = bP_0/a$, we also have

$$V_{\max} = (2N\beta/\gamma) [\exp(\gamma A) - 1] = 2N\beta A [\exp(\gamma A) - 1]/(\gamma A). \quad (5.2.21)$$

In short, we have derived the empirical force-velocity curve from a particular model of crossbridge dynamics, and we have determined the relationship between the microscopic properties of the model crossbridges and the macroscopic constants of the muscle. The formula for the isometric force can be explained by the observation that all of the attached crossbridges have the configuration $x = A$ when $v = 0$. Thus, the force exerted by each of these crossbridges is $p(A) = p_1(\exp(\gamma A) - 1)$. Since the number of attached crossbridges is $\alpha n_0/(\alpha + \beta)$, P_0 is given by equation (5.2.18).

The maximum velocity of shortening is determined by setting $P = 0$. This means that the negative forces exerted by the crossbridges with $x < 0$ balance the positive forces exerted by the crossbridges with $x > 0$. The velocity at which this occurs is independent of the overall size of the crossbridge population. Thus, $V_{\max}$ is independent of αn_0 .

It is interesting to contrast the behavior of P_0 and $V_{\max}$ with respect to the parameter β , which is the rate of detachment of crossbridges. Increasing β decreases P_0 by decreasing the number of attached crossbridges. When β increases, $V_{\max}$ increases, however, because rapid detachment prevents most crossbridges from sliding into negative values of x , where they would oppose shortening. A muscle with high β is fast but weak; a muscle with low β is slow but strong.

5.3 Computer Simulation of Crossbridge Attachment and Detachment

In this section we introduce a kind of computer simulation that is different from those discussed in the previous chapters. Instead of working with

the differential equations that describe the crossbridge population, we shall simulate directly the processes that underlie these equations. That is, we shall write a computer program that follows the individual crossbridges as they undergo attachment, detachment, and sliding. Because attachment and detachment occur at random, we shall need to introduce random numbers into the simulation. Computer simulation with random numbers is often called Monte Carlo simulation.

We consider a model half-sarcomere containing n_0 crossbridges. At any given time, each crossbridge may be in one of two states: attached or detached. We keep track of the state of the i -th crossbridge with a variable $a(i)$ such that $a(i) = 1$ when the crossbridge is attached and $a(i) = 0$ when it is detached, for $i = 1, \dots, n_0$. An attached crossbridge i has a displacement $x(i)$ from its equilibrium configuration. When $x(i) > 0$, the crossbridge exerts positive force, tending to shorten the muscle. When $x(i) < 0$ the crossbridge exerts negative force, opposing shortening, and when $x(i) = 0$, the crossbridge exerts no force at all. The displacement $x(i) = A > 0$ at the moment of attachment, and $dx(i)/dt = -v(t)$ thereafter, until detachment occurs. Here $v(t)$ is the shortening velocity of the half-sarcomere (proportional to the shortening velocity of the muscle), which we assume is a known function of the time, t . Thus we consider the problem of calculating the force generated by a muscle whose shortening velocity is some prescribed function of time. (The dual problem of calculating the motion when the force is prescribed will be considered in the suggested computing projects.)

The force generated by the model half-sarcomere will be calculated in our simulations simply by adding the forces generated by all of the attached crossbridges. Let $p(x)$ be the function that gives the force of an attached crossbridge with displacement x . Recall that $p(0) = 0$ by definition, since the displacement is measured from the equilibrium (zero-force) configuration of the crossbridge. Since a crossbridge which is not attached also generates no force, it is both natural and convenient to say that such a crossbridge has $x = 0$. With this convention, the total force P generated by the model half-sarcomere is simply $\sum_{i=1}^{n_0} p(x(i))$.

We still have to describe the random process by which $a(i)$ changes from $0 \rightarrow 1$ (attachment) or from $1 \rightarrow 0$ (detachment). An important assumption of our model is that these random processes are independent for the different crossbridges. This means that the state of one crossbridge has no influence on the attachment or detachment of another crossbridge. Let α be the probability per unit time (defined below) for attachment, and let β be the probability per unit time for detachment.

To say that an event happens with probability per unit time ν means that in a short interval of time Δt the probability that the event occurs is $\nu \Delta t$. This is an approximate statement, but the approximation gets better as the time interval gets shorter. To make this precise, let $\mathcal{P}(\Delta t)$ be the probability that the event happens in the time interval of duration Δt . Then the limit as $\Delta t \rightarrow 0$ of $\mathcal{P}(\Delta t)/\Delta t = \nu$.

Thus, in our model, in a short interval of time of duration Δt , we shall assume that a detached crossbridge has a probability $\alpha\Delta t$ of attaching, and that an attached crossbridge has a probability $\beta\Delta t$ of detaching. These will be good approximations only if $\alpha\Delta t \ll 1$ and $\beta\Delta t \ll 1$.

Let us now consider how to make an event happen (e.g., the computer says "YES!") with probability *prob* in a computer program. The first step is to choose a random number *R* which is uniformly distributed on the interval (0, 1). This means that the probability of *R* falling into any subinterval of (0, 1) is equal to the length of that subinterval. For example, the probability is 1/4 that *R* falls between 0 and 1/4, and the probability is 3/4 that it falls between 1/4 and 1. Most computer languages provide a function that generates such a random number (a different one each time the function is called, independent of the previous choices, see below for further discussion of this point, however.) In Matlab, this function is **rand**, and the way to choose a random number *R* from the uniform distribution on (0, 1) is simply to say *R=rand*. Having chosen *R*, we need to compare it to the assigned probability *prob*. The probability that *R* < *prob* is equal to the length of the interval from 0 to *prob*, which, of course, is equal to *prob*, so the test *if*(*R* < *prob*) will be satisfied with probability *prob* and unsatisfied with probability 1-*prob*.

As an example of the foregoing, the following computer program asks you to formulate a question to which the computer will answer "YES!" or "NO!" randomly, with probability *prob* that it will answer "YES!". The answer, alas, has nothing to do with the question, but hey, you probably know people like that, too. Let's just be happy to get an answer, and not the same one every time.

```
%filename: ask_the_computer.m
nquest=20    % number of questions
prob=.4      % probability the computer will answer 'YES!'
for count=1:nquest
    x=input('Please type a YES/NO question followed
            by <return>:\n','s');
    R=rand;   % choose a random number from uniform
              distribution on (0,1)
    if(R<prob)% test whether it falls below prob
        'YES!'
    else
        'NO!'
    end
end
end
```

When you are working with random numbers generated by a computer, there are a few things to keep in mind. First, they are not truly random. Instead, they are (usually) generated in a definite sequence by an algorithm that is cleverly constructed to produce numbers that pass all kinds of sta-

tistical tests for randomness. Numbers generated in this way are called pseudo-random numbers, and that is what most computer languages give you when you ask for a random number. This has some important consequences in terms of what happens when you run the same program twice. In Matlab, if you run the same program twice within the *same* Matlab session, you will get *different* random numbers, so the different runs are like independent trials of an experiment. But if you quit Matlab after each run of your program, and start it up anew for the next run, then you will get exactly the same results every time, since the random number generator always starts off in the same state. There are ways to overcome this by setting the state of the random number generator yourself. For more information about this, execute the command `help rand`. But the easiest way to simulate independent trials is to run all trials within the same Matlab session.

We now have all the ingredients needed for Monte Carlo simulation of crossbridge attachment, sliding, and detachment. The main program is as follows:

```
%filename: crossbridge.m
%units:
% length: nm = nanometers (nano=10^-9)
% time: s = seconds
% velocity: nm/s
% force: pN = picoNewtons (pico=10^-12; Newton=Kg m/s^2)
clear all
clf
global p1 GAMMA; %constants in p(x)
global Tstart V; %constants in v(t)
n0=10000 %number of crossbridges
alpha=14 % /s probability per unit time for attachment
beta=126 % /s probability per unit time for detachment
klokmax=3000 %total number of time steps
dt=0.01/(alpha+beta) % s duration of time step
A=5 % nm displacement of newly attached crossbridge
%constants used in the fuction p(x):
p1=4 % pN
GAMMA=0.322 % /nm
%test the function p(x) to make sure p(0)=0
%to within 10*eps, where eps=machine precision:
if(abs(p(0)/p(A))>10*eps)
    error('p(0) is not zero')
end
%constants used in the function v(t):
Tstart=10/(alpha+beta) % s time at which motion begins
V=500 % nm/s shortening velocity for t>Tstart
```



```

%all crossbridges are initially detached:
a=zeros(1,n0);
x=zeros(1,n0);
%number of bins for population density function:
nbins=round(sqrt(n0*alpha/(alpha+beta)));
%
for klok=1:klokmax           %loop over time steps
    t=klok*dt;               %current time
    dx=-v(t)*dt;             %change in half-sarcomere length
%Note that the following 'for' loop is commented out.
%It explains the logic of the program but would execute slowly.
%The code actually used is a vectorized version which appears
%immediately below the 'for' loop.
% for i=1:n0                 %loop over crossbridges
%     if(a(i))                %if crossbridge is attached
%         x(i)=x(i)+dx;        %crossbridge displacement
%         if(rand<beta*dt)      %if crossbridge detaches
%             a(i)=0;          %record its state as detached
%             x(i)=0;          %and its displacement as zero
%         end
%     else                    %crossbridge is detached
%         if(rand<alpha*dt)     %if crossbridge attaches
%             a(i)=1;          %record its state as attached
%             x(i)=A;           %and its displacement as A
%         end
%     end
% end
%The following vector statements are equivalent
%to the above commented-out 'for' loop:
x(find(a))=x(find(a))+dx; %displace only attached crossbridges
prob=(beta*dt)*a+(alpha*dt)*(1-a);%probability of changing state
change=rand(1,n0)<prob; %decide which crossbridges change state
a=xor(change,a); %change the state of those crossbridges
x(find(a&change))=A; %for newly attached bridges, set x=A
x(find(~a))=0; %for all detached bridges, set x=0
%store results for future plotting:
tsave(klok)=t; %current time
Usave(klok)=sum(a)/n0; %fraction attached
Psave(klok)=sum(p(x)); %force
P=Psave(klok) %write force on screen
end
figure(1)
subplot(2,1,1),plot(tsave,Usave)
subplot(2,1,2),plot(tsave,Psave)
figure(2)

```

```
[u,xc]=popdens(x,a,nbins); %evaluate population density
plot(xc,u,'*')           %plot population density
```

The explanation of this program is as follows:

First, the parameters are initialized. This program uses the nanometer (10^{-9} meters) as its unit length, the picoNewton (10^{-12} Newtons) as its unit force, and the second as its unit of time. The stated parameters are realistic except for $n0$, the number of crossbridges in a half-sarcomere, which doesn't have any standard value since it would depend on the cross-sectional area of the muscle, which of course varies widely from one muscle to another. The particular number used here, $n0=10000$, is chosen for convenience. It is large enough that the crossbridge noise does not obscure the underlying regularity of the results, yet small enough that the noise can be seen. One of the suggested computing projects is to experiment with $n0$ and observe the effect on the noisiness of the results.

After initialization of parameters, the main loop over time steps begins. At the beginning of this loop, the current time t is evaluated, and then the change in length dx of the model half-sarcomere is computed based on the current shortening velocity $v(t)$, a function which will be provided and discussed below. Nested within the loop over time steps is a loop over crossbridges. Two versions of this loop are given. The first one, commented out in the program but included for its explanatory value, is an explicit for loop. The other, more efficient version uses vectorized statements to deal with all crossbridges "at once." We describe the explicit for loop first.

For each crossbridge there are two cases, depending on whether it is attached or detached. If attached, the crossbridge changes its displacement by the same amount dx as the half-sarcomere changes length, and then a random number is chosen and compared to $\text{beta} \cdot dt$ to see whether the crossbridge has detached. If it has, its state $a(i)$ is set to 0, meaning "detached," and its displacement is also set to 0 so that it will not contribute to the force calculation. If the crossbridge was detached at the beginning of the time step, then the only action that has to be taken is to see whether it attaches. This is done by choosing a random number and comparing it to $\text{alpha} \cdot dt$. If this results in attachment, then the state of the crossbridge is set to $a(i)=1$, meaning "attached," and its displacement is set to $x(i)=A$, which is the displacement of newly attached crossbridges.

The vectorized version of the above described loop over crossbridges makes use of two Matlab features, a built-in function called `find`, and a mechanism for accessing part of a vector. The function `find` applied to a vector returns a list of indices of the nonzero elements of the vector. In our context, `find(a)` returns a list of indices of all attached crossbridges. This list can then be used to address only those crossbridges, as in the statement `x(find(a))=x(find(a))+dx`, which moves only the attached crossbridges. After this, the line `prob=(beta*dt)*a+(alpha*dt)*(1-a)` evaluates the probability of changing state on the current time step for all of the cross-

bridges at once. This probability is either $\beta \Delta t$ if the crossbridge is attached the beginning of the time step, or $\alpha \Delta t$ if the crossbridge is detached at the beginning of the time step. The factors of a and $(1-a)$ select the right probability in each case. Then a whole list of random numbers (one for each crossbridge) is generated by the expression `rand(1,n0)`, which creates a matrix of independent random numbers with 1 row and $n0$ columns. These random numbers are then compared, element-by-element, to the corresponding probabilities of changing state. The operation `a < b` in Matlab produces a numerical result: 1 if a actually is less than b , and 0 otherwise. Thus the line `change=rand(1,n0)<prob` makes the decision whether to change state for all crossbridges at once (but an independent decision for each of them) and records the result as a 1 in the elements of the array `change` corresponding to crossbridges that actually change state during the current time step, and as a 0 in the elements of `change` corresponding to crossbridges that maintain their state during the current time step. The array `change` is then used to bring the array `a` up to date by recording the changes in state that have occurred. The key here is the built-in Matlab exclusive-OR function `xor`, where `xor(a,b)` is 1 if a or b , but not both, are nonzero, and where `xor(a,b)` is 0 otherwise. If `change` and `a` can only take on the values 0 or 1, then `a=xor(change,a)` actually changes the state of a if `change` is equal to 1, and leaves the state of a alone if `change` is equal to 0. In our case, `change` and `a` are vectors (of the same length), and the above is done element-by-element. Finally, with the state of the crossbridges updated, we have to remember to set the displacement of the newly attached crossbridges equal to A , and we have to set the displacements of the newly detached crossbridges equal to 0 (but in the latter case, it doesn't matter if we instead set the displacements of all of the detached crossbridges equal to 0, since that is the value they are supposed to have.) We can locate *newly* attached crossbridges with the expression `find(change&a)`, and we can locate all of the detached crossbridges with the expression `find(~a)`. Here `&` means "AND" and `~` means "NOT." Thus we execute the commands `x(find(a&change))=A` and `x(find(~a))=0` to set the crossbridge displacements of these particular crossbridges. The displacements of the other crossbridges were updated earlier.

After all crossbridges have been processed in this way, various data are saved for future plotting. These are the time, the fraction of attached crossbridges, and the force generated by the model half-sarcomere.

Finally, when the loop over time steps is complete, various plots are generated. These include the history of the fraction of attached crossbridges and the history of the force generated by the model half-sarcomere. Also, a separate figure is generated showing the crossbridge population density at the end of the computer experiment. See below for further explanation of how this is done. The plot of population density is for comparison with the theory developed earlier in the chapter. Note that such a plot of population density could be generated at any time during the experiment, and you

might want to do this in your own version of the program to see how the population density evolves over time. We save it for the end here because the theory with which we aim to compare it is a steady-state theory, so we want to give the crossbridge population time to equilibrate to a constant shortening velocity before evaluating the crossbridge population density.

We now consider the functions needed by `crossbridge.m`. The first of these is the shortening velocity $v(t)$. This can be anything you like, so you can play with it to see how the model half-sarcomere behaves when constrained to change length in various ways. Some possibilities will be outlined in the suggested computing projects, below. For now we consider a simple case, in which $v(t)=0$ until the time T_{start} , and $v(t)=V$ thereafter. This simulates a muscle which is held at a fixed length (isometric) until a certain time and then allowed to shorten at a controlled velocity. The purpose of the isometric phase is to allow time for the crossbridge population to equilibrate (recall that we start with all crossbridges detached) to its equilibrium fraction of attached crossbridges. You can judge whether this has happened by looking at the graph of the fraction of attached crossbridges and checking that it has leveled off by the time T_{start} . Does anything interesting happen to the fraction of attached crossbridges when the half-sarcomere starts to shorten? Why, or why not? Similarly, does anything interesting happen to the force at this time? If these two variables behave differently, can you explain the difference?

```
function vv=v(t)
%filename v.m
%prescribed velocity of shortening at time t
%isometric contraction for t<Tstart
%shortening velocity V for t>Tstart
global Tstart V;
if (t<Tstart)
    vv=0;
else
    vv=V;
end
```

The next function we need is $p(x)$, which gives the force generated by an attached crossbridge whose displacement is x . The function we specify here is the one used earlier in the chapter, which was chosen because (together with our other assumptions) it gives the observed relationship between force and velocity for the muscle as a whole. One of the ways you can use the program `crossbridge.m` is to experiment with different $p(x)$ to see what effect the form of this function has on the force-velocity relationship. Although you have a lot of freedom in the choice of $p(x)$, the one restriction that it has to satisfy is $p(0)=0$, since this property was used in writing `crossbridge.m` to make sure that detached crossbridges do not generate

any force. In fact, `crossbridge.m` tests $p(x)$ for this property and will halt with an error message if it is not satisfied.

```
function pp=p(x)
%filename p.m
%crossbridge force as a function of crossbridge displacement
%x is vector of crossbridge displacements
%pp=p(x) is a vector of corresponding forces
global p1 GAMMA;
pp=p1*(exp(GAMMA*x)-1);
```

The remaining function `popdens` is needed to generate the crossbridge population density function for comparison with the theory developed earlier in this chapter.

The function `popdens` counts how many attached crossbridges fall into each of several bins. The number of bins, `nbins`, is defined by the user. Then `popdens` looks at the subarray `x(find(a))` which contains the displacements of attached crossbridges only. It sends this array to the Matlab function `hist` which sets up `nbins` bins and counts how many of the attached crossbridges fall into each bin. An array of bin centers `xc` is also returned by `hist`, which is useful for plotting the results. Finally, the numbers of attached crossbridges in each bin have to be normalized properly to obtain the crossbridge population density function. First, they are divided by the total number of crossbridges (whether attached or detached) to convert them to fractions. Then, they are divided by the bin width so that the integral of the crossbridge population density over any interval will give the fraction of crossbridges which are attached and have their displacements within that interval.

```
function [u,xc]=popdens(x,a,nbins)
%filename popdens.m
%find population density function of attached crossbridges
%input variables:
% x=array of crossbridge displacements
% a=array of crossbridge states (0=detached,1=attached)
% nbins=number of bins
%output variables:
% xc=array of (equally spaced) bin centers
% u=array of crossbridge population densities
n0=length(x); %total number of crossbridges
Nabridges=length(x(find(a))); %number of attached crossbridges
%nbbridges=number of attached crossbridges in each bin;
%xc=centers of the bins:
[nabridges,xc]=hist(x(find(a)),nbins);
%normalize nabridges to find crossbridge population density
xstep=xc(2)-xc(1);
```

```
u=(nabridges/Nabridges)/xstep;
```

Once `popdens` has been called, the crossbridge population density can be plotted by executing `plot(xc,u,'*')`, as is done at the end of `crossbridge.m`. This particular command will plot a symbol for each bin rather than a continuous line, thus emphasizing the bin structure of the plot. It may take some experimentation to find a good choice of `nbins`. It is tempting to use a large number of bins to get good spatial resolution, but if too many are used then there will be too few attached crossbridges in each bin for statistical reliability, and the plot will show a lot of randomness. An attempt at a reasonable compromise has been made in `crossbridge.m` where `nbins` has been set equal to the square root of the expected number of attached crossbridges. You should experiment with smaller and larger values of `nbins` to see the effect.

5.4 Suggested Computing Projects on Crossbridge Dynamics

The first project is to compare the theory developed earlier in the chapter to the computational results generated by `crossbridge.m` and the functions that it calls. Run `crossbridge.m` for several different shortening velocities V and plot a few points of a force-velocity curve. On the same plot, show the theoretical curve, the parameters of which can be calculated from the parameters used in `crossbridge.m`. (In doing this calculation, though, note that we have only one half-sarcomere, so skip the multiplication by $2N$ that was used earlier in the chapter to convert sliding velocity within a half-sarcomere to muscle velocity.) Also compare the theoretical crossbridge population densities for each V with those of the computational experiment. Enough parameters are given to make these comparisons quantitative. The only sources of error should be the noise that comes from the random nature of crossbridge attachment and detachment (and you can reduce this noise by increasing `n0`), and the numerical error associated with the finite size of the time step (and of course this source of error can be reduced by making `dt` smaller).

By experimenting with different `n0`, study the effect of the number of crossbridges in a half-sarcomere on the behavior of the muscle. In particular, study the effect of `n0` on the noisiness of the results and on the relationship between force and velocity.

Next, try some different $v(t)$ and see what happens. Here is one interesting example, and you may think of others: Set $v(t)=0$ for $t<Tstart$, and $v(t)=V*\sin(2*\pi*f*(t-Tstart))$ for $t>Tstart$. The parameter `Tstart` has the same meaning as before. It is the amount of time during which the crossbridge population is allowed to equilibrate under isometric conditions. The parameter V now has a different meaning. It is the amplitude of an

imposed sinusoidal oscillation in velocity. The frequency of that oscillation is a new parameter f , which has units of Hz (cycles per second). Be sure to let the simulation run long enough for the computed force waveform to settle down to a periodic steady state. Once this happens, does the force waveform look sinusoidal, or does it show evidence of distortion? It should be sinusoidal for small V and become distorted as V gets larger. See if you can demonstrate this. With V small enough that the force waveform looks sinusoidal, see if you can fit the force waveform to a function of the form $P(t) = P_0 + P_1 \sin(2\pi f(t - T_{\text{start}}) - \phi)$ where P_0 is the average force (which should be close to the isometric force in this computer experiment), P_1 is the (positive) amplitude of the force oscillation, and ϕ is the phase shift between the imposed velocity oscillation and the resulting force oscillation. Do not try to fit the whole force waveform to this relationship, just the part after it has settled down to a periodic steady state. How do P_1 and ϕ depend on f ? Be sure to try a range of frequencies, some large and some small in comparison to $(\alpha + \beta)/(2\pi)$. Plot P_1 against f on a log-log plot. Also plot ϕ against f on a semi-log plot, with the logarithm applied to f , not ϕ . See if you can explain the low-frequency and high-frequency behavior of the model half-sarcomere.

Other projects involve changing the properties of the muscle in various ways. For example:

Try some different force functions $p(x)$ and see what difference it makes for the force-velocity curve. What if $p(x) = k \cdot x$, for example, see also Exercise 5.1. A reasonable choice of k would be such that $p(A)$ has the same value as before.

What happens if crossbridges instantaneously detach when they reach displacements greater than some value B , where $B > A$? (See also Exercise 5.4.) Of course, this will not affect the force-velocity curve of shortening, but it will affect the force-velocity curve of lengthening ($V < 0$ in our notation). Modify `crossbridge.m` to include this effect by inserting the line `change=change|(x>B)` right after the line on which `change` is first defined. The symbol “|” means “OR,” and the effect of this is to say that all crossbridges which find themselves in the region $x > B$ will change state, i.e., detach (since they would have to be in the attached state to get into that region).

Finally, recall that the program `crossbridge.m` is designed to calculate the force when the motion of the muscle is prescribed. What about the dual problem of calculating the motion when the force is prescribed, either as a constant or as some given function of time? For this, you will need a new program, similar in some respects to `crossbridge.m`, but differing especially in the manner in which the displacement dx is computed during each time step. Now dx will have to be such that the force generated by whatever crossbridges happen to be attached equals the prescribed force. See if you can figure out how to solve for dx . This computation is facilitated by the exponential form of $p(x)$.

5.5 Annotated References

There are many aspects of muscle mechanics that could not be covered in this introductory chapter. The reader who would like a more comprehensive treatment of the subject will find it in:

Huxley, A.F.: *Reflections on Muscle* (The Sherrington Lectures XIV). Princeton University Press, Princeton, NJ, 1980.

The classic paper on the force-velocity curve of muscle is that of A.V. Hill:

Hill, A.V.: The heat of shortening and the dynamic constants of muscle. *Proceedings of the Royal Society B* **126**, 136-195, 1938.

The idea that the mechanical behavior of muscle could be explained in terms of the dynamics of the crossbridge population was introduced by A.F. Huxley:

Huxley, A.F.: Muscle structure and theories of contraction. *Progress in Biophysics* **7**, 255-318, 1957.

The particular crossbridge theory used in this chapter is similar to Huxley's but not quite the same. It was introduced in the Ph.D. thesis of H.M. Lacker and is further discussed in the following reference:

Lacker, H.M. and Peskin, C.S.: A mathematical method for the unique determination of crossbridge properties from steady state mechanical and energetic experiments on macroscopic muscle. *Lectures on Mathematics in the Life Sciences* **16**, American Mathematical Society, Providence, RI, 1986, 121-153.

Exercises

5.1. Linear Force-Velocity Curve

Show that the assumption $p(x) = kx$ leads to a linear force-velocity curve. Do this in two ways:

1. By substituting $p(x) = kx$ in (5.2.13) and evaluating the integral. Try integration by parts.
2. By noticing that $p(x) = kx$ is a limiting case of an exponential spring. That is,

$$p_1(\exp(\gamma x) - 1) = p_1 \gamma x (\exp(\gamma x) - 1) / \gamma x \rightarrow kx$$

as $\gamma \rightarrow 0$, $p_1 \rightarrow \infty$, but with $p_1 \gamma = k$. Having made this observation, take the corresponding limit in (5.2.17) through (5.2.21) and check that your result agrees with part 1 of this exercise.

5.2. Derivation of the Force-Velocity Curve

Write out the derivation of (5.2.15).

5.3. Superisometric Force

When a force greater than the isometric force is applied to a muscle, the muscle stretches at a velocity that depends on the applied force. Calculate the force-velocity curve for stretch, making the same assumptions that were made in this chapter. Remarks: Treat stretch as a *negative* velocity of shortening ($v < 0$). Crossbridges still form at $x = A$, but now they are carried into the region $x > A$ by the sliding process. Thus, $u = 0$ for $x < A$, and the equation that replaces (5.2.4) is

$$\alpha(1 - U) = \beta \int_A^{x_0} u(x) dx - v u(x_0).$$

Starting from this equation, repeat the derivation that led to the force-velocity curve and see what you get. Sketch the behavior of the force-velocity curve for positive and negative velocities of shortening on a single graph.

5.4. Why Muscles Yield at Large Superisometric Forces

The force-velocity curve for stretch computed in Exercise 5.3 predicts infinite force at a finite velocity of lengthening. This is very unrealistic. In fact, muscles *yield* when the applied force is greater than $1.8 P_0$. That is, there is a maximum force that the muscle can sustain. To explain this kind of behavior, modify the crossbridge model by assuming that crossbridges inevitably break if they reach a configuration $x = B$, where $B > A$. This has no effect on the force-velocity curve for shortening, but it does influence the force-velocity curve of lengthening, since all of the integrals now extend over the interval (A, B) instead of (A, ∞) .

1. Derive a formula for the force-velocity curve of lengthening for this model. Express your result in terms of the dimensionless variables $P^* = P/P_0$, $v^* = \gamma v/\beta$ and in terms of the dimensionless parameters $r^* = \alpha/\beta$, $A^* = \gamma A$, and $\omega = \gamma(B - A)$.
2. Show that your result reduces to the force-velocity curve calculated in Exercise 5.3 when $B = \infty$.
3. Make a quantitative plot of v^* as a function of P^* for both shortening and lengthening when $r^* = A^* = \omega = 1$.
4. Explain why the muscle has a finite maximum force when B is finite and why this force becomes infinite as $B \rightarrow \infty$.
5. On your graph, it should look as though the force-velocity curves of lengthening and shortening join up smoothly at $v^* = 0$, even though these curves are described by different formulae. How many derivatives of the two curves actually match at $v^* = 0$?

5.5. Quick Stretch

A muscle is shortening at velocity V and force P . Suddenly, the muscle is stretched by an amount ΔL . The stretch is too sudden for any crossbridges to break or for any new crossbridges to form during the stretch.

1. What happens to the crossbridge population $u(x)$ during the stretch?
2. What happens to the force P ?
3. How could such experiments be used to check the hypothesis that $p(x) = p_1(\exp(\gamma x) - 1)$?

6

Neural Systems

Studying the brain has been one of the most active and interesting recent adventures in science. Ingenious experiments and exquisite experimental devices are moving us toward understanding how a brain works. To move beyond knowledge of molecular aspects of the brain, we can use methods such as those developed and used by engineers to study and design large complex systems. Their approaches are based on mathematical models of electronic devices, and in this chapter we explore similar approaches to study brain systems.

This knowledge is beginning to be used to engineer various devices that can interface the brain and other neural networks directly to the outside world. Such devices are used for driving prosthetics, such as artificial limbs, and for treatment of neurological diseases, such as Parkinson's disease.

A particular problem in neural engineering is how to find suitable models of brain structures. In the physical sciences there are laws that guide modeling. For example, Newton's second law tell us how to model small oscillations of a pendulum: If the pendulum has mass m and the line from the support point to its center of mass makes an angle x with the straight down direction, then $F = ma$ implies that

$$\Omega - gmx - f\dot{x} = m\ddot{x},$$

where the first term on the left accounts for the torque applied to the support point of the pendulum, the second for the force of gravity, and the third for the dissipation of energy due to friction.

Similar equations appear in studies of the electrical circuits in a brain, even though there are no known laws similar to Newton's for these biological problems. Our models are based on the logic of Ockham's razor: If we observe a change in a system's response due to a small change in the stimulus, we compare the observation to a catalog of canonical mathematical structures that exhibit similar behavior, and select one that reproduces the observations. The result is a mathematical model that exhibits the observation. The next step is to compare predictions made using the canonical model with further experiments to validate the model, to identify parameters, etc.

For example, A.L. Hodgkin (1948) applied a constant voltage across a neural membrane. He observed that as the stimulus was increased, the voltage measured across the membrane began to oscillate due to ionic currents passing through the membrane, as described earlier, in Chapter 3. The oscillations were of two kinds. First (Class 1), the membrane voltage began to oscillate with (roughly) a constant amplitude, but with a frequency that increased with the stimulus. In the second (Class 2), the membrane voltage began to oscillate with a small amplitude that increased with the stimulus, but the frequency remained (roughly) constant. Both kinds of neural membranes occur *in vivo*; e.g., Class 1 behavior is observed in sensors and Class 2 behavior is observed in motor neurons.

There are two canonical models that describe such phenomena: In each, there is one parameter (corresponding to the stimulus strength), and in each there is a threshold of the parameter beyond which the system switches from being at rest to oscillating. The first, called a saddle-node on limit cycle bifurcation model, is the one we study in this chapter. The second is called the Andronov-Hopf bifurcation model, and it is described in Exercise 6.8.

Another approach that has been useful in neuroscience and in neural engineering is the use of state-of-the-art electronic circuits as analogues of neural structures. This was done by L. Lapique (1907) using a low-pass filter together with a circuit breaker, by B. van der Pol (1926) using vacuum tube circuits, by A.L. Hodgkin and A.F. Huxley (1952) using variable-conductance shunting circuits, by R. FitzHugh and J.S. Nagumo (1960) using tunnel diodes, and by F.C. Hoppensteadt (1980) using phase locked loops (PLL). These analog circuits are widely used for interpreting and understanding data from neurophysiological experiments and for suggesting further investigations.

In the sense described by Hodgkin, neurons are oscillators that are controlled by an applied voltage, and we say that they are voltage-controlled oscillators. There is an important integrated circuit, called a voltage-controlled oscillator, that we will use in this chapter to model neurons. We will see in this chapter that the canonical model for Class 1 neurons and the mathematical model for an electronic self-excited voltage-controlled oscillator are the same! For this reason and the fact that the model is

tractable within the scope of this book, we use it here to study networks of neurons.

Two neural systems are investigated in this chapter: First, a model of how the thalamus can sort out signals to pass on to the neocortex is formulated and simulated. Second, a model is derived for how patterns of theta-rhythm activity can be created in the hippocampus, based on inputs from the medial septum and the entorhinal cortex. The models are formulated using the voltage-controlled oscillator (VCO) methodology.

Designing circuit networks that are relevant to the brain relies on experiments and ordinary language models created by neuroscientists and on mathematical models formulated by mathematicians and engineers. The usefulness of this approach to understanding the brain remains to be seen, but at least it has guided the development of parallel engineering systems for prosthetics and the control of neural diseases such as Parkinson's disease.

We do not attempt to model a complete living neuron. A typical neuron includes an input structure (the dendritic tree), a cell body that maintains the cell's metabolism and shapes action potentials as described in Chapter 3, a transmission structure (the axons), and an output structure (the synapses) that releases chemical neurotransmitters in response to an action potential arriving from its axon. The voltage-controlled oscillator models can be related to biological neurons as describing the electrical activity in the hillock region lying at the proximal end of an axon where action potentials are generated. While such models are far from describing all the complexity of a complete neuron, we still refer to them as neurons.

This chapter begins with some experimental results from neuroscience. We describe a model for these, and then use it to synthesize, analyze, and simulate models for two important brain systems: interaction between the thalamus and the cortex and pattern formation on the hippocampus.

There are four key facts from neurophysiology that we use in modeling:

1. There are frequency relations between inputs to and outputs from neurons.
2. The timing of cell firing is important in communication between neurons.
3. Inhibitory and excitatory connections between neurons are important.
4. Several time scales can be involved in neuron firing.

The models we study here illustrate these features, and we use them to investigate networks.

6.1 Guttman's Experiments on Phase Locking

Some experimental results from neuroscience lay a basis for our modeling. These include work of Hodgkin on various classes of excitability that

was described in the preceding section, and work by R. Guttman and her colleagues on phase locked responses of neural tissue to oscillatory stimulation.

Phase locking is an amazing phenomenon that is observed in many physical and biological systems. People have been aware of it since at least the time of Aristotle, when lyre strings of approximately the same physical dimensions were heard to vibrate at the same frequency. This phenomenon is a key design element in many engineering systems that control timing in computers and vibrations in vehicles. It is also important in how our bodies work.

How do we describe phase locking? The simplest oscillatory signals are of the form $\cos(\omega t + \phi)$, where t is time in some units, perhaps milliseconds (msec), ω is the frequency (this signal has period $2\pi/\omega$), and ϕ is the phase deviation. This is essentially the form of the signal that a radio receives. If the broadcast is frequency modulated (FM), then ω is the address of the transmitter to which a receiver is tuned, and ϕ , which changes more slowly than ωt , carries the music that we listen to. In FM radio, the transmitter frequency is usually in the range of megahertz, and the phase deviation changes on a much slower time scale of kilohertz. The reason FM radio is stable in the presence of noise (e.g., you rarely hear static on FM radio) is that even though ω is not tuned exactly to the station, the receiver can still decode the phase correctly, because of the phenomenon of phase locking.

Phase locking has also been observed to occur in neural tissue. Guttman and her colleagues stimulated a patch of squid axon membrane, similar to Hodgkin and Huxley's preparation described in Chapter 3, with oscillatory voltages of various amplitudes and frequencies. A stimulus of frequency μ and amplitude A was applied, and the response of the patch was recorded. The response was usually observed to be a regular voltage oscillation having an easily determined frequency, say ω . The results are presented in Table 6.1 in terms of the frequency response ratio ω/μ .

The set over which $\omega/\mu = 1$ describes combinations of stimulation amplitudes and frequencies for which the neuron is locked to the forcing frequency! This is referred to as 1:1 phase locking. Note that there are significant combinations of stimulus parameters for which other interesting fractional responses are observed, especially $\frac{1}{3}$, $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, etc.

There are other neuronal systems that are known to phase-lock. These include the abdominal ganglion of *Aplysia californicus* (sea slug), the thoracic stretch receptors in *Procambarus clarkii* (crayfish), the auditory nerve fibers of the squirrel monkey, the lateral eye of *Limulus* (horseshoe crab), breathing and stride in humans, and seasonal foraging behavior in *Microtus montanus*, the montane vole. The model that we derive next will have phase locking properties similar to Guttman's observations. The phenomenon of phase locking and how it arises in neural networks are studied further in the references in Section 6.4.

Table 6.1. Guttman's Data: Frequency Response of Squid Axon (Data rounded to nearest reduced fraction)

Stimulus Intensity ($\mu A/cm^2$)	Stimulus Frequency (Hz)									
	20	40	60	80	100	120	140	160	180	
34.2	0	0	1/9	1/2	1/9	0				
37.8	0	0	1/2	1/2	1/2	2/5	0			
41.4	0	7/8	3/4	1/2	1/2	2/5	0			
45.0	0	1	1	1/2	1/2	1/2	1/3	1/9	0	
50.4	0	1	1	2/3	1/2	1/2	2/5	1/3	1/9	
57.6	0	1	1	3/4	1/2	1/2	1/2	1/3	1/3	
62.1	2/3	1	1	1	1/2	1/2	1/2	1/3	1/3	
70.2	1	1	1	1	2/3	1/2	1/2	1/2	1/3	
72.9	1	1	1	1	2/3	1/2	1/2	1/2		
77.4	1	1	1	1	3/4	1/2	1/2	1/2		
82.8	1	1	1	1	1	2/3	1/2	1/2	1/2	
86.4	1	1	1	1	1	2/3	1/2	1/2	1/2	
90.0	1	1	1	1	1	3/4	1/2	1/2	1/2	
113.4	1	1	1	1	1	1	2/3	1/2	1/2	
148.5	1	1	1	1	1	1	1	3/4	1/2	
220.5	1	1	1	1	1	1	1	1	1	
351.0	1	1	1	1	1	1	1	1	1	

6.2 Biological Rhythms

Oscillatory signals have been studied over the past two millennia by astronomers, engineers, mathematicians, and philosophers, leading to an understanding of time and the development of clocks to track it.

Clocks are devices that we all have grown up with, but we do not think much about them. We learn to tell time by the location of the big and little hands on a clock face, and we ignore the fact that the numbering system is discontinuous. We also expect that the hands proceed at a uniform rate throughout a day.

There are many timing devices in our lives that distort these features. Biological clocks move rapidly through parts of a day or month and slowly through other parts. A similar effect on a wall clock would be to accelerate the hands during the morning and to slow them down during the afternoon. Or equivalently, we could use unequal spacings of the marks around the edge of a 24-hour clock to tell biological time, for example, morning times spaced closer and afternoon ones farther apart, while the hands move at uniform rates.

6.2.1 Modulating the Clock on the Wall

Time can be measured using an arbitrary scale around a clock face's edge: Minutes, as on a wrist watch, or radians, as shown in Figure 6.1. We choose

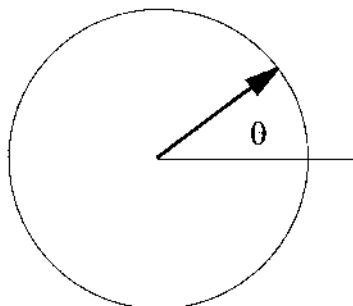


Figure 6.1. A clock face that has a single hand with time measured in radians.

radians for mathematical convenience. If θ denotes the angle ("time") shown on the clock, then θ might increase in proportion to solar time, or it might change at some other fixed rate. For example, we could have $\theta = \omega t + \phi$, where t indicates Greenwich mean time, say measured in seconds, ω denotes the number of radians moved per second, and ϕ denotes the longitude. It is more convenient for our purposes to write this mathematical description of a clock as a differential equation:

$$\dot{\theta} = \omega. \quad (6.2.1)$$

This equation indicates that the rate of change of θ is ω . Note that the units of θ are radians and those of ω are radians/time. Because of this, we refer to ω as a *frequency*, and θ is referred to as a *phase variable*. Given the value of θ , the coordinates of the tip of the hand are $x = R \cos \theta$, $y = R \sin \theta$, where R is the hand's length. Note that this clock goes counterclockwise (unless $\omega < 0$).

Model (6.2.1) is referred to as a simple clock. A simple harmonic oscillator gives further insight to time. Consider the equation

$$\ddot{u} + \omega^2 u = 0,$$

which models small oscillations in a frictionless pendulum. All solutions have the form $u = A \cos(\omega t + \phi)$, where A and ϕ are constants determined by the initial conditions of the solution. If we set $u = A \cos \theta$, where $\theta = \omega t + \phi$, then we get that $d\theta/dt = \omega$. Thus, the harmonic oscillator is equivalent to a simple clock, since the problem can be reduced to a single equation for the phase θ , provided that $A \neq 0$. In fact, most oscillators can be reduced to phase equations except in certain singular, but important, cases.

While a simple clock is driven by a motor that moves its hand at a constant rate ω , the human body does not always perceive time in this

way. In fact, it happens that the driving frequency of most natural clocks can be *modulated* by many things, among them solar time brought into the body through the senses. The mathematical model

$$\dot{\theta} = \omega + f(2\pi t/1440),$$

where f has period 2π , shows how a simple clock could be modulated by an external signal (f). When t increases through one day by minutes (1440 minutes per day), then the modulation f passes through one full cycle. This equation indicates that the hand moves in an irregular way throughout the day. For example, this clock is accelerated ($f > 0$) or slowed down ($f < 0$) in response to the time of day, perhaps due to sunlight or temperature. In this case the clock hand is the vector $(\cos \theta, \sin \theta)$, where now $\theta = \omega t + F(t)$, where $dF/dt = f(2\pi t/1440)$.

Clocks can also be modulated by external forces such as gravity acting on the hand's position on a mechanical clock. For example, if the clock hand is moving from top to bottom, gravity will try to move it faster than when it moves up from bottom to top, when gravity slows it down. Therefore, if χ measures time and $\chi = 0$ corresponds to noon, we might have

$$\dot{\chi} = \alpha + \varepsilon \sin \chi,$$

where $\chi = \pi$ corresponds to 6 o'clock and $\varepsilon \sin \chi$ describes the influence of gravity on the clock's hand. This is an example of feedback: The rate at which the hand moves depends on its present position.

6.2.2 Biological Clocks

A clock has three main parts: an oscillating system, such as a pendulum, spring, or electrical circuit that is a source of energy; a trigger mechanism or escapement that connects the energy source to the output (face); and a face that presents the oscillator's output in some useful way.

Biological clocks have the same ingredients. For example, a nerve cell's membrane can sustain oscillations: The energy is supplied by the cell's metabolism, and the trigger mechanisms are controlled ionic channels in the membrane (See Chapter 3). The result is a cycle of membrane voltage whose phase (like χ above) can be viewed as a hand moving on a clock face.

Natural rhythms of approximately one day, called circadian rhythms, were observed in animals by Aristotle, and it is now known that all living things have a variety of rhythms. These rhythms are sometimes driven by external signals, like the interval of light seen by an animal each day (photoperiod), or by internal mechanisms, like circuits in the brain stem that control heart beat or breathing. In addition, internal and external timers can interact to cause interesting behavior, like that observed in foraging activity by voles, who are nocturnal when the daily light interval

is long, diurnal when the light interval is short, and crepuscular (i.e., being active at dawn and dusk) in spring and fall.

One cannot tell time by looking only at the center of the clock. This rather obvious statement actually has some substance for biological clocks. In most biological clocks, where there is never a rigid hand describing “time,” it is possible to stimulate the clock’s mechanism at a special time that moves the whole timing mechanism to the center. All timing ability is then lost by the system until it is corrected.

The timers studied here are neurons. They set the beat. Muscles and glands act like filters and actuators that scale time by averaging neuron output, and the result is a physiological observable, like regular respiration or regular hormone cycles.

Biological clocks are rarely isolated. They all operate in a common chemical bath, and there are many electrical, chemical, and mechanical connections between them. It is difficult to grasp the complications of living with many clocks running at different rates, but the mathematical methods derived in this chapter can help. We will see how one clock can modulate another and how networks of neurons can interact.

Neurons can transmit information by modulating their firing frequency and phase, in the spirit of FM radios, and we focus on this aspect in our discussion of neurons as timers.

6.2.3 *Electronic Clocks: Voltage-Controlled Oscillators*

Now that we have some experience with feedback in phase-based models, we can describe a phase locked loop (PLL) model. Electrical engineers and physicists did a brilliant thing in designing a circuit whose mathematical model is in terms of a phase variable! A PLL is described by the simple model

$$\dot{\theta} = \omega + \cos \theta,$$

where ω is a voltage-controlled oscillator’s (VCO) center frequency and $\cos \theta$ is the VCO output. Thus, the model is a VCO whose output voltage is fed back as its controlling voltage. This also happens to be exactly the canonical model for Hodgkin’s Class 1 neurons that were described at the start of this chapter.

6.3 Model Neural Networks

We first describe network notation and how we model networks using the methodology just described.

The graph

$$N_1 \rightarrow^+ N_2$$

indicates a network of two neurons in which N_1 has an excitatory synapse impinging on N_2 . If the model membrane potentials of the cells are $\cos \theta_1$ and $\cos \theta_2$, respectively, then the network is modeled by the differential equations

$$\begin{aligned}\dot{\theta}_1 &= \omega + \cos \theta_1, \\ \dot{\theta}_2 &= \omega + \cos \theta_2 + A \cos(\theta_1 + \phi_{12}),\end{aligned}$$

where the first equation describes N_1 and the second one N_2 . A (> 0) describes the connection strength, and ϕ_{12} describes time delays in the connection and those due to external sources.

The network

$$N_1 \rightarrow^- N_2$$

describes an *inhibitory synapse* from N_1 to N_2 . The model is the same as above, except that now A is a negative constant.

To model a network of M neurons, we write

$$\dot{\theta}_j = \omega_j + \cos \theta_j + \sum_{k=1}^M C_{k,j} \cos(\theta_k + \phi_{k,j}) + E_j \quad (6.3.1)$$

for $j = 1, \dots, M$. The factor $C_{k,j}$ describes the strength of connection from network element k to j , and $\phi_{k,j}$ describes the phase deviation due to the connection. These might be constant or they might change with time. The term E_j describes an external signal applied to the system. The next two examples illustrate such networks.

6.3.1 A Thalamocortical Circuit.

In numerous places in the brain two cells interact, one inhibitory and the other excitatory, in such a way that one fires a rapid burst of action potentials followed by the other firing a slow pulse. Thus, two time scales are involved. The model described in this section reproduces this phenomenon.

Most sensory input to the brain is eventually processed in the thalamus, which sorts out the signals to be passed on for further processing in other brain structures like the entorhinal cortex and the hippocampus. This involves a circuit between excitatory cells in the thalamus and inhibitory cells in a parallel structure called the reticular complex. The thalamic cells fire rapidly, and the reticular complex cells fire (relatively) slowly. The network here is shown in Figure 6.2.

We model a single channel from stimulus to TH to RC to NC using a pair of circuits described by the equations, called the *atoll model*,

$$\begin{aligned}\dot{x} &= 5.0(1 + \sigma + \cos x - \cos y), \\ \dot{y} &= 0.04(1 + \cos y + 10 \cos x).\end{aligned}$$

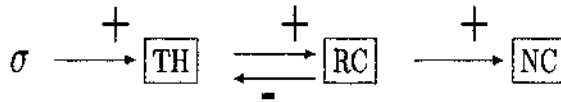


Figure 6.2. Wiring diagram for the atoll model. Here σ denotes stimulation to the thalamic cell (TH), which interacts with a cell in the reticular complex (RC), which in turn eventually excites a neocortical structure.

The ratio of time scales is $5/.04 = 125$, which corresponds to recorded observations in the gastric mill controller in lobsters, and this system bursts repetitively when $\sigma > 0$. The solution of this model is shown in Figure 6.3.

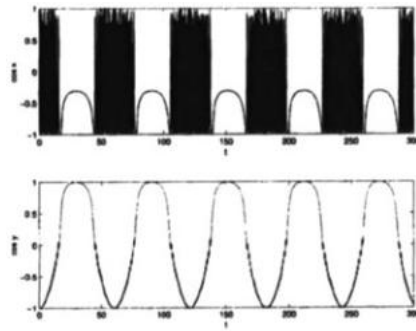


Figure 6.3. Output of the atoll oscillator with $\sigma = 0.5$.

It has been proposed that the brain focuses attention on the most persistent among many competing stimuli using a circuit in the thalamus. We will model this using an electronic circuit that builds on the atoll oscillator. When stimuli of various intensities arrive in the thalamus, it passes only the most demanding stimulus to the neocortex for further processing. An excited thalamic cell excites a reticular complex cell, which feeds back inhibition to the thalamic cell in such a way that repetitive bursts are observed as in the atoll model. The activity of the pair effectively inhibits other (nearby) RC cells, and it excites a single neocortical structure. The full network is shown in Figure 6.4.

The voltage phases are $x_1, \dots, x_N$ and $y_1, \dots, y_N$, for TH and RC cells, respectively. The phases of the neocortical projections are $z_1, \dots, z_N$. A model for this network is

$$\begin{aligned}
 \dot{x}_j &= 5.0(1 + \cos x_j - \cos_+ y_j + \sigma_j(t)), \\
 \dot{y}_j &= 0.04(1 + \cos y_j + \tanh(2 \cos_+ x_j - 10L(t))), \\
 \dot{z}_j &= 10(0.1 + \cos z_j - \cos_+ y_j),
 \end{aligned} \tag{6.3.2}$$

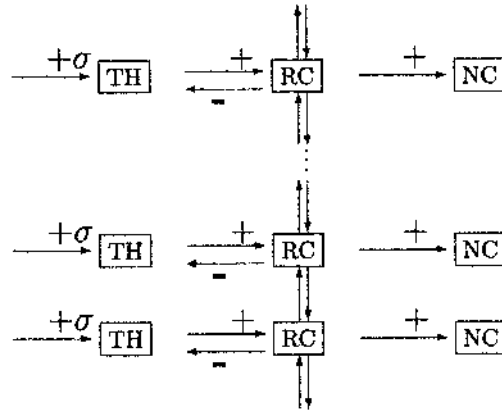


Figure 6.4. Wiring diagram for the model thalamocortical system. Here σ denotes the external stimulus to a thalamus cell (TH). TH cells have excitatory outputs to and receive inhibitory inputs from the reticular complex cells (RC). All vertical connections are inhibitory. NC denotes a neocortex structure that is eventually excited by input from an RC cell.

for $j = 0, \dots, N$, where L accounts for lateral inhibition in the reticular complex. (Recall that the notation $a_+ = (a + |a|)/2$ is the positive part of a .) The constants in this network are chosen to have a ratio of time scales approximately 100:1. The terms $\cos_+ x(t)$, etc., mimic action potentials, and the sigmoid-shaped function $\tanh$ scales the total inputs to the reticular complex to lie in the interval $[-1, 1]$: Positive terms are excitatory and negative ones are inhibitory.

In Figure 6.5 we take

$$L(t) = \frac{1}{N} \sum_{k=1}^N \cos_+ y_k,$$

which represents the lateral inhibition in the reticular complex.

We refer to the dynamics in Figure 6.5 as “winner take all in the frequency domain.” That is, the channel with highest frequency input to TH is the one that eventually dominates NC activity, suppressing other signals.

6.3.2 Hippocampus

The hippocampus is a highly complex system, and what it does is still mostly unknown, although it is intensively studied by neuroscientists and neuroengineers. Stimulus-response experiments and emerging connection charts are uncovering its roles in navigation, memory, and novelty detection. These suggest that we begin to study analogous mathematical systems and their responses to similar inputs. In particular, we formulate here a network

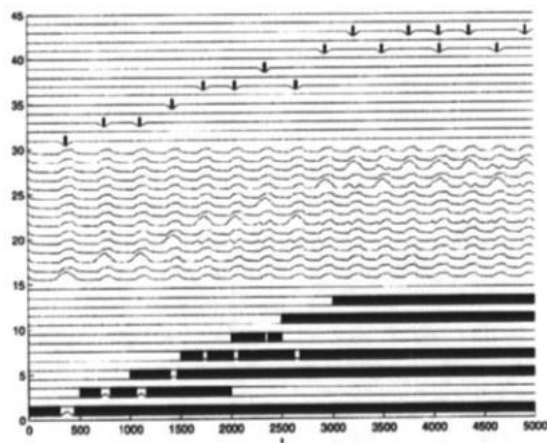


Figure 6.5. Simulation of the thalamocortical model based on 15 channels. The top 15 lines plot $\cos_+ z_j(t)$, the middle 15 lines plot $\cos_+ y_j(t)$, and the bottom 15 lines plot $\cos_+ x_j(t)$. Stimulation is applied to the first at time $t = 0$ ($\sigma_1 = 0.1$). A higher-frequency stimulus is applied at time $t = 500$ ($\sigma_3 = 0.3$). Only this last stimulus gets through to the neocortex. At time $t = 1000$, a higher frequency is applied ($\sigma_5 = 0.5$), etc. In every case the highest frequency stimulus is the only one to be expressed in the neocortex.

that comprises elements firing at gamma rhythms (approximately 40 Hz) but that can be entrained into patterns of theta rhythms (approximately 5–12 Hz) by particular input signals, in analogy with known responses of the hippocampus to inputs from both the medial septum and from the entorhinal cortex.

Two major frequencies observed in neural activity are the theta and gamma frequencies:

- Theta (Θ) frequency (5–12 Hz);
- Gamma (Γ) frequency (40+ Hz).

In this section we investigate distinctive patterns of gamma and theta rhythms in a model hippocampus.

Segment Model of the Hippocampus

Physiological and anatomical evidence shows the hippocampus to be a highly complex brain structure. One can imagine the hippocampal formation as having a long axis and two axes perpendicular to it. Each two-dimensional slice perpendicular to the long axis contains three main regions, called the dentate gyrus, the CA1 field, and the CA3 field. These regions include both excitatory and inhibitory neurons. Experimental evidence suggests that slices can sustain oscillations without external input,

and the natural frequencies of these oscillations are in the gamma range (40–70 Hz). As a first step toward modeling this, we suppose that interactive excitatory and inhibitory neural populations in the thin slice form an oscillator that we model using a single PLL. We model the hippocampus as a chain of such oscillators distributed along the long axis, as depicted in Figure 6.6.

Two primary inputs to the hippocampus are from afferents of the entorhinal cortex (EC) and of the medial septum (MS).

Morphological and neurophysiological data suggest that both inputs distribute their signals among the hippocampal oscillators with appropriate delays in phase because of different propagation distances from their sources. For example, oscillators nearer EC will have EC inputs nearly the same as the driving EC signal, while MS inputs to the same oscillators will be substantially delayed behind the MS source because of its distance from the MS source. The inputs both supply low-frequency theta-rhythm stimulation. We show here that the model's behavior depends on the deviation in timing between the MS and EC inputs. In addition, the frequencies of the input signals and their phase relations determine which oscillators of the hippocampal model receive sufficient synchronized input to entrain them to the low frequency. As a result, a pattern of theta and gamma rhythms in the array of oscillators will emerge. The input phases from each are described in the caption to Figure 6.6.

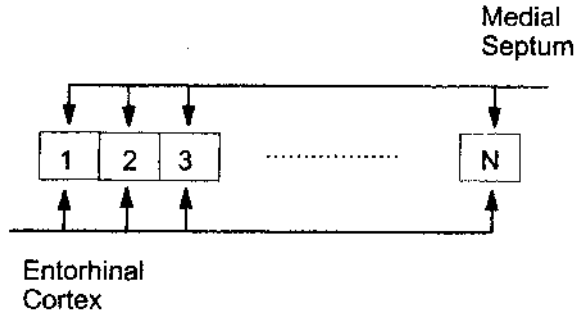


Figure 6.6. An oscillator model of the hippocampus comprising N oscillators that have inputs from the medial septum (left) and from the entorhinal cortex (right). The signals at the MS and EC have phases $\Theta t + \phi_S$ and $\Theta t + \phi_C$, respectively. The inputs to the j th oscillator have phases $\theta_S = \Theta t + \phi + (N - j)\Delta\Phi$ and $\theta_C = \Theta t + \psi + j\Delta\Phi$, respectively, where $\Delta\Phi$ is the time lag for the signal to propagate from one segment to the next.

Pattern Analysis of the Hippocampus Model

We omit the oscillator index n below to simplify formulas. Let θ be the phase of a segment oscillator, and let θ_C and θ_S denote the phases of inputs

I_C and I_S , respectively. The model of this configuration is

$$\dot{\theta} = \Gamma + K(\cos(\pi/2 + \theta - \theta_C) + \cos(\pi/2 + \theta - \theta_S)). \quad (6.3.3)$$

Here Γ is the nominal natural frequency of a hippocampal oscillator ($\Gamma \approx 40$ Hz). We suppose that phases of the EC and MS inputs are

$$\theta_C = \Theta t + \phi_C, \quad \theta_S = \Theta t + \phi_S,$$

respectively, where $\Theta \approx 5$ Hz is the frequency of a theta rhythm. Substituting the last formulas gives

$$\begin{aligned} \dot{\theta} &= \Gamma + 2K \sin((\theta_C + \theta_S)/2 - \theta) \cos((\theta_C - \theta_S)/2) \\ &= \Gamma + 2K \sin(\Theta t + \bar{\phi} - \theta) \cos(\psi/2), \end{aligned}$$

where $\psi = \phi_C - \phi_S$ and $\bar{\phi} = (\phi_C + \phi_S)/2$. Substituting $\Phi = \theta - \Theta t - \bar{\phi}$ results in the equation

$$\dot{\Phi} = \Gamma - \Theta - 2K \sin \Phi \cos(\psi/2).$$

This equation will have a stable rest point, say $\Phi = \Phi^*$, if the condition

$$\left| \frac{\Gamma - \Theta}{2K \cos(\psi/2)} \right| \leq 1 \quad (6.3.4)$$

is satisfied, namely,

$$\Phi^* = \sin^{-1} \left(\frac{\Gamma - \Theta}{2K \cos(\psi/2)} \right).$$

If Φ does approach the equilibrium Φ^* , then we have that $\theta/t = \Theta + \Phi/t + \bar{\phi}/t \rightarrow \Theta$ as $t \rightarrow \infty$. In this case, the oscillator will exhibit a theta rhythm. The condition for stability depends on three parameters $K, \Gamma - \Theta$, and ψ .

Simulations

The spatiotemporal pattern of neural activity depends on ψ , which is a phase deviation between two inputs. We can corroborate this calculation with computer simulations of the model equation (6.3.3) using the stability condition (6.3.4) as a guide for selection of parameters. The results of a simulation are shown in Figure 6.7, which demonstrates that the oscillatory neural network, using the parameters indicated in the figure caption, does phase lock to a theta rhythm for some phase deviations ψ , and it oscillates near a gamma rhythm for the others. The frequency pattern is shown in Figure 6.7, where the output frequency of the oscillator $\bar{\Omega} = \lim_{t \rightarrow \infty} (\theta(t)/t)$ is plotted for each phase deviation $\Delta\phi$ between the input signals. Since all of the segment oscillators are identical except for their input signals, which differ from their neighbors by an amount $\Delta\phi$, this figure also depicts the actual pattern that is observed along the long axis of the model when there are $N = 32$ segments.

This analysis of the PLL model and computer simulations of it show that the timing of inputs to an oscillator can lead to a dramatic change in its

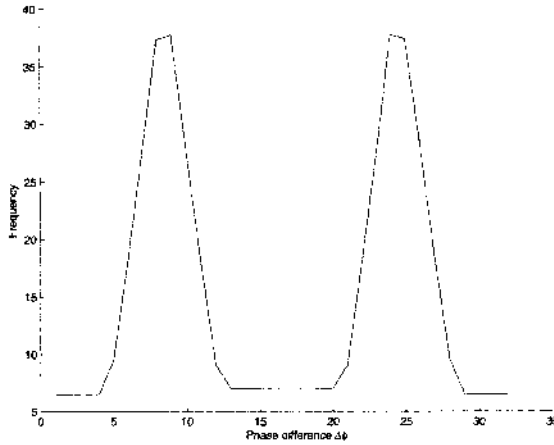


Figure 6.7. Frequency output of oscillators as a function of input phase. This shows a pattern of gamma rhythms (≈ 40 Hz) and theta rhythms (≈ 5 Hz).

frequency of firing. In particular, various timing combinations can create characteristic patterns of theta-rhythm activity in the array of oscillators. Different choices for ψ can create other patterns.

This work reminds us of the caution expressed in the Introduction. We have pruned away features of neurons that are not needed to produce the kinds of outputs we seek. In doing so, we have dropped many important aspects of neurons. In this case, we have lumped complex neural structures into single segment oscillators. This model represents a small step toward investigating the hippocampus. At least, the results demonstrate that a real electronic circuit can exhibit the desired behavior.

The circuits described here are widely used electronic circuits, which have a mathematical model that is easy to understand, and they lend themselves to computer simulation. This work should enable the reader to digest novel developments in neuroscience as they emerge by reproducing new observations using such analog circuits.

6.4 Annotated References

Many of the concepts used in this chapter are presented in greater detail in the books:

Hoppensteadt, F.C.: *An Introduction to the Mathematics of Neurons: Modeling in the Frequency Domain*, 2nd edition. Cambridge Univ. Press, Cambridge, UK, 1997.

Hoppensteadt, F.C., Izhikevich, E. M.: *Weakly Connected Neural Networks*. Springer-Verlag, New York, 1997.

More can be found about interesting aspects of biological time and pattern generators in the book:

Winfree, A.T.: *The Geometry of Biological Time*. Springer-Verlag, New York, 1980. (2nd edition, 2001.)

The experimental work on phase-locking in neural tissue by Rita Guttman and her collaborators is published in the article:

Guttman, R., Feldman, L., and Jakobsson, E.: Frequency entrainment of squid axon membrane. *J. Membrane Biol.* **56**, 9–18, 1980.

The searchlight hypothesis is described in the article:

Crick, F.: Function of the thalamic reticular complex: the searchlight hypothesis. *Proc. Natl. Acad. Sci.(USA)*, **81**, 4586–4590, 1984.

A useful reference for electronic circuits and background material in electronics is given in the book:

Horowitz, P., and Hill, P.: *The Art of Electronics*, 2nd edition. Cambridge Univ. Press, New York, 1989.

Exercises

6.1. Clocks

1. Solve the equation $d\theta/dt = \alpha$ for $\theta(t)$ and plot the points using polar coordinates $x = \cos\theta(t)$, $y = \sin\theta(t)$ in the xy -plane.
2. Consider two independent clocks having phases θ and ϕ . Solve the two equations $\dot{\theta} = \alpha$ and $\dot{\phi} = 1$.

Plot the solutions using toroidal coordinates

$$\begin{aligned}x &= (a + b \cos \phi) \cos \theta, \\y &= (a + b \cos \phi) \sin \theta, \\z &= b \sin \phi,\end{aligned}$$

where a, b are constants with $0 < b < a$ (you choose them) in three-dimensional xyz -space. Show using a computer simulation that this solution describes the torus when $\alpha = \sqrt{2}$ (an irrational number). Next, plot the solution when $\alpha = \frac{1}{3}$. The result is called a trefoil torus knot.

3. Consider a system of differential equations in polar coordinates ($r^2 = x^2 + y^2$ and $\theta = \tan^{-1} y/x$),

$$\begin{aligned}\dot{r} &= r(1 - r), \\ \dot{\theta} &= \alpha.\end{aligned}$$

Describe the solution of this system that starts at the point $r(0) = 0.01$, $\theta(0) = 0.0$. Find the simple clock among the solutions of this system for various choices of $r(0)$ and $\theta(0)$.

6.2. Modulated Clocks

Plot the values of $v(t) = \cos(\alpha t + \Phi(t))$, where

$$d\Phi/dt = \cos(2\pi t/100.0),$$

for each of the values $\alpha = 1.0$ and $\alpha = 2\pi/100.0$.

6.3. Phase Resetting: A Rubber-Handed Clock

Phase-resetting experiments have helped in many ways to uncover biological rhythms. For example, a hamster kept in a cage under controlled conditions of light and dark, say having 16 hours of light and 8 hours of darkness each day (which we denote by L:D = 16:8), will use its exercise wheel at regular times during the dark period. When the light-dark conditions are changed to L:D = 0:24, the hamster continues to run at regular intervals as before, but slightly shifted to a shorter "day." The timing of the exercise period can be changed, either slowed down or accelerated, by briefly shining a bright light on the animal at various times during its activity period.

Consider a *rubber-handed clock* that has the following unusual properties:

1. Left undisturbed, the hand's motion is described by the differential equation
2. If the hand is pulled or compressed and then released, it returns instantly to its original shape, but it points in a new direction, namely, the direction it points when released.
3. If the hand is crushed to the center of the clock, it stays there.

If the clock is at phase θ , then the hand extends from the origin (center of the clock) to the point $(\cos \theta, \sin \theta)$ on the clock's edge (we take the length of the hand at rest to be 1). Moving the clock's hand horizontally A units changes it into a hand extending from the origin to the point $(A + \cos \theta, \sin \theta)$, where we release it. The change of angle is $\Delta(A, \theta) = \theta - \phi$, and show that

$$\cos \Delta = \frac{1 + A \cos \theta}{\sqrt{A^2 + 2A \cos \theta + 1}}.$$

Plot the results of this by plotting the points (θ, A, ϕ) . Hint:

$$\phi = \theta - \cos^{-1} \left(\frac{1 + A \cos \theta}{\sqrt{A^2 + 2A \cos \theta + 1}} \right).$$

6.4. Phase-Locking Analysis*

1. Consider the system of two equations

$$\begin{aligned} dx/dt &= \omega + \cos(x - y), \\ dy/dt &= \mu. \end{aligned}$$

Show that if $|\omega - \mu| < 1$, then $x - y \rightarrow C$, a constant, as $t \rightarrow \infty$. Deduce that $x \rightarrow \mu t + C$ and that $x(t)/y(t) \rightarrow 1.0$ for all such values of ω and μ . This is the simplest example of phase-locking.

2. Consider the system of two equations

$$\begin{aligned} dx/dt &= \omega + \cos(x - y) + \cos(2x - y), \\ dy/dt &= \mu. \end{aligned}$$

Show that if $|\omega - \mu| \ll 1$, then $x/y \rightarrow 1.0$, and if $|2\omega - \mu| \ll 1$, then $x/y \rightarrow 0.5$. This can be done either using computer simulations as in the preceding exercise or by mathematical analysis.

6.5. The Atoll Oscillator

The following program solves the atoll model. It shows how to solve the differential equations using a very useful Matlab package called `ode15s`. Execute this program by typing `atoll` on the command line.

```
function atoll
%atoll.m
tf=50*pi;
Fs=32;
tt=0.01:1/Fs:tf;
options=odeset('AbsTol',1e-8,'RelTol',1e-6);
tic
x0=zeros(2,1); x0(2,1)=0*pi;
[t x]=ode15s('atlrhs',tt,x0,options);
toc
figure(1), clf
subplot(3,1,1),plot(mod(x(:,1),2*pi),mod(x(:,2)-pi,2*pi),'.')
axis([0 2*pi 0 2*pi]);
xlabel('x MOD 2 pi'),ylabel('y MOD 2 pi')
subplot(3,1,2),plot(t,cos(x(:,1)))
xlabel('t'), ylabel('V_1')
subplot(3,1,3),plot(t,cos(x(:,2)))
xlabel('t'), ylabel('V_2')

function xp=atlrhs(t,x)
%atlrhs.m
xp=x; xp(1)=5*(1+.5*cos(x(1))-cos(x(2)));
xp(2)=0.04*(1+cos(x(2))+10*cos(x(1)));
```

6.6. Matlab Programs for the Hippocampus

The following Matlab program solves the hippocampus segment model described in this chapter. Execute this program by typing `hippo` on the command line. For help with any of the functions in it, use Matlab Help.

```
function hippo
%hippo.m
global K DEL gamma tau theta NN aa bb cc dd
```

```

gamma=40*2*pi; tau=.5;
theta = 8; NN=32; K=90; tf=10; Fs=64;
tt=1/Fs:1/Fs:tf;
y0=ones(2*NN,1); y=y0;
aa=kron(eye(NN),[-1/tau 0;1 0]);
bb=kron(ones(NN,1),[1/tau;0]);
cc=kron(ones(NN,1),[0;1/tau]);
e=ones(2*NN-1,1);
dd=diag(e,1);
DEL=zeros(2*NN,1);
for i=1:(2*NN), DEL(i)=pi*i/NN; end
% The evens are phases and the odds are activities.
options=odeset('AbsTol',1e-8,'RelTol',1e-6);
tic
[t y]=ode15s('hiprhs',tt,y0,options);
toc
[ntf mm]=size(y);
figure(1), clf
hold on
zz=zeros(NN,1);
for jj=1:NN
    zz(jj)=y(ntf,2*jj)/(2*pi*tf);
end
plot(zz)
xlabel('Phase difference \Delta\phi')
ylabel('Frequency');

function yp=hiprhs(t,y)
%hiprhs.m
global K DEL gamma tau theta NN aa bb cc dd
yp=y;
yp=aa*y+gamma*bb+2*K*dd*(cc).*cos(DEL).*(sin(theta*t-y))/tau;

```

6.7. Winner Take All in the Frequency Domain*

1. The following program was used to generate Figure 6.5. Execute this program by typing `atn` on the command line.

```

function atn
%atn.m
global N N2 N3 input s A

N=15;
N2=2*N;
N3=3*N;
input=zeros(N,1);
s=input;
input(1)=0.01; %input(5)=0.5;

A=zeros(N);

```

```

A=A+diag(ones(N-1,1),1)+diag(ones(N-1,1),-1);
A(1,2)=2;
A(N,N-1)=2;

x0=0.5*pi*ones(N3,1);
x=x0;
tf=5000;
Fs=32;
tt=0:1/Fs:tf;
options=odeset('AbsTol',1e-8,'RelTol',1e-6);

tic
[t x]=ode15s('atn2rhs',tt,x0,options);
toc

figure(1),
clf
hold on;
for j=1:N3
    plot(t,cos(x(:,j))/2+j);
end;
xlabel('t'), ylabel('0.5*cos x_j(t)')

function xp=atn2rhs(t,x)
global N N2 N3 input s A

z=cos(x);
zp=max(z,0.0);

nnn=2*floor(t/500);
if(nnn<=12)
    input(min(nnn+1,N),1)=0.1+.1*nnn;
else
    input(min(nnn+1,N),1)=0.;
end
if(nnn>=8)input(3,1)=0;end

if(nnn>=10)input(9,1)=0;end

xp=x;
xp(1:N)=5*(1+z(1:N)-zp(N+1:N2)+input);
xp(N+1:N2)=0.04*(1+z(N+1:N2)+1*tanh(2*zp(1:N)-10*sum(zp(N+1:N2)))));
xp(N2+1:N3)=10*(0.1+z(N2+1:N3)+zp(N+1:N2));

```

2. Simulate this system using $N = 15$, but instead of only two separate inputs, apply input stimuli of increasing frequency. Describe the results of your simulation. Try other combinations of stimuli to determine the response of this system to the highest frequency inputs.

6.8. The Andronov-Hopf bifurcation

Consider the equation

$$\ddot{x} + (x^2 + \dot{x}^2 - \lambda)\dot{x} + x = 0.$$

This is called Robinson's equation in view of work by the physicist Neville Robinson. Show that if $\lambda < 0$, then $x \rightarrow 0$, but if $\lambda > 0$, an oscillation appears having amplitude $\sqrt{\lambda}$ and frequency ω . (Hint: Convert this single second-order differential equation to two first-order differential equations using polar coordinates: $x = r \cos \theta, \dot{x} = r \sin \theta$.)

7

Population Dynamics

Knowing about how populations live and respond to external signals is important in a society where government, economics and commerce are based on population projections and descriptions; and in medicine, science, and agriculture, where cell populations are targets for the treatment of diseases and for production in biotechnology. We study here two important realms of biology: microbial populations and animal populations. Mathematical aspects of all population phenomena usually involve descriptions of population sizes and ways to extrapolate present knowledge to information about the future, and they involve descriptions of population structures and ways to determine and project them into the future.

It is important to uncouple aging from the progress of time. In humans, we measure age as chronological time since birth, using some convenient units like years. But for bacteria, aging can be counted as the percentage of progress made through the mitotic cycle toward splitting into two daughter cells, and this will generally not be proportional to chronological time.

If each individual in a population more than replaces himself, the population will grow, but sorting out how fast the population grows is complicated. In the easiest case, if the population comprises bacterial cells whose life cycles are synchronized, then at each reproduction time the cell population will double, and we would say that the population during the n th cycle is 2^n times the initial population. That is, the population grows geometrically with common ratio, or growth rate, 2. On the other hand, if births of individuals are not synchronized, as in human populations, counting population size is more difficult. If stratification of the population by age or disease or genetic type is to be accounted for, things are even more complicated.

We begin our study of these problems with a description of synchronized bacterial growth.

7.1 Bacterial Cultures

Bacteria are important to us in many ways. There are approximately 10–100 bacterial cells in your body for each “normal” cell. These perform many important tasks, ranging from digesting food to occupying likely places where infections can invade and so protecting us from illness. Bacteria also are mainstays of biotechnology. For example, they are used to create drugs to treat a variety of infections.

A (typical) bacterium is made up of a cell wall that encloses a variety of structures that enable the cell to live (e.g., energy metabolism and production of significant by-products) and to reproduce. Of particular importance to us here is the chromosome, which encodes information for the cell’s activities on segments of DNA called *genes*, and extrachromosomal DNA elements called *plasmids*, that carry genes, but can be passed horizontally from cell to cell. We consider the dynamics of a population of cells in this chapter, and certain aspects of its genetics in the next chapter.

The life of a (typical) bacterium has four phases:

1. GAP1, an interval during which the cell accumulates and produces all the materials it needs for synthesis and replication of its structures.
2. SYNTHESIS, an interval in which the cell replicates its chromosomes and metabolic structures.
3. GAP2, an interval during which the cell accumulates and produces materials needed for splitting into two daughters.
4. MITOSIS, an interval in which the cell splits into two daughter cells, each receiving half of the materials of the mother (with the possible exception of plasmids which might be distributed unequally between the daughters).

This life cycle is depicted in Figure 7.1.

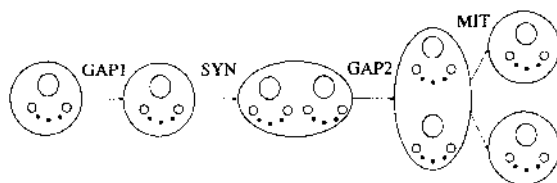


Figure 7.1. Bacterial cell cycle.

For our purposes here, we suppose that the two gap intervals are of variable length depending on the amounts of nutrients available, that the synthesis interval is always of fixed length (that is, cells do not begin synthesis unless they can complete it), and that mitosis occupies a negligible amount of time compared to the other phases.

7.1.1 Batch Culture Growth

Batch cultures of cells are grown in containers having the needed growth media; these materials are usually referred to as *minimal media*. The solution is inoculated with a known number of cells, and placed in a test environment, for example in an incubator. At various times a sample is drawn from the batch, and the number of cells is counted to determine the concentration of cells (usually in units of cells per 100 milliliters) in the batch at that time. This counting can be done by spreading the sample out on a grid etched on a microscope slide and counting the cells. Then statistical methods can be used to estimate the density of cells in the batch. Or the optical density of the sample can be determined using a colorimeter that is calibrated to yield the concentration of cells in it.

If the cells are healthy and ready to grow, the cell population initially increases geometrically, but it eventually levels out, and after a long time interval, it starts to decrease. The first phase is called the *exponential growth* phase, the second is the *stationary phase*, and the third phase is that of culture degradation. The preliminary stage during which cells, for example those just removed from storage, become growth-competent is called the *refractory* phase. Several methods for growing bacteria have been devised for research and for industrial production. The simplest method that maintains cells at comparable growth competence involves serial dilutions of cells. In this, the batch is allowed to grow until the midst of the exponential phase, when a sample is drawn and used to inoculate a fresh batch culture, and so on. On the other hand, there are continuous growth devices that maintain various parameters. Sometimes these are called chemostats, pH-stats, gradostats, etc. We will discuss them later. Finally, cells are often grown on plates containing a supportive matrix, such as agar, and minimal growth media. An inoculum of cells is spread over the surface of the plate, and certain of them will grow into colonies having various morphologies. These can be selected for further experiments or transferred to other plates for further analysis. Some plating experiments are described in Chapter 10.

Geometric Growth of Bacterial Batch Cultures

Large batches of bacteria are important for many scientific and industrial uses. A typical procedure is to inoculate a flask containing growth media with a sample of cells taken from storage. The inoculum might contain 1.0×10^8 cells/ml. The inoculated mixture is then placed in an incubator and

Table 7.1. Observed Bacteria Numbers

Sample Time (Hours)	Observed Density (Cells/ml)
1	1.9×10^8
2	3.6×10^8
3	6.9×10^8
4	1.3×10^9
5	2.5×10^9
6	4.7×10^9
7	8.5×10^9
8	1.4×10^{10}

shaken for approximately 8 hours. After this time, there are approximately 1.0×10^{10} cells/ml. The batch culture can be sampled at regular time intervals, for example by drawing a sample of 1 ml and measuring its optical density. In this way, the concentration of cells in the growth chamber can be monitored. Typical data are presented in Table 7.1.

It is interesting to derive a mathematical model to “explain” these data. In particular, we would like to give some simple description of the population’s growth that could perhaps be used to characterize the cell strain. Let B_n denote the cell numbers observed at the n th sampling time: For example, suppose that $B_0 = 1.0 \times 10^8$, the initial cell concentration, $B_1 = 1.9 \times 10^8$, etc. We expect that during the growth phase the number of cells at one time will be some multiple of the previous measurement and that this multiple will be constant over several sampling times. In this hope, we write the equation

$$B_{n+1} = rB_n. \quad (7.1.1)$$

The constant r is called the *growth rate*, and it is this number that we will attempt to determine from the observations. This model is referred to as Malthus’s model, since it is of the same form that Malthus used in a different context in 1798. If the cells divide every hour and our sampling is done every hour, then we should observe that $r = 2$, and we say that the cell’s doubling time is 1 hour.

Doubling times

When a bacterium is growing in a suitable environment, its time to mitosis is fixed over many generations, and so its offspring can be counted at fixed time intervals that are related to their reproductive biology. We did not do this in our earlier calculations, where we simply measured the total population every hour. But if we could determine the time to mitosis, which is referred to as the *doubling time* of a cell, then we would get useful information about the cells. From our calculation in the preceding section it is clear that our choice of measuring every hour is not necessarily matched to the cell’s doubling time, although it was close in that case.

A mathematical formula for the doubling time can be found by studying the sampling problem using small (infinitesimal) sampling intervals. Let $P(t)$ denote the population size at time t , and as before, suppose that the change observed over a time interval, say of length h , is proportional to the population size. Since the percent change of the population is small if the sampling interval h is small, we write $r = 1 + hb$: Then the equation $P(t+h) = rP(t)$ can be rewritten as

$$P(t+h) - P(t) = hbP(t).$$

The number b is called the infinitesimal birth rate, or *intrinsic growth rate* of the population. We view $P(t)$ as a smooth function of the continuous time variable t . Then dividing by h and passing to the limit $h \rightarrow 0$ gives the differential equation

$$\frac{dP}{dt} = bP, \quad (7.1.2)$$

which we must solve for the population size $P(t)$. This is referred to as the overlapping-generations, or continuous-time, version of Malthus's model for discrete generations in equation (7.1.1). Note that no assumption is made about synchrony of the population in this derivation, only about its instantaneous changes. Equation (7.1.2) can be solved using the method of separation of variables: Dividing both sides of the equation by P and multiplying by dt gives

$$dP/P = b dt.$$

Integrating this equation from $t = 0$ to $t = T$ (Thus, the left-hand side from $P = P(0)$ to $P = P(T)$), we have

$$\log P(T) - \log P(0) = bT,$$

where here and below $\log$ denotes the natural logarithm (base e). Equivalently,

$$P(T) = P(0)e^{bT}.$$

So the population grows exponentially, which is the continuous-time analogue of geometric growth in synchronized populations.

The model of bacterial growth in equation (7.1.1) is one based on population size at distinct time steps, $n = 0, 1, 2, \dots$, and the result is referred to as a difference equation. There are many advantages in converting, when possible, a difference equation to one based on continuous time, as we have done here. In fact, calculus was introduced in large part to facilitate solving difference equations, and it often enables us to calculate a solution in a convenient form and to analyze solutions more efficiently than is possible for difference equations.

The solution of the discrete-time model $B_n = r^n B_0$ and the solution of the overlapping generation model $P(t) = e^{bt} P(0)$ are easy to connect: If

we sample $P(t)$ at times nh , where h is the length of the sampling interval, the resulting numbers should be related to the sequence B_n . Because of this, we set $P(nh) = B_n$, and then $r^n = e^{bhn}$. Equivalently, $b = (1/h) \log r$. In the bacterial sampling data in Table 7.1, $h = 1.0$ hours.

This observation enables us to derive the cell doubling time from the data presented at the start of this section. We ask, What time T is needed so that $P(T) = 2P(0)$? Thus, $e^{bT} = 2.0$, or

$$T = \log 2/b.$$

This formula also relates the doubling time T to the intrinsic growth rate r :

$$T = h \log 2 / \log r,$$

although we do not make further use of this formula. We can use this calculation to replace b and get the interesting formula

$$P(t) = e^{bt} P(0) = e^{(\log 2/T)t} P(0) = 2^{t/T} P(0).$$

So, every time t increases through an integer multiple of T , the population again doubles. Since $r = 1.9$ from Table 7.1, we have that the doubling time is $T = 1.08$ hours, which is different from the sampling time interval but close to it. If we sampled the population every minute and again used least-squares estimation methods (described later), we would improve our estimate of b .

These same methods give remarkably good predictions of Sweden's population numbers for nearly a century (the percent error does not exceed 10%!) (See Exercise 7.1). Similar calculations for Wales (1860-1960) are good to 1920, but lead to errors of more than 25% thereafter. The model is acceptable for some purposes over certain phases of growth, but it must eventually become invalid as a population reaches its limits of growth. Causes of this might lie in size-dependent changes in the physical or biological environment, or in the age structure of the population. In 1760, Euler suggested that populations grow geometrically, but as a result of later work by Malthus (1798); equation (7.1.1) is now named for him. In the years following 1800, data were collected and analyzed to test such theories, and like the examples, populations were observed to grow geometrically during certain phases. We will investigate the limits of growth later.

In order to estimate r from the data, we use the statistical method called the *method of least-squares estimation*. Estimating growth of a batch culture of bacteria is not straightforward. In fact, there are many causes for noise to enter into our observations, and deriving a number, say the growth rate, in the presence of random noise poses several interesting challenges. There are even difficulties in defining randomness (See Appendix B).

7.1.2 Least-Squares Estimation of the Growth Rate

The size of a population can be viewed as a random variable that is observed at each census. We measure this at fixed times, and the observed sizes form a sequence of observations of this random variable.

The method of least squares is effective in making sense of such a sequence of observations. Suppose that data are observed, say

$$(x_1, y_1), \dots, (x_N, y_N),$$

and that a theory predicts that these data are linearly related, say

$$y = ax + b.$$

In our application, x_n will be the time of the n th observation, and y_n will be the observed size at that time, so $\{y_n\}$ can be viewed as being a random process. We use a method for finding the parameters a and b that will "best fit" the data. First, we introduce a measure of how far the data deviate from the theoretical straight line. Let

$$L(a, b) = \sum_{i=1}^N (y_i - ax_i - b)^2.$$

The parameters a and b are chosen to minimize this quantity. If a and b do give a minimum, then

$$\frac{\partial L}{\partial a} = 0 \text{ and } \frac{\partial L}{\partial b} = 0.$$

So it is from among the solutions of these two equations that we seek a and b . These two equations can be written as

$$\begin{aligned} a\langle x^2 \rangle + b\langle x \rangle &= \langle xy \rangle, \\ a\langle x \rangle + b &= \langle y \rangle, \end{aligned}$$

where

$$\langle x \rangle = \frac{1}{N} \sum_{i=1}^N x_i, \quad \langle y \rangle = \frac{1}{N} \sum_{i=1}^N y_i,$$

and

$$\langle x^2 \rangle = \frac{1}{N} \sum_{i=1}^N x_i^2, \quad \langle xy \rangle = \frac{1}{N} \sum_{i=1}^N x_i y_i.$$

The solutions of these equations are

$$a = \frac{\langle xy \rangle - \langle x \rangle \langle y \rangle}{\langle x^2 \rangle - \langle x \rangle^2}, \quad b = \frac{\langle x^2 \rangle \langle y \rangle - \langle x \rangle \langle xy \rangle}{\langle x^2 \rangle - \langle x \rangle^2}.$$

Using these constants in the straight-line formula gives us the least-squares approximation to the data.

From the probabilistic and statistical points of view, L is a random variable that takes on a different value for each experiment $\{x_n, y_n\}$ that we perform. Moreover, the formulas we have just derived for a, b define them also as random variables that are different for each experiment we run.

Statistical methods enable us to use a finite set of observations to estimate the distributions of all these random variables.

The least-squares method can be used to estimate the bacterial growth rate r , but not directly. It works only if there is a linear relation between the data and the sampling times, which is apparently not the case for equation (7.1.1). However, the model in equation (7.1.1) can be solved for B_n . Starting with an arbitrary time, say n , we successively substitute backwards using the equation, and we get that

$$B_n = rB_{n-1} = r(rB_{n-2}) = \cdots = r^n B_0.$$

That is,

$$B_n = r^n B_0. \quad (7.1.3)$$

This is a convenient form for the solution of the model, and it shows that the number of cells (B_n) at any time (n) depends only on n, r , and B_0 . This formula also shows that the population grows geometrically with *common ratio* r . It is apparent from the solution that B_n and n are not linearly related. However, if we take logarithms of the solution, we get

$$\log B_n = n \log r + \log B_0.$$

Thus, the theory predicts that there is a linear relation between $\log B_n$ and n . Therefore, $x_1 = 1, y_1 = \log 1.9, x_2 = 2, y_2 = \log 3.6$, etc., from Table 7.1. The method of least squares gives numbers a and b , where a is an estimate of $\log r$. In this case, $\log r = 0.642$, or $r = 1.9$.

In this way, we estimate the expected value of the growth rate. Further statistical work shows how the variance of this random variable can be found using other methods.

7.2 Age Structures

It is important to consider other kinds of aging of organisms than chronological aging. This is most easily done in bacteria, where age can be taken to be the proportion of the mitotic cycle that a cell has completed. Thus, the age of a bacterium might be 0.25 mitotic cycle units (1 corresponds to complete mitosis), while the cell's chronological age might be 1000 years! Cells growing under optimal conditions might double in 20 minutes, but many cells in less pleasant environments double only after months or years.

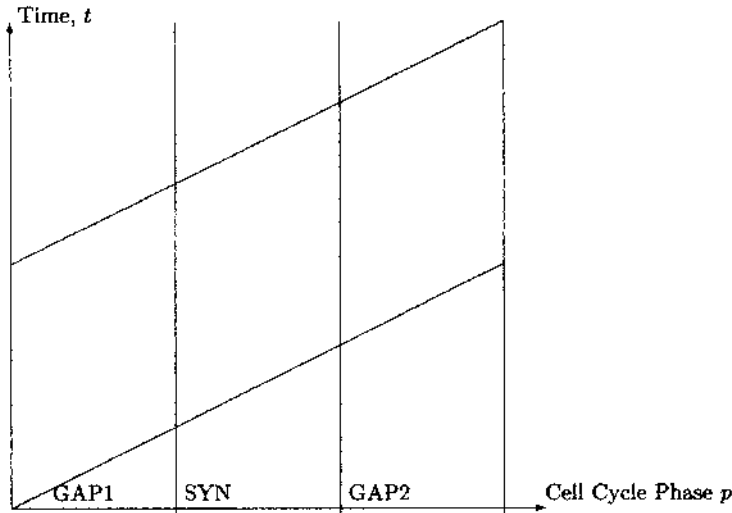


Figure 7.2. Cohort of cells growing under optimal conditions.

7.2.1 Cohort Analysis

The phase of the cell's mitotic cycle has the dimensions of time, but it is connected to chronological time in a complicated way that involves the availability of nutrients and environmental conditions. As noted before, the cell's mitotic cycle has four parts: Its progress through GAP1 depends on what nutrients are taken up; its progress through SYNTHESIS is at a fixed rate (evidently, cells do not start synthesis until they are able to complete it); its progress through GAP2 depends what is taken up; and we suppose that its progress through mitosis is at a fixed rate that is fast compared to the other phases.

Let p denote the phase of a cell in its mitotic cycle. While p depends on chronological time, the relationship is not necessarily linear. It is useful for visualizing cell aging to plot t and p simultaneously for a given cell. The result is referred to (here) as a cohort diagram. Each of the vertical lines denotes a boundary between reproductive intervals: GAP1 is terminated at the first vertical line, SYNTHESIS at the second, and GAP2 and MITOSIS at the third vertical line, while the newborns emerge from the t -axis where $p = 0$.

The cell-cycle phase variable p changes with chronological time and with environmental parameters. How this happens depends on the particular cell and its setting.

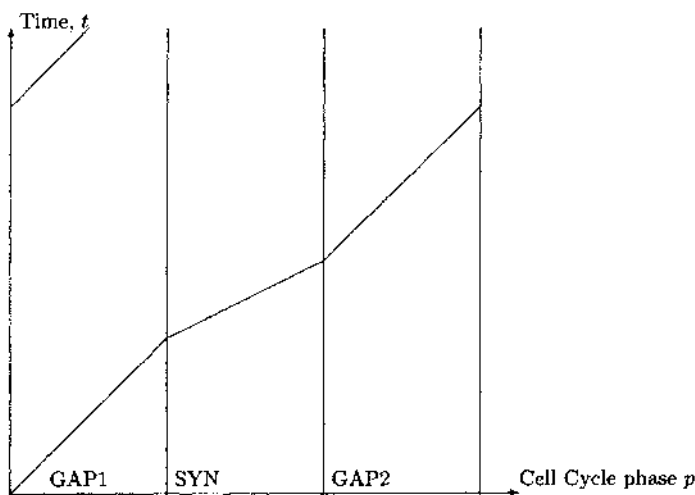


Figure 7.3. Cohort of cells growing under less than optimal conditions. Aging slows in the gap intervals, but not in synthesis.

A common situation is that of cells grown under ideal conditions. In this case, we take $\dot{p} = \omega$ (here and below we use the notation $\dot{p} = dp/dt$), where ω is a constant having units of number of mitotic cycles completed per unit of chronological time. If chronological time is measured in hours, then optimally, $\omega = 3$. This is shown in Figure 7.2.

Mathematics is effective in dealing with complicated situations partly because quite simple models can be shown (sometimes proven!) to represent essential parts of a phenomenon. Such models are referred to as canonical, and one such model is shown in Exercise 7.3, which we adapt to the cell-aging problem with no further justification.

A typical situation is shown in Figure 7.3, where the aging slows dramatically during the two gap intervals. The diagonal lines in each of Figures 7.2 and 7.3 that cross the interval boundaries describe the progress of a cohort of daughter cells through the mitotic cycle.

These ideas and notation will be used later to determine how cell cultures can be synchronized.

7.2.2 Birth Rates

We begin with a very old story that describes age structure in a synchronized population. The ideas are later generalized to study populations that have synchronized generations and then to populations with overlapping generations.

Fibonacci's Bacteria

Age structure of a population is more difficult to model and analyze than total population size, but the following example, which is a variation of Fibonacci's model, introduces the basic ideas of age structure in a simple way. We consider an age-like attribute of cells, somewhat like vertical transmission of a lethal pathogen.

We consider an unknown strain of bacteria that can carry a lethal marker. We begin with a single cell. If the cell has no markers, it leaves two daughter cells, one marked and the other clean. If it has one marker, it leaves two cells, one with two markers, the other clean. If it has two markers, it leaves no daughters. Therefore, at each mitotic event there is one unmarked daughter and the other gains a marker. The graph in Figure 7.4 illustrates the reproductive scheme.

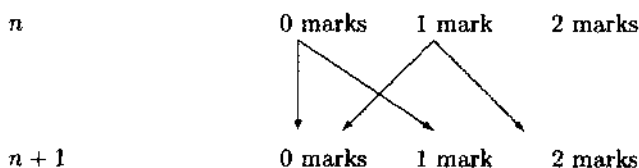


Figure 7.4. Fibonacci's bacteria.

Let B_n^0 denote the number of newborn cells having no marker, B_n^1 the number with one marker, and B_n^2 the number with two markers at the n th measurement. We take $B_0^0 = 1$, $B_0^1 = 0$, $B_0^2 = 0$. The population's dynamics are described in Table 7.2, which can also be viewed as a cohort diagram for the population.

Table 7.2. Fibonacci Bacteria

n	B_n^0	B_n^1	B_n^2
0	1	0	0
1	1	1	0
2	2	1	1
3	3	2	1
4	5	3	2
$\vdots$	$\vdots$	$\vdots$	$\vdots$

We see from this that the number of newborns in the n th census is

$$B_n^0 = B_{n-1}^0 + B_{n-1}^1$$

for $n > 0$, and since $B_{n-1}^1 = B_{n-2}^0$, the sequence of clean births, say $B_n = B_n^0$, satisfies Fibonacci's equation

$$B_n = B_{n-1} + B_{n-2}. \quad (7.2.1)$$

The time sequence of newborn cells will then be 1, 1, 2, 3, 5, 8, 13, ..., but we quickly tire of listing these numbers.

The analysis of this model has intrigued mathematicians for centuries. We could continue as earlier to list the Fibonacci sequence, where each term is the sum of the preceding two, but this quickly becomes pointless. Fortunately, it is possible to find a convenient solution of the problem. Experience with Malthus's model suggests that we ask whether the term B_n might be a power of some common ratio:

$$B_n = r^n.$$

Substituting this into the equation, we see that r must satisfy the equation

$$r^2 - r - 1 = 0,$$

which has two solutions:

$$r = \frac{1 \pm \sqrt{5}}{2}.$$

The theory of discrete-time Laplace transforms (called z -transforms in the engineering literature) shows that the solution B_n rests on these two roots. This approach shows that the terms in the sequence are given by the formula

$$B_n = ar_1^n + br_2^n,$$

where

$$r_1 = \frac{1 + \sqrt{5}}{2}, \quad r_2 = \frac{1 - \sqrt{5}}{2}.$$

The coefficients a and b are determined to satisfy the initial conditions $B_0 = 1, B_1 = 1$, so

$$a = \frac{1 + \sqrt{5}}{2\sqrt{5}}, \quad b = \frac{\sqrt{5} - 1}{2\sqrt{5}}.$$

There are two surprising results of this formula:

1. Although complicated, this expression for B_n always gives an integer value!
2. The ratio B_n/B_{n-1} approaches $(1 + \sqrt{5})/2 \approx 1.618 \dots$ as n increases.

The second point shows that Fibonacci's sequence is a sequence of integers that grows (approximately) geometrically.

Fibonacci's interest in 1202 was not bacteria (they had not yet been discovered), but rather the growth of money in commerce. Its applications described the compounding of money invested at interest. Still, the example nicely illustrates important methods used to study more realistic age-structured populations. [See Exercises 7.2, 7.13, 7.14].

Euler's Renewal Theory

Fibonacci's model suggests how we could describe human populations by relating current births to earlier ones. In fact, Euler described the age structure of human populations in a similar way in 1760. Consider a population that we count at specific census intervals, but we keep track only of the newborn females. Let B_n denote the number of females born during the n th census interval.

The population is divided into age classes, each a census interval long. Those in the k th age class are survivors of those born k periods earlier, and they can contribute to births in the next census period. B_{n-k} denotes the number of females born in census $n-k$, so the survivors among them at time n have (census) age k . Their chronological age is k times the census interval, which is usually 5 or 10 years. We denote by λ_k the proportion of those born k census periods ago who survive. Thus, λ_k is the probability of a newborn surviving from birth to the k th census. We denote by b_k the fertility of those in their k th census interval. Therefore, the individuals born k censuses ago are expected to produce $b_k \lambda_k B_{n-k}$ daughters during the k th census. Finally, there might be some births in the n th census interval from those who were present when we started counting everything, and we lump all of these into one number h_n .

Adding up all of these numbers, we get that the number of births in the n th census interval is

$$B_n = h_n + b_1 \lambda_1 B_{n-1} + b_2 \lambda_2 B_{n-2} + \cdots + b_n \lambda_n B_0. \quad (7.2.2)$$

This equation is referred to as the *renewal equation*. In Fibonacci's case, $\lambda_1 b_1 = \lambda_2 b_2 = 1$, $\lambda_j b_j = 0$ for all $j \geq 3$, and $h_n \equiv 0$ for all n .

The numbers $\{\lambda_k\}$ have been tabulated for many human populations, and the results are summarized in what is called a *life table*. Birth-rate data for these populations give the fertilities $\{b_k\}$. A wealth of data and interesting examples are given in the reference book by Keyfitz and Flieger.

There are various ways of extracting information from the renewal equation. Fibonacci's model demonstrates the usual method, so we take the same point of view toward solving equation (7.2.2), where the data for the life table and the fertilities are assumed to be known. We suppose that fertility is zero for large ages, say $b_k = 0$ for $k > M$, so that $h_n = 0$ for $n > M$.

For example, a census interval might be 5 years, and since human female fertility stops by age 55 (approximately), $M = 11$.

Let us suppose that $B_n \approx r^n$. Then, for $n > M$,

$$r^n = b_1 \lambda_1 r^{n-1} + \cdots + b_M \lambda_M r^{n-M}.$$

Multiplying by r^{M-n} gives

$$r^M = b_1 \lambda_1 r^{M-1} + \cdots + b_M \lambda_M.$$

This is an M th order polynomial for r . It is analogous to the second-order polynomial for r that we derived for Fibonacci's model. There will be M roots for r , say $r_1, \dots, r_M$, and Laplace's method shows that the solution for B_n will have the form

$$B_n = A_1 r_1^n + \dots + A_M r_M^n$$

for $n = 0, 1, \dots$, and for some constants $A_1, \dots, A_M$. This formula generalizes the one we found for B_n in Fibonacci's model.

The following result is very important for such problems. Although we do not prove it here, its validity is quite plausible. First, we define the maternity parameter

$$p = \sum_{k=1}^M b_k \lambda_k.$$

This is the expected number of female offspring that each newborn will produce in her entire life. We state the results as a theorem:

Renewal Theorem: Let the fertilities $b_1, \dots, b_M$ and the life table $\lambda_1, \dots, \lambda_M$ be given, and let p be the expected number of daughters each newborn will bear. Then:

1. If $p < 1$, $B_n \rightarrow 0$ as $n \rightarrow \infty$. That is, if each newborn less than replaces herself over her lifetime, the population will die out.
2. If $p \geq 1$ and if there is a unique root of the characteristic equation

$$r^M - b_1 \lambda_1 r^{M-1} - \dots - b_M \lambda_M = 0,$$

say $r = r^*$, such that all other roots satisfy $|r| < r^*$, then

$$\lim_{n \rightarrow \infty} r^{*-n} B_n = C,$$

where C is a constant. If $r^* = r_1$, then $C = A_1$.

The value $p = 1$ is a threshold value for the population's maternity parameter. If each newborn will more than replace herself ($p > 1$), then the number of newborns grows geometrically, since

$$B_n \approx C r^{*n}$$

for large values of n . The constants C and r^* can be estimated from population data using the least-squares estimation method described earlier (see Exercise 7.15). The parameter r^* describes the rate at which the population will grow.

7.2.3 Age Distributions

It is possible to describe the entire age distribution of the population at one time just from its birth rate B_n , life table, and fertilities. To do this,

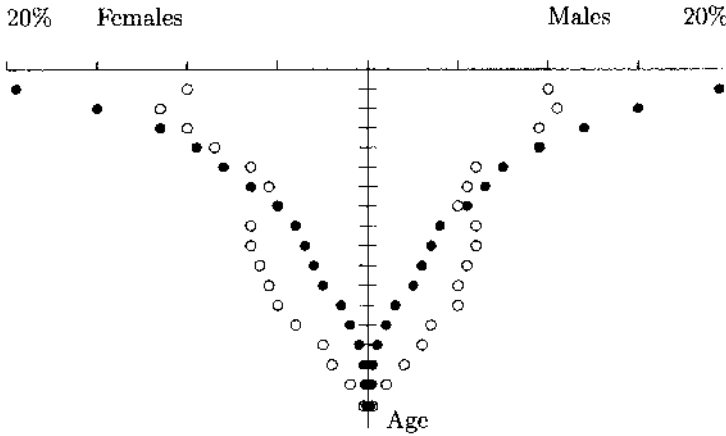


Figure 7.5. Age Pyramid: • Mexico, 1966; ○ USA, 1966.

we define new variables

$$\begin{aligned} u_{0,n} &= B_n, \\ u_{1,n} &= \sigma_1 u_{0,n-1} \\ u_{2,n} &= \sigma_2 u_{1,n-1}, \dots, \\ u_{M,n} &= \sigma_M u_{M-1,n-1}, \end{aligned}$$

where the σ 's indicate the proportion of one age group that will survive to the next census. Since $B_n \approx Cr^{*n}$, we have that

$$u_{j,n} = \sigma_j \sigma_{j-1} \cdots \sigma_1 B_{n-j} \approx \sigma_j \cdots \sigma_1 r^{*(n-j)}.$$

As a result, we have that the ratio of two age classes approaches a stable limit, called the *stable age distribution*:

$$u_{j,n}/u_{k,n} \approx r^{*(k-j)} \sigma_j \cdots \sigma_1 / (\sigma_k \cdots \sigma_1).$$

See Exercise 7.14.

Human populations have two sexes, and our model so far has accounted only for females. There is a complicated problem for demographers, called the coupling problem. Namely, if you know the age distribution of females and the age distribution of males, what is the distribution of reproducing couples? The simplest assumption is that any female who wants to be fertilized can be, and so consideration can be reduced to one sex. This is a reasonable assumption in large populations.

However, it is important to keep track of both sexes in describing a population, and this is done using an age pyramid, where for each age group both the numbers of males and numbers of females are plotted at once. Two examples are shown in Figure 7.5.

7.3 Microbial Ecology

The preceding models describe populations that can grow without bound. They do not account for some essential element limiting growth. This might be a nutrient, insufficient space to avoid stress, an infectious disease, etc.

We first consider consumption of a limiting nutrient by bacteria, and then later some more complicated population phenomena.

It is generally viewed that only one aspect of a cell's environment will limit growth at any time. This might be different at different times, but at any one time, there is one limiter to growth. This can be described using a continuous-time model of bacterial growth,

$$\dot{P} = bP,$$

but now the growth rate b depends on another quantity, which we refer to as the *substrate* and which we denote by $S(t)$.

7.3.1 Uptake of Nutrients

Population growth can be limited by levels of some limiting nutrient. In this case, the growth rate b will change with the availability of the nutrient $S(t)$. We write $b(S) = VS/(K + S)$ to denote this. Jacob and Monod considered how the growth rate should change, and the calculations in this section describe their model.

They described the uptake of nutrients (more generally of any substrate) by cells by the differential equation

$$\dot{S} = -\frac{VS}{K + S} \frac{P}{Y},$$

where S denotes the concentration of substrate and P denotes the concentration of cells. Here Y , V , and K are constants: V is called the *maximum uptake rate*, and K is called the *saturation constant*. Note that when $S = K$, the uptake is $V/2$, half its maximum. For this reason, the value K is often taken to be the level of substrate below which cells do not grow because they cannot locate the substrate. The constant Y is called the *yield*, and it measures the number of cell units produced per unit of substrate taken up. Y is often ignored in models of this kind, since it can be scaled out of the problem by setting $\tilde{P} = P/Y$, and then erasing the tildes.

The complete model is

$$\begin{aligned}\dot{S} &= -\frac{VS}{K + S} \frac{P}{Y}, \\ \dot{P} &= \frac{VS}{K + S} P.\end{aligned}\tag{7.3.1}$$

Adding the two equations shows that

$$\frac{d}{dt}(YS + P) = 0,$$

so the total cell mass $YS + P$ is conserved throughout the growth process.

These two variables are most easily described using a two dimensional plot called the *phase portrait* of solutions. Plotting the two coordinates (S, P) and using the conservation law, we see that solutions of this system of two equations will lie on a straight line

$$P = -YS + (P(0) + YS(0))$$

having slope $-Y$ and S -intercept $P(0) + YS(0)$. The solution S decreases, and when it is close to zero, we have

$$\dot{S} \approx -\frac{V(P(0) + YS(0))}{KY} S,$$

which tells us that S actually approaches 0 as $t \rightarrow \infty$.

A computer simulation of this model is described in Figure 7.6. The bottom plot in this figure reflects the conservation law in the model: The solutions in the SP phase plane move up and to the left along the straight line toward $S = 0$.

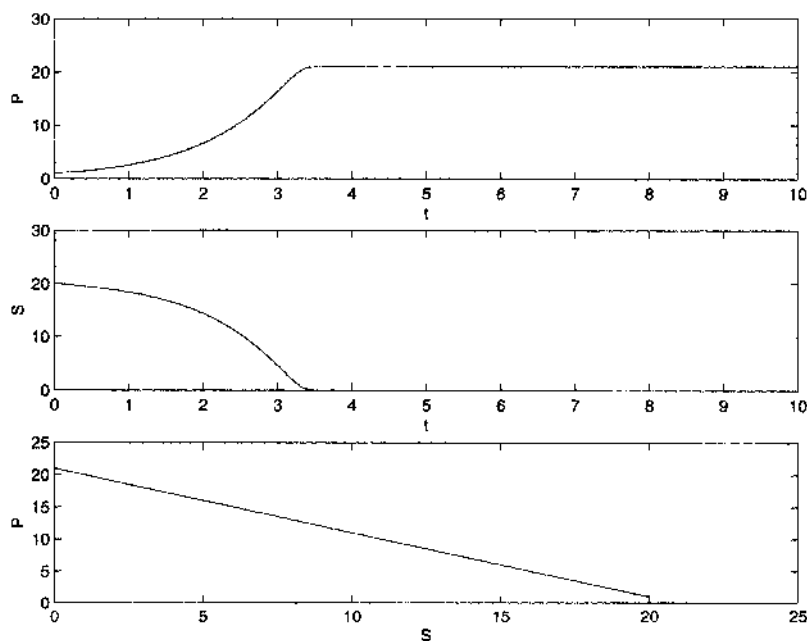


Figure 7.6. Computer simulation of batch culture in equations (7.3.1). ($P(0) = 1$, $S(0) = 20$.)

This model illustrates two interesting phenomena: First, when $P(0)$ is small and S is large, S is slowly depleted and $P(t)$ grows like e^{Vt} , which

reflects the exponential phase of growth for the culture. Second, when S is small, $\dot{P} \approx 0$, which is the stationary phase of growth.

One of the variables can be eliminated by using the conservation of biomass. For example, $S = (C - P)/Y$, where $C = YS(0) + P(0)$. Then

$$\dot{P} = \frac{V(C - P)}{KY + C - P}P.$$

This is often approximated by the simpler equation

$$\dot{P} = b^*(1 - P/K^*)P,$$

where b^* is the *intrinsic growth rate* and K^* is referred to as the *carrying capacity*. This equation is referred to as the *logistic equation*.

When one substrate is depleted, cells can sometimes use an alternative one, but one that requires more energy to process and use. That is, the cell can switch to an alternative metabolic pathway, but at higher cost. There are various mechanisms for doing this, including using an existing pathway or creating a new one by turning on a gene, which we discuss in Chapter 8.

7.3.2 Growth and Quiescence

When a limiting substrate is depleted, cells can change from a form in which they are growth-competent but vulnerable, to a quiescent, but better-protected, form. They can exist in quiescent form for a very long time. In order to describe this, we introduce a second population of bacteria: the population of quiescent cells. Now, S denotes the limiting substrate concentration, P denotes the size of the growth-competent bacterial population, and Q denotes the size of the quiescent cell population. In addition to describing the uptake of substrate, we must describe the rates at which growth-competent cells become quiescent and the reverse. The model is

$$\begin{aligned}\dot{S} &= -\frac{VS}{K+S}\frac{P}{Y}, \\ \dot{P} &= \frac{VS}{K+S}P - \alpha(S)P + \beta(S)Q, \\ \dot{Q} &= \alpha(S)P - \beta(S)Q,\end{aligned}\tag{7.3.2}$$

where the functions α and β describe the rates of inactivation and returning, respectively. If S is high, $\alpha(S) \approx 0$ and $\beta(S) > 0$, so cells move from Q to P . If S is low, $\alpha(S) > 0$ and $\beta(S) \approx 0$, and cells will move from P to Q . In Figure 7.7 we use

$$\alpha(S) = \frac{1}{1+S}, \quad \beta(S) = \frac{S}{1+S}.$$

The figure depicts a simulation of the model with growth and quiescence.

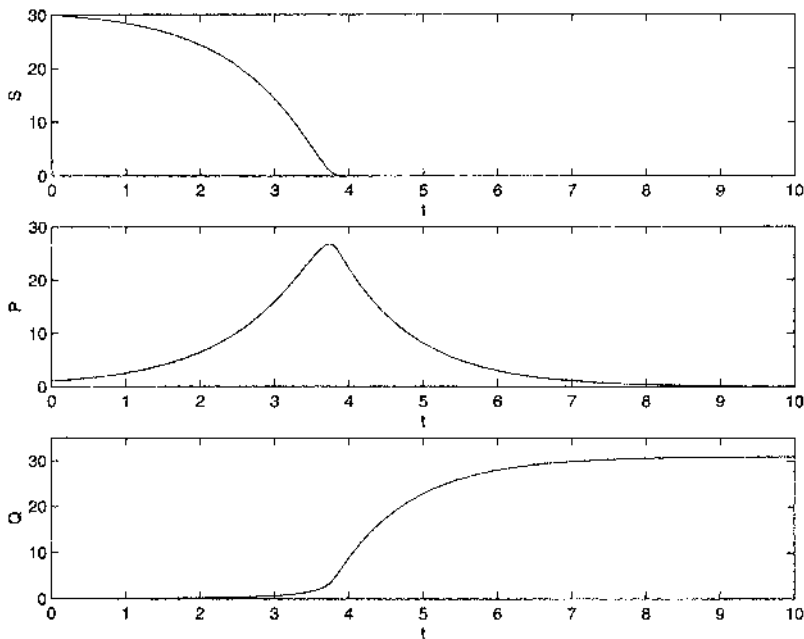


Figure 7.7. Simulation of SPQ model equation (7.3.2).

Interesting hysteresis phenomena can be observed in such systems for other choices of α and β . For example, suppose that $\alpha(S) > 0$ for $0 \leq S < S_1$ and $\alpha(S) \equiv 0$ for $S > S_1$, and $\beta(S) \equiv 0$ for $S < S_2$ but $\beta(S) > 0$ for $S > S_2$, where $S_1 < S_2$. If there are no quiescent cells, then the net growth rate of P is $VS/(K+S) - \alpha(S)$. We can perform an experiment of serial dilutions where we begin a batch culture with $S \gg S_2$, and after an interval of growth, we transfer some of the cells to another flask containing a lower concentration of substrate. For each dilution, the net growth rate decreases slowly like $VS/(K+S)$ until S_1 , at which time cells start moving from P to Q . So for each such dilution we will observe fewer P cells in a fixed time interval.

Now if we increase S in serial titrations, we see that in the inoculum most of the cells are in Q , perhaps none in P . This situation will not change appreciably until $S > S_2$ in our titration. Then cells will return from Q and we will see growth begin again. The net growth rate in this experiment describes a hysteresis loop like that simulated in Figure 7.8.

This phenomenon has been observed in cultures of *Salmonella typhimurium* that were titrated to reduce pH. Some bacteria grow well when the pH is near 6.5, but they rapidly become quiescent as the pH drops below 4.0.

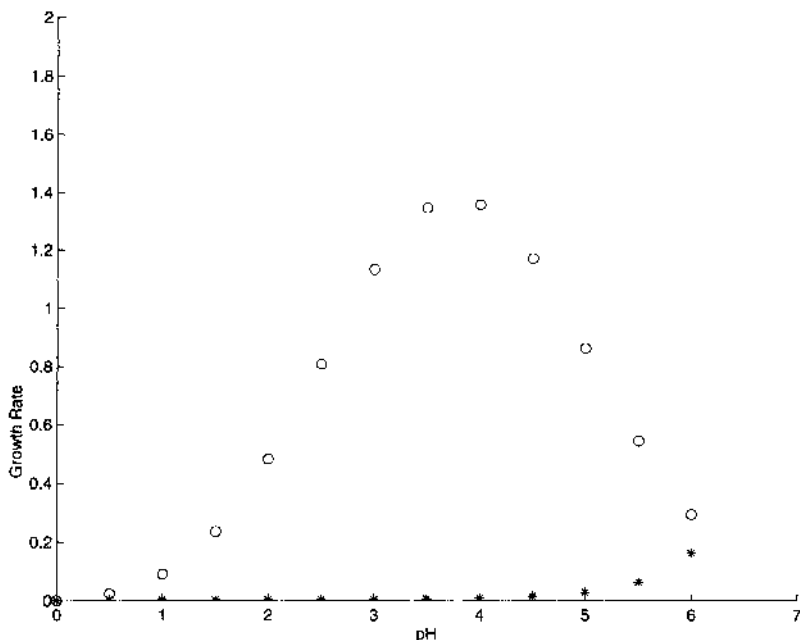


Figure 7.8. Serial dilutions in the SPQ model and hysteretic dependence on pH. o dilutions with decreasing pH, * with increasing pH.

7.3.3 Biofilms and Growth in a Continuous Flow

It is possible to grow cells in a fixed volume that is constantly renewed by the flow of media through it. Here such a device is referred to as a *chemostat*. In this, a fixed volume (say V) is occupied by a mixture of growth media and cells. The mixture is pumped out at the same rate as fresh media are pumped in, whereby a constant volume is maintained. The flow (dimensions are volume/time) divided by the volume V is called the washout rate w . The chamber is assumed to be well stirred, so that concentrations of solutes are the same throughout. We model this system by balancing changes in masses: the result is the following system of equations, which describes the concentration of substrate S and of cells P :

$$\begin{aligned}\dot{S} &= -\frac{VSP}{(K+S)Y} + w(S_0 - S), \\ \dot{P} &= \frac{VSP}{K+S} - wP,\end{aligned}\tag{7.3.3}$$

where S_0 is the concentration of substrate in fresh media.

In this case,

$$\frac{d(YS + P)}{dt} = w(YS_0 - YS - P),$$

so $YS + P \rightarrow YS_0$ as $t \rightarrow \infty$. However, this can occur in two different ways: It is possible that cells wash out of the chamber, so $S \rightarrow S_0$, or a dynamic equilibrium can be achieved. The outcome can be predicted using the observation that

$$\frac{\dot{P}}{P} = \frac{VS}{K+S} - w.$$

A dynamic equilibrium is reached when $(V - w)S = Kw$. This is possible only if $V > w$. Thus, if the uptake velocity is greater than the washout rate, the cells could establish themselves in the flow, but if the washout rate is larger than the uptake velocity, the cells will not survive.

Such systems are used in drug production and in a variety of important experimental procedures. This model is simulated with $w = 0.1$ and the other data taken to be 1.0 in Figure 7.9.

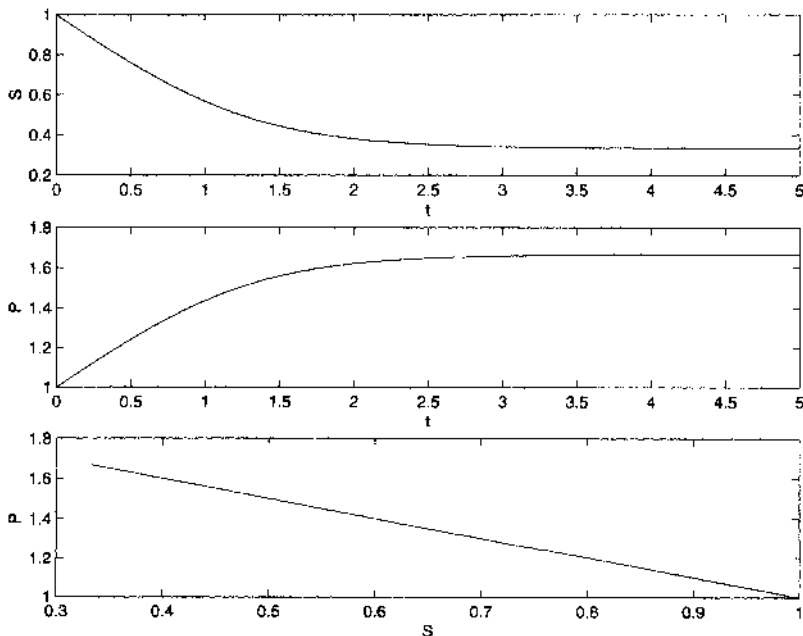


Figure 7.9. Simulation of a chemostat with $S_0 = 2$, $P(0) = S(0) = 1$, $w = 0.5$, $V = 2.0$. All other data are taken to be 1. This system approaches a dynamic equilibrium with $S = \frac{1}{3}$, $P = \frac{2}{3}$.

A slight variation of this model is useful in studying *biofilms*. In this case, we have cells that are freely floating in the growth chamber and cells that are attached to the chamber walls. The model is essentially a combination of the chemostat model and the previous one, but here we denote by Q the population of cells that make up the biofilm on the growth chamber's walls (here and below we suppose that $Y = 1$):

$$\begin{aligned}\dot{S} &= -\frac{VSP}{K+S} - \frac{V_{\text{wall}}SQ}{K_{\text{wall}}+S} + w(S_0 - S), \\ \dot{P} &= \frac{VSP}{K+S} - wP - \alpha(S)P + \beta(S)Q, \\ \dot{Q} &= \frac{V_{\text{wall}}SQ}{K_{\text{wall}}+S} + \alpha(S)P - \beta(S)Q,\end{aligned}\tag{7.3.4}$$

where α is the rate of attachment of free-floating cells to the wall and β is the rate at which cells leave the biofilm to float free. Note that there is no washout of cells in the biofilm. Also, cells on the wall and those in the chamber have different uptake velocities and saturation constants for the substrate.

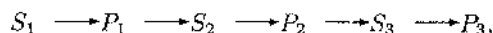
Biofilms line most ducts that carry fluids, from water pipes to digestive and transport systems in the body. They have many industrial uses involving artificially created films of specific compositions. Many interesting mathematical problems arise in studying such systems, including detailed descriptions of nutrient flow into and through biofilms and their interactions with flows in the free chamber. This model is not discussed further here, but it provides an opportunity for interesting student projects.

7.3.4 A Solitvory Food Chain

The chemostat model leads us naturally to modeling more complicated microbial communities in which several species of microorganisms interact. For example, a metabolic by-product of one species can serve as a limiting substrate for another.

We consider a simple food chain for bacteria that is similar to methanogenesis in a swamp. We suppose that as nutrients become unavailable for each species, cells move from active growth to quiescence, but return to growth competence when adequate nutrition is available.

Our model food chain consists of three species of bacteria and three substrates. It is described by the graph



where S 's denote substrate concentrations and P 's denote bacterial population densities. This model follows the same format as the quiescence

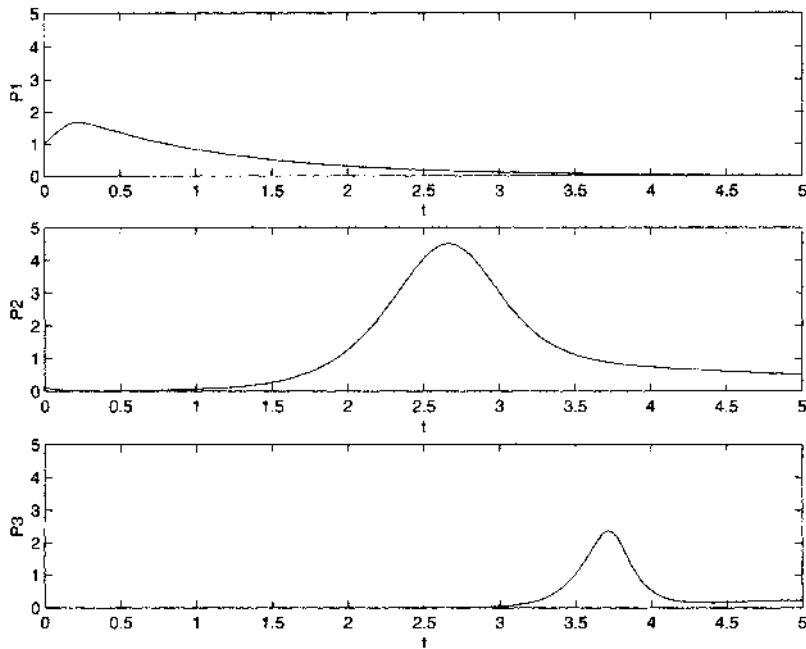


Figure 7.10. Simulation of three species solutivory chain model: $P_1 \rightarrow P_2 \rightarrow P_3$.

model (all yields Y_1 , etc., are taken to be 1 here):

$$\begin{aligned}
 \dot{S}_1 &= -\frac{V_1 S_1 P_1}{K_1 + S_1}, \\
 \dot{P}_1 &= \frac{V_1 S_1 P_1}{K_1 + S_1} - \alpha_1(S_1)P_1 + \beta_1(S_1)Q_1, \\
 \dot{Q}_1 &= \alpha_1(S_1)P_1 - \beta_1(S_1)Q_1, \\
 \dot{S}_2 &= -\frac{V_2 S_2 P_2}{K_2 + S_2} + m_{1,2}P_1, \\
 \dot{P}_2 &= \frac{V_2 S_2 P_2}{K_2 + S_2} - \alpha_2(S_2)P_2 + \beta_2(S_2)Q_2, \\
 \dot{Q}_2 &= \alpha_2(S_2)P_2 - \beta_2(S_2)Q_2, \\
 \dot{S}_3 &= -\frac{V_3 S_3 P_3}{K_3 + S_3} + m_{2,3}P_2, \\
 \dot{P}_3 &= \frac{V_3 S_3 P_3}{K_3 + S_3} - \alpha_3(S_3)P_3 + \beta_3(S_3)Q_3, \\
 \dot{Q}_3 &= \alpha_3(S_3)P_3 - \beta_3(S_3)Q_3.
 \end{aligned}$$

The difference here is that P_1 produces substrate S_2 at rate $m_{1,2}$, etc.: The rates of substrate production are indicated by $m_{i,j}$, which denotes the rate at which species i produces substrate j . The functions α and β are as before.

A pulse of nutrient S_1 is introduced, and the chain's dynamics are shown in Figure 7.10. (The data for this are shown in Exercise 7.9.) The chain's output is reflected in the profiles of P_1 , P_2 , and P_3 .

The models in this section are expressed in terms of differential equations, and we use methods of calculus to gain information about their solutions. The next section takes an entirely different approach to population modeling, based on discrete time observations.

7.4 Nonlinear Reproduction Curves

Careful experiments with bacteria and with insects, and careful observations of human populations, suggest that Malthus's model equation (7.1.1) is often valid during a certain (exponential) phase of a population's growth. But more complicated relationships eventually emerge. Crowding of a population can suppress reproduction in a variety of ways: Increased stress or reduced nutrition can reduce fertility, and depletion of essential nutrients can cause more deaths. Therefore, as the population grows, its growth rate should eventually decrease. This happens even in the data in Table 7.1, where the predicted population sizes and the sizes actually observed diverge at $n = 7, 8$.

In this section we study discrete-time models of nonlinear population phenomena. Rather than explicitly accounting for changes in some limiting substrate as we did earlier, we proceed in the spirit of the logistic equation. We begin with a reproduction rule more general than Malthus's having the form

$$B_{n+1} = r(B_n)B_n,$$

where the *intrinsic growth rate* $r(B)$ now can change with changes in the population size B . This model should describe the population over a wider range of sizes, but the problem of estimating $r(B)$ requires substantially more data, and the least squares estimation method does not apply directly to finding this function of B . In this section we study three nonlinear models that illustrate analytical, geometrical, and computer simulation methods.

7.4.1 Compensatory Growth: Verhulst's Model

Consider a strain of bacteria, such as a strain of histidine auxotrophs, that is, cells that cannot produce their own histidine, which is an amino acid needed to construct proteins and enable cell reproduction. When these

bacteria are grown in media containing sufficient histidine, they double in size and divide every 40 minutes. Let us begin with a known biomass (dry weight) of cells, say B_0 , at time zero and monitor the population size at successive division times. Let B_1 = number of cells after 40 minutes, B_2 = biomass of cells after 80 minutes, etc. The simple formula

$$B_{n+1} = 2B_n$$

describes the population measurements when there is adequate histidine and other necessary nutrients and when the sampling and doubling times are both 40 minutes.

When the biomass of cells becomes large, competition for the limited amount of histidine ensues, and the bacteria get an inadequate supply to sustain division every 40 minutes. Thus, observations every 40 minutes no longer are coordinated with the division cycle, as discussed earlier.

When the population is large, it less than reproduces itself each interval. We describe this by setting $r(B) = 2/(1 + B/K)$, so

$$B_{n+1} = \frac{2}{1 + B_n/K} B_n. \quad (7.4.1)$$

Studies of histidine uptake by growing cells suggest this form, where K is comparable to the saturation constant in the Jacob-Monod model of equation (7.3.1). When $B = K$, reproduction is only one-half its maximum. In this model, the larger the cell mass becomes, the smaller will be its increase every 40 minutes. Once B_0 is known (or estimated), the rest of the population sequence can be determined by straightforward iteration of this recursion.

A model similar to this one was presented by P.F. Verhulst in 1845 for studies of certain human populations. For this reason, equation (7.4.1) is called *Verhulst's model*. It has also appeared in the fishery literature where it is called the *Beverton-Holt model*.

This model is not as easy to solve as Malthus's, but we can find its solution in a number of interesting ways.

Analytic Iterations

It is actually possible to derive a closed form solution to Verhulst's model by transforming it into an equation where successive back substitutions can be carried out. In this case, we define new variables $R_0, R_1, \dots$ as the reciprocals of the B 's:

$$R_n = \frac{1}{B_n}.$$

Then the sequence $\{R_n\}$ satisfies the linear relationship

$$R_{n+1} = \frac{R_n}{2} + \frac{1}{2K}.$$

Successive back substitutions show that

$$\begin{aligned} R_{n+1} &= \frac{R_n}{2} + \frac{1}{2K} \\ &\approx \frac{R_{n-1}}{2^2} + \frac{1}{4K} + \frac{1}{2K} \\ &= \frac{1}{2^{n+1}} R_0 + \frac{1}{2K} \{1 + \cdots + \frac{1}{2^n}\}. \end{aligned}$$

The last term, here enclosed in braces $\{\}$, is a geometric progression, and it can be evaluated directly using the formula

$$1 + \cdots + \frac{1}{2^n} = \frac{1 - (1/2)^{n+1}}{1 - (1/2)}.$$

Therefore, $R_{n+1} \rightarrow 1/K$ as $n \rightarrow \infty$, or equivalently, $B_n \rightarrow K$, so K is apparently the system's carrying capacity for this strain of bacteria.

Since the equation for the reciprocal masses R_n is linear, we can also use the least-squares method to estimate the carrying capacity K from experimental data.

Being able to find an explicit solution for B_n is unusual. Another approach to this problem uses a geometrical method that is sometimes useful when an explicit solution cannot be found.

Geometric iterations

We can plot the graph of the reproduction function

$$F(B) = r(B)B = \frac{2B}{1 + B/K},$$

as shown in Figure 7.11. The reproduction curve crosses the one-to-one reference line at $B = K$. This graph tells us what the next size will be from the present size. For example, if $B_0 = 10$, we see from the graph that $B_1 \approx 16.7$. By reflecting this back to the horizontal axis using the one-to-one line, we can find $B_2 \approx 25$, etc. This process is shown in Figure 7.12, and it can be repeated as indicated there.

This shows that B_n increases up toward K in successive samples. The graphical method described here is called *cobwebbing*, or Picard's method; it provides an easy way to sketch population dynamics. But this method is of limited use in practice: First, it is difficult to estimate a reproduction function from data to a sufficient degree of accuracy to be confident in its predictions. Second, we will see later that there are reproduction curves similar to this one whose cobwebs are too complicated to draw. Still, it is useful in simple cases.

Verhulst's Model with Predation

In ecological systems, say where there are prey and predators, or in the human immune system, where there are macrophages that respond to bac-

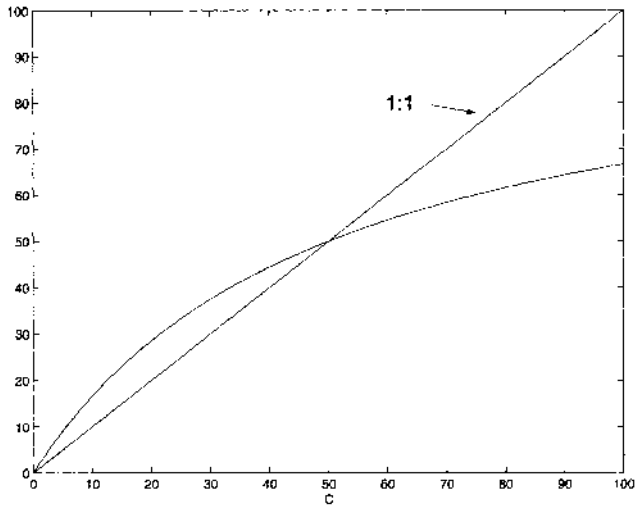


Figure 7.11. Verhulst's reproduction curve ($F(B) = 2B/(1 + B/50)$).

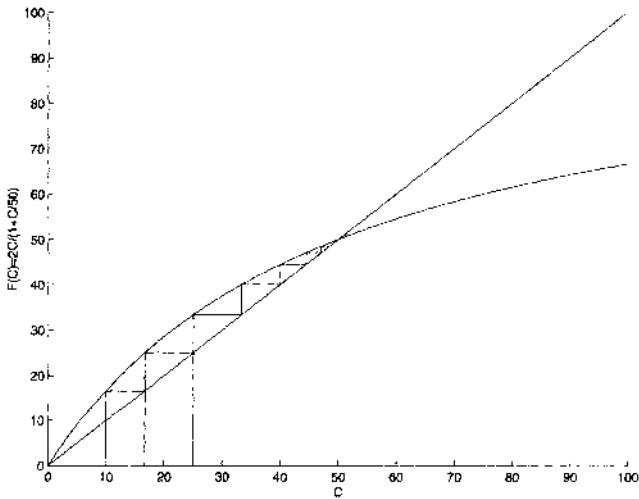


Figure 7.12. Geometric Iteration of Verhulst's model.

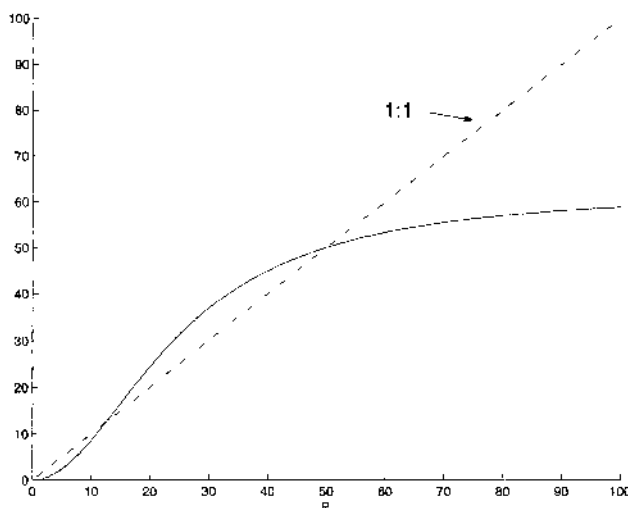


Figure 7.13. Verhulst's model with predation.

terial infections, there is a threshold size below which the prey or bacteria, respectively, are, for practical purposes, completely eliminated. This can be accounted by modifying Verhulst's model: Let

$$F(B) = \frac{rB^2}{1 + (B/K)^2}.$$

The resulting model is

$$B_{n+1} = \frac{rB_n^2}{1 + (B_n/K)^2}.$$

Now, however, we cannot solve this equation in closed form using the back-substitution method. In this case, we rely on the cobwebbing method to sketch solutions for the various choices of r and K , but we do not attempt to estimate these parameters from data. The reproduction curve is drawn in Figure 7.13.

Static population sizes for this model are those for which

$$F(B) = B.$$

These occur when

$$B = 0, \text{ or } B = K \left(\frac{rK}{2} \pm \sqrt{\left(\frac{rK}{2} \right)^2 - 1} \right).$$

Let us denote the middle of these three roots by T , and the larger one by K^* . In Figure 7.13, $T = 12$ and $K^* = 50$. A population starting below T will less than reproduce itself, so it dies out. Above T , the population satisfies

the predator's demands and will eventually establish itself at the stable carrying capacity, $P = K^*$. The reader should verify this with cobwebbing.

The interval $0 < B < T$ is a *pit of extinction*, since once a population is forced into it, for example by a significant change in its environment, there is no return.

7.4.2 Chaos

Some models have no randomness in them, yet their solutions are highly irregular, and for these the geometric cobwebbing method is not useful. Such irregularities were known in the nineteenth century, but some new ideas emerged in the 1970s that reignited interest in them and in particular, in the study of such systems that occur in nature. The term "chaos" was coined to describe such a situation. One of the simplest descriptions of chaos was given in the context of ecological systems and in terms of reproduction functions. Insight into the behavior of reproduction processes like those described here can be gained using computer simulations.

In this section we first study iterations of chaotic reproduction curves. Then, we consider a particular reproduction curve that was derived and used for the management of fisheries.

Iterating Reproduction Function Graphs

An interesting way to study a reproduction function is to iterate it using a computer. For example, suppose that $B_{n+1} = F(B_n)$. Then the population two generations ahead is given by

$$B_{n+2} = F(B_{n+1}) = F(F(B_n)) = F^{[2]}(B_n),$$

and the population three generations ahead is given by

$$B_{n+3} = F(B_{n+2}) = F(F(B_{n+1})) = F(F(F(B_n))) = F^{[3]}(B_n).$$

The population in generation 10 is described by the iterated reproduction function

$$F^{[10]}(B) = F(F(\cdots F(B))),$$

where there are 10 F 's on the right-hand side.

Such iterations are difficult for humans to carry out, but computers love them. Figure 7.14 shows the first, second, third, and seventh iterates of a simple quadratic polynomial

$$F(B) = 4B(1 - B).$$

It may seem surprising that such few iterations of a simple parabola can create such a complicated graph. However, it has long been known to mathematicians that *any* continuous function (even really wild ones) can be approximated closely by a polynomial of sufficiently high degree. We next

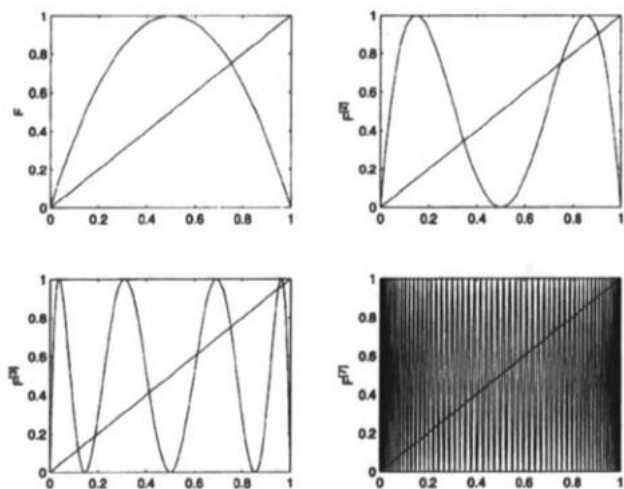


Figure 7.14. Iteration of a quadratic polynomial. Upper Left F ; Upper Right $F^{[2]}$; Lower Left $F^{[3]}$; Lower Right $F^{[7]}$.

consider reproduction curves that are more useful in population studies than this quadratic function.

Cascading Bifurcations: Ricker's Model

An interesting reproduction function, called Ricker's model, was used to study fisheries in the 1950s. It happens that reproduction can actually decrease as population sizes become large and crowding occurs. For example, adult guppies will feed on their young, so the offspring of an extremely large adult population will face little chance of survival. Although populations of intermediate size may more than reproduce themselves, large ones may not.

Ricker's model illustrates this phenomenon. Consider the reproduction function

$$F(B) = r \exp(-B/K)B,$$

where r gives the population's geometric growth rate when the population is small (viz., $\exp(-B/K) \approx 1$ when $B/K \approx 0$). When $B = K$, the reproduction rate is $r \exp(-B/K) \approx r/2$, so K is like a saturation constant for this population (see Exercise 7.4). The iterates of this function are shown in Figure 7.15.

Ricker's model is

$$B_{n+1} = r \exp(-B_n/K)B_n. \quad (7.4.2)$$

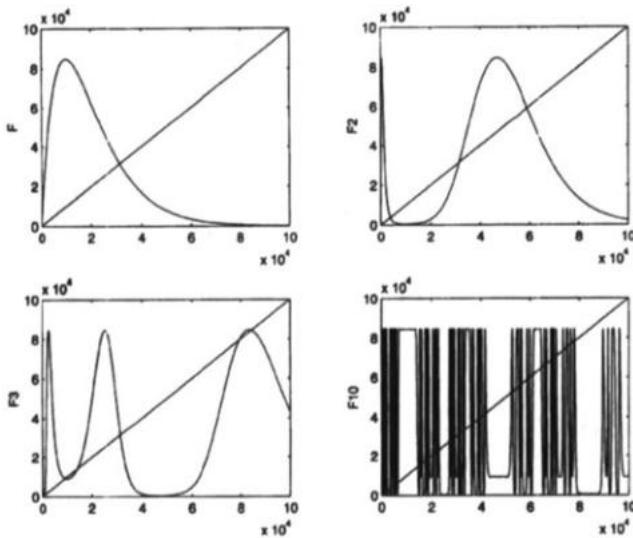


Figure 7.15. Iterates of Ricker's reproduction function: Upper left, F ; Upper right, $F^{[2]}$; Lower left $F^{[3]}$; Lower right $F^{[10]}$.

His computer simulations showed that for some values of r and K the iterates of this function are highly irregular.

If $r < 1$, then the population will less than replace itself, and so will die out. However, if $r > 1$, a small population will more than replace itself, and the equilibrium $B^* = K \log r$ (where $F(B^*) = B^*$) is a likely eventual outcome for the population. This might or might not be the case depending on the value of r . If $r - 1$ is small, then the iterates will approach B^* . However, for larger values of r , the solutions can do other things. Table 7.3 illustrates four cases.

These sample paths are also drawn in Figure 7.16. When $r = 0.4$, the population dies out; when $r = 2.0$, it approaches $B_n \rightarrow B^* = 3466$. When $r = 10.0$, the population numbers settle to a simple oscillation between two numbers 4673 and 18353, which is called an orbit of period two. However, when $r = 20.0$, the population numbers seem to be chaotic.

A histogram method provides a way for capturing this strange behavior. For example, for each value of r in Table 7.3, we can describe where the iterates lie by breaking the interval $0 \leq P \leq 40,000$ into 400 equal cells and putting a marker in a cell each time an iterate hits that cell. The results in this case are shown in Figure 7.17 for the four cases in Figure 7.16.

For each r -value an initial point (5000 there) is iterated many times as in Table 7.3. Figure 7.17 plots the results where each time a cell is hit, a marker is placed in it, and at the end of the simulation the numbers of accumulated markers are plotted. The histogram describes how often the iterates visit

Table 7.3. Four sample paths generated by Ricker's model for four different values of r . Population numbers are $\times 10^4$.

n	$B_n(r = 0.4)$	$B_n(r = 2.0)$	$B_n(r = 10.0)$	$B_n(r = 20.0)$
0	0.5000	0.5000	0.5000	0.5000
1	0.0736	0.3679	1.8394	3.6788
2	0.0254	0.3525	0.4645	0.0469
3	0.0097	0.3484	1.8345	0.8544
4	0.0038	0.3471	0.4678	3.0944
5	0.0015	0.3467	1.8354	0.1270
6	0.0006	0.3466	0.4672	1.9704
7	0.0002	0.3466	1.8353	0.7658
8	0.0001	0.3466	0.4673	3.3111
9	0.0000	0.3466	1.8353	0.0881
10	0.0000	0.3466	0.4673	1.4774
11	0.0000	0.3466	1.8353	1.5390
12	0.0000	0.3466	0.4673	1.4174
13	0.0000	0.3466	1.8353	1.6649
14	0.0000	0.3466	0.4673	1.1921
15	0.0000	0.3466	1.8353	2.1974
16	0.0000	0.3466	0.4673	0.5424
17	0.0000	0.3466	1.8353	3.6663
18	0.0000	0.3466	0.4673	0.0479
19	0.0000	0.3466	1.8353	0.8712

each cell. For example, in Table 7.3 in the case $r = 10.0$, the cells containing 9346 and 36706 would eventually receive all of the hits, so after many iterations the total collection of iterates would be equally divided between these two, except for the first 10 transient iterates. Therefore, we eliminate the first few iterates to suppress transients and plot the rest in a histogram as in Figure 7.17.

A full-scale simulation of this model involves exactly the same procedure, but for many values of r . The results could be plotted as a surface where each cross-section for a fixed r value would be a histogram like above. The reader should try this, as shown in Exercise 7.4. (However, it is interesting to plot only the projection of this histogram onto the (r, B) -plane.) This is described in Figure 7.18. There, for each value of r between 0 and 20, the same experiment as described above is performed. Fifty initial iterates at each r value are ignored, and thereafter whenever a cell is hit, it is darkened, so dark cells indicate at least one hit. The results in Figure 7.18 show how the collection of iterates of Ricker's model depend on r . This striking diagram shows that as r increases, the behavior of iterates becomes successively more complicated. For example, as r increases through a value near 7.5 (75 on the scale in the figure), a stable equilibrium splits into an orbit of period two. Near 12.5 (125 there), that splits into an orbit of

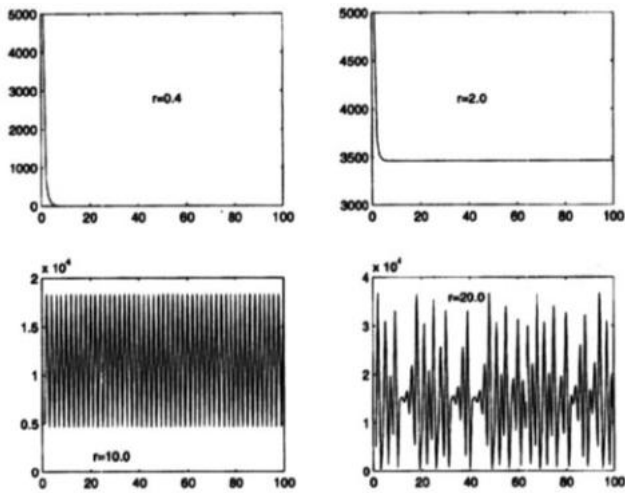


Figure 7.16. Graphs of sample paths described in Table 7.3. Upper left $r = 0.4$; Upper right $r = 2.0$; Lower left $r = 10.0$; Lower right $r = 20.0$.

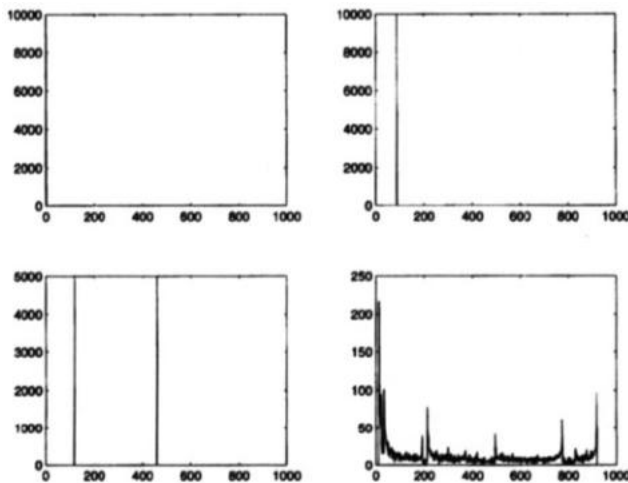


Figure 7.17. Histograms of iterates described in Table 7.3. Upper left $r = 0.4$ (all hits are at $x = 0$); Upper right $r = 2.0$ (all hits are at $x = 3466$); Lower left $r = 10.0$; Lower right $r = 20.0$.

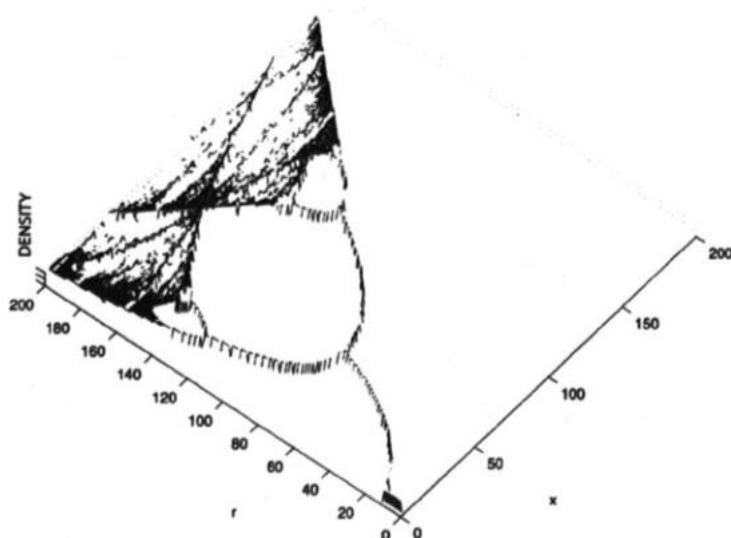


Figure 7.18. Bifurcation diagram for Ricker's model. The horizontal scale is $20 \times r$.

period four, and so on. These splittings are referred to as *bifurcations*, and the diagram is called a bifurcation diagram. It shows to some extent how chaotic behavior emerges as r increases.

The theory of chaos and its applications are still being developed. There are interesting connections between chaotic population dynamics and random processes; in some senses, random processes, like Brownian motion, are more random than chaotic processes.

We have shown in this section that nonlinear aspects of populations near their limits of growth can lead to some surprising outcomes. Some nonlinear models can be solved using simple back-substitution methods, some can be sketched using the cobwebbing method, but some can be neither solved nor sketched. The three examples here illustrate all of these possibilities.

7.5 Controlling Populations

Reproduction of organisms is sometimes synchronized perhaps because a population gains a selective advantage or perhaps in response to environmental signals. For example, some plants (e.g., bamboo), insects (e.g., cicadas) and fishes (e.g., Pacific salmon) appear only rarely, and in so doing overwhelm predators and leave a large set of viable offspring. This synchronization is due to the ecological system in which the organisms live, and it was probably shaped by natural selection. On the other hand, it is possible

for reproduction to be synchronized by external signals, like photo-period or weather. For example, certain montane voles's reproduction is triggered by the appearance of specific chemicals in growing grass.

We will consider examples of these modes of synchronization. First, we study a laboratory method used to synchronize mitosis in bacterial cultures, and then we study an ecological system in which competition between cohorts can lead to coexistence of all cohorts or to the exclusion of some. This section ends with a discussion of some strategies for maximizing yields from populations.

7.5.1 *Cell Synchronization*

At the beginning of this chapter we saw that the mitotic cycle of a bacterium contains two gap intervals and an interval for synthesis of DNA. Observations suggest that cells do not initiate synthesis until they have sufficient resources to complete it. This suggests a control law for synchronizing cells based on alternating starvation and nutrition:

1. Raise the cells under optimal conditions.
2. Then switch them to starvation media. Those cells that are in GAP1 or GAP2 will cease to progress through the mitotic cycle, but those in synthesis will carry through to completion and accumulate at the phase boundary between SYN and GAP2.
3. Next, switch the cells back to optimal conditions. The newly formed cohort of cells that accumulated at the end of synthesis continue through the cycle as do others.
4. Switch the culture back to starvation when the new cohort enters synthesis again. Then other cells that are in synthesis will be added to this cohort.
5. Continue this switching between starvation and nutrition until all cells belong to one cohort.

There are a few interesting problems to consider:

1. How should the switching be scheduled to synchronize the population in minimum time?
2. How long will synchrony last?

The first of these questions is studied further in Exercise 7.10, but the persistence of synchrony depends on the effects of random noise on the cell cycle. This can be described using methods from probability theory, which is not done here. However, in practice synchrony for 6-12 generations is observed.

7.5.2 Cohort Dynamics near the Limits of Growth

Insects such as cicadas that have a long life cycle remain as immature young for many years, but are adults capable of reproducing only in their last year. Adult cicadas reproduce in trees by laying fertilized eggs on branches. Their offspring drop to the ground and hatch as nymphs, who then enter the ground and attach to tree rootlets for nourishment. They may change form several times while underground. At the right time, they emerge from the ground, change to a flying form, and fly to the branches to reproduce. Then the reproduction cycle repeats. One interesting observation made of cicada populations is that short-lived ones (7-year life spans or less) seem to appear in comparable numbers every year, while longer-lived ones (say 13 years) appear only every 13 years. The cohorts are distinct from year to year, but somehow they can become synchronized. The first situation is referred to as *balanced emergences*, meaning balance between the cohorts, and the second is referred to as *periodical emergences*, for which the off-year cohorts vanish.

Some insight into this state of affairs can be gained by considering a theoretical organism that has a 2-year life cycle where after 1 year, newborns become immature youths and after 2 years mature adults. There are species of insects, plants, and fish, Pacific salmon among them, that have such life cycles.

Let Y_n denote the number of immature young in year n , and let A_n be the number of mature adults. The number of adults in year $n + 1$ will be some proportion of the young, namely those that survive to reproductive age:

$$A_{n+1} = \sigma Y_n, \quad (7.5.1)$$

where $\sigma \leq 1$, called the survival probability, is the proportion of youths who survive to maturity. The number of young next year will depend on the number of adults this year. We write

$$Y_{n+1} = f(A_n), \quad (7.5.2)$$

where f describes the reproduction relation between breeding adults and next year's young. If predators must be satisfied during the reproduction periods, then f will have the form already studied in Section 7.4.1 (Figure 7.13). We suppose that σ and the graph of f do not change over the generations. These relations can be plotted, as shown in Figure 7.19

A modification of the cobwebbing method helps in determining the dynamics of such a structured population. This modification is based on the observation that in cobwebbing the key ingredients are the horizontal axis, the one-to-one reference line, and the reproduction curve. The vertical axis plays no essential role in the procedure because we always reflect in the one-to-one line. Therefore, we can combine the two figures in Figure 7.19,

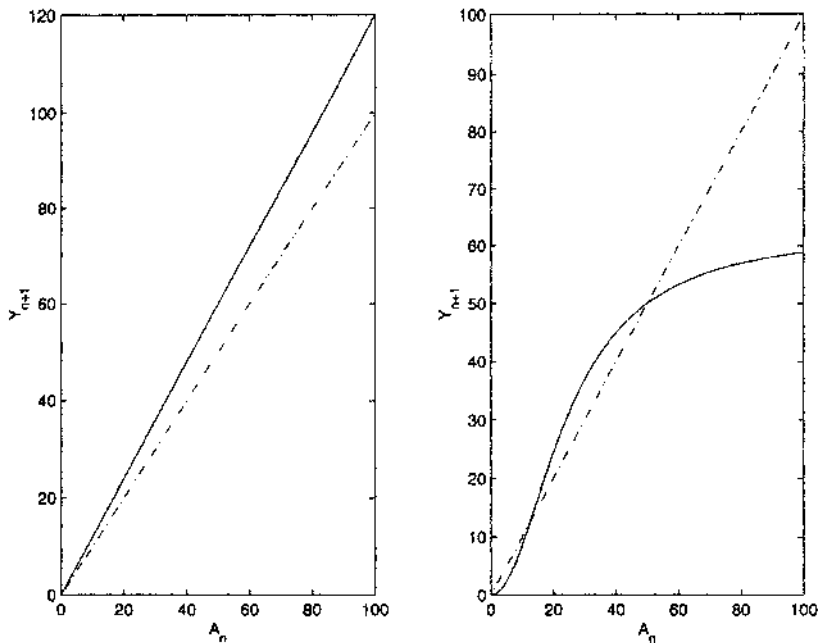


Figure 7.19. Reproduction curves for two age classes.

as shown in Figure 7.20. The iteration begins at an initial point (A_1, Y_1) and determines the next point (A_2, Y_2) , etc.

This iteration method illustrates an interesting feature of cohort competition: There are three different kinds of behavior, depending on the initial distribution of adults and juveniles. These are described in Figure 7.21 by the regions I, II/II', and III. Populations starting near $(0, 0)$ die out. Those starting near (A^*, Y^*) , where $\sigma Y^* = f(A^*)$, approach (A^*, Y^*) . Those starting with one cohort small and the other large eventually lose the smaller cohort, and so exhibit periodic emergences, say with $(A_{2n}, Y_{2n}) \rightarrow (A^*, 0)$ if (A_0, Y_0) is near $(A^*, 0)$.

In fact, it is easy to convince yourself that if (A_1, Y_1) lies in region I, then $(A_n, Y_n) \rightarrow (0, 0)$ in successive years, and the population becomes extinct. Populations beginning in region III approach the distribution (A^*, Y^*) and so equilibrate to a static state or balanced emergence. Finally, populations that begin in either part of region II eventually alternate between the points $(0, Y^*)$ and $(A^*, 0)$. Such alternating behavior indicates that one of the cohorts becomes extinct while the other persists, and the population exhibits a synchronized emergence pattern. In particular, adult breeding stock will appear only every other year. Thus, we see that three quite different results

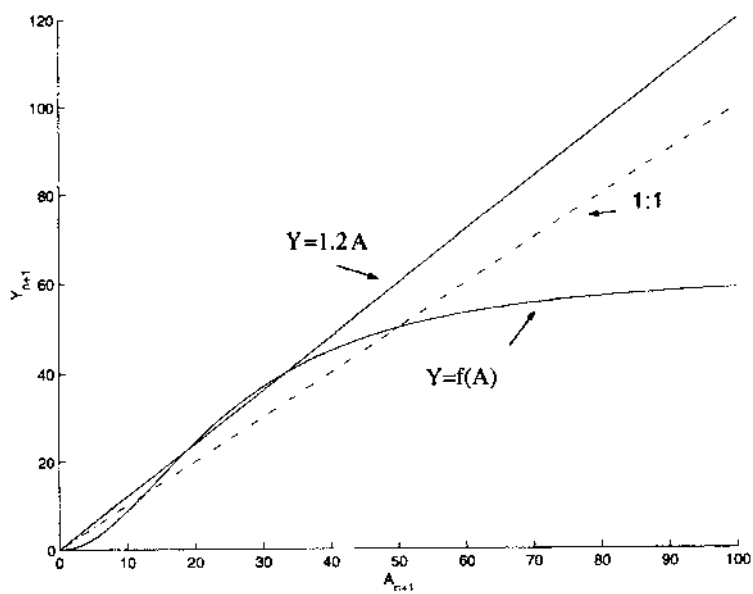


Figure 7.20. Combined reproduction curves and AY cobwebbing.

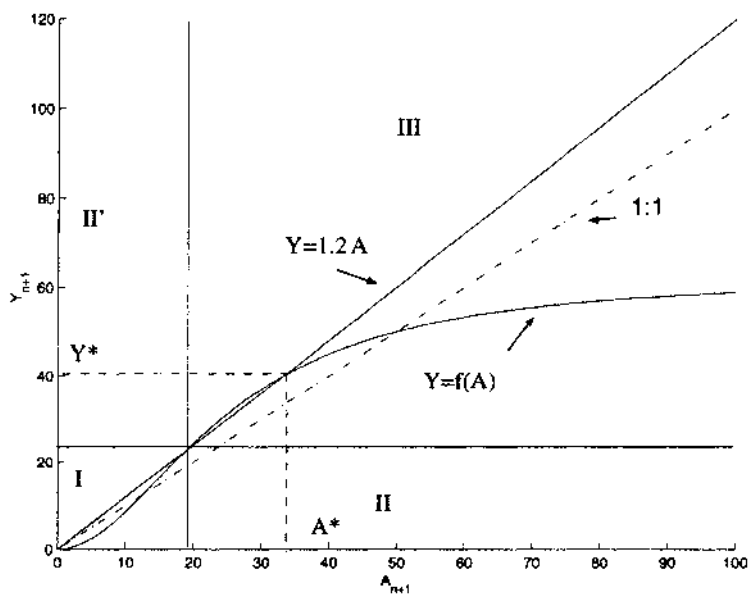


Figure 7.21. AY stability regions.

occur depending initially on only the population distribution among ages and their sizes.

Region I is referred to as the extinction region, region III as the balanced emergence region, and region II as the synchronized emergence region. Exercise 7.11 studies the cases of 7-, 13-, and 17-year cicadas.

7.5.3 *Harvesting: Economics of Populations*

Food is produced by harvesting crops and animals, and we can produce new biotechnology products by harvesting microorganisms and byproducts of their metabolisms. In this section we see how efforts and yields from harvesting can be described using the methodologies of this chapter.

To fix ideas, we might consider the population to be one of fish, and the number B_n describes the fish population biomass at the end of the n th harvesting season. Whatever the underlying biological system, we will consider one brief harvest at the end of each interval. At the beginning of the $(n+1)$ st interval the biomass is B_n , and at the end of the interval the biomass is $f(B_n)$ less the mass of the harvest, where f describes the population dynamics of the fish population. For example, the population we are modeling might be the mature adult population, and $f(B)$ may denote the number of new recruits to this population based on the adult population at the end of the previous interval. Therefore, we write

$$B_{n+1} = f(B_n) - h_n,$$

where h_n is the harvest taken at the end of the $(n+1)$ th interval, and B_{n+1} denotes the population at the end of this interval after harvesting.

Our first question is, What is the largest sustained harvest that this population will produce? Let us denote the yield by Y . If the population is at equilibrium size B and steadily producing a yield Y , then

$$Y = f(B) - B.$$

The maximum of this function occurs where

$$\frac{df}{dB} - 1 = 0.$$

Therefore, a population B^* will produce a maximum sustained yield if $(df/dB)(B^*) = 1$.

To formulate a harvesting policy, we might adjust the population size to reach B^* , and then it will maintain itself as we collect its yield $Y^* = f(B^*) - B^*$. Unfortunately, many populations that we deal with are neither observable nor directly controllable, such as wild fish populations. In those cases, we must take an indirect approach.

Two observables in fisheries are the effort expended in harvesting and the yield that it produces. For example, effort might be measured in days a fishing boat spends on a fishing site, and yield might be measured in tons

of fish harvested. We make the following assumption that relates effort and yield:

Effort-Yield Relationship: The effort required to harvest one unit from a population of size B is $1/(qB)$. The constant q is called the *coefficient of catchability*.

The effort needed to sustain yield Y in the equilibrium model can be determined by adding the effort needed to reduce the population from $f(B)$ to B , one unit at a time (we suppose that $f(B) > B$ here). The effort-yield relationship leads to the formula

$$E = \frac{1}{qf(B)} + \frac{1}{q(f(B)-1)} + \cdots + \frac{1}{qB} = \frac{1}{q} \sum_{k=B}^{k=f(B)} \frac{1}{k} \approx \frac{1}{q} \int_B^{f(B)} \frac{dx}{x}.$$

This sum can be approximated by the integral shown. As before, we write $f(B) = r(B)B$. The result is

$$E \approx \frac{1}{q} \log \frac{r(B)B}{B} = \frac{1}{q} \log r(B).$$

The integral is the area under the curve $(1/q) \log r(B)$, and this is the sum plus an error that is no larger than $2/(qB)$, which is small if qB is large. (See Exercise 7.12.)

By introducing the effort E , we have three variables, but only two equations:

$$\begin{aligned} Y &= r(B)B - B, \\ qE &= \log r(B). \end{aligned}$$

If it is possible to eliminate B from these equations, then a direct relationship between effort and yield will be found. The second equation gives

$$r(B) = \exp(qE).$$

Putting this in the first equation gives

$$B = \frac{Y}{\exp(qE) - 1},$$

and substituting this in the second equation gives

$$qE = \log r(Y/(\exp(qE) - 1)),$$

or

$$\exp(qE) = r \frac{Y}{\exp(qE) - 1}. \quad (7.5.3)$$

To make use of this result, we must know the population's growth rate r . We consider three cases:

1. *Malthus's model.* In this case $r(B) = r$ is a constant, so $Y = (r - 1)B$ and $E = (1/q) \log r$. There is no maximum sustained yield in this case. If $r > 1$, then the yield will be proportional to the standing stock B , and it will require an effort $(1/q) \log r$ to produce it. If $r < 1$, then there is no sustained yield possible, reflecting the fact that the population is dying out.
2. *Beverton-Holt model.* As mentioned earlier, Verhulst's model was discovered separately by fishery managers, and it is referred to there as the Beverton-Holt model. In this case

$$r(B) = r/(1 + B/K),$$

so

$$(rK \exp(-qE) - K) = Y/(\exp(qE) - 1),$$

or

$$Y = K(r \exp(-qE) - 1)(\exp(qE) - 1).$$

Any effort that exceeds $(1/q) \log r$ would eliminate the population, since then $Y < 0$. Moreover, the maximum sustained yield is obtained when $E_{\text{MSY}} = (1/2q) \log r$. These yields are plotted in Figure 7.22.

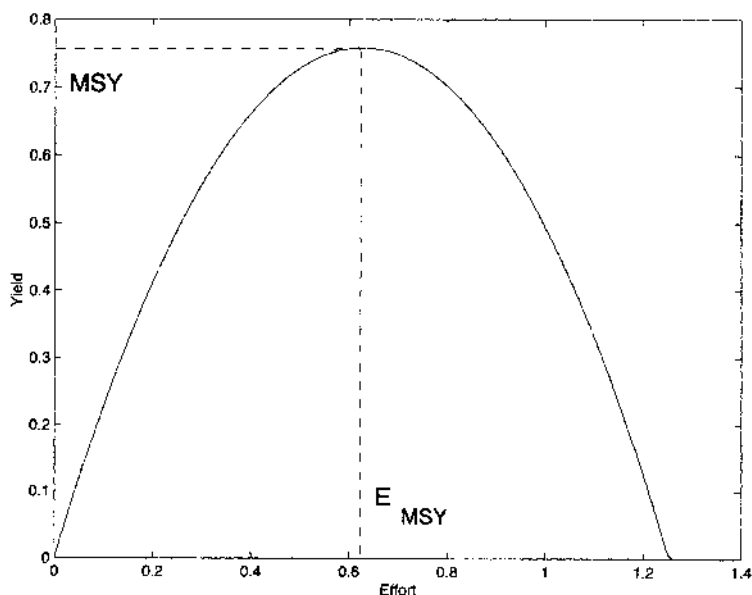


Figure 7.22. Maximum sustained yield (MSY at effort E_{MSY}).

This calculation was carried out for fisheries by Beverton and Holt in 1957, and it led to the following policy for controlling a fishery: Effort is allowed to increase until a decrease in yield is observed. This means that E_{MSY} has been passed. Then effort is reduced until another decrease in yield is observed. This procedure is followed iteratively until the maximum sustained yield is reached.

3. *Ricker's model.* In this case $r(B) = r \exp(-B/K)$, and

$$Y = (\log r - qE)(\exp(qE) - 1)K.$$

4. *Predator satiation model.* Consider $r(B) = rB \exp(-B/K)$. Then the effort-yield relation is given parametrically by the formulas

$$\begin{aligned} Y &= B(rB \exp(-B/K) - 1), \\ E &= (1/q) \log(rB \exp(-B/K)). \end{aligned}$$

As we have seen, the first two examples are straightforward. Ricker's model is included because we see that there is an optimal choice for effort to maintain a sustained maximum yield, but we know this model to have chaotic solutions when r is large. Because of this, it is certain that the sustained maximum will be unstable, and almost surely not realizable.

The predator model illustrates a major problem with the Beverton-Holt control law. With $B = Y/(\exp(qE) - 1)$, we get

$$\exp(qE) = rB \exp(-B/K).$$

The maximum of the right-hand side is rK/e , so for each E up to $E_{\text{MSY}} = (1/q) \log(r/e)$, there are two roots of the equation for z , say $z_1(E) < z_2(E)$. They come together at $z = K$, so $z_1(E^*) = z_2(E^*) = K$. For larger values of E , there are no sustained yields possible. The important point here is that if we allow E to increase until we see yields decline, the population can be headed for irreversible collapse before decreases in Y are detected. Therefore, we run a significant risk of destroying the resource if we follow the Beverton-Holt policy while the correct population model involves predator satiation.

7.6 Annotated References

Many of the facts about bacteria used in this chapter can be found in the excellent reference books:

Escherichia Coli and Salmonella (Neidhardt, et al., eds.): Amer. Soc. Microbiol., Washington, 1994.

Gerhardt, P., et al.: *Manual of Methods for General Bacteriology*, Am. Soc. Microbiol., Washington, 1981.

There is a great deal of data available regarding populations of humans and their age structures. In particular, the work of Nathan Keyfitz presents mathematical methods of demography and useful data for studying many countries in the world.

Keyfitz, N., and Flieger, W.: *Populations: Fact and Methods of Demography*, W.H. Freeman, San Francisco, 1971.

Many of the calculations and methods presented in this chapter were derived in the books:

Hoppensteadt, F.C.: *Mathematical Methods of Population Biology*, Cambridge University Press, 1982.

Hoppensteadt, F.C.: *Mathematical Theories of Populations: Demographics, Genetics, and Epidemics*, SIAM Publics, Philadelphia, 1975.

Clark, C.W.: *Bioeconomics: Modeling and Fishery Management*, Wiley-Interscience, New York, 1985.

The theory and introduction to the applications of chemostats is presented in the book

Smith, H., Waltman, P.: *The Mathematical Theory of Chemostats*, Cambridge University Press, 1997.

References for most of the topics in probability theory and its applications that are used in this book are given in Appendix B and the book

Feller, W.: *The Theory of Probability and its Applications*, J. Wiley, New York, 1969.

A popular account of recent developments in the area of chaos is given in the book

Gleick, J.: *Chaos: The Making of a New Science*, Viking Press, New York, 1987.

Exercises

7.1. Estimation of Human Population Growth Rate.

Census data for Sweden collected at 20 year intervals are presented in Table 7.4.

Let P_n denote the population size in year n , and suppose that the model for this population is $P_{n+20} = rP_n$.

Use the least-squares method to show that $r \approx 1.16$.

Thus, the model becomes

$$P_{n+20} = 1.16P_n.$$

Table 7.4. Swedish Population Data 1780-1860

Year	Population (1000s)
1780	2104 = P_{1780}
1800	2352 = P_{1800}
1820	2573 = P_{1820}
1840	3123 = P_{1840}
1860	3824 = P_{1860}

This can be solved by starting with P_{1780} and marching forward: In general, K census intervals after 1780, we predict that the population will be

$$P_{1780+20K} = (1.16)^K P_{1780}.$$

Compare the predicted values with those actually observed by completing Table 7.5 Here % error = 100 (observed - predicted)/predicted.

Table 7.5. Swedish Population Data Projections 1780-1960.

Year	K	Observed	Predicted	% error
1780	0	2104	2104	0
1800	1	2352	2441	4
1820	2	2573	2831	10
1840	3	3123	3284	5
1860	4	3824	3810	0.5
1880	5	4572		
1900	6	5117		
1920	7	5876		
1940	8	6356		
1960	9	7480		

Plot the observed Swedish population data in Table 7.5 in pairs (P_n, P_{n+20}) . Use the least-squares method to fit a straight line to these data. Compare this answer with your earlier result.

7.2. Generalization of Fibonacci's Equation

Consider the recursion

$$R_{n+1} = aR_n + bR_{n-1},$$

with $R_0 = 1$ and $R_1 = 1$. For what values of a and b does $R_n \rightarrow 0$? What happens in the other cases?

7.3. Mitotic Cycle

Suppose that a strain of bacteria ages according to a simple rule: Consider the equation for mitotic phase $p(t)$:

$$\dot{p} = \omega + \cos 4\pi p.$$

The solution's behavior is easy to determine: If $\omega < 1$, then p gets hung up where $\cos 4\pi p = -\omega$. This would correspond to the cell being stuck in either of the two gap intervals. If $\omega > 1$, then p progresses in an orderly way throughout the cycle over and over again. (If $\omega \gg 1$, then $p \approx \omega t$.) Every time p crosses the value $p = 1$ corresponds to a cell division. Draw cohort diagrams for each of these cases.

Use ω as a control variable in the alternate starvation-nutrition cycles to synchronize the cohorts. What is the minimum number of cycles needed to synchronize cohorts in a population that is initially uniformly distributed with respect to age?

7.4. Ricker's Model

1. Reproduce the results in Table 7.3
2. Carry out Table 7.3 using the same program, but for 300 intervals and 30,000 iterations. Plot the histogram for $r = 10.0$ and $r = 20.0$.
3. Repeat the calculation for several values for r to investigate various parts of Figure 7.18.

7.5. Other Chaotic Mappings

1. Consider the reproduction function

$$P_{n+1} = rP_n/(1 + P_n^M).$$

Describe the solutions of this iteration for any fixed values of the constant integer M .

2. Consider the reproduction function

$$P_{n+1} = rP_n(1 - P_n/K)$$

with $P_0 = 0.5$ and $0 \leq P_n \leq K$. Describe the behavior of its solutions for various values of r , $0 \leq r \leq 4$, by calculating and plotting iteration histograms.

3. Show that when $r = 4$, a solution for part 2 can be found. [Hint: Let $P_n = \sin^2 \theta_n$. Then

$$\sin^2 \theta_{n+1} = 4 \sin^2 \theta_n \cos^2 \theta_n = \sin^2 2\theta_n.$$

Therefore, $\theta_n = 2^n \theta_0$ (modulo 2π).]

7.6. Entropy

Consider a histogram having frequencies $[f_1, \dots, f_N]$, meaning that the numbers f_n are all nonnegative and sum to 1. We define the entropy of the histogram to be

$$H(f_1, \dots, f_N) = \frac{1}{N} \sum_{k=1}^N f_k \log f_k,$$

where if one of the frequencies is zero, that term in the sum is taken to be zero too.

1. Show that H is a smooth function of its arguments.
2. Show that $0 \leq H(f_1, \dots, f_N) \leq 1.0$ for any histogram $[f_1, \dots, f_N]$.
3. Show that if all data are in one cell, say in cell J so $f_J = 1$, then $H = 0.0$, and this is the only situation when $H = 0.0$.
4. If the histogram is uniformly distributed (i.e., $f_j = 1/N$ for all j), then $H = 1.0$, its maximum value. In fact, $H < 1.0$ for all other distributions. Therefore, the smaller H is, the more organized is the histogram.

Consider $N = 3$. Find all distributions (p_1, p_2, p_3) for which $H = 1.0$ and for which $H = \log 2 / \log 3$, and plot them on triangular coordinates. That is, draw an equilateral triangle having altitude 1. Label the vertices 1, 2, 3. A point (p_1, p_2, p_3) with $p_j \geq 0$ and $p_1 + p_2 + p_3 = 1$ can be located at a unique point on this triangle by moving p_1 units away from the side opposite vertex 1, p_2 units away from the side opposite vertex 2, and p_3 units from the side opposite vertex 3. These lines intersect at one point, since the vertical distances add up to the altitude of the triangle. (Triangular coordinates are also useful later, in our discussion of genetics.)

7.7. Strange attractors

Rectangular coordinates (x, y) can be written in polar coordinates:

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arctan(y/x)$$

for $0 \leq r < \infty$, $0 \leq \theta < 2\pi$. The inverse relationship is $x = r \cos \theta$, $y = r \sin \theta$.

Consider the following mapping of the plane into itself:

$$r_{n+1} = r_n e^{1-r_n}, \quad \theta_{n+1} = 2\theta_n \text{ modulo } 2\pi.$$

Show that all iterates approach the unit circle $r = 1$. (That is, show that $r_n \rightarrow 1$ as $n \rightarrow \infty$ if $r_0 \neq 0$.) The unit circle is invariant under this mapping in the sense that if the iteration starts on it, it stays on it. On the unit circle, we need only study the mapping $\theta_{n+1} = 2\theta_n \text{ (modulo } 2\pi)$, of the circle into itself. Show that there is a solution of period 2^N for any integer N . (Hint: Since $\theta_n = \theta_0 \text{ (modulo } 2\pi)$, find $\theta_0 \neq 0$ such that $\theta_N = 2^N \theta_0 = \theta_0 \text{ (modulo } 2\pi)$.)

Draw the graph $\theta_{n+1} = 2\theta_n \text{ (modulo } 2\pi)$ by plotting θ_{n+1} vs. θ_n . Use the histogram method to study this mapping of the circle into itself.

Show that there are initial points θ_0 that lead to iterations that never repeat themselves.

These calculations show that the unit circle is an attractor, and that behavior of iterates on the attractor is quite irregular, having periodic and aperiodic iterations hopelessly intertwined. Such a set is called a strange attractor.

7.8. Logistic Equation

Show that if $r = 1 + bh$ and h is small, then the difference equation

$$P_{n+1} = rP_n / (1 + P_n/K)$$

and the differential equation

$$\frac{dp}{dt} = bp(1 - p/K)$$

are closely related.

Show that a differential equation

$$\frac{dp}{dt} = F(p),$$

where F is a continuously differentiable function and p and F are scalars, must have all solutions monotone (that is, either constant, strictly increasing, or strictly decreasing). Conclude that it cannot have chaotic, or even periodic, solutions.

7.9. Solutivory chain

Show that a pulse of nutrients propagates through the solutivory food chain when

- $m_{1,1} = 0.5$, $m_{2,2} = 1.0$,
- $V_{2,1} = V_{3,2} = V_{1,3} = 0.9$,
- $Y_{2,1} = Y_{3,2} = Y_{3,1} = 0.9$,
- $K_{2,1} = K_{3,2} = K_{1,3} = 2 \times 10^{-4}$,
- $F_{1,3} = F_{2,1} = F_{3,2} = 1$, other F and G values are zero,
- $T_0 = T_1 = 0.05$,
- $S_3(0) = 1.5$, and the other substrates are initially zero,
- The initial population sizes are all taken to be 10^{-5} ,

by carrying out a simulation of the model (see Ex. 7.16). Next investigate the chain's dynamics for other values. Find conditions that stop propagation.

7.10. Starvation-Nutrition Cycles

Demonstrate the starvation nutrition mechanism for synchronizing a culture. Begin with a culture whose cells are uniformly distributed among phases in their mitotic cycles.

- Simulate this culture growing under optimal conditions ($\dot{p} = 1$).
- Simulate this culture growing under starved conditions ($\dot{p} = 0$ when $p \in \text{GAP1}$ or $p \in \text{GAP2}$, but $\dot{p} = 1$ when $p \in \text{SYN}$).
- Feed the culture on the time interval $0 \leq t < 1$; starve it on the time interval $1 \leq t < 2$; feed it on $2 \leq t < 3$; etc.
- Determine how many starvation-nutrition cycles are needed to get all cells into one cohort.

[Hint: You may use the age-distribution methods based on Leslie's matrix (Exercise 7.14).]

7.11. Periodic Cicadas

Consider a population of cicadas that have a life-span of L years. Let X_n denote the number of newborns that become established underground in year n . Let α denote the probability of survival underground per year, and let P denote the predator satiation threshold. Then the number of cicadas that emerge L years later is $X_n \alpha^L$. The remainder of this emergence after predation can be denoted by

$$E_n = (X_n \alpha^L - P)_+.$$

Here $(h)_+ = h$ if $h \geq 0$, but $(h)_+ = 0$ if $h < 0$. If each of these survivors bears b viable offspring, then $H_n = bE_n$ gives the number of newborns that could become established underground in year $n + L$.

However, the environmental carrying capacity might limit this number. If K is the carrying capacity for developing cicadas, the residual carrying capacity is

$$K_n = \left(K - \sum_{j=1}^{L-1} X_{n-j} \alpha^j \right)_+.$$

Therefore, the number of offspring becoming established underground in year $n + L$ is

$$X_{n+L} = \min\{H_n, K_n\}.$$

Choose data so emergences are balanced for $L \leq 7$ and periodic for $L \geq 10$.

Suppose that L is a composite integer ($L = pq$ where p and q are integers greater than 1). Determine conditions on b , α , P , K , and L that ensure that there can be a p -periodic emergence pattern. That is show that under certain conditions on the data there is a solution of the form $X_{N+p} = X$ for $N = 1, 2, \dots$, but $X_j = 0$ if j is not a multiple of p .

7.12. Effort-Yield Curves for Harvesting

Plot the effort-yield curves for the two cases of Ricker's model and the predator satiation model as derived in Section 7.5.3. Use $K = 3.5$, $r = 2.0$, $q = 1$. Discuss the possible results of the maximum sustained yield management policy in these two cases.

7.13. Project on Population Simulations

Write and execute a computer program for solving the renewal equation (7.2.2), and so calculate the sequence $\{B_n\}$ using the data $\lambda_n = 0.95^n$, $b_n = 0$ for $n \leq 15$ and $b_n = 1.1$ for $16 \leq n \leq 30$, and beyond that $b_n = 1.1(0.8)^n$. Let $h_n = 0$ for all n , but $B_0 = 1.0$.

Calculate the numbers $B_1, B_2, \dots, B_{150}$.

Plot successive age histograms (see Figure 7.5) for the total female population size every fifth sampling interval by calculating the number of individuals of each age group.

Calculate the value of r^* as described in the Renewal Theorem.

Plot the successive age histograms for $r^{*-n}B_n$. These converge to the population's stable age distribution. Explain what this means.

7.14. Project on Leslie's Matrix.

A matrix formulation of Fibonacci's model is possible. Let $v_n = (B_n, B_{n+1})$ for $n = 1, 2, \dots$ and let

$$E = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Show that Fibonacci's model is equivalent to the iteration

$$v_{n+1} = v_n E.$$

Find the eigenvalues of E . Use the solution for B_n in Section 7.2.2 to find v_n . Rewrite your answer in terms of the eigenvalues of E . This method illustrates the spectral method for solving matrix iterations.

7.15. Laplace's Method for Solving the Renewal Equation

Consider the continuous version of the discrete renewal equation (7.2.2), when there are overlapping generations:

$$B(t) = h(t) + \int_0^t m(a)B(t-a) da,$$

where $m(a) = b(a)\lambda(a)$ is described in Section 7.2.2. Laplace's transform of a function $f(t)$ is defined to be

$$\hat{f}(s) = \int_0^\infty e^{st} f(t) dt.$$

Apply Laplace's transform to the renewal equation and show that

$$\hat{B}(s) = \hat{h}(s)/(1 - \hat{m}(s)).$$

Methods of complex variables show that a function can be found from its Laplace transform using the formula

$$B(t) = \frac{1}{2\pi} \int_C e^{st} \hat{B}(s) ds \approx B_0 e^{s^* t},$$

where C is an appropriate contour in the complex plane, s^* is the largest solution of the characteristic equation

$$1 = \hat{m}(s),$$

and B_0 can be determined from a formula that is not needed here. The characteristic equation is generally difficult to solve. However, suppose that

$$m(a) = M \text{ is a constant for } 15 \leq a \leq 30,$$

but is otherwise zero. Show that the characteristic equation in this case is

$$1 = M \frac{e^{-15s} - e^{-30s}}{s}.$$

Describe the stable age distribution of the population in this case.

7.16. Project on Microbial Ecology

We suppose that there are N species and M substrates. The model for this microbial ecology is

$$\dot{S}_j = \sum_{i=1}^N \left(m_{i,j} - \frac{V_{i,j} S_j}{Y_{i,j}(K_{i,j} + S_j)} \right) P_i, \quad (7.6.1)$$

$$\dot{P}_i = \left(\sum_{j=1}^M \frac{V_{i,j} S_j}{K_{i,j} + S_j} - \alpha_i(S_1, \dots, S_N) \right) P_i + \beta_i(S_1, \dots, S_N) Q_i,$$

$$\dot{Q}_i = \alpha_i(S_1, \dots, S_N) P_i - \beta_i(S_1, \dots, S_N) Q_i,$$

for $i = 1, \dots, N$. The subscript i, j denotes i for species, j for substrate, where appropriate.

In these equations, the inactivation rates α and the activation rates β are certain functions that describe turning off and on of growth competence, respectively. If all (relevant) S_j are small, then α is large, and β is large if any of the (relevant) S_j are large. Relevance refers to the ability of the species to take up a specific substrate, and the numbers $F_{i,j}$ and $G_{i,j}$ in the following formulas are either zero or one to indicate whether or not a given substrate (j) is usable by (i.e., relevant to) a given species (i). These rates are defined by the formulas

$$\alpha_i = \max \left(\prod_{j=1}^M \frac{K_{i,j}}{K_{i,j} + F_{i,j} S_j} - T_0, 0 \right)$$

and

$$\beta_i = \max \left(\sum_{j=1}^M \frac{G_{i,j} S_j}{K_{i,j} + S_j} - T_1, 0 \right).$$

Here T_0 and T_1 are cutoff values for switching to or from quiescence.

Consider the population described by the graph



This graph is encoded in the model through the matrices V and m : In this case,

$$V = \begin{pmatrix} 0 & 0 & + \\ + & 0 & 0 \\ 0 & + & 0 \end{pmatrix}$$

and

$$m = \begin{pmatrix} + & 0 & 0 \\ 0 & + & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

We have depicted these rates using qualitative notation to indicate which reactions are active. Keep in mind that the rows correspond to species and the

columns to substrates, so a '+' in position i, j indicates that species i consumes (V) or produces (m) substrate j . Figure 7.10 illustrates this phenomenon.

Build a food web involving 4 species and 5 substrates. Describe it using the matrices V and m , and simulate your model.

8

Genetics

Genetics is the study of heredity and variation among organisms. The initial breakthroughs in genetics came with observations made by Darwin and Mendel in the nineteenth century, and mathematical methods played important roles in bringing us to our present-day understanding of genetics. Work in the first half of the twentieth century by J.B.S. Haldane, S. Wright, R.A. Fisher, and others put this important area of biology on a firm mathematical basis that helped in the design and interpretation of experiments. Many think that genetics still represents the outstanding application of mathematics in the life sciences.

It is known that reproduction of organisms involves passing from one generation to the next a genetic code that determines basic physical attributes of the offspring. This code is carried on *chromosomes*, which are large molecules made up of DNA (deoxyribonucleic acid), and carry templates for building all cellular components. The human genome is made up of twenty-three matched pairs of chromosomes, except for reproductive cells, which have twenty-three single chromosomes. The pairs are two copies of the same chromosome, though they are not perfect copies, and small differences give rise to immense variability between individuals and species. Other organisms, including plants, can have chromosomes in matched sets of three or four or more. When chromosomes occur singly, the cell is called a *haploid* cell. It is a *diploid* cell if the chromosomes appear as matched pairs, and a *polyploid* cell otherwise. Human sperm and ova are haploid cells having twenty-three single chromosomes, and other human cells are diploids, having their 23 chromosome pairs arranged like strands of thread stuck

together near their centers. Generally, bacteria are haploid cells having a single chromosome.

Genes are segments of a chromosome that code for some specific (identifiable) cell function, such as the production of a specific protein molecule. The location of a gene on its chromosome is called its *locus*. A gene may appear in a population in several slightly different forms that are detectable experimentally. These variants are called *alleles* of the gene. For example, the two matched copies of a single chromosome in a diploid cell can have different alleles at a gene locus, one on each strand. The genetic code is expressed in terms of a sequence along the chromosome of chemicals, called bases, that are labeled A, C, G and T. These sequences have been decoded to an extent that shows how they correlate with the synthesis of proteins.

The Human Genome Project began in 1990 and was completed in 2001, and it resulted in a complete listing of the genetic code for humans. Using this information, experimenters will uncover what information genes carry. The genetic makeup of an organism is called its *genotype*, and how this is eventually expressed in the organism is called its *phenotype*. The mapping from genotype to phenotype is not invertible; for example, each organism can be further shaped by its physical and even cultural environments.

Genetics encompasses all the studies of genes in all living things. It has been an active and interesting area of work over the past century. Genetic studies have given us tools to understand and reduce diseases. They have resulted in important ethical, social, political, and legal decisions and policies; made possible the development of entirely novel organisms that are important in agriculture, biotechnology and medicine; and stimulated the growth of new sciences. For example, the study of genetics contributed to the foundations of probability theory and mathematical statistics as derived by Fisher, Haldane, and Wright. The genome project having been undertaken by the United States government and commercial interests underscores the importance of this area to our lives.

We consider two major aspects of genetics: population genetics and genetic methods in biotechnology. In the section on population genetics we first consider Mendel's laws, which summarize observations of the reproduction of plants made by Gregor Mendel around 1850. These laws can be used to predict the evolution of a population, including the effects of natural selection, which is the varying ability of organisms to survive and to leave offspring. Then we describe how the genetic structure of a population can change due to random events, which can cause some genes to disappear while others thrive.

The second section studies some mathematical methods used in biotechnology. We begin with *plasmids*, which are extrachromosomal DNA elements that can carry important genes like those governing antibiotic resistance. Plasmids are primary tools used in genetic engineering, since their genes can be copied along with the cell's chromosome, which is called vertical transmission of genes, and they can be passed directly into other

cells, which is called horizontal transmission. An important issue in biotechnology is to determine when a plasmid introduced into a cell line will or will not persist in the cell line. We study this using Markov chain methods. We also consider three important chemical reactions in cells: the control of plasmid copy number, cell mechanisms used to switch specific genes on and off, and complex metabolic pathways in cells.

The mathematical methods used in studying these problems include iterations of functions that describe gene pool dynamics, random processes that describe random genetic drift, differential equations that describe genetic control circuits, and graphs that describe metabolite-flux pathways.

8.1 Population Genetics

Genes carry the code elements that enable living organisms to survive and reproduce. Molecular methods have been very successful in uncovering how to find genes and how they work. The way in which a gene ends up being expressed in a larger system, like a human body, remains an area of active investigation. This mapping from genotype to phenotype is an area of current research called *functional genomics*. Genes are also expressed as population phenomena, and population genetics is the discipline that studies them.

We study here two basic aspects of population genetics. First, how can a population's genetic structure be described using mathematics, and how can mathematics be used to extrapolate the population's evolution in the presence of selective pressures. This is carried out by studying Mendel's laws of inheritance and building a mathematical model based on them.

Second, a population's genetic structure can change even if there are no selective pressures acting on it. In fact, there are several places in the cycle of survival and reproduction where random sampling occurs, and randomness can have profound impact, especially in small populations. This phenomenon is referred to as *random genetic drift*.

8.1.1 Natural Selection

Let us consider a diploid population and denote two possible alleles at a certain locus by **a** and **b**. Individuals can be classified by their genes at this locus: They are either **aa**, **ab**, or **bb**. (Note that **ab** and **ba** are indistinguishable.) These are called genotypes. In the case of the blood disease sickle cell anemia, the **b** gene might code for the anemic trait, and those of type **bb** will have the disease. Those of types **aa** and **ab** do not have the disease, although the offspring of those of type **ab** are at risk of inheriting two **b** genes from an **ab** $\times$ **ab** mating, and so inherit the disease while neither parent has it.

The whole population carries a pool of genes at the locus under consideration, and by studying the gene pool we can make certain predictions about the fate of, say, the **b** gene in the population. This is done in the following way. First, to avoid the (minor) complications of overlapping generations, we consider a synchronized population, and we attempt to describe the changes that will occur in the gene pool with each reproduction.

The gene pool is described from one generation to the next by the sequence $\{g_n\}$, where g_n denotes the proportion of the gene pool that is of type **a** immediately before the n th reproduction. The reproduction process creates a new population whose genes are selected from the parent's gene pool at random, but according to Mendel's laws. The laws are summarized in Table 8.1, where, for example, one parent of type **aa** mated with a parent of type **bb** will produce only offspring of type **ab**, but the offspring of an **ab** mated with an **ab** will be **aa** with probability $\frac{1}{4}$, **ab** with probability $\frac{1}{2}$, and **bb** with probability $\frac{1}{4}$.

Table 8.1. Mendel's Laws of Inheritance

Mating	Offspring		
	aa	ab	bb
aa $\times$ aa	1	0	0
aa $\times$ ab	1/2	1/2	0
aa $\times$ bb	0	1	0
ab $\times$ ab	1/4	1/2	1/4
ab $\times$ bb	0	1/2	1/2
bb $\times$ bb	0	0	1

Creation of a new generation by its parents can be viewed as a random process. A population of N individuals carries $2N$ genes in its pool, and the distribution of their reproductive haploid cells (eggs in females and sperm in males) at reproduction time gives the probabilities of various ones being used in reproduction. (We suppose that the female and male populations are identically distributed.) For example, the proportion of the offspring gene pool that is of type **a** will be

$$g_n^2 + g_n(1 - g_n), \quad (8.1.1)$$

since the probability of two **a** genes being selected for reproduction will be $g_n \times g_n$, and the probability of one **a** and one **b** gene being selected will be $2g_n(1 - g_n)$, half of which will be **a** genes.

The fitnesses of various offspring to survive and reproduce are like the numbers in the renewal equation in Chapter 7. They describe the probability of survival to the next reproduction time and the fertility. These will be denoted by the numbers R for **aa**, S for **ab**, and T for **bb** types of offspring. Therefore, the total gene pool at the next reproduction time will

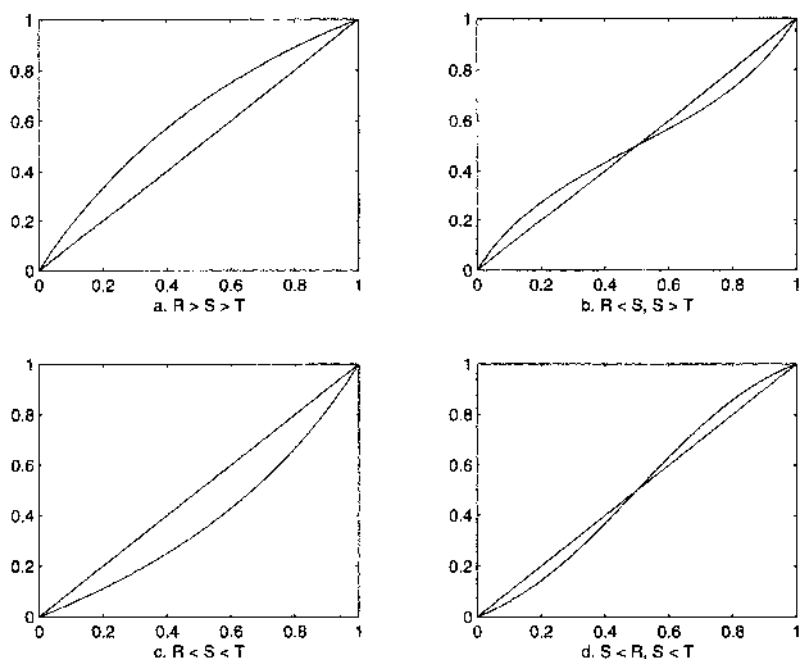


Figure 8.1. a. **a** dominates; b. Heterosis; c. **b** dominates; d. Disruptive selection.

be proportional to

$$w_n = Rg_n^2 + 2Sg_n(1 - g_n) + T(1 - g_n)^2$$

and those of type **a** will be proportional to

$$Rg_n^2 + Sg_n(1 - g_n),$$

as in equation (8.1.1). Therefore, the gene pool proportion of type **a** at the next reproduction is given by the formula

$$g_{n+1} = \frac{Rg_n^2 + Sg_n(1 - g_n)}{w_n}. \quad (8.1.2)$$

The changes in the gene pool through the generations can be determined by iterating this relation as described in several examples in Chapter 7. Four typical cases are shown in Figure 8.1.

The four cases described in Figure 8.1 require a little bit of mathematical analysis to uncover, but they have simple interpretations. In case a, $R > S > T$, and the **a** gene is favored. It will eventually dominate the gene pool, since $g_n \rightarrow 1$. This is the situation for sickle cell disease in the United States, so **b** should die out of the population, but very slowly. In case c, $R < S < T$, and **b** wins. The remaining two cases are of particular interest.

Heterosis occurs if the mixed type **ab** has a higher probability of surviving and reproducing than the two pure types **aa** and **bb**. For example, in sickle cell disease in a tropical climate the **aa** types are at higher risk of contracting malaria than the other two types. The **bb** type has the sickle cell trait and so is at risk from anemia. The **ab** types are apparently at lower risk from malaria and do not suffer from fatal anemia. Figure 8.1, case b, therefore shows that the gene b, which is the deleterious one in an environment having no malaria, can be maintained by the disease in another climate. The final case (d) is difficult to document in the field, but it is believed to be an important mechanism in the process of speciation and evolutionary divergence of species.

The model (8.1.2) was derived by Fisher, Wright, and Haldane in the 1930s, and it has been widely used to study genetic traits. The model is based on the gene pool that is available at reproduction time, but the gene pool is not observable in most populations. Moreover, the genotype of an individual is often difficult to determine, since its visible expression (its phenotype) might not distinguish between **aa** and **ab** types, for example as in sickle cell. Differences between genotypes might be visible to us only through chemical testing.

If there is no selective difference between the genotypes, then $R = S = T$, and we have

$$g_{n+1} = \frac{g_n^2 + g_n(1 - g_n)}{g_n^2 + 2g_n(1 - g_n) + (1 - g_n)^2} = g_n. \quad (8.1.3)$$

Therefore, the genotype proportions in the n th generation satisfy

$$\mathbf{aa} : \mathbf{ab} : \mathbf{bb} = g_n^2 : 2g_n(1 - g_n) : (1 - g_n)^2 = g_0^2 : 2g_0(1 - g_0) : (1 - g_0)^2,$$

and the population is said to be in Hardy-Weinberg proportions. This observation was purportedly made by the mathematician G.H. Hardy during a tennis game and independently by the German physician W. Weinberg (See Exercise 8.5).

8.1.2 Slow Selection

In many situations, selection occurs only over many generations: We say that selection is acting on a much longer time scale than birth and death. This idea is captured by our model when all of the selection parameters are nearly the same. We write

$$R = b(1 + \varepsilon\rho), S = b(1 + \varepsilon\sigma), T = b(1 + \varepsilon\tau),$$

where ε is a small number. As a result,

$$\begin{aligned}w_n &= Rg_n^2 + 2Sg_n(1 - g_n) + T(1 - g_n)^2 \\&= bg_n^2 + 2bg_n(1 - g_n) + b(1 - g_n)^2 + b\varepsilon(\rho g_n^2 + 2\sigma g_n(1 - g_n) + \tau(1 - g_n)^2) \\&= b + b\varepsilon(\rho g_n^2 + 2\sigma g_n(1 - g_n) + \tau(1 - g_n)^2).\end{aligned}$$

Substituting these changes into the equation for g_{n+1} gives

$$\begin{aligned}g_{n+1} &= \frac{Rg_n^2 + Sg_n(1 - g_n)}{w_n} \\&= \frac{g_n + \varepsilon(\rho g_n^2 + \sigma g_n(1 - g_n))}{1 + \varepsilon(\rho g_n^2 + 2\sigma g_n(1 - g_n) + \tau(1 - g_n)^2)}.\end{aligned}$$

Taylor's theorem from calculus enables us to expand the right-hand side of this expression in powers of ε :

$$g_{n+1} = g_n + \varepsilon g_n(1 - g_n)(\rho g_n + \sigma(1 - 2g_n) - \tau(1 - g_n)) + O(\varepsilon^2).$$

This is a difference equation trying to be a differential equation! If we view g_n as samples of a smooth function, say $p(t)$, at regular intervals of length ε , so $g_n = p(\varepsilon n)$, then we have

$$\frac{p(t + \varepsilon) - p(t)}{\varepsilon} = p(1 - p)((\rho - 2\sigma + \tau)p + \sigma - \tau) + O(\varepsilon),$$

and passing to the limit $\varepsilon \rightarrow 0$, we have

$$\dot{p} = p(1 - p)((\rho - 2\sigma + \tau)p + \sigma - \tau). \quad (8.1.4)$$

Similar results to those shown in Figure 8.1 hold for this equation in the four cases $\rho > \sigma > \tau$, etc.

8.1.3 Genotype Frequencies

Our work so far has focused on the dynamics of the gene pool. What happens to the genotype frequencies?

Let P denote the total population size, and let x denote the size of the **aa** population, $2y$ the size of the **ab** population, and z the size of the **bb** population. Suppose that there is no selection and that each subpopulation has the same birth rate, say b , and death rate, say d . The change in the population is described by the equations (here $q = 1 - p$)

$$\begin{aligned}\dot{x} &= pbx + pby - dx, \\2\dot{y} &= qbx + (p + q)by + pbz - 2dy, \\\dot{z} &= qbz + qby - dz,\end{aligned}$$

where the first term on the right indicates that the proportion of all the births to **aa**'s that are fertilized by an **a** allele is p ; all of these will be **aa**'s.

The second term indicates the number of **aa**'s born to **ab**'s. Note that z doesn't appear, since **bb**'s cannot have **aa** offspring. The third term is for death. Similar equations result for y and z .

The total population size is $P(t) = x(t) + 2y(t) + z(t)$, and from the equations for the subpopulations we have that

$$\dot{P} = bx + b2y + bz - dx - 2dy - dz = (b - d)P.$$

Then $p = (x + y)/P$ and $q = 1 - p = (y + z)/P$.

Next, we consider the population proportions that are described by the variables $D = x/P$, $H = y/P$, $R = z/P$ (for historical reasons the notation represents dominant, heterozygote and recessive phenotypes although here neither is advantaged when there is no selection). Then

$$\begin{aligned}\dot{D} &= b(p^2 - D), \\ \dot{H} &= b(pq - H), \\ \dot{R} &= b(q^2 - R).\end{aligned}$$

Note that $\dot{p} = \dot{D} + \dot{H} = 0$, which is consistent with equation (8.1.4) when $\rho = \sigma = \tau$.

There is an interesting method for visualizing the three genotype frequencies simultaneously. Observe that $D(t) + 2H(t) + R(t) \equiv 1$ for all t . Therefore, the point $(D, 2H, R)$ in 3-dimensional space lies on a plane in the first octant having direction numbers $(1, 1, 1)$ (recall that the variables are D , $2H$ (not H), and R). The intersection of this plane with the first octant is an equilateral triangle, so we are motivated to use triangular coordinates to plot solutions. Recall that given a point in the interior of an equilateral triangle of altitude 1, the sum of the vertical distances from the point to the sides of the triangle is 1. Thus, we locate the point $(D, 2H, R)$ in the triangle by first labeling the vertices, starting from the lower left and proceeding clockwise: **aa**, **ab**, **bb**. We come D units away from the side opposite the vertex **aa**, away from the side opposite **ab** $2H$ units, etc. This locates a unique point in the triangle, which we denote by $(D, 2H, R)$. The set of points $D + H = c$, a constant is a straight line in the triangle, perpendicular to the base, and intersecting it $2c/\sqrt{3}$ units from **bb**. These calculations are illustrated in Figure 8.2, which in genetics is referred to as being a DeFinetti diagram and in other contexts as triangular or areal coordinates.

The parabolic curve in this figure is the trace of points

$$(p^2, 2p(1-p), (1-p)^2)$$

for $0 \leq p \leq 1$, which we referred to earlier as the set of Hardy-Weinberg equilibria. In fact, since $\dot{p} = 0$, the solution will start at the point $(D(0), 2H(0), R(0))$ and move vertically to the point on this curve $(p(0)^2, 2p(0)q(0), q(0)^2)$, where $p(0) = D(0) + H(0)$ and $q(0) = 1 - p(0)$.

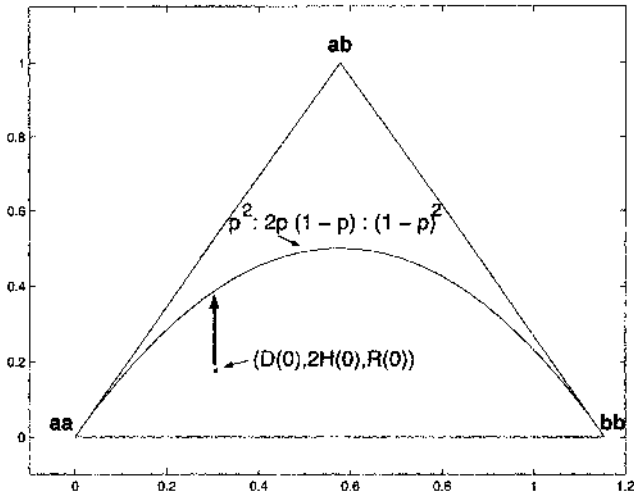


Figure 8.2. DeFinetti's diagram for a one-locus, two-allele trait. The parabolic curve describes the Hardy-Weinberg equilibria. A population distribution beginning at $(D(0), 2H(0), R(0))$ quickly approaches a Hardy-Weinberg equilibrium if there is no selection.

Finally, if there are only small differences in birth rates between the genotypes, then p will change slowly according to equation (8.1.4), and we can depict this on a DeFinetti diagram. In that case, we would see a rapid approach of the solution to the Hardy-Weinberg curve and slow movement along it, depending on the relative sizes of ρ, σ, τ . (See Exercise 8.5.)

8.1.4 Random Genetic Drift

As we have seen, creation of one generation by its parents can be viewed as a random process. For a one-locus, two-allele genetic trait, a population of N humans carries $2N$ genes, and the distribution of their gametes at reproduction time gives the probabilities of various genes being used to construct the offspring. For example, the offspring of a population comprising one male and one female that produce two offspring is summarized in Table 8.2.

Table 8.2 shows that with probability $1/8 = (1/16 + 1/16)$, one of the alleles is entirely lost from the population! Moreover, with probability $1/4$ there are no heterozygotes in the population. Therefore, random genetic drift can cause quite important changes in the population's genetic structure. This is especially apparent in small populations.

The gene pool of the next generation is formed by sampling the adult gamete pool. In the first section of this chapter we considered a large pop-

Table 8.2. Two offspring of two ab's.

One Offspring	The other one	Probability
aa	aa	1/16
ab	aa	1/8 + 1/8 = 1/4
ab	ab	1/4
bb	ab	1/4
bb	bb	1/16
aa	bb	1/8

ulation, and calculated expected numbers to formulate our model. But, for a small population such an average will not capture the dynamics as described in Table 8.2; we must account for all possible events and their probabilities. We do this using a Markov chain.

Let α_n denote the number of **a** genes in the n th generation. Then $p_n = \alpha_n/(2N)$ gives the frequency of **a** genes in the gene pool. The population is assumed to mate at random and to be synchronized with nonoverlapping generations. Furthermore, the population size is assumed to remain constant through the generations.

The sequence of random variables $\{\alpha_n\}$ describes the offspring gene pool immediately after reproduction in successive generations. If α_n is known, then α_{n+1} has a *binomial distribution* (also known as sampling with replacement) with parameters $p_n = \alpha_n/(2N)$ and index $2N$. In particular, at the end of the n th reproductive period, the gene pool consists of α_n **a** genes and $2N - \alpha_n$ **b** genes. The probability that $\alpha_{n+1} = j$ in the next generation given that $\alpha_n = i$ is

$$\begin{aligned}
 P_{i,j} &= Pr[\alpha_{n+1} = j | \alpha_n = i] = \binom{2N}{j} p_n^j (1 - p_n)^{2N-j} \\
 &= \binom{2N}{j} (i/(2N))^j (1 - (i/2N))^{2N-j}
 \end{aligned}$$

for $j = 0, \dots, 2N$, and where $\binom{2N}{j}$ is the binomial coefficient

$$\binom{2N}{j} = \frac{(2N)!}{j!(2N-j)!}.$$

The matrix $P = (P_{i,j})$ defines a Markov chain for simple human genetic traits. This is called the Fisher-Wright chain with no selection, and we can use it to simulate how a population's genetic distribution changes through the generations. Rather than pursuing that here, we consider a similar problem in epidemics in Chapter 9, in Appendix A, and in Exercise 8.2.

8.2 Biotechnology

Biotechnology is a vast enterprise. For example, most of agriculture involves biotechnology through the use of plants and animals to produce food and products for human health and nutrition. In addition, industries from mining to pharmaceuticals use microbial processes in research, development and production. In this section we study four mathematical topics in biotechnology: First, we consider the stability of genes that are carried by plasmids. That is, once a plasmid is introduced to a cell line, how long will it persist? Second, we study the control of DNA replication. How do cells keep from choking on DNA. Third, a genetic control circuit for turning genes on and off is studied. Finally, we study a methodology used to keep track of metabolic pathways in cells.

8.2.1 *Plasmids*

The biological problems described in this section are taken from genetic engineering, and they are concerned with a mechanism used to introduce new genes into a bacterium to cause it to perform a specific function, such as the production of a particular enzyme. In this section we derive a model that accounts for random sampling of the plasmid pool at each reproduction. Obviously, since the reproduction involves randomness at each step, we cannot predict exactly how the population will evolve. Instead, we derive the statistical distribution defined by many similar populations. The model that results is a Markov chain (See Appendix A). Several examples of Markov chains will be studied in this and the following two chapters.

The genetic system that we study here is simpler than the diploid model, since bacteria are haploid organisms. Bacteria are single-celled organisms that are enclosed within a cell wall; the interior is made up of cytoplasmic material containing the various biological machinery needed for cell life and reproduction. In particular, the chromosome is a circular loop or a single linear strand of DNA that carries a code for all cell functions, as opposed to pairs of linear chromosomes described earlier for diploids. A gene is a segment of the chromosome that codes for production, such as of an enzyme.

Many types of bacteria have additional genetic material called extrachromosomal DNA elements or *plasmids*. Plasmids are small circular pieces of DNA that also carry genes. However, plasmids can pass from cell to cell, and some genes can jump from plasmids to chromosomes, and in this way become permanently incorporated in the cell's code.

The cell cycle begins with a newborn daughter. All of the cell's components, including the chromosome and plasmids (if any are present), are copied during the synthesis phase. The replication is followed by division of the cell into two daughters, each receiving one replicate of the chromosome.

In this section we study how drug resistance carried by genes on plasmids is passed vertically through generations of bacteria by the mechanism of random plasmid drift. We consider a plasmid $\mathbf{P}$ that appears in two forms, say $\mathbf{P}'$ and $\mathbf{P}''$. These two forms differ perhaps in a small proportion of their DNA, but otherwise, they are identical. For example, $\mathbf{P}'$ could carry a gene that codes for resistance of the cell to the antibiotic ampicillin, and $\mathbf{P}''$ could code for resistance to tetracycline.

The purpose here is to predict the fate of the two plasmid types $\mathbf{P}'$ and $\mathbf{P}''$ in the cell line after many generations.

The plasmid $\mathbf{P}$ is assumed to appear with N copies in all newborn cells and to replicate according to a cycle depicted by the graph

$$N \rightarrow 2N \rightarrow N.$$

Typically, the number of copies that each daughter receives is not N , but a random number. However, to capture the impact that genetic drift can have, we consider only the case where each daughter receives exactly N copies. Furthermore, we suppose for simplicity that the division cycle of all cells is synchronized throughout the population, and that the copy number N is constant for all cells.

At each cell division there is a random allocation of plasmids among the two daughters, N to each. As a result of this, there can be a great number of possible evolutions of a cell population. This illustrates random genetic drift, since cell genotypes change, but as a result of random sampling rather than natural selection or other forces.

A model of plasmid dynamics is formulated in the following way: Let p_i be the proportion of newborn cells having $i\mathbf{P}'$ -plasmids and $(N-i)\mathbf{P}''$ plasmids. After replication, there will be $2N$ plasmids, $2i$ of which will be of type $\mathbf{P}'$ and $2N-2i$ will be of type $\mathbf{P}''$.

There are a number of ways that these $2N$ plasmids can be distributed among the two daughters, and the theory of probability tells how to count the possibilities. We suppose that each daughter receives N copies. Of these, j copies of $\mathbf{P}'$ can be given to one daughter in

$$\binom{2i}{j}$$

different ways. The total number of ways that the set of $2N$ plasmids can be broken into two sets of size N is $\binom{2N}{N}$, and the number of ways that one daughter can have $j\mathbf{P}'$ and $(N-j)\mathbf{P}''$ plasmids is

$$\binom{2i}{j} \binom{2N-2i}{N-j},$$

if the mother originally had i plasmids of type $\mathbf{P}'$ and $(N - i)$ plasmids of type $\mathbf{P}''$. Therefore, the formula

$$P_{i,j} = \frac{\binom{2i}{j} \binom{2N-2i}{N-j}}{\binom{2N}{N}}$$

gives the probability of one daughter having j type $\mathbf{P}'$ plasmids and $(N - j)$ type $\mathbf{P}''$ plasmids given that the mother cell had i type $\mathbf{P}'$ plasmids and $(N - i)$ of type $\mathbf{P}''$.

Based on these calculations, we can determine the evolution of the plasmid distribution in an interesting way. The population in a given generation can be described by a vector $p = (p_0, \dots, p_N)$, which tells the proportions of the various cell types appearing in the population. Let us denote these numbers in the next generation by $q_0, \dots, q_N$, respectively. Then

$$q_j = p_0 P_{0,j} + \dots + p_N P_{N,j}.$$

This states that the proportion of cells having j type $\mathbf{P}'$ plasmids in the next generation is the sum of the proportion having no plasmids of type $\mathbf{P}'$ in the parent generation times the probability of leaving j plasmids starting from none, the proportion of cells having one plasmid of type $\mathbf{P}'$ times the probability of leaving j starting from one, etc. Using matrix notation (See Appendix A and Appendix B), we write this more concisely as

$$q = pP,$$

where we set $q = (q_0, \dots, q_N)$. This recursion scheme is not difficult to analyze and use. If $p = p^{[n]}$ is the distribution in the n th generation, then $q = p^{[n+1]}$, and so using the back-substitution method we have that

$$p^{[n]} = p^{[0]} P^n, \quad (8.2.1)$$

which is a recursion of a Markov chain. Markov's ideas from the nineteenth century were taken up in the twentieth century by R.A. Fisher and others, who applied them to many problems in genetics. In fact, Fisher's work laid the foundations for the modern mathematical theory of genetics in plants, animals, and microbes.

The numbers in Table 8.3 show the evolution of proportions for a population that starts out having two of four plasmids of type $\mathbf{P}'$. These were generated by a computer simulation that simply evaluates the sequence of vectors $p^{[0]} P^n$ for $n = 1, 2, \dots, 40$.

Note that the model (8.2.1) has no randomness in it although it accounts for a random phenomenon. Once $p^{[0]}$ is known, all the vectors $p^{[n]}$ are uniquely determined by the formula. The components of $p^{[n]}$ give the proportions of the population having various plasmid combinations at the n th sampling. That is, $p_{j,n}$, which is the j th component of $p^{[n]}$, gives the probability that at the n th sampling a cell having j type $\mathbf{P}'$ plasmids would be selected as a sample from the population. Methods of probability theory

Table 8.3. Plasmid dynamics, $N = 4$.

Time n	$p^{[n]}$				
	$i = 0$	$i = 1$	$i = 2$	$i = 3$	$i = 4$
0	0	0	1	0	0
1	0.01	0.235	0.51	0.235	0.01
2	0.065	0.25	0.37	0.25	0.065
	.	.	.	.	.
40	0.5	0	0	0	0.5

show that

$$p^{[0]}P^n \rightarrow p^{[0]} \left(\begin{pmatrix} 0 \\ 1/N \\ 2/N \\ \vdots \\ \frac{N-1}{N} \\ 1 \end{pmatrix} (0, \dots, 0, 1) + \begin{pmatrix} 1 \\ \frac{N-1}{N} \\ \vdots \\ 1/N \\ 0 \end{pmatrix} (1, 0, \dots, 0) \right)$$

as $n \rightarrow \infty$. If we take $N = 4$ and $p^{[0]} = (0, 0, 1, 0, 0)$, then

$$p^{[0]}P^n \rightarrow \frac{2}{4}(0, 0, 0, 0, 1) + \frac{2}{4}(1, 0, 0, 0, 0) = (0.5, 0, 0, 0, 0.5),$$

as shown in Table 8.3.

These data demonstrate an important phenomenon. After many generations, the population is divided (almost) equally between those having all P' plasmids and those having all P'' plasmids, and neither is resistant to both antibiotics, as was the original population! It was thought for many years that there was an interaction between plasmids of different types that drove them apart so that only one could be present in a cell at one time in the absence of external selective forces. This phenomenon was referred to as *plasmid incompatibility*. However, we see from the calculation just done that it can be simple random sampling at each division time that is responsible for segregation of plasmids. In rough terms, if a cell is purely one type or the other, all of its daughters will also be of that type, and we say that the cell is in an *absorbing state* with respect to this plasmid type. Any other combination of the two plasmids has a chance to change at each division, and so the absorbing states always win!

As noted earlier, changes in the genetic profile of a population due simply to random sampling of genes at reproduction is referred to as random genetic drift, and it is believed to play an important role in the evolution of species. The mathematical analysis of Markov chain models provides an explanation for these facts, which are actually observed to occur in bacterial experiments.

The issue of stability of plasmids in cell lines is quite an important one, since in biotechnology cells can be programmed to yield certain products by insertion of a plasmid carrying the instructions as genetic code. It is important that the plasmid stay in the cell line to ensure that production continues. The next section develops another approach to stability of plasmids based on a continuous-time model.

8.2.2 Control of Plasmid Replication

Rather than counting the number of plasmids at each replication for each cell, we can describe changes in the overall plasmid content of a cell population that is growing. Consider a population of bacteria occupying a total volume V . We suppose that a plasmid is present in the population, uniformly distributed throughout it with a concentration $\mathbf{P}$. Then $V\mathbf{P}$ gives the total mass of plasmids in the population.

We suppose that the volume of the population is growing at rate α , so

$$\dot{V} = \alpha V.$$

The total plasmid mass will change at some other rate, say α_p :

$$(d/dt)(V\mathbf{P}) = \alpha_p(V\mathbf{P}).$$

Therefore,

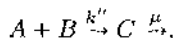
$$\dot{\mathbf{P}} = (\alpha_p - \alpha)\mathbf{P}.$$

The plasmid will not be stable in the population unless $\alpha_p = \alpha$.

If the plasmid is stable in the population, something must control α_p to maintain this value. The purpose of this section is to describe a mechanism that accomplishes this.

Chemical Reactions in an Expanding Volume

First, consider a chemical reaction involving three chemical species, say A, B, C :



occurring in a changing volume V where

$$\dot{V} = \alpha V.$$

Here k'' is the mass-action reaction rate between A and B , and μ is the rate at which C degrades. If x , y , and z denote the masses of A , B , and C , respectively, then

$$\dot{z} = k''xy - \mu z.$$

We write the chemical concentrations as $A = x/V$, $B = y/V$, and $C = z/V$. Then

$$\dot{C} V + C \dot{V} = k'' AV BV - \mu CV.$$

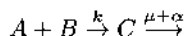
Setting $k = k''V$ and dividing both sides by V gives

$$\dot{C} + \alpha C = k \frac{A}{V} \frac{B}{V} - \mu C.$$

We say that the reaction rate k'' is pressure (volume) dependent, and we write

$$k'' = k/V.$$

The reaction rate k'' is like the probability of two molecules meeting to react effectively, and as the volume increases, the reaction rate decreases. This calculation shows that in addition to the two reaction rates k and μ , dilution can be accounted for with rate α ; namely,



Michaelis-Menton Approximation

In many important biochemical reactions, a chemical C is removed by a reaction with an enzyme E_C . The result is a complex $\mathbf{C}$ that can uncouple into the enzyme and a product chemical X . We denote these reactions by the graph



In mathematical terms we have

$$\begin{aligned}\dot{C} &= k_1 C E_C - k_2 \mathbf{C} - k_3 \mathbf{C}, \\ \dot{E}_C &= -k_1 C E_C + k_2 \mathbf{C} + k_3 \mathbf{C}, \\ \dot{\mathbf{C}} &= -k_1 C E_C + k_2 \mathbf{C}.\end{aligned}$$

Note that the total mass of E_C is conserved:

$$E_C + \mathbf{C} = E_C(0),$$

since the sum of the first two equations is zero, and the initial concentration of $\mathbf{C}$ is zero. The complex reactions are often very fast compared to others, and so we assume that $\mathbf{C}$ is in a quasi-equilibrium:

$$0 = k_1 C E_C - k_2 \mathbf{C} - k_3 \mathbf{C}.$$

Using these two equations to eliminate $\mathbf{C}$ and E_C , we have

$$\dot{C} = -\frac{V_C C}{K_C + C},$$

where $V_C = k_3 E_C(0)$ and $K_C = (k_2 + k_3)/k_1$. This is the widely used Michaelis-Menten approximation, and it was introduced and studied early

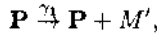
in the twentieth century. It was also the motivation for the Jacob–Monod model of substrate uptake by cells that we used in Chapter 7. These calculations are used next to analyze a replication control reaction.

Replication Control Reactions

Replication of plasmids is controlled by chemical reactions that involve the creation and destruction of intermediate molecules and both positive and negative feedback loops.

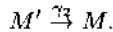
The replication reactions are:

1. The plasmids $\mathbf{P}$ can be copied to a segment of ribosomal nucleic acid (RNA), say M' , at rate γ_1 :

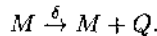


where M' is a segment of RNA in a form that is vulnerable to clogging.

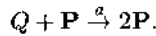
2. M' folds into a stable form M at a rate γ_2 :



3. M holds a template for a protein Q that is synthesized at rate δ :

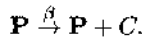


4. The protein Q regulates the production of $\mathbf{P}$ at rate α :

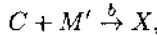


The inhibition reactions are:

1. A segment of $\mathbf{P}$ is copied to a controlling molecule C at rate β :



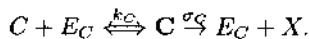
2. The molecule C can interact with the unfolded M' to inactivate it at rate b :



where X is our generic notation denoting an inactive molecule that we need no longer consider.

The degradation reactions are:

1. There is an enzyme E_C that catalyzes the degradation of C into a complex $\mathbf{C}$, which can break up into E_C and another molecule that plays no further role in these reactions. We write



This is a Michaelis–Menten reaction.

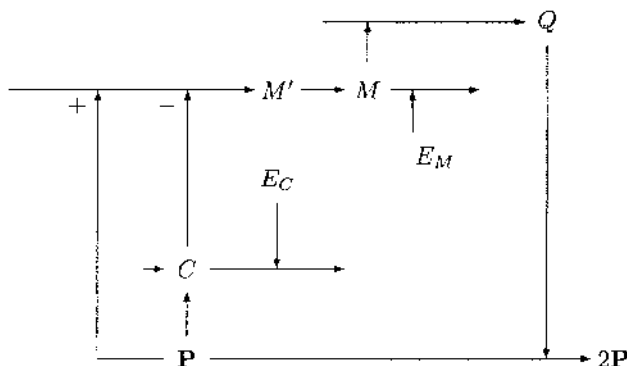
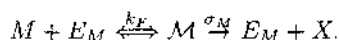


Figure 8.3. Plasmid replication control reaction. Vertical lines indicate catalytic or regulatory reactions. Horizontal lines indicate mass fluxes.

2. Similarly for M . There is an enzyme E_M that degrades M :



We will consider these chemical reactions in the context of a growing population, that is, in a diluting medium.

The control reactions can be visualized in an interesting way using the notation of genetic control circuits, as in Figure 8.3, where horizontal arrows indicate the mass flux of material from one form to another and vertical arrows indicate the passage of information through enzymes or other regulating chemical reactions.

Equilibrium Analysis of the Control Circuit

We now combine all of this work. First, we introduce the dilution rate α into each of the reactions. Then, we apply the Michaelis-Menton approximation where appropriate. As a result, the model describing the reactions summarized in Figure 8.3 is

$$\begin{aligned} \alpha P + \dot{P} &= aQP, \\ \alpha C + \dot{C} &= \beta P - bCM' - \frac{V_C C}{K_C + C}, \\ \alpha M' + \dot{M}' &= \gamma_1 P - bCM' - \gamma_2 M', \\ \alpha M + \dot{M} &= \gamma_2 M' - \frac{V_M M}{K_M + M}, \\ \alpha Q + \dot{Q} &= \delta M - aQP. \end{aligned}$$

This is a complicated set of rate equations, but we will try to determine the relative sizes of equilibrium values.

Experimental observations suggest that:

1. The growth rate α is much smaller than the chemical reaction rates. This is reasonable, since cell growth is on the scale of minutes and the chemical reactions are much faster.
2. The levels of Q are small.
3. The system is at equilibrium (so all time derivatives are zero).
4. $\mathbf{P}$ is constant and independent of α . (Remember that this is our goal.)

Using these assumptions, we will investigate the equilibrium balance equations.

$$\begin{aligned}
 \alpha \mathbf{P} &= aQ\mathbf{P}, \\
 \alpha C &= \beta \mathbf{P} - bCM' - \frac{V_C C}{K_C + C}, \\
 \alpha M' &= \gamma_1 \mathbf{P} - bCM' - \gamma_2 M', \\
 \alpha M &= \gamma_2 M' - \frac{V_M M}{K_M + M}, \\
 \alpha Q &= \delta M - aQ\mathbf{P}.
 \end{aligned}$$

For example, from the assumptions, Q and α are small, so αQ is really small. So, when we ignore such terms, the last equation gives

$$M = \alpha Q \mathbf{P} / \delta.$$

Then the first equation gives

$$M = \alpha \mathbf{P} / \delta.$$

This is our first result: M is proportional to α , which is small. Therefore, we can ignore the terms αM .

The second from last equation then gives

$$M' = \frac{MV_M}{\gamma_2 K_M}.$$

The result here is that M' is small, too:

$$M' = \alpha \gamma' \mathbf{P},$$

where $\gamma' = V_M / (\gamma_2 K_M \delta)$.

From the third equation we have that

$$bC = \frac{\gamma_1 \mathbf{P}}{M'} - \gamma_2 = \frac{\gamma_1 - \alpha \gamma' \gamma_2}{\alpha \gamma'},$$

so C is large and proportional to $1/\alpha$. We determine from the second equation that

$$(\beta - \gamma_1) \mathbf{P} = V_C + \alpha C = V_C + \gamma_1 / (b\gamma').$$

which gives the equilibrium value for $\mathbf{P}$.

There is a mathematical theory, called singular perturbation theory, that can be used to carefully (and rigorously) go through these steps.

This analysis tells us that in fact, $\mathbf{P}$ does not change as the growth rate α changes, so we see that the replication reactions can control α_P to keep $\dot{\mathbf{P}} = 0$. They accomplish this by tuning C and M' in a reciprocal relation: The slower the cell growth, the larger the RNA inhibitor pool (C) and the smaller the messenger RNA pool (M, M').

8.2.3 Regulation of Gene Expression

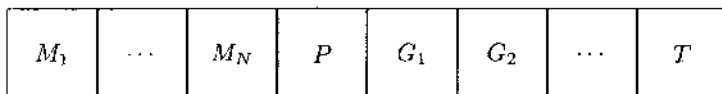
Recent discoveries in biology have shown that bacteria have the ability to turn on specific genes to create products in response to external signals received at their membrane. One such mechanism is the histidine protein kinase (HPK) system, which creates a two-component switch that enables cells to express selected genes.

When a certain protein on the cell membrane senses a specific signal (e.g., pH change or antibiotic), it changes its configuration inside the cell to become a kinase, which is an enzyme that catalyzes transfer of phosphate from ATP to a substrate. In this case, the substrate is usually a protein inside the cell that in accepting a phosphate becomes capable of participating in regulation of gene expression.

Once initiated, the control of gene products proceeds (roughly) as in Figure 8.4, which describes the production and control circuit in terms of chemicals that are created and used in the process.

Activation of a Gene

An *operon* is a segment of the chromosome that includes binding sites for several different chemical elements. First, there are modulator sites that must be occupied for copying of genes to begin. Then there is the promoter site where copying starts, followed by sequences carrying the genes themselves. These are followed by a termination sequence where copying halts. The following diagram shows these elements.



where each of the letters denotes a segment of DNA. The M 's are modulator sites where specific chemical elements, referred to as regulator proteins, can bind. These are needed for the strand to be copied. P is a promoter segment that indicates where copying should start. $G_1, G_2, \dots$, are segments coding for genes, and T is a terminator sequence.

When the modulating sites are occupied, then the replication process starts at the promoter site for the genes by opening the door for mRNA polymerase to begin making RNA copies of the genes. When completed,

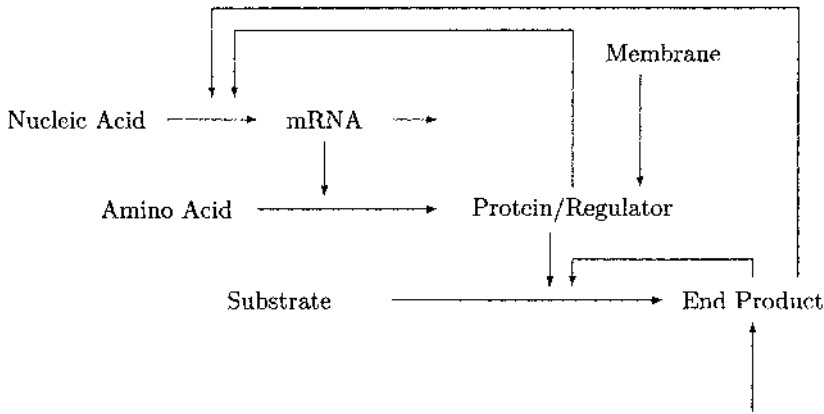
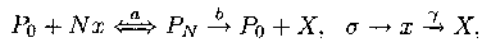


Figure 8.4. Gene expression control. Nucleic acids are copied with mRNA, which provides the template for linking together amino acids into proteins, usually enzymes. The protein regulates production of an end product from substrate. External regulators might influence transcription, and external end products might enter the pool.

mRNA serves as a template for the ribosomes to link amino acids together to form a protein, often an enzyme for catalyzing reactions that convert various substrates into end products.

Let us suppose that the regulator proteins are the same, say x . The source of x is the HPK system described earlier, and we denote the source by σ . In addition, x is removed from the resource pool at a rate γ .

The modulator sites act as substrates for the regulators, and we have the reaction



where

1. x denotes the regulator element,
2. P_0 denotes the segment with open modulator sites,
3. P_N is the complex with its N modulating sites fully occupied,
4. X describes inactivated regulators.

The forward rates (to the right in this diagram) are a and b , respectively, and the backward rate from P_N is c .

The chemical kinetics are described by rate equations. The first reaction is reversible, and it involves N regulators and the open operon combining to form a complex P_N that is ready for transcription to RNA. This complex can reverse to release the regulators, or it can move forward releasing the regulators in inactive form. In particular, P_0 is removed by binding with regulators and is freed by P_N . The rate equation for this is

$$\dot{P}_0 = -ax^N P_0 + (b+c)P_N.$$

The rate equation for P_N is similar:

$$\dot{P}_N = ax^N P_0 - (b+c)P_N.$$

The amount of P_N reflects the readiness of the operon, but we do not pursue that direction further here. Because of conservation of sites, we have that

$$P_0 + P_N = K,$$

which we can use to eliminate P_0 . Here K is the total of promoter sites, those occupied and those not.

In the same way, we have the following equation for x :

$$\dot{x} = \sigma - Nax^N P_0 + NcP_N - \gamma x.$$

After we eliminate P_0 , the system becomes

$$\begin{aligned}\dot{P}_N &= ax^N(K - P_N) - (b+c)P_N, \\ \dot{x} &= \sigma - \gamma x + N(cP_N - ax^N(K - P_N)) \\ &= N(c + ax^N)P_N - Nax^N K + \sigma - \gamma x.\end{aligned}$$

Again, we will consider the equilibrium balance equations where $\dot{P}_N = 0$ and $\dot{x} = 0$:

$$P_N = \frac{ax^N K}{b+c+ax^N}.$$

The equilibrium equation for the regulator x becomes

$$0 = \sigma - \gamma x - \frac{NabKx^N}{b+c+ax^N}. \quad (8.2.2)$$

As a result, the equilibrium distribution of the regulator is found by solving the equation

$$\gamma x^{N+1} + (NbK - \sigma)x^N + \frac{(b+c)}{a}\gamma x - \sigma \frac{(b+c)}{a} = 0.$$

We rewrite this equation as

$$x^{N+1} - \alpha x^N + \beta x - \delta = 0$$

where $\alpha = (\sigma - NbK)/\gamma$, $\beta = (b+c)/a$, and $\delta = \sigma(b+c)/a\gamma$. If we write $h(x) = x^{N+1} - \alpha x^N$ and $g(x) = \delta - \beta x$, then the solutions for x lie at the points where $h(x) = g(x)$. This can be determined by plotting both of

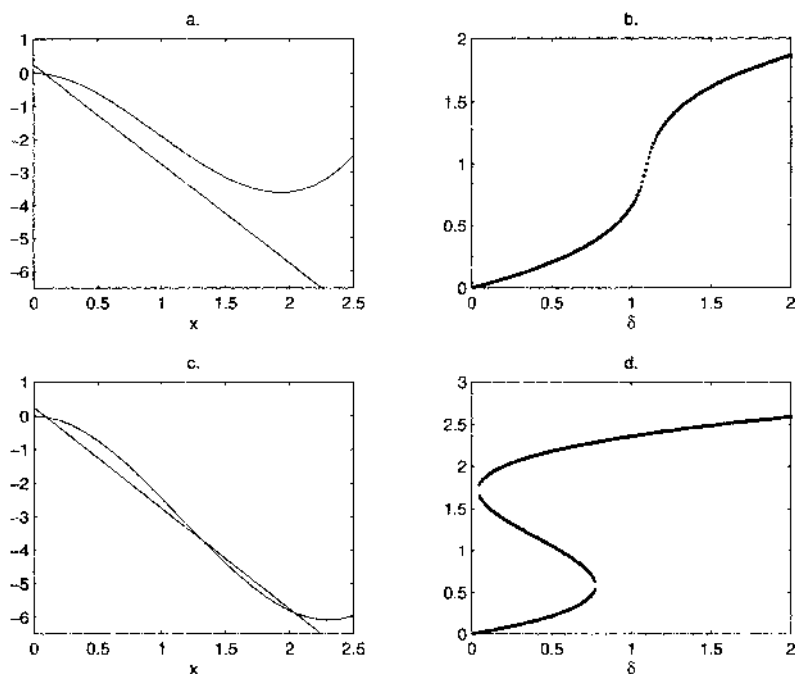


Figure 8.5. In these simulations $N = 2$ and $\beta = 3.0$. a. $\alpha = 2.9, \delta = 0.25$. b. In this case, there is only one static state for each $0 \leq \delta \leq 2.0$. c. $\alpha = 3.45, \delta = 0.25$. d. In this case, there are 3 static states for small values of δ , but only one for larger values.

these functions and seeing where they cross. We consider only two cases in Figure 8.5: a, b, $\alpha = 2.9$ and c, d, $\alpha = 3.45$.

This calculation illustrates that the regulatory system for expression of an operon can be quite complicated. In fact, in the first case, we expect only one level of regulator molecules, but in the second, there are three possible levels. In the former case, there is a critical value of δ near 1 at which the concentration of x changes rapidly. In the latter case, the largest and smallest are stable states for the x rate equations, so the system exhibits bistability!

The process described in this section is for activation or positive regulation of transcription, also called up-regulation. There is also an inactivation or negative regulation of transcription, also called down-regulation, by which occupation of the modulator sites prevents RNA polymerase from making mRNA.

In addition, there is a hierarchy of regulatory control circuits above the operon level that are known. These include the regulons and modulons, and they describe multigene networks as stimulus-response systems. These

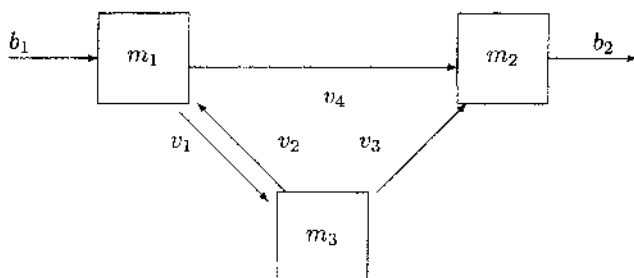


Figure 8.6. Graph of a hypothetical metabolic pathway.

systems and the HPK switches enable a cell to control a large number of cellular processes, including osmosis; levels of nitrogen, carbon, phosphate, and oxygen; pH; and responses to thermal and mechanical shocks. Regulons and modulons are controlled by HPKs and by other control mechanisms.

8.2.4 Stoichiometry of Metabolic Pathways

A cell proceeds with its life by carrying out necessary processes to maintain its metabolism and by responding to changes in its environment. For example, when a certain substrate, such as histidine, is required by the cell, but it is not available, then the cell can turn on genes for the production of histidine, and allocate some of its energy to this process. This can be done through activation of appropriate operons as described in the preceding section. In this case, the cell is said to have switched on a metabolic pathway.

Possible pathways in a cell's chemical processing can be mapped out using a stoichiometric counting process. This is a highly complex process, but it does give a systematic way to proceed. In this, the cell's metabolites, say m_j , are listed, as well as fluxes between them, say v_i . Underlying this approach is the assumption that the metabolic system is operating in an optimal steady mode. Fluxes from or to the outside of the system are denoted by b_j . Then a graph is formed whose nodes are the metabolites and whose edges represent the fluxes. We illustrate this process with a simple situation where there are three metabolites and two external fluxes, as shown in Figure 8.6

This graph can be summarized in a *stoichiometric matrix* where the rows are labeled by the metabolites and the columns by the fluxes. We place ± 1 in the matrix depending on the flux orientation (in or out). If none, we place a zero. A 1 is entered if the flux into that metabolite is positive, -1 if it is negative, and 0 if the flux does not relate to that metabolite. The

matrix for the graph in Figure 8.6 is

$$S = \begin{matrix} & v_1 & v_2 & v_3 & v_4 & b_1 & b_2 \\ \begin{matrix} m_1 \\ m_2 \\ m_3 \end{matrix} & \begin{pmatrix} -1 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 \\ 1 & -1 & -1 & 0 & 0 & 0 \end{pmatrix} \end{matrix}.$$

In this example we have a 3×6 matrix, and it has rank 3 (See Appendix A). Therefore, there are 3 linearly independent right null vectors; that is, there are three vectors ϕ_1, ϕ_2, ϕ_3 such that $S\phi = 0$, and they are linearly independent. For example,

$$\phi_1 = [1, 1, 0, 0, 0, 0]^{\text{tr}}, \quad \phi_2 = [0, 0, 0, 1, 1, 1]^{\text{tr}}, \quad \phi_3 = [1, 0, 1, 0, 1, 1]^{\text{tr}}.$$

The significance of these vectors is that they describe the possible pathways through this system of reactions. Here ϕ_1 indicates that energy can pass continually back and forth between metabolites m_1 and m_3 ; ϕ_2 that energy can enter metabolite m_1 , pass to m_2 and leave; and ϕ_3 that energy can enter m_1 , pass to m_3 , then to m_2 , then out. The idea here is that a graph of a complicated chemical process is drawn that indicates possible fluxes into, out of, and between metabolites. Then a stoichiometry matrix is formed. It is analyzed for its null vectors, and these indicate possible pathways for energy flow in the process. The applicability of this methodology to real metabolic pathways requires a great deal of knowledge and computational effort. However, it provides a useful methodology for bookkeeping and for systematically identifying viable pathways for energy flow through a system. A cost for each null vector can be assigned, and so optimal pathways under various conditions can be calculated.

8.3 Annotated References

The history of how the study of genes was started and how it progressed through early stages is described in the book

Judson, H.F.: *The Eighth Day of Creation*. Simon and Schuster, New York, 1979.

Population genetics methods are described in the books

Fisher, R.A.: *The Genetical Theory of Natural Selection*. Dover Publics., New York, 1958.

Crow, J.F., and Kimura, M.: *An Introduction to Population Genetics Theory*. Harper & Row, New York, 1970.

Hoppensteadt, F.C.: *Mathematical Methods of Population Biology*. Cambridge University Press, 1982.

Many genetic control problems in biotechnology discussed here are described in greater detail in the book

Escherichia Coli and Salmonella. (Neidhardt, et al.). Amer. Soc. Microbiol. Publications, Washington, 1994.

Exercises

8.1. Selective Pressure in the Gene Pool

Consider the model (8.1.2) for the dynamics of the gene pool determined by a one-locus, two allele trait. Describe the behavior of g_n in the following cases: $R = S = T$; $R > S > T$; $R > S, T > S$; $S > R, S > T$; and $R = S, T = 0$.

8.2. Fisher-Wright Chain Simulation

Using a hand calculator or computer, simulate the Fisher-Wright chain in Section 8.2 with $N = 2$ and $p_1^{(0)} = 1$. Present the answer in tabular form as in Table 8.2.

8.3. Plasmid Cure Rates

A simple assay used by plasmid biologists counts the number of cells in each generation of two antibiotic-resistant strains that have both plasmid types. This gives the rate at which cells are "cured," that is, the rate at which immunity to both antibiotics is lost from the population when it is in nonselective media.

The cure rate can be determined theoretically by using certain properties of the transition probability matrix P . Namely, we iterate P a number n of times and call the result P^n . The norm of this matrix is denoted by $|P^n|$. Then the cure rate is approximately $|P^{n+1} - P^n|/|P^n - P^{n-1}|$ for n large.

Show that for the data in Table 8.3 the cure rate is 0.86.

8.4. Gene Pool Dynamics: Infinitesimal Sampling

We saw in Section 8.1 that the gene pool proportions each generation, say g_n , are determined by the recursion formula

$$g_{n+1} = (\rho g_n^2 + \sigma g_n(1 - g_n))/w_n,$$

where $w_n = \rho g_n^2 + 2\sigma g_n(1 - g_n) + \tau(1 - g_n)^2$.

If we consider the same problem when there are infinitesimal sampling intervals, then we consider a function $p(t)$ such that for some small step size, say h , $p(nh) = g_n$.

Suppose that $\rho = 1 + \alpha h$, $\sigma = 1 + \beta h$, and $\tau = 1 + \gamma h$. Show that

$$\dot{p} = p(1 - p)((\alpha - 2\beta + \gamma)p - \alpha + \beta)$$

in the limit $h \rightarrow 0$. This is the equation describing the continuous time analogue of the model gene pool.

Under what conditions on the fitnesses will there be a stable polymorphism, that is, a stable equilibrium for this equation where $0 < p(\infty) < 1$? This ensures that both alleles are permanently established in the population.

8.5. Hardy-Weinberg Equilibria

Plot $(g_n^2, 2g_n(1-g_n), (1-g_n)^2)$ on a DeFinetti diagram. That is, use triangular coordinates to plot any point, say (x, y, z) , in E^3 for which $x \geq 0, y \geq 0, z \geq 0$ and $x + y + z = 1$. To do this, begin with an equilateral triangle whose altitude is 1. Then show that for any point in the interior of the triangle, the sum of the distances to the three sides is 1. Label the vertices X, Y , and Z . Plot the point (x, y, z) by moving x units away from the side opposite X , y units from the side opposite Y , and z units away from the side opposite Z . What is the locus of points $(g^2, 2g(1-g), (1-g)^2)$ as g increases over the interval $0 \leq g \leq 1$?

8.6. Markov Chain Analysis*

A Markov chain is defined by the probabilities of moving among its possible states at each time. Let the states be labeled by $i = 1, \dots, M$, and let $P_{i,j}$ be the probability that the process moves from state i to state j in one time step. The probability that the process is in state i at time n is denoted by $p_{i,n}$. The state probabilities at the next sampling time will be

$$p_{j,n+1} = \sum_{i=1}^M p_{i,n} P_{i,j}$$

for $j = 1, \dots, M$, the number of possible states. Setting $\mathbf{P} = (P_{i,j})$, and $\mathbf{p}_n = (p_1, \dots, p_M)$, we can describe these changes by the vector equation

$$\mathbf{p}_{n+1} = \mathbf{p}_n \mathbf{P}.$$

Therefore, back-substitutions gives

$$\mathbf{p}_n = \mathbf{p}_0 \mathbf{P}^n.$$

This exercise shows us how to evaluate this formula.

The matrix $\mathbf{P}$ can be written in its spectral decomposition

$$\mathbf{P} = \sum_{j=1}^M \lambda_j \mathbf{P}_j,$$

where λ_j is the j th eigenvalue of $\mathbf{P}$, and $\mathbf{P}_j$ is a projection matrix. That is, for each j , the equation $\mathbf{p}\mathbf{P} = \lambda_j \mathbf{p}$ has a nonzero solution for $\mathbf{p}$, and for each $i \neq j$, $\mathbf{P}_i \mathbf{P}_j = 0$ and $\mathbf{P}_i^2 = \mathbf{P}_i$. Using these facts, show that

$$\mathbf{P}^n = \sum_{j=1}^M \lambda_j^n \mathbf{P}_j.$$

All of the eigenvalues of $\mathbf{P}$ satisfy $|\lambda_j| \leq 1$. If the eigenvalues satisfy $\lambda_1 = \dots = \lambda_k$ and $|\lambda_j| < 1$ for $j = k+1, \dots, M$, then show that

$$\mathbf{p}_n \rightarrow \mathbf{p}_0(\mathbf{P}_1 + \dots + \mathbf{P}_k).$$

So, this simple formula gives the final distribution of the process among the possible states.

Apply this result to the Fisher-Wright chain for $N = 5$. What is the population's distribution after 10 generations?

8.7. Metabolic Pathway

Show that the eigenvectors of the metabolic flux matrix in Section 8.2.4 list all possible metabolic pathways through the network shown there.

8.8. Cusp Surface and Three Static States

Identify the sets of parameters for which the HPK system described in Section 8.2.3 has at least two stable operating states. That is, plot the solutions of the differential equation

$$-\varepsilon \dot{x} = \gamma x^3 - (\sigma + 2cK)x^2 + \frac{(b+c)}{a}\gamma x - \sigma \frac{(b+c)}{a}.$$

8.9. Control of Plasmid Copy Number

Simulate the full plasmid regulation model with $\beta = 2$, $\alpha = 0.01, 0.05$, and other data equal to 1.

8.10. A One-Locus, Three-Allele Genetic Trait

Consider a one-locus, three-allele trait having alleles **a**, **b**, **c**. Now, the gene pool has three parts, and so it can be modeled using the frequencies p, q, r , where p is the proportion of the gene pool that is of type **a**, q that of type **b**, and r that of type **c**. Derive a model for how these frequencies change with time, say p_n, q_n, r_n . Solve the model in the cases where there is slow selection and plot your answers for (p_n, q_n, r_n) using triangular coordinates (note that $p_n + q_n + r_n \equiv 1$).

9

A Theory of Epidemics

The spread of a contagious disease involves interactions of two populations: the susceptibles and the infectives. In some diseases these two populations are from different species. For example, malaria is not passed directly between animals but by anopheline mosquitoes, and schistosomiasis is passed from animal to animal only through contact with water harboring snails that can incubate the disease-causing helminths. In other diseases, the infection can be passed directly from infectives to susceptibles: Viral diseases like chickenpox, measles, and influenza, and bacterial diseases like tuberculosis can pass through a population much as fire spreads through a forest.

There are useful analogies between epidemics and chemical reactions. A theory of epidemics was derived by W.O. Kermack, a chemist, and A.G. McKendrick, a physician, who worked at the Royal College of Surgeons in Edinburgh between 1900 and 1930. They introduced and used many novel mathematical ideas in studies of populations. One important result of theirs is that an infection determines a threshold size for the susceptible population, above which an epidemic will propagate. Their theoretical epidemic threshold is observed in practice, and it measures the extent to which a real population is vulnerable to spread of an epidemic. At roughly the same time, V.I. Semenov derived a theory of combustion that identified explosion limits beyond which combinations of pressure and temperature cause chemicals to begin explosive chain-branched reactions. The two calculations are quite similar. Thresholds of population size beyond which there might be an epidemic are derived and discussed in this chapter.

Propagation of infection is modeled to determine what aspects of a population might be controlled to reduce the risk of an epidemic. We begin with the Reed-Frost model, which describes the spread of infection in a small population based on random sampling (See Appendix B). The second model is due to Kermack and McKendrick; it is a nonrandom model that describes propagation in large populations. The connections between these models are described in the second section, and a calculus-based version of the Kermack McKendrick model is derived and solved in the third section.

These models can serve as building blocks to study other diseases, for example those having intermediate hosts, and diseases in stratified populations where there are mixing groups, like families, preschools, schools, and age, gender and social groups, that have various contact probabilities. Some of these extensions are described in the exercises.

9.1 Spread of Infection Within a Family

What happens when an infectious disease is introduced into a small family? What is the likelihood of spread within the family? We can answer these questions by carefully counting the possibilities.

Consider a fixed population and assume that three kinds of individuals in it are defined by a disease: susceptibles, infectives, and removals. Susceptibles can acquire the infection upon effective contact with an infective, infectives have the disease and are capable of transmitting it, and removals are those who have passed through the disease process but are no longer susceptible or infective. The flow of the disease is described by the graph

$$S \rightarrow I \rightarrow R,$$

indicating that susceptibles might become infectives, and infectives might be removed, but there is no supply of new susceptibles to the process, etc.

There are problems with taking such a simple view of a disease. For example, there are great variations in the level of susceptibility, infectiousness, and immunity among individuals in populations. Also, these definitions may depend on stratifying attributes such as age groups, genetic type, or mixing groups. There are many diseases whose transmission mechanisms are not known, as well as diseases for which the latency period, the time between becoming infective and the appearance of symptoms, is not known.

Despite all of these limitations, we proceed to model the spread of a disease. The population is monitored at fixed sampling times, and the numbers in these subpopulations at the n th sampling time are labeled S_n , I_n , and R_n , respectively. The population is assumed to be constant, say with size N , during the course of the disease, so

$$S_n + I_n + R_n = N.$$

The population is also assumed to be thoroughly mixing, so that each individual has equal probability to make effective contact with any other individual. Let p denote the probability of effective contact between any susceptible and any infective during one sampling interval. Finally, we suppose that the sampling intervals are exactly the same length as the duration of infectiousness in infectives. At the end of the n th sampling interval, the population is examined and new numbers S_{n+1} , I_{n+1} , and R_{n+1} are observed and recorded.

The probability that a susceptible avoids contact with any one infective is $1 - p$ and with all I_n of the infectives during the sampling interval is

$$q_n = (1 - p)^{I_n}.$$

Therefore, the probability that $S_{n+1} = k$ is

$$\binom{S_n}{k} q_n^k (1 - q_n)^{S_n - k},$$

which is a simple *binomial distribution*. That is, the probability that k survive the sampling interval as susceptibles is the number of ways k can be selected from among S_n candidates times the probability that k avoid effective contact and the probability that $S_n - k$ have effective contact. If $S_{n+1} = k$, then $I_{n+1} = S_n - k$ and $R_{n+1} = R_n + I_n$.

This calculation is summarized by the formula

$$\Pr[S_{n+1} = k \mid S_n \text{ and } I_n] = \binom{S_n}{k} q_n^k (1 - q_n)^{S_n - k}, \quad (9.1.1)$$

which we read as “the probability that $S_{n+1} = k$ given S_n and I_n is” This is called the *Reed-Frost model*. This is a Markov chain model, but it is more complicated than those studied in Chapter 8, since the probability q_n depends on I_n and so changes from one sampling time to the next.

It is not possible to solve the Reed-Frost model, but some experiments using a hand calculator or computer give indications of how a disease propagates. The approach described here is the Monte Carlo simulation of a random process.

For example, suppose that $S_0 = 4$, $I_0 = 1$, and $p = 0.05$. Table 9.1 shows the probabilities of various outcomes at the first sampling time. From Table 9.1 we see that it is most likely that the infection will die out without recruiting any new infectives, but there is a very small chance that all susceptibles will become infective during this first interval.

One possible evolution of the disease can be simulated using these probabilities. First, plot the probabilities in Table 9.1

	a	b	c	d	e
0	0.81	0.98	0.999	0.9995	1.0

Figure 9.1. Probability intervals (not to scale) for the data in Table 9.1.

Table 9.1. Disease in a Family of Five: The First Two Weeks ($S_0 = 4$, $I_0 = 1$, $p = 0.05$, Period of Infection is Two Weeks)

	Observation			Probability of this Observation
	S_1	I_1	R_1	
<i>a</i>	4	0	1	$\Pr[S_1 = 4, I_1 = 0] \approx 0.814$
<i>b</i>	3	1	1	$\Pr[S_1 = 3, I_1 = 1] = 0.172$
<i>c</i>	2	2	1	$\Pr[S_1 = 2, I_1 = 2] = 0.0135$
<i>d</i>	1	3	1	$\Pr[S_1 = 1, I_1 = 3] = 0.000475$
<i>e</i>	0	4	1	$\Pr[S_1 = 0, I_1 = 4] = 0.000005$

Table 9.2. Disease in a Family of Five after Two Periods ($S_1 = 3$, $I_1 = 1$)

	Observation			Probability of this Observation
	S_2	I_2	R_2	
<i>a</i>	3	0	2	$\Pr[S_2 = 3, I_2 = 0] = 0.857$
<i>b</i>	2	1	2	$\Pr[S_2 = 2, I_2 = 1] = 0.135$
<i>c</i>	1	2	2	$\Pr[S_2 = 1, I_2 = 2] = 0.00712$
<i>d</i>	0	3	2	$\Pr[S_2 = 0, I_2 = 3] = 0.000125$

on a unit interval as shown in Figure 9.1. Next, select a random number from a hand calculator, a table of random numbers, or from a computer. Plot the number on the unit interval. It will fall within one of the five intervals (most probably in *a* or *b*), and so select the letter of that interval as the outcome of the experiment. For example, the number might be 0.96, so interval *b* is selected, and we say that

$$S_1 = 3 \quad \text{and} \quad I_1 = 1.$$

The process can be continued: First, we recalculate the probabilities of possible outcomes as shown in Table 9.2. Again we split the unit interval into subintervals corresponding to these probabilities. Now there are four intervals. Suppose that we draw 0.9. This will fall in interval *b*, so we have

$$S_2 = 2 \quad \text{and} \quad I_2 = 1.$$

Table 9.3 shows the possibilities if we begin with these values. This time we draw 0.25 (interval *a*), so the infection has died out of the family after infecting three, while two survived the 6 weeks in which the infection was present.

We have just simulated one sample path of the Reed-Frost process using a methodology called Monte Carlo simulation. An improved way to do this is at each step to select a random number first and then calculate the lengths of the intervals *a*, *b*, ..., until the random number is hit. This saves computation of all probabilities in most cases.

Table 9.3. Disease in a Family of Five after Three Periods ($S_2 = 2, I_2 = 1$)

	Observation			Probability of this Observation
	$S_3,$	$I_3,$	R_3	
<i>a</i>	2	0	3	$\Pr[S_3 = 2, I_3 = 0] = 0.902$
<i>b</i>	1	1	3	$\Pr[S_3 = 1, I_3 = 1] = 0.095$
<i>c</i>	0	2	3	$\Pr[S_3 = 0, I_3 = 2] = 0.0025$

It is interesting to create many sample paths and then to calculate statistics of the results using this population of sample paths. In this way, we can calculate the average duration of the disease in a family, the expected number of people infected, etc. To illustrate this, we consider a disease in a family of five under three different assumptions about its infectiousness.

We consider the Reed-Frost model in three cases: $p = 0.1$, $p = 0.3$, and $p = 0.5$. In each case, we repeat the simulation, as in Tables 9.1-9.3, 40 times. That is, we consider the spread in 40 identical families as predicted by the Reed-Frost model. We will score the infection's success in each case by determining the number of susceptibles who have not been infected at the time when the infection disappears from the population. We call this S_∞ to indicate that it is the ultimate value of this number, although in fact, the disease always drops out in no more than five steps. $S_\infty = 2$ in the simulation in Table 9.3.

The results of these simulations are presented in Table 9.4.

Table 9.4. Final Size for 40 Sample Paths of the Reed-Frost Model for Each of Three Choices for p .

S_∞	$p = 0.1$	$p = 0.3$	$p = 0.5$
0	1	12	23
1	1	8	12
2	5	3	1
3	11	3	1
4	22	14	3
	—	—	—
	40	40	40

When $p = 0.1$, one of the 40 simulations resulted in no surviving susceptibles ($S_\infty = 0$), but 22 resulted in no further infectives appearing. When $p = 0.5$, 23 of the simulations resulted in all becoming infectives ($S_\infty = 0$). This table can be constructed in less than an hour using a hand calculator, or much more quickly, even using many more iterations than 40, with a programmable calculator or a computer.

Table 9.4 illustrates several important points. When the disease is highly infectious ($p = 0.5$), there are three cases where an epidemic did not run

($S_\infty = 4$). However, in 23 cases there are no survivors. In the case of low infectiousness ($p = 0.1$), only two sample paths result in severe epidemics where $S_\infty = 0$ or 1.

Note that as p increases, the distribution of survivors changes. It moves from one whose mode (highest frequency cell) is at $S_\infty = 4$ for ($p = .1$), to one having two modes ($S_\infty = 0$ and $S_\infty = 4$), to one that has a single mode at $S_\infty = 0$. In all cases there are some simulations where the infection dies out without recruiting any new infectives. This becomes unlikely as p increases, but it is as likely as a severe epidemic in the middle case. Is $p = .3$ near a threshold value beyond which an epidemic will propagate? It is difficult to answer this question directly from the Reed-Frost model, but we answer it in the next section by taking a different point of view.

Simulations of this kind have been important in planning immunization scheduling in large populations, as was done for polio vaccines. Also, they provide the only practical way for discovering the behavior of Markov chains that are as complicated as the Reed-Frost model.

As the population becomes large, we can turn away from the random model and get similar useful information about thresholds from a nonrandom model.

9.2 The Threshold of an Epidemic

In order to avoid confusion with the notation used in the preceding section, where the variables S_n , I_n , and R_n denote random variables whose distributions were prescribed by the Reed-Frost model, we let x_n denote the number of susceptibles, y_n the number of infectives, and z_n the number of removals, so $x_n + y_n + z_n \equiv N$. These numbers will change according to the Kermack-McKendrick model that we derive next.

It might be expected that the number of susceptibles in the next sampling interval would be

$$x_{n+1} = (1-p)^{y_n} x_n, \quad (9.2.1)$$

since the factor $(1-p)^{y_n}$ is the proportion of susceptibles who avoid effective contact with all infectives. In fact, $x_n = E\{S_{n+1} | S_n, I_n\}$, the expected value of S_{n+1} given $S_n = x_n$ and $I_n = y_n$. However, this formula does not tell how the expected values of the susceptibles $E\{S_n\}$ evolve; those numbers are given by the formulas

$$E\{S_{n+1} | S_n, I_n\} = (1-p)^{I_n} S_n,$$

but they involve I_n . So the connection between $E\{S_{n+1}\}$ and $E\{S_n\}$ does not follow directly from this formula. Still, we will compare the implications of the model (9.2.1) with the results obtained from the Reed-Frost model (9.1.1) to understand how the two models are related.

Let $a = -\log(1-p)$, so that e^{-a} is the probability that a given susceptible will successfully avoid contact with each infective during the sampling interval. We will now let go of the assumption that the sampling interval is the same as the interval of infectiousness. Suppose that those leaving the susceptible class enter directly into the infectious class, and that a proportion, say b , of the infectives remains infective at the end of each sampling interval. Then

$$\begin{aligned}x_{n+1} &= e^{-ay_n} x_n, \\ y_{n+1} &= (1 - e^{-ay_n}) x_n + b y_n.\end{aligned}\tag{9.2.2}$$

We do not keep track of the removals, since the formula

$$z_{n+1} = z_n + (1 - b)y_n$$

determines their numbers, and they play no further role in the disease process. The model (9.2.2) is the *Kermack-McKendrick* model.

Will an infective more than replace itself? Let us look at the first step assuming that $y_0 = 1$:

$$y_1 - 1 = b - 1 + (1 - e^{-a})x_0.$$

This is positive if

$$x_0 > (1 - b)/(1 - e^{-a}) \equiv x^*.\tag{9.2.3}$$

If the susceptible population size exceeds the critical number x^* , then an epidemic will propagate, since the infectives gain new recruits. The number $1 - b$ is the probability of an infective not surviving to the next sampling period, so improved scrutiny and treatment of the population by public health officers can increase this number. The number $1 - e^{-a}$ is the probability of being infected. This can be decreased by reducing contacts between individuals, for example by quarantine.

Formula (9.2.3) gives x^* as the threshold value of the population. It gives a useful criterion for controlling a disease. We can increase the threshold level in either of the two ways just discussed or we can reduce the susceptible population below x^* . All of these techniques have been used in various disease outbreaks over time. Of course, the best solution to epidemics would be its eradication or at least the discovery of cures for the disease.

Table 9.5 presents the results of three simulations using the Kermack-McKendrick model: These are for a large threshold, an intermediate threshold, and a small threshold.

The numbers x_∞ and y_∞ denote the (approximate) limit values of x_n and y_n as $n \rightarrow \infty$. In the first case, the infectives never replace themselves, and the disease does not propagate. In the second case, most of the susceptible population is eventually infected, so a severe epidemic takes place over five sampling intervals. Note that some susceptibles survived the outbreak. In the last case, a short and intense outbreak occurs with all of the susceptible population being infected within three sampling intervals.

Table 9.5. Solutions of the Kermack-McKendrick Model in Three Cases: ($b = 0$, $x_0 = 25$, and $y_0 = 5$ in each Case)1. $a = 0.02$. The threshold population size is $x^* \sim 50$.

Sampling Interval	x_n	y_n
n		
0	25	5
1	22.6	2.379
2	21.6	1.051
3	21.1	0.4
∞	$x_\infty = 20.8$	$y_\infty = 0$

2. $a = 0.07$. The threshold population size is $x^* \approx 15$.

Sampling Interval	x_n	y_n
n		
0	25	5
1	17.6	7.4
2	10.5	7.1
3	6.4	4.1
4	4.8	1.6
5	4.3	0.5
∞	$x_\infty = 4.1$	$y_\infty = 0$

3. $a = 0.149$. The threshold population size is $x^* \approx 10$.

Sampling Interval	x_n	y_n
n		
0	25	5
1	11.9	13.1
2	1.7	10.2
3	0.4	1.3
∞	$x_\infty = 0.3$	$y_\infty = 0$

The simulations in Table 9.5 can be related to those in Table 9.4 by observing that if we scale the susceptible and infective populations, then the value of a scales by the inverse factor, since the model for $u_n = x_n/5$ and $v_n = y_n/5$ is

$$\begin{aligned}u_{n+1} &= e^{-5av_n}u_n, \\v_{n+1} &= (1 - e^{-5av_n})u_n + b v_n.\end{aligned}$$

In the three cases in Table 9.5, $a = 0.02$, 0.07 , and 0.149 , respectively. Evaluating the formula $p = 1 - e^{-5a}$ in each case gives 0.1 , 0.3 , and 0.5 ,

respectively, which are the values used in Table 9.4. As in Table 9.4, the first and third cases resulted in no epidemic and a severe one, respectively, and an epidemic leaving some survivors resulted in the second case. The value x_∞ for the Kermack-McKendrick model gives the approximate value of the Reed-Frost model of S_∞ corresponding to the epidemics that *do* propagate. Although x_∞ does not give the expected value of S_∞ in each case, it does indicate the size of most of the epidemics that do run. In this way, we see a connection between the Reed-Frost model, which is based on random sampling, and the Kermack-McKendrick model, which involves no randomness.

9.3 Predicting the Severity of an Epidemic

If an epidemic runs, how severe will it be? That is, what will x_∞ be? Although this calculation can be carried out directly for the model (9.2.2), it is interesting to consider the case of infinitesimal sampling intervals to take advantage of calculus. Suppose that there are smooth functions $S(t)$, $I(t)$, and $R(t)$ and a small sampling interval h such that $x_n = S(nh)$, $y_n = I(nh)$, and $z_n = R(nh)$. Since the sampling interval h is small, we must rescale a and b : Let $a = rh$ and $b = 1 - h\sigma$. This is plausible, since when $h = 0$ there should be no change in the numbers. Then setting $t = nh$, we have

$$\begin{aligned} S(t+h) &= e^{-rhI(t)}S(t), \\ I(t+h) &= (1-h\sigma)I(t) + (1-e^{-rhI(t)})S(t), \\ R(t+h) &= R(t) + (1-(1-h\sigma))I(t), \end{aligned}$$

and so

$$\begin{aligned} S(t+h) - S(t) &= (e^{-rhI(t)} - 1)S(t) = -rhI(t)S(t) + O(h^2), \\ I(t+h) - I(t) &= -h\sigma I(t) + (1 - e^{-rhI(t)})S(t), \\ &= -h\sigma I(t) + rhI(t)S(t) + O(h^2), \\ R(t+h) - R(t) &= h\sigma I(t). \end{aligned}$$

Dividing these equations by h and passing to the limit $h = 0$ gives three differential equations for $S(t)$, $I(t)$, and $R(t)$. We write these as

$$\begin{aligned} \frac{dS}{dt} &= -r I S, \\ \frac{dI}{dt} &= r I S - \sigma I, \\ \frac{dR}{dt} &= \sigma I. \end{aligned} \tag{9.3.1}$$

This system of equations is the continuous-time version of Kermack and McKendrick's model (9.2.2). Incidentally, the calculation just completed gives a neat derivation of the *law of mass action* in chemistry, in which the rate at which two chemical species having concentrations S and I interact, is proportional to the product SI . Thus, the law of mass action can be derived from the binomial distribution of random interactions, since the expected number of interactions occurring in a specified (short) time interval is

$$(1 - q_n)S_n = (1 - \exp(-rhI_n))S_n \approx rhI_nS_n.$$

Obviously, this law and the two models derived for epidemics depend on the assumption that the populations are thoroughly mixing as the process continues.

We can solve these differential equations and so determine the severity of an epidemic. Taking the ratio of the first two equations gives

$$\frac{dS}{dI} = -\frac{rIS}{rIS - \sigma I} = \frac{-rS}{rS - \sigma}.$$

Therefore,

$$dI = \left(\frac{\sigma}{rS} - 1 \right) dS.$$

Integrating this equation gives

$$I = (\sigma/r) \log S - S + C,$$

where C is a constant of integration that is determined by the initial conditions:

$$C = I_0 - (\sigma/r) \log S_0 + S_0.$$

Typical trajectories are shown in Figure 9.2, where $S^* = 1$, $I_0 = 0.1$, and $S_0 = 5S^*, 1.5S^*, S^*$, respectively.

In the infinitesimal sampling process, the threshold level of S^* is

$$S^* = (1 - b)/(1 - e^{-a}) \approx \sigma/r,$$

which is the value of S for which $dI/dt = 0$. We see in Figure 9.2 that trajectories starting near but above this value describe epidemics that end at a comparable distance below this value. Trajectories that start well above S^* end up near $S = 0$. However, in each case the final size of the susceptible population, $S(\infty)$, is where the trajectory approaches the S axis. Therefore, solving the equation

$$S - (\sigma/r) \log S = S_0 - (\sigma/r) \log S_0$$

for the smaller of its two roots gives the final size. This is not possible to do in a convenient form; however, it is easy to do using a computer. In this way, we can estimate an epidemic's severity once we have estimated the infectiousness (a or r) and the removal rate (b or σ) (see Exercise 9.3).

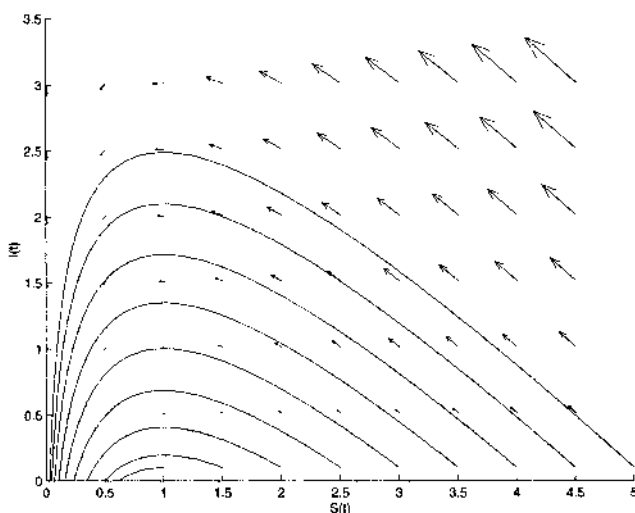


Figure 9.2. I vs. S in nine cases: $S_0 = S^* = 1$, $S_0 = 1.5 S^*$, ..., $S_0 = 5 S^*$. $I_0 = 0.1$ in each case. The vector field shown in this figure was produced using the `quiver` command in Matlab. It shows the local directional behavior of solutions at several initial points in the SI -plane.

Recurrent diseases. There are diseases in which removals can eventually become susceptible again. This is the case for a variety of sexually transmitted diseases, for example gonorrhoea. The flow of such a disease is depicted by the graph

$$S \rightarrow I \rightarrow S.$$

Without further discussion, we can write down a model of such a disease:

$$\begin{aligned} dS/dt &= -rIS + \sigma I, \\ dI/dt &= rIS - \sigma I. \end{aligned}$$

Since $I + S$ is constant (its derivative is zero), we can reduce these equations to the single equation

$$dS/dt = (\sigma - rS)(I_0 + S_0 - S).$$

We see that if $S^* = \sigma/r < I_0 + S_0$, then $S \rightarrow S^*$! When $S \rightarrow S^*$, the disease is endemic. In the other case, $S \rightarrow I_0 + S_0$, and the infection dies out of the population.

Stratification of the population, latency periods of the disease, hidden carriers, seasonal cycling of contact rates, and many other factors confound the study of epidemics, but the simple models derived here provide useful and interesting insights that have guided disease control policies.

9.4 Annotated References

The following classical papers in the theory of epidemics are still worth reviewing.

Kermack, W.O., and McKendrick, A.G.: A contribution to the theory of epidemics, *Proc. Roy. Soc.(A)* **115** (1927), 700–721, **139** (1932), 55–83.

McKendrick, A.G.: Mathematics applied to medical problems, *Roy. Soc. Edinb.*, 1926.

Resources for modeling and data of epidemics can be found in the following sources:

Bailey, N.T.J.: *The Theory of Infectious Diseases and Its Applications*, Charles Griffin, London, 1975.

May, R.M., and Anderson, R.M.: Transmission dynamics of HIV infections, *Nature* **326** (1987), 37–42.

Hoppensteadt, F.C.: *Mathematical Methods of Population Biology*, Cambridge University Press, 1982.

An excellent introduction to computer simulation of random processes is the book

Hammersley, J.M., and Handscombe, D.C.: *Monte Carlo Methods*, Methuen, London, 1964.

Exercises

9.1. Reed–Frost Simulation in a Family

Carry out the Reed–Frost simulation as described in Tables 9.1 through 9.3.

9.2. Final Size of Reed–Frost Epidemics

Reproduce the results in Table 9.4 by carrying out the Reed–Frost simulation described there.

9.3. Kermack–McKendrick Threshold

Consider the sequence x_n, y_n that is generated by the Kermack–McKendrick model (9.2.2) with $z_0 = 0$.

1. Show that the following limit exists:

$$x_\infty = \lim_{n \rightarrow \infty} x_n.$$

2. Show that

$$\lim_{n \rightarrow \infty} y_n = 0.$$

(Hint: Use the first equation in (9.2.2).) Deduce that $x_\infty = N - x_\infty$.

3. Show that $x_n = x_0 \exp(-az_n/(1-b))$.
 4. Let $F = x_\infty/x_0$. F measures the final size of the susceptible population, and $1 - F$ is the severity of the epidemic. Show that

$$F = \exp\left(-\frac{ax_0}{1-b} \left(1 + \frac{y_0}{x_0} - F\right)\right).$$

5. Let $T = ax_0/(1-b)$ and $y_0/x_0 = i = 0.01$. Solve the equation in part 4 of this exercise, for F for values of T , $0 \leq T \leq 3.0$, by steps of 0.01 using Newton's method. Plot your answers by plotting F vs. T .
 6. Explain what happens at $T = 1$.

9.4. Final Size of Kermack-McKendrick Epidemics

Reproduce the results in Table 9.5.

9.5. Asymptomatic Carriers

Consider a population in which there are infectives who cannot be detected. We denote their number by C . In that case, the Kermack-McKendrick model becomes

$$x_{n+1} = x_n e^{-a(y_n + C)}.$$

Describe what happens to the solution in this case.

9.6. The Law of Mass Action

Begin with the binomial distribution and derive the law of mass action. In particular, $E[S_1 - S \mid S, I] = S(1-p)^I - S$. Show that if p is near 0, then this is (approximately) proportional to IS .

9.7. Solutions of the Kermack-McKendrick Model

We obtained the equation $I = (\sigma/r) \log S - S + C$ by solving system (9.3.1). Set $I = 0$ and solve for S using Newton's method. Plot your results as a function of rS_0/σ . Use $r/\sigma = 1$. Compare this result to the one you obtained in Exercise 9.3 using F .

9.8. Epidemics with Intermediate Classes

Some diseases place new recruits from the susceptible class into an "exposed, but not yet infectious" class, which we denote by E . The graph $S \rightarrow E \rightarrow I \rightarrow R$ describes the process.

1. Derive a mathematical model for this disease using either difference equations or differential equations. Consider two cases:

- (a) A certain proportion of the exposed group becomes infective at each sampling time ($dE/dt = -r_0 E$).
- (b) Individuals remain in the exposed class for a fixed time, say T , ($dE/dt = r_0 I(t)S(t) - r_0 I(t-T)S(t-T)$).

2. Derive a threshold result for this system in either of the cases in part 1.

9.9. Spread of an Infection in Two Small Families

Formulate a Reed–Frost model for the spread of a disease within and between two identical families of five each. Assume that there is a different contact rate between families than within them. Simulate the spread of an infection in the combined group beginning in one family. Estimate the severity of the epidemic and present your results in tabular form as in Table 9.4.

9.10. Schistosomiasis

Schistosomiasis is a disease caused by a wormlike parasite (a helminth). Male and female helminths must mate in a host (e.g., humans, ducks, or swine). Thereafter, some of the fertilized eggs leave the host in its feces. When an egg comes in contact with fresh water, it hatches and attempts to find a snail that it can penetrate. Once a snail is infected, a large number of larvae are produced. They swim freely in search of a host in which to reproduce, and they might penetrate the skin of a host or be ingested with water or food grown in the water. We sample the populations at regular intervals, and we denote by H_n the mean number of worms infecting each host, I_n is the number of infected snails. Then

$$\begin{aligned} H_{n+1} &= (1 - \mu)H_n + c I_n, \\ I_{n+1} &= (1 - \delta)I_n + b \frac{(S - I_n)H_n^2}{1 + H_n}, \end{aligned}$$

where μ and δ are death probabilities of hosts and snails, c is the number of helminths becoming established in a host due to each infected snail, S is the number of snails (fixed) and the term $bH_n^2/(1 + H_n)$ gives the mean number of paired worms per infected host and so is proportional to the number of eggs produced. The last term describes the interaction between eggs and susceptible snails.

Consider the steady-state distributions of helminths and infected snails: $H_n = H$ and $I_n = I$. The equations for these values are

$$\begin{aligned} H &= (1 - \mu)H + c I, \\ I &= (1 - \delta)I + b \frac{(S - I)H^2}{1 + H}. \end{aligned}$$

1. Describe the static states of the schistosomiasis model.
2. Which of these solutions are stable?

9.11. Recurrent Diseases: Relapse–Recovery Model

Relapse–recovery diseases are denoted by the graph

$$S \rightarrow I \rightarrow S.$$

Suppose that after a fixed length of time, say τ , an infective returns to being fully susceptible again. Then instead of the model in Section 9.3, we have

$$\begin{aligned}\frac{dS}{dt} &= -r I(t)S(t) + r I(t-\tau)S(t-\tau), \\ \frac{dI}{dt} &= -r I(t-\tau)S(t-\tau) + r I(t)S(t).\end{aligned}$$

Since $I + S = N$, we have the single equation

$$dI/dt = r(I(t)(N - I(t)) - I(t-\tau)(N - I(t-\tau))).$$

1. It is known that $I \rightarrow I^*$, a constant, as $t \rightarrow \infty$. Determine the value of I^* .
2. Test the stability of I^* .

Patterns of Population Growth and Dispersal

Bacteria in our gut, insects in a field, trees in a forest, and plants borne by air and water around the world are among the many examples of spatial distributions of organisms that are important and interesting to study. Mechanisms of pattern formation and dispersal operate on a wide range of spatial scales from microscopic random movements to long-distance transport by air or water. On the other hand, organisms can also reflect uneven distributions of nutrients or toxins in their growth patterns. These are important, for example, in paleontology, where the distributions of microbial fossils are used to find oil, and they provide places to look for new species of microbes, such as petroleum-digesting bacteria that are now used to clean oil tanks.

While patterns can be formed by organisms through random or directed movement, growth or accretion patterns can also be formed by temporal fluctuations in their environment. When fluctuating patterns are frozen in time, like fossils, they provide records of the history of fluctuations through accretion patterns. On the other hand, immobilized organisms, like some bacteria attached to plant roots, can find themselves in habitats where nutrients are washed through the soil in regular or irregular cycles. In these cases, temporal patterns of growth can result that are important for plant life.

Patterns of living organisms can be important in how they influence their environment or how they survive adversity. For example, there are foci of infectious diseases in human populations, like those for measles in city neighborhoods, and foci of infestation insects in forests, like the spruce budworm in Canadian forests, that cause recurring outbreaks. Identifying

these foci can lead to eradication of the infection or infestation. In another case, patterns of bacterial growth in habitats having gradients of nutrients, antibiotics, or mutagens are used as diagnostic tools in medicine, like the Ames test, and as engineering tools in biotechnology, as in biofilms used in waste remediation. In addition, the geographic distributions of blood types in humans have provided useful data for anthropologists and historians: These distributions are records of major events like migrations or long periods of isolation. Finally, patterns of bacteria in our intestines create microbial ecologies that block invasion by pathogens, digest nutrients, and create products that are essential to us: By cell count, there are 10-100 bacteria in our bodies for every "normal" cell, and they reside in complex, structured ecosystems.

Many of the ideas in the preceding chapters can be applied to understanding aspects of patterns. We begin with a study of Brownian motion by using a discrete-time version called *random walks*. The result is a Markov chain model that we can analyze in a number of useful and interesting ways. In particular, this gives us a way to describe how nutrients and cells can diffuse through a habitat, and so it enables us to model and simulate several important processes. Three experiments are studied in detail here: First, we consider diffusion of a drop of nutrient in an agar pour plate. Then, we describe how accretion patterns of cells can form as a result of interactions between cell growth and the quality of the environment. Finally, we model how patterns can form as a result of interactions between cells and two required nutrients.

10.1 Random Walks and the Process of Diffusion

We have seen how random phenomena can sometimes be described using Markov chains; for instance, the evolution of genetic traits and the spread of an infection in a family were modeled in Chapters 8 and 9, respectively, using probabilities of the various possible outcomes and a matrix of transition probabilities over each sampling interval. These discrete-time, discrete-state models can be studied using computer simulations, but it is usually difficult to "solve" them. Of course, the word "solution" has various meanings to various people. For example, a computer simulation might be exactly what is needed to interpret or design an experiment. On the other hand, if one wishes to determine how sensitive or robust a phenomenon is to changes in environmental parameters, a formula that shows explicitly how the solution depends on the parameters would be desirable.

Most random processes are closely related to normal distributions (See Appendix B). This important fact is used by statisticians, and the result is usually summarized in a central limit theorem that shows how to scale random variables to relate them to normal distributions. Once a connection

has been established between a problem and a normal distribution, many useful statistical methods, like the least-squares method in Chapter 7, can be applied to estimate parameters and to establish confidence intervals. Although we do not go into this statistical theory here, we do derive an important connection between random-walk Markov chains and normal distributions. In particular, we show in this chapter that the random-walk Markov chain can be approximated by a differential equation, the *diffusion equation*, whose solutions are normal distributions. Similar approaches can be used to study the genetics problems described in Chapter 8 and the epidemics in Chapter 9.

Observations made by the botanist R. Brown in the early nineteenth century described microscopic particles in suspensions that could follow highly irregular paths. It took nearly a century of work by physicists and mathematicians to create a theory that described this phenomenon in mathematical terms that could be used to describe where such particles would go. Among those involved in this work were Fourier, Einstein, and Wiener. The theory is now referred to as the theory of Brownian motion, and it involves some of the best and most interesting mathematical methods.

In this section we study a discrete-time, discrete-space version of Brownian motion in more detail and derive useful mathematical methods to describe it. We will see how measurement of a cell's position in a solution defines a random variable, how the random variable is related to a random walk, and how this can be used to interpret and precisely describe experimental observations.

10.1.1 Random Walks

Brownian motion can be described in terms of random walks. Consider a line of trees around a lake, some of which are infested with an insect, like the fruit fly *Drosophila*, and count the number of insects in each tree at certain sampling times. We keep track of how these numbers change over many sampling times, and so build up a profile of the insect population's distribution.

We can simulate this process in the following way. Suppose that in each sampling interval the numbers in each tree change according to the following rules:

1. A proportion q of insects in one tree jumps to the tree on the left.
2. A proportion q jumps to the tree on the right.
3. A proportion $1 - 2q$ remains in the tree. (We suppose that $q < \frac{1}{2}$.)

Such insects are said to be performing a random walk.

This is enough to formulate a very interesting model. First, a labeling of the trees is needed. We pick one tree, marked with 1, to set a frame of

Table 10.1. Random Walk with $q = 0.05$

$n \backslash m$	4	3	2	1	M	M - 1	M - 2
0				1			
1			0.05	0.9	0.05		
2		0.0025	0.09	0.815	0.09	0.0025	
3	0.0001	0.0058	0.12285	0.7425	0.12285	0.0058	0.0001

reference, and we locate tree addresses from that site; say m is the m th tree counterclockwise from 1 around the lake. The labeling goes around until the last tree, M , which stands next to tree 1. The proportion of insects in the m th tree at the n th sampling is denoted by the numbers $U_{m,n}$ = proportion of the population that is in tree m at time n . Therefore, the total value of the U 's at any time is 1:

$$\sum_{m=1}^M U_{m,n} = 1$$

for each n , since these numbers represent proportions of the total population.

We suppose that the number of insects greatly exceeds the number of trees, but we ignore the wind, habitat preference, and many other features. Also, we suppose that there is no reproduction or death during the time period we are monitoring.

The number in the m th tree at sampling time $n+1$ will be the proportion that remain there from the previous time plus those that immigrate from one side plus those that immigrate from the other. This is described by

$$U_{m,n+1} = (1 - 2q)U_{m,n} + q U_{m+1,n} + q U_{m-1,n} \quad (10.1.1)$$

for $m = 1, \dots, M$. When necessary in this formula, we replace $M + 1$ by 1 and 0 by M to ensure that the implied periodicities are met. This model provides a qualitative description of the infestation. It shows how the numbers change in time among the trees surrounding the lake. Table 10.1 shows a simulation of this process when only tree 1 is occupied at first and when $q = 0.05$.

10.1.2 Boundary Conditions

We have imposed a constraint on the model by assuming that it is around a lake, so the neighbors of tree $m = 1$ are $m = 2$ and $m = M$, etc. This is referred to as a periodic boundary condition, because it indicates what happens at the ends of the labeling domain that we are studying. There are other boundary conditions we might be interested in, such as those in which the two end trees are like any others, except that they each have only one neighbor (called reflecting boundary conditions), or the two end trees

are controlled to be zero (called absorbing boundary conditions). These are considered in the exercises, but we restrict attention here to problems that have periodic boundary conditions. All three of these boundary conditions arise naturally in the study of various random processes.

Since we have normalized the counts in each tree by the total population size, we can consider the vector $\vec{p}_n = (U_{1,n}, \dots, U_{M,n})$ to be a probability vector that describes the distribution of the insects among the trees. With this, we can rewrite the model using concise matrix notation as we did earlier for Markov chains.

In the case of periodic boundary conditions, we have that

$$\vec{p}_{n+1} = \vec{p}_n \begin{pmatrix} 1-2q & q & 0 & \dots & 0 & 0 & q \\ q & 1-2q & q & \dots & 0 & 0 & 0 \\ 0 & q & 1-2q & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1-2q & q & 0 \\ 0 & 0 & 0 & \dots & q & 1-2q & q \\ q & 0 & 0 & \dots & 0 & q & 1-2q \end{pmatrix}.$$

The matrix of probabilities of possible changes reflects the periodic boundary conditions in this case, and this model is referred to as a *periodic Markov chain*. This matrix has an interesting structure that we use later in this chapter. In particular, we can write it as $I + qT$ where I is the $M \times M$ identity matrix and T has the special form

$$T = \begin{pmatrix} -2 & 1 & 0 & \dots & 1 \\ 1 & -2 & 1 & \dots & 0 \\ 0 & 1 & -2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & -2 \end{pmatrix}.$$

With this the equation can be written simply as

$$\vec{p}_{n+1} = \vec{p}_n(I + qT).$$

Similar formulations can be made for the cases of reflecting and absorbing boundaries and for the unlimited domain.

Another way to look at this tree model is to straighten out the line of trees and place many exact copies of itself end to end in a line. So, in analogy, along a highway there are identical stretches of trees, each of length M . We can use the integers to identify trees, and each interval of M trees will be identical with any other that is M units away. So along the highway we see the numbers $1, \dots, M$, over and over.

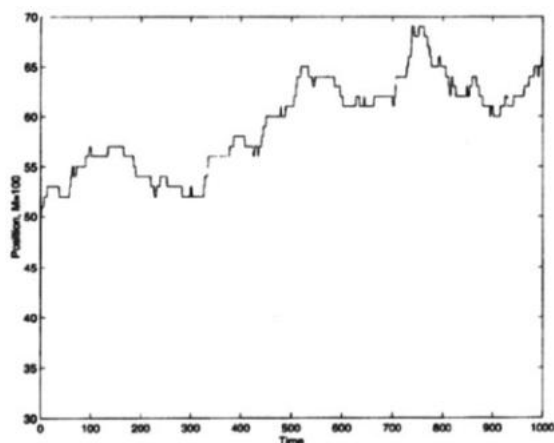


Figure 10.1. One-dimensional random walk sample path generated by a Monte Carlo simulation.

10.1.3 Sample Paths

It is possible to describe typical sample paths of individual insects using a Monte Carlo simulation, like the ones in Chapter 9. We do this using the simple equation

$$x_{n+1} = x_n + z_n \text{ modulo } M,$$

where $x_0 \in [0, \dots, M-1]$. (Recall that any integer, say k , can be written uniquely in the form $k = mM + r$ for some integers m, r where $0 \leq r < M$. The integer M is called the base or radix, m is called the quotient, and r the remainder or residue of k modulo M , and we define new notation by writing $k \equiv r$ modulo M , or simply $k = r$ modulo M .) The number $z_n \in \{-1, 0, 1\}$ is a random number whose distribution is $P(z_n = -1) = q$, $P(z_n = 1) = q$, and $P(z_n = 0) = 1 - 2q$.

The simulation in Figure 10.1 proceeds in the following way. The unit interval is divided into three subintervals, two of length q and one of length $1 - 2q$. These intervals are labeled "Left," "Right," and "Stay," respectively. A number is selected using a random number generator, as in Chapter 9, and if the number is in the first interval, we take $z_0 = -1$; if in the second, $z_0 = 1$; otherwise, we take $z_0 = 0$.

Take $q = 0.05$ as in Table 10.1. The three intervals are described by

Left: $0 \leq r < 0.05$, Right: $0.05 \leq r < 0.10$, Stay: $0.10 \leq r \leq 1$,

respectively. We select a sequence of random numbers, and list the corresponding moves as shown in Figure 10.1. This insect doesn't get far from home even in 1000 steps. It is possible, though improbable, that every step is to the right, etc.

10.1.4 Diffusion Approximation

Computers can be used to simulate random walks and related random processes in even more sophisticated ways. We show in this section that the random walk can be described by a differential equation that we can solve. Suppose that $u(x, t)$ is a smooth function of x and of t such that the numbers $U_{m,n}$ are just evaluations of u at sampling sites $x = m\Delta x$ and sampling times $t = n\Delta t$ where Δx and Δt are spacings between trees and between sampling times, respectively. That is, if the distance between trees is Δx and between sampling times is Δt , then

$$U_{m,n} = u(m\Delta x, n\Delta t).$$

Putting this into equation (10.1.1), we have that

$$u(x, t + \Delta t) = u(x, t) + q(u(x + \Delta x, t) - 2u(x, t) + u(x - \Delta x, t)),$$

where we have set $x = m\Delta x$, $t = n\Delta t$, and we have ignored the periodic boundary conditions for the moment. Dividing both sides by Δt and by Δx^2 gives

$$\frac{u(x, t + \Delta t) - u(x, t)}{\Delta t} = \frac{q\Delta x^2}{\Delta t} \frac{u(x + \Delta x, t) - 2u(x, t) + u(x - \Delta x, t)}{\Delta x^2}.$$

This is called the diffusion approximation to the Markov chain. A. Einstein argued that it is correct for Brownian motion to take the limit $\Delta t, \Delta x \rightarrow 0$ in such a way that the ratio $q\Delta x^2/\Delta t$ approaches a constant, say D :

$$q \frac{\Delta x^2}{\Delta t} \rightarrow D, \text{ as } \Delta t \rightarrow 0, \Delta x \rightarrow 0.$$

D is a constant called the *diffusivity*. This assumption has been corroborated in many experiments.

We will pass to the limit as $\Delta t, \Delta x \rightarrow 0$ holding $D = q\Delta x^2/\Delta t$ constant and holding the circumference of the lake $L = \Delta x M$ constant as well, so in this process the number of trees and the number of sampling times go to infinity (i.e., $M \rightarrow \infty$) while L stays fixed. The circle of trees becomes like a circular hedge in this limit.

A short calculus computation shows that the left side is approximately $\partial u/\partial t$ and the righthand side is approximately $D\partial^2 u/\partial x^2$, so the process should be described by a function that satisfies the equation

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}.$$

Such a function is

$$u(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right),$$

which can be verified by simply substituting this function into each side of the equation and checking that everything balances. This function is

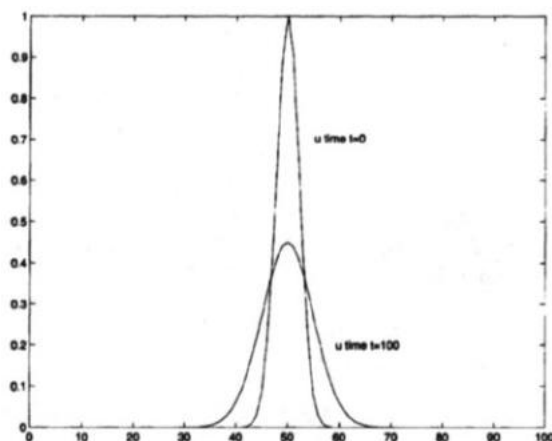


Figure 10.2. Diffusion of a Gaussian (normal) drop in 1 dimension. Shown here are the initial distribution and the distribution after 100 diffusion time units.

referred to as the *Gaussian or normal density function*. It is a normal distribution for a random variable (the position x) having mean 0 and variance $2Dt$ (See Appendix B). So, the distribution is bell shaped, but it spreads out as t increases. Two samples of this are shown in Figure 10.2.

Even though this function does not satisfy periodic boundary conditions, which are that $u(x + L, t) = u(x, t)$ for all x and t , we will see that its shape appears repeatedly in our computer simulations. There are ways to construct a solution of the equation that does satisfy the correct boundary conditions using this function, but we do not pursue them here.

10.1.5 Two-Dimensional Diffusion

These same kinds of calculations can be made for two-dimensional domains, which we must consider to study bacterial growth in petri plates. Here is the analogy: Consider a forest that extends over the whole county, but suppose that each section (e.g., acre, hectare, or square mile) in the forest is exactly identical: the same kinds of trees in the same relative locations. Thus, the forest is a periodic array of forestlets, each looking exactly like the other. So, for this system, we need only consider what happens in one section. The trees at the boundary of a section have neighbors that are the same as those in their same section but across it. We do as above and get a function $u(x, y, t)$ where the coordinates of points in the forest are (x, y) , and a single section is described by the square $0 \leq x, y \leq L$. The diffusion equation is the same (we take $\Delta x = \Delta y$) except for having an

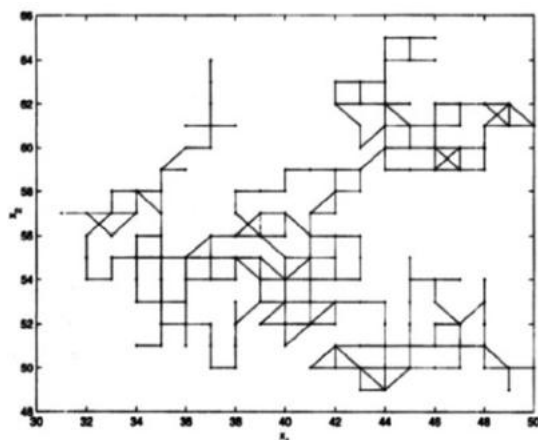


Figure 10.3. A sample path for a random walk in the plane where the process begins at the point $(50, 50)$.

added dimension:

$$\frac{\partial u}{\partial t} = D \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right).$$

The function

$$u(x, y, t) = \frac{1}{4\pi Dt} \exp \left(-\frac{x^2 + y^2}{4Dt} \right)$$

solves this equation, but as for the 1-dimensional case, it does not satisfy the periodic boundary conditions. This is the 2-dimensional Gaussian density function. It describes a bell-like surface that spreads out as t increases as described earlier.

10.1.6 Sample Paths in Two-Dimensions

The random walk of insects in a two-dimensional forest can be described as in the case around the lake, and sample paths of individual movements can be described using a difference equation: If (x_n, y_n) denotes the position in the forest of an individual at the n th time sample, then

$$(x_{n+1}, y_{n+1}) = (x_n, y_n) + \sigma(z_1, z_2),$$

where at each step z_1 and z_2 are identical, statistically independent random variables that are ± 1 or 0 with probabilities q and $1 - 2q$, respectively. A sample path is shown in Figure 10.3

We can simulate this process many times and construct histograms of where insects actually go, but further thought gives us a way to determine this directly using diffusion equations as we did in one-dimensional systems.

10.1.7 Numerical Solutions: The Method of Lines

Why not use the Markov chain to simulate these diffusion processes using the random-walk approach? If the spatial step sizes are small and the time steps are small, we must take $q\Delta x^2/\Delta t < \frac{1}{2}$ to ensure that our computed solution is near the real one. This turns out to be a serious constraint on Δt . But we can take advantage of fast, sophisticated numerical solvers for ordinary differential equations to avoid this restriction. We will compute the solutions of diffusion equations using the method of lines: We will discretize only the space variable and leave time as a continuous variable. This makes it possible to simplify the computations, but still take advantage of highly accurate differential equation solvers.

Because we are interested in eventually applying our methods to populations of bacteria on petri pour plates, the relevant boundary condition is that the walls of the plate are reflecting; that is, there is no flux of material through them. However, it is similar to this and somewhat faster in computation to use periodic boundary conditions.

Let $U_{k,j}(t)$ denote the proportion of insects in the tree at position (k, j) on our patch of forest at time t . The model is now a system of ordinary differential equations

$$\frac{dU_{k,j}(t)}{dt} = (D/\delta x^2)(U_{k+1,j}(t) + U_{k-1,j}(t) + U_{k,j+1}(t) + U_{k,j-1}(t) - 4U_{k,j}(t))$$

for $k, j = 1, \dots, M$. Therefore, in a domain with 20×20 grid points, we must solve these 400 ordinary differential equations. This is easy to do using Matlab.

To solve diffusion equations with periodic boundary conditions using this approach, we begin with a function $u(x, y, t)$ defined over a square $0 \leq x, y \leq L$. Next, we select a grid size for this square, say $\Delta x = \Delta y = 1/M$, where M is an integer, and then for $k, j = 1, \dots, M$ we define

$$U_{k,j}(t) = u(k\Delta x, j\Delta x, t).$$

So, for each t the matrix $U(t)$ represents samples of the function $u(x, y, t)$ over the grid, with spatial sampling intervals of size $1/M$. The result is a system of differential equations for the matrix U like that derived in the preceding paragraphs. To account for periodic boundary conditions and at the same time make the notation concise, we introduce the matrix T as in the last section: The result is

$$\dot{U} = (D/\delta x^2)(TU + UT).$$

Recall that the matrix T is the tridiagonal matrix whose rows are zeros except for 1, -2, 1 centered at the main diagonal:

$$T = \begin{pmatrix} -2 & 1 & 0 & \dots & 1 \\ 1 & -2 & 1 & \dots & 0 \\ 0 & 1 & -2 & \dots & 0 \\ 0 & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & -2 \end{pmatrix}.$$

The term TU accounts for the derivatives in the x direction, and the term UT accounts for derivatives in the y direction. The periodic boundary conditions in each of the x and y directions are built into this matrix. Exercise 10.14 describes a Matlab program for solving this system quickly.

10.2 Bacterial Growth on a Petri Plate

It is difficult to grow bacteria without getting patterns of some sort. There are several mechanism for irregular growth in response to a fluctuating environment and movement toward or away from chemical or physical signals.

10.2.1 Diffusion of a Drop of Substrate

The ideas of Brownian motion and diffusion can help to describe how a quantity of substrate will move through a supportive matrix, like an agar gel.

A petri pour plate consists of a shallow dish containing some matrix to support a solution. Agar is frequently used for this. Since we are interested in the spread of chemicals in a pour plate, we should consider diffusion in three dimensions. However, since the dish is shallow, we make the assumption (which is usually supported by observations) that these processes are adequately described by the diffusion equation in two dimensions (x, y) . In particular, we let $S(x, y, t)$ denote the concentration of substrate at position (x, y) at time t , and we suppose that this quantity satisfies the diffusion equation

$$\frac{\partial S}{\partial t} = D \left[\frac{\partial^2 S}{\partial x^2} + \frac{\partial^2 S}{\partial y^2} \right],$$

where D is the coefficient of diffusivity of the substrate in this matrix. For example, we suppose that the dish is square. So, we suppose that $0 \leq x \leq L$ and $0 \leq y \leq L$. A drop of solute placed at the center of the dish can be described by the function

$$S(x, y, 0) = A \exp \left(-\alpha((x - L/2)^2 + (y - L/2)^2) \right),$$

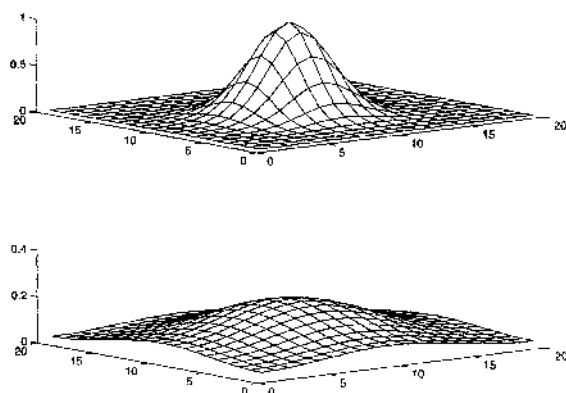


Figure 10.4. Diffusion of a Gaussian drop in 2 dimensions. Top: Initial distribution of U . Bottom: Distribution of U at time $t = 20$.

which describes a bell-shaped distribution of substrate at the start of the experiment. Its maximum concentration is A , and α describes how spread out the drop is.

We will do a numerical simulation of how the concentration changes by solving the diffusion equation using the method of lines and periodic boundary conditions.

The numerical problem is

$$\dot{U} = (D/\Delta x^2)(TU + UT),$$

where the initial values at time $t = 0$ are

$$U_{k,j}(0) = A \exp(-\delta((k - M/2)^2 + (j - M/2)^2)),$$

which describes the concentration of substrate at the start of the process, and T is the tridiagonal matrix described earlier. Here $\delta = \alpha\Delta x$. Figure 10.4 shows the initial configuration and the distribution of a drop at time $t = 20$, where the sets of points $(k, j, U_{k,j}(0))$ and $(k, j, U_{k,j}(20))$ are plotted as surfaces. In this way we can plot the distribution of substrate.

Since the initial drop resembles a normal distribution, we expect that at the beginning the concentration will resemble the Gaussian distribution

$$S(x, y, t) \approx \frac{C_0}{4\pi Dt} \exp\left[-\frac{x^2 + y^2}{4Dt}\right].$$

As long as this is the case, that is, before the drop interacts with the boundary, we can estimate how a line of constant concentration of S will change: In fact, if $S(x, y, t) = C$, then $r^2 \equiv x^2 + y^2$ will satisfy

$$r^2 \approx 4Dt \log[(4\pi Dt)C/C_0].$$

This shows that the effective coverage of substrate will spread out at approximately the rate dr/dt , which is proportional to $1/\sqrt{t}$. So, because of radial dilution the front of constant concentration of S slows down.

10.2.2 Bacterial Growth Due to Diffusing Nutrition

A lawn of bacteria is spread uniformly on the surface of the pour plate and a drop of limiting nutrient that the cells need to grow is placed at the center. What will happen?

We can describe the growth of cells on the plate using the Jacob-Monod model of cell growth from Chapter 7. The bacteria on the plate are fixed, and they grow as they take up substrate. The accretion pattern that develops can be described by a function $P(x, y, t)$ that denotes the number of cells at position (x, y) on the plate at time t . Then

$$\frac{dP}{dt} = \frac{VSP}{K + S},$$

where V is the maximum uptake rate and K is the saturation constant of substrate. If S is near zero, growth is slow. If S is near K , the cells are growing at half their maximum. If S is much larger than K , the cells are growing at their limiting rate V . Thus, we say that K is the concentration needed for cell growth.

The dynamics between the cells and the substrate are described by the mathematical model

$$\begin{aligned}\frac{\partial S}{\partial t} &= D\left(\frac{\partial^2 S}{\partial x^2} + \frac{\partial^2 S}{\partial y^2}\right) - \frac{VSP}{K + S}, \\ \frac{\partial P}{\partial t} &= \frac{VSP}{K + S}.\end{aligned}$$

(The yield Y from Chapter 7 is taken here to be $Y = 1$.)

The full two-dimensional model is studied further in the Exercises 4, 5, 13, and 14. Such a system arises when the bacteria are histidine auxotrophs and the substrate supplied by the drop is histidine. We will describe a slightly more complicated case next, so this is not illustrated here.

10.2.3 Growth Patterns Due to Two Nutrients

As we saw in Chapter 7, cells require nutrition, supplied here by the substrate and a favorable environment, especially a favorable pH. Yet, cells produce acids as a byproduct of metabolism, so they influence their local pH when growing. Chemicals called buffers are used to maintain constant pH even in the face of titration by acids. Therefore, we add an interesting and useful feature to the model of the previous section; namely, we account for the buffer concentration (G) as describing the local pH of the medium.

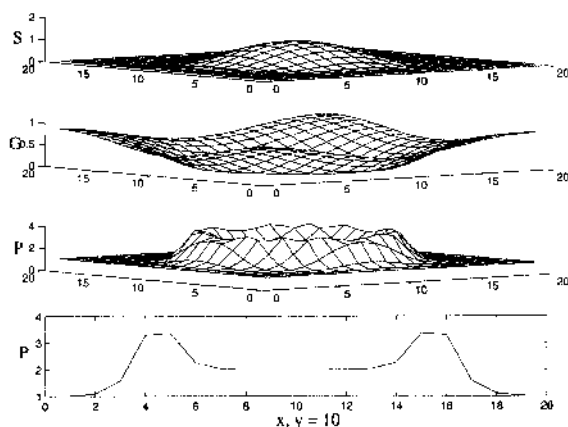


Figure 10.5. Simulation of the SPG model. The patterns of substrate and buffer at time $t = 100$ are shown in the first two plots. The third plot shows the pattern of bacteria at this time, and the last plot shows a transect of the population taken through the middle of the plate, parallel to the x -axis. In this simulation $D_S = D_G = 0.1$, $\delta = 0.1$, $P_0 = G_0 = 1$, and $A = 10$, $V = K = 1$. A drop of nutrient, $S(x, y, 0)$, is placed at the center as described in Section 10.2.2.

The mathematical model now is

$$\begin{aligned}\frac{\partial S}{\partial t} &= D\nabla^2 S - VSGP/(K + SG), \\ \frac{\partial P}{\partial t} &= VSGP/(K + SG), \\ \frac{\partial G}{\partial t} &= D\nabla^2 G - VSGP/(K + SG).\end{aligned}$$

where we use the more concise notation $\nabla = (\partial/\partial x, \partial/\partial y)$ and $\nabla^2 = (\partial/\partial x)^2 + (\partial/\partial y)^2$. Here we have modified the Jacob-Monod term to account for cells needing acceptable levels of both buffer and substrate. This is done by allowing the saturation constant to vary as $1/G$ does, so if G is small, the growth is small, etc. With this, $VS/((K/G) + S) = VGS/(K + S)$. We refer to this as the SPG model.

The computer solution of this model using the method of lines is shown in Figure 10.5.

This figure shows that the interaction between substrate and buffer that is mediated by the cells creates a central core of growth bounded by a band of high growth. If this process is allowed to continue, more bands of growth will appear until the concentrations of both S and G are low everywhere. Interaction among nutrient, pH, and the cells provides this mechanism for pattern formation.

10.3 Concluding Remarks

The discussion of the diffusion approximation completes our introduction to the mathematics of populations. We have seen in the preceding chapters that discrete-time and continuous-time population phenomena can be modeled using nonrandom difference and differential equations and random models such as Markov chains.

An important question is, What is the relation between the nonrandom and the random models of the same phenomenon? This is not easy to answer in all cases, but the calculations in this chapter give some further insight into such questions.

It is always possible to derive a corresponding nonrandom version of the random model, although care is needed in how the results are interpreted. For example, in Chapter 9, the Reed-Frost model of disease is closely related to the Kermack-McKendrick model, which describes the mean of the epidemics in the Reed-Frost model that actually do run when the population is large. But, the Kermack-McKendrick model does not estimate the probability that a disease will drop out of a population, which is possible even when it is super-threshold. If the population is small, random sampling can cause the disease to drop out in a few steps. For this the Kermack-McKendrick model is not useful.

Markov chain models are attractive because they are easy to derive and to simulate on a computer. Analyzing them can be "tough as nails." Fortunately, associated with each Markov chain is a diffusion approximation whose solution describes the probability distribution of the states of the process. Finding the diffusion approximation is often useful, but it may not always be possible. In many population problems, the variance of the random process, which measures how spread out the probabilities are, is inversely proportional to population size. Therefore, the variance is reduced to near zero when large populations are considered, and it is possible to show, using stochastic differential equations, how the random models pass over into the nonrandom ones as population size increases.

There is a close connection between the theory of chemical reactions and the theory of population problems, especially epidemics, but also in biogeography. Reaction-diffusion systems in chemistry have the form

$$\frac{\partial u}{\partial t} = D \nabla^2 u + g(t, x, u),$$

where u is a vector of concentrations of chemicals, D describes their diffusivities, and $g(t, x, u)$ describes chemical reactions at position x at time t . Similar equations can be derived for the biogeography of genetic traits, ecological systems, and epidemics. Analysis of such problems leads to understanding more complicated aspects of spatial patterns of growth. An interesting example of this is a model of genetic clines that was formulated and studied by R.A. Fisher (See Exercise 10.10).

10.4 Annotated References

Several important works on population biology have involved diffusion processes. The following is a brief list giving starting points to the literature in diverse areas of population biology.

Mourant, A.E. , Kopec, A.C., and Domaniewicz-Sobczak, K.: *The Distribution of Blood Groups and Other Polymorphisms*, 2nd ed. Oxford Univ. Press, New York, 1976.

Murray, J.D.: *Mathematical Biology*, Springer-Verlag, New York, 1989.

Hoppensteadt, F.C.: *Mathematical Methods in Population Biology*, Cambridge University Press, Cambridge, 1982.

An excellent introduction to stochastic differential equation modeling and analysis is given in

Oksendal, B.: *Stochastic Differential Equations*, Springer-Verlag, New York, 1992.

Finally, the following book by Einstein is interesting to look through. It sets up the ideas of Brownian motion on firm physical principles.

Einstein, A.: *Investigation on the Theory of Brownian Movement*, Dover Publics., New York, 1956.

Exercises

10.1. Random Walks

Use the random walk model with $q = 0.25$ and $M = 2$ to determine the evolution of the initial conditions shown below using the reflecting, absorbing, and periodic boundary conditions in three separate simulations.

Trec	-2	-1	0	1	2
# insects	0	0	100	0	0

Construct sample paths of these random walks by using the methods in Chapter 9 for Tables 9.1 through 9.3. Present your results by listing or plotting the location of the sample individual at each sampling time. That is, construct a sample path that an individual might follow.

10.2. Diffusion Equation in an Unlimited Region

Consider the diffusion equation for $-\infty < x < \infty$:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

Show that $u(x, t) = \exp(-x^2/4t)/\sqrt{4\pi t}$ solves this equation, and that for each t , the area under the graph of this function is equal to 1.0. Because of this, we refer to u as a probability distribution.

10.3. Statistics of the Diffusion Equation

Evaluate the mean, variance, and entropy of the distribution $u(x, t)$ in Exercise 10.2. That is, evaluate

$$\begin{aligned}\mu &= \int_{-\infty}^{\infty} x u(x, t) dx, \\ \sigma^2 &= \int_{-\infty}^{\infty} (x - \mu)^2 u(x, t) dx,\end{aligned}$$

and

$$H = - \int_{-\infty}^{\infty} u(x, t) \log u(x, t) dx,$$

respectively.

10.4. Diffusion Equation in a Two-Dimensional Region

1. Derive the diffusion equation in two dimensions (x and y) from the random-walk model as shown in Section 10.2.
2. Show that

$$u = \frac{\exp\left[-\frac{x^2+y^2}{4Dt}\right]}{4\pi Dt}$$

solves this equation.

3. Convert both the equation and the solution to polar coordinates by setting $x = r \cos \theta$ and $y = r \sin \theta$. Verify that u solves the resulting equation.

10.5. Bacterial Growth on a Spread Plate

Consider the model of growth on a spread plate derived in Section 10.2.1. The distribution of nutrient is described by $A u(r, t)$, where u is given by the formula

$$u(r, t) = \frac{e^{-\frac{r^2}{4Dt}}}{4\pi Dt},$$

and the bacterial population density $B(r, t)$ can be determined by solving the equation

$$\frac{dB}{dt} = \frac{\alpha A u(r, t)}{K + A u(r, t)} B(r, t).$$

Solve this problem for $B(r, t)$ by substituting the expression for $u(r, t)$ into this formula. Let $A = K/2$, $A = K$, and $A = 2K$. For each of these choices of A , plot the profile of bacteria density as a function of dish radius r at various times, say t near zero, $t = 1.0$, $t = 10.0$. How large will the region of growth be in each case? The profile you find is the record of the nutrient's movement under the bacterial population.

10.6. Stochastic Matrices

The matrix of probabilities given in Section 10.1.1, which we call $\mathbf{P}$ here, is for a random walk on a periodic domain. $\mathbf{P}$ is doubly stochastic; that is, the sum of each column is one and the sum of each row is one. (A matrix is called stochastic if its row sums are all equal to 1.0.) Show that the probability distribution

$$\mathbf{p}^* = \frac{1}{M+1}(1, 1, \dots, 1)$$

is a stationary distribution for this Markov chain. That is,

$$\mathbf{p}^* = \mathbf{p}^* \mathbf{P}.$$

Conclude that 1.0 is an eigenvalue for the matrix $\mathbf{P}$. Show that all other eigenvalues satisfy $|\lambda| < 1$. Conclude that $\mathbf{p}_n \rightarrow \mathbf{p}^*$ as $n \rightarrow \infty$.

10.7. Fourier Series for Diffusion Equations with Boundaries

If $f(x)$ is a continuously differentiable function that has period X (that is, $f(x+X) = f(x)$ for all x), then it can be written in terms of a Fourier series:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{2\pi nx}{X} + b_n \sin \frac{2\pi nx}{X} \right).$$

Use this fact to find Fourier series solutions of the diffusion equation in the three cases of periodic, reflecting, and absorbing boundary conditions. (Hint: Let

$$u(x, t) = \frac{a_0(t)}{2} + \sum_{n=1}^{\infty} \left(a_n(t) \cos \frac{2\pi nx}{X} + b_n(t) \sin \frac{2\pi nx}{X} \right)$$

and find equations for the unknown functions $a_n(t)$ and $b_n(t)$ in each case.) Conclude that $u(x, t)$ approaches a constant in the first two cases and zero in the third.

10.8. The Solution of the Diffusion Equation, Given $u(x, 0)$

Let $f(x)$ be a differentiable function. Show that the formula

$$u(x, t) = \frac{1}{\sqrt{4\pi Dt}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x-y)^2}{4Dt}\right) f(y) dy$$

defines a solution of the diffusion equation by verifying that it solves the equation.

Show that

$$\lim_{t \rightarrow 0^+} u(x, t) = f(x).$$

10.9. Normal Random Variables

Show by simulation that a normal distribution can be obtained from the random number generator in your hand calculator by the following method: At each step draw 12 random numbers from a calculator, add them up and subtract 6.0.

10.10. A Genetic Cline

Consider a long line of sites, where at each site is located a population that carries a one-locus, two-allele genetic trait. Let $g_{m,n}$ denote the frequency of the gene

pool at site m that is of type **a**. If the sites are isolated, the reproduction of the population would be described by the recursion

$$g_{m,n+1} = f(g_{m,n}),$$

where $f(g) = (\rho g^2 + \sigma g(1-g))/(\rho g^2 + 2\sigma g(1-g) + \tau(1-g)^2)$. We suppose that at each sampling time the population reproduces and then does a random walk to neighboring sites. Thus,

$$g_{m,n+1} = \lambda(f(g_{m+1,n}) + f(g_{m-1,n})) + (1-2\lambda)f(g_{m,n}).$$

Write a computer program to solve this problem for 100 sites using periodic boundary conditions. Suppose that at some sites $\rho > \sigma > \tau$ and at others $\rho < \sigma < \tau$. Using computer simulation, describe the population's final distribution of genotypes. An irregular distribution of gene pool proportions is called a *genetic cline*.

We showed in Exercise 8.4 that iteration of the function f can be rewritten as a differential equation for the frequency of the gene pool that is of type **a**, say $p(t)$, at time t :

$$dp/dt = p(1-p)((\alpha - 2\beta + \gamma)p - \alpha + \beta),$$

where α , β , and γ are the Malthusian fitnesses of the genotypes. Let us denote the right-hand side of this equation by $F(p)$. Then as in 8.4 we have

$$f(g) = g + hF(g).$$

1. Show that if we replace $g_{m,n}$ by a smooth function $g(x, t)$ for which $g(m\delta x, n\delta t) = g_{m,n}$ for small sampling steps δx and small times δt , then the recursion scheme is approximated by the diffusion equation

$$\frac{\partial g}{\partial t} = D \frac{\partial^2 g}{\partial x^2} + F(g).$$

2. If $D = 1$ and $F(g) = g(1-g)(4-8g)$, then show that

$$g(x, t) = (\tanh x + 1)/2$$

is a solution of the diffusion approximation. Therefore, we have a convenient mathematical example of a cline. Explain how this cline is formed and maintained by selective pressures.

10.11. The Game of Life

A topographical map can be divided into small segments, and the numbers of organisms inhabiting each segment can be recorded. The numbers might be placed at the midpoint of each segment and updated at regular sampling-time intervals. The result is a sequence of maps that describes the dynamic changes of the population's spatial distribution—like a motion picture where each frame is one of the updated maps. It is possible to describe both these spatial and temporal movements as the solution of a mathematical model that involves both space and time variables.

The game of life is an idealized version of this, and it nicely illustrates the underlying ideas. The game was introduced by J. Conway in a *Scientific American* article in 1970, and it has captured the interest of mathematicians and students of computer science.

We consider a grid on the plane labeled by integer pairs. We denote by $S_{n,i,j}$ the number of organisms observed at site (i,j) at the n th sampling time. Resolution as fine as the number of organisms might not be needed, or even possible, and the only information collected might be a score that tells whether a site is occupied or vacant at each sampling. For example, we could write $S_{n,i,j} = 1$ if site (i,j) is occupied and $S_{n,i,j} = 0$ otherwise.

We consider the latter case. In the game of life these numbers will change at each tick of a timing clock according to the following rules:

1. An occupied site having two or three neighboring sites occupied will continue to be occupied. (There are *eight* neighbors to each site.)
2. Each occupied site having no, one, or four or more occupied neighbors becomes vacant.
3. Each vacant site becomes occupied if it has exactly three occupied neighbors.

Determine the evolution of the game of life for the following initial distributions:

a.

```

1  1
1  1
1  1
1  1

```

b.

```

0  1  0
0  0  1
1  1  1

```

10.12. Cellular Automata

Another interesting cellular automaton was introduced by Norbert Wiener in 1947. It allows more than two states at each site. We now consider a distribution of sites, each having three possible states. To fix ideas, we use the analogy of a forest infested with defoliating insects. The three possible states of a site are green, infested with defoliating insects, and defoliated. These states are indicated, respectively, by $S = 0$, $S = 1$, and $S = -1$, and the following rules describe their dynamics. In each sampling period:

1. An infested site becomes defoliated ($1 \rightarrow -1$).
2. A defoliated site becomes green ($-1 \rightarrow 0$).
3. A green site becomes infested if any of its four nearest neighbors is infested ($0 \rightarrow 1$ if so).

Describe what happens in each of the following cases. (Determine the period of each configuration and describe the pattern generated by it.)

1.

$$\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \end{array}$$

2.

$$\begin{array}{ccc} 1 & 0 & 0 \\ -1 & -1 & 0 \\ 0 & 1 & 0 \end{array}$$

3. If a period-three focus and a period-four focus are started at different places on the same grid, what happens after a long time?

(Show that if a habitat is seeded with a period-three focus and elsewhere with a period-four focus, then eventually the habitat becomes synchronized with period three.)

Now consider the same system distributed over the surface of a sphere. Suppose each site represents a muscle cell that is at rest (0), contracting (1), or in a refractory state (-1) after contraction. If the same rules for changing the state of sites are used, then a single contracting cell at one pole of the sphere will cause a wave of contraction to spread uniformly around the sphere to the antipode. The refractory wave that follows the wave of contraction will quench activity at the antipode, and the array will return to rest. If at the pole there is a periodic array, then regular waves of contraction will be created, and they will propagate to the antipode in a regular progression, roughly similar to heart contraction. Illustrate this with a computer simulation.

10.13. Fick's Law

Fick's law states that if a solute has concentration $c(x, y, t)$ at position (x, y) in a solution at time t , then the concentration changes to reduce gradients: Conservation of mass is summarized by the statement that changes in the concentration of a mass in solution are due only to the flux of material. That is,

$$\partial c / \partial t = -\nabla \cdot \text{flux} + \text{sources} - \text{sinks}.$$

Fick's Law states that in a solution the flux is

$$\text{flux} = -D\nabla c + fc,$$

where D is the diffusivity of the solute (a quantity that biochemists can measure, having dimensions of L^2/T) and f is the advective velocity, the velocity (units of L/T) at which solute is carried along by movement of the solution.

Therefore,

$$\partial c / \partial t = D\nabla^2 c - \nabla \cdot (fc).$$

Simulate solutions of this equation for $f = 1$.

Simulate the 2-dimensional random walk

$$(x_{n+1}, y_{n+1}) = (x_n, y_n) + \sigma(z_1, z_2),$$

where at each step, z_1 and z_2 are independent random variables that have a normal distribution with mean zero and variance 1. We take σ such that $\sigma^2/2 = D$.

Simulate the solutions of Fick's equation with $f = 0$ and so relate the random walk simulation to a diffusion simulation.

10.14. The SPG Diffusion Model.

Figure 10.5 was generated using the following Matlab program. First, reproduce that figure. Then experiment using this program by defining initial distributions of nutrient and buffer that are not symmetric in the plate.

```
function spgdif
%spgdif.m
global T M D

M=20;
tt=0:0.9:100;
T=-2*eye(M)+diag(ones(M-1,1),1)+diag(ones(M-1,1),-1);
T(1,M)=1;T(M,1)=1;

del=0.1;D=0.1;

s0=zeros(M);s=s0;

for k=1:M, for j=1:M
    s0(k,j)=10*exp(-(k-M/2)^2+(j-M/2)^2);
end, end

p0=ones(M);g0=ones(M);

figure(1),
subplot(3,1,1), mesh(s0),
subplot(3,1,2), mesh(g0),
subplot(3,1,3), mesh(p0)

S0=reshape(s0,M^2,1); P0=reshape(p0,M^2,1);G0=reshape(g0,M^2,1);
v0=[S0;P0;G0]; v=v0;

tic
[t v]=ode45('spgrhs',tt,v0);%options);
toc

[m n]=size(v);
s=reshape(v(m,1:M^2),M,M); p=reshape(v(m,M^2+1:2*M^2),M,M);
g=reshape(v(m,2*M^2+1:3*M^2),M,M);

figure(2),
subplot(4,1,1), mesh(s), subplot(4,1,2), mesh(g)
subplot(4,1,3), mesh(p), subplot(4,1,4),plot(1:M,p(M/2,:))

function vp=spgrhs(t,v)
%spgrhs.m
```

```

global T M D

vp=v;
S=v(1:M^2,1); P=v(M^2+1:2*M^2,1); G=v(2*M^2+1:3*M^2,1);
s=reshape(S,M,M);p=reshape(P,M,M);g=reshape(G,M,M);

sp=-s.*p.*g./(1+s.*g); pp=-sp; gp=sp;

sp=D*(T*s+s*T)+sp; gp=D*(T*g+g*T)+gp;

Sp=reshape(sp,M^2,1);Pp=reshape(pp,M^2,1);Gp=reshape(gp,M^2,1);
vp=[Sp;Pp;Gp];

```

Appendix A

Getting Started with Matrices and MATLAB

Matrices. A *matrix* is a rectangular array of numbers, which we usually write in the form

$$A = \begin{pmatrix} A_{1,1} & \dots & A_{1,N} \\ \vdots & \ddots & \vdots \\ A_{M,1} & \dots & A_{M,N} \end{pmatrix}. \quad (\text{A.0.1})$$

This indicates an array having M rows and N columns, and we say that A is an $M \times N$ matrix.

The transpose of the matrix A is denoted by A^{tr} , or in Matlab by A' , and it is the $N \times M$ matrix B for which

$$B = \begin{pmatrix} A_{1,1} & \dots & A_{M,1} \\ \vdots & \ddots & \vdots \\ A_{1,N} & \dots & A_{N,M} \end{pmatrix}, \quad (\text{A.0.2})$$

which we write as

$$B_{n,m} = A_{m,n}$$

for $m = 1, \dots, M$, $n = 1, \dots, N$.

Much of our work and the computer package Matlab involve matrix calculations, and the purpose of this section is to define some of those operations and to give examples of them using Matlab. ($>$ will denote a Matlab command line. *Do not write $>$ in on a command line!*)

$> A = [1, 2, 3; 4, 5, 6]$ or
(here and below cr denotes carriage return) gives

$A =$

1 2 3

4 5 6

This is a 2×3 matrix where $A_{1,1} = 1$, $A_{2,1} = 4, \dots$, etc.

$> A = \text{ones}(2,5)$ or

gives

$A =$

1 1 1 1 1

1 1 1 1 1

$A_{2,4}$ is found by writing

$> A(2,4)$ or

This gives

$ans =$

1

$> A = \text{zeros}(3,1)$ or

gives

$A =$

0

0

0

Note that if we end a statement with a semicolon $;$, then nothing is printed. For example, $>a=1;$ or returns nothing, but the variable “a” is defined to be 1.

Vectors are matrices that have only one column or row. In the preceding Matlab example A is a column vector.

$> x = [1; 2; 3]$ or

gives a column vector having components 1, 2, 3, respectively, and

$> x = [1 \ 2 \ 3]$ or

gives a row vector having components 1, 2, and 3, respectively.

Matrix Arithmetic. Matrices of the same size can be added by adding their respective components, resulting in a matrix of the same size. A scalar (i.e., a 1×1 matrix) times a matrix is a matrix of the same size having all of its components multiplied by the scalar. Two matrices can also be multiplied if certain of their dimensions agree. There are several cases of interest.

Dot product: (dot). Let the vectors x and y be $N \times 1$ matrices. Then the *dot product* of these two is defined to be

$$x \cdot y = \sum_{n=1}^N x_n y_n.$$

The dot product of two vectors has a geometric interpretation: It is the product of their magnitudes times the cosine of the angle between them. So, $x \cdot y = |x||y| \cos \theta$, where the magnitude of a vector is $|x| = \sqrt{x \cdot x} = \sqrt{\sum_{k=1}^N x_k^2}$. This is also called the Euclidean length of the vector.

```
> x = [1 2]; y = [3 4]; dot(x,y) cr
gives
ans =
11.
```

Two vectors are orthogonal if their dot product is zero.

Matrix multiplications: $(*, \text{kron}, *)$. Let A be an $M \times N$ - matrix and let B be an $N \times K$ matrix. Then $C = AB$ is an $M \times K$ matrix whose m, k -component is

$$C_{m,k} = \sum_{n=1}^N A_{m,n} B_{n,k}$$

for $m = 1, \dots, M$ and $k = 1, \dots, K$. The number $C_{m,k}$ is equivalent to the dot product of the m th row of A with the k th column of B .

For example, consider

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix},$$

$$B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

and their product

$$C = A * B,$$

which in Matlab is accomplished by the command

```
> A = [1 2; 3 4]; B = [5 6; 7 8]; C = A * B cr
resulting in
C =
```

$$\begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

and

```
> D = B * A cr
gives
D =
```

$$\begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix}$$

If x and y are row vectors of the same length, then $\text{dot}(x, y) = x * y'$. The product of an $M \times N$ matrix and an $N \times K$ matrix is an $M \times K$ matrix. We say the inner dimensions (N) are the same. A frequent error in using Matlab involves trying to multiply matrices that are the wrong sizes for multiplication to be defined.

Kronecker product: $\text{kron}(A, B)$. This multiplies each component of A by the matrix B . If A, B are as above, then

```
> kron(A, B) cr
```

gives
`ans =`

5	6	10	12
7	8	14	16
15	18	20	24
21	24	28	32

The Kronecker product of A and B is a matrix whose size is the product of the sizes of A and B .

Component-wise product: (`.*`). The matrix operation $A.*B$ multiplies each component of A by the corresponding component of B having the same row number and column number. If A, B are as above, then

`> A.*B` or
 gives
`ans =`

5	12
21	32

Matrix division: (`/`). Dividing one matrix by another is more complicated than dividing one scalar by another. If we were interested in finding the quotient of two numbers, say a divided by b , we would require that $b \neq 0$ in order to define a/b . There is a comparable requirement for matrices, but it is more complicated.

A *null vector* x of an $M \times N$ matrix A is an $N \times 1$ matrix for which $Ax = 0$, where here 0 denotes the $M \times 1$ matrix of zeros. Two null vectors, say x and y , are said to be *linearly independent* if the only constants a, b for which $ax + by = 0$ are $a = b = 0$. This is a difficult concept to grasp at first, but if you visualize two vectors in space, they are linearly independent if the lines through them are not parallel to each other.

The *nullity* of a matrix A is the largest number of linearly independent null vectors.

`> q = [1 2 3]; null(q)` or
 gives
`ans =`

-0.9636	0
0.1482	-0.8321
0.2224	0.5547

The two (column) null vectors given here are calculated so that they are normal (i.e., have length 1). Obviously, these are not unique as null vectors since any linear combination of null vectors is also a null vector.

Square matrices. A matrix with the same number of rows and columns is called a *square matrix*. The *rank* of an $N \times N$ matrix A is its dimension less its nullity; in other words, $\text{rank} + \text{nullity} = N$. If a matrix has *full*

rank (i.e., $\text{rank}(A) = N$), then its nullity is zero, and so there are no null vectors.

```
> A = eye(3) cr
```

gives

```
A =
```

```
1 0 0
0 1 0
0 0 1
```

```
> rank(A) cr
```

gives

```
ans =
```

```
3.
```

```
> B = ones(3) cr
```

gives

```
B =
```

```
1 1 1
1 1 1
1 1 1
```

```
> rank(B) cr
```

gives

```
ans =
```

```
1.
```

In this case, there are two null vectors; e.g., $(1, 0, -1)'$ and $(1, -2, 1)'$.

Inverse of a matrix. At last we can define matrix division. Suppose that A is an $N \times N$ matrix whose rank is N . Then there is a matrix B such that $AB = BA = I$. This matrix is usually written as $B = A^{-1}$, and it is called the *inverse* of A .

A square matrix has full rank if and only if its determinant is not zero. The determinant is defined in a complicated way that we do not present here, but Matlab can often compute it quickly.

With $A = \text{eye}(3)$ and $B = \text{ones}(3)$ as above, we have

```
> det(A) cr
```

gives

```
ans =
```

```
1
```

```
> det(B) cr
```

gives

```
ans =
```

```
0
```

If $\det(A) \neq 0$, then the inverse of A can be defined. For example,

```
> A = [1 2; 3 4]; B = inv(A) cr
```

gives

$B =$

$$\begin{bmatrix} -2.0000 & 1.0000 \\ 1.5000 & -0.5000 \end{bmatrix}$$

Here's what happens if a matrix has no inverse:

`> A = ones(2); B = inv(A) cr`

gives (here A is a 2×2 matrix all of whose components are 1)

Warning: Matrix is singular to working precision.

`ans =`

`Inf Inf`

`Inf Inf`

This message indicates that the matrix A has no inverse. Note that its rank is 1, but its dimension is 2.

If A and B are $N \times N$ matrices and $\det(B) \neq 0$, then we can divide A by B : We write

`> A/B cr`

With this, Matlab will calculate $\text{inv}(B)$ and give the result $A * \text{inv}(B)$.

In practice, it is good to know the following facts about 2×2 matrices. Consider

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

where a , b , c , and d are some numbers. If its determinant, $ad - bc \neq 0$, then the rank of this matrix is 2. So, it has an inverse, and the inverse is given by the formula

$$\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The eigenvalues of this matrix are determined by solving the characteristic equation

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0.$$

The term $a + d$ is the sum of the main diagonal elements, which is called the *trace* of the matrix. Since the term $ad - bc$ is the determinant of the matrix, this equation is often written as

$$\lambda^2 - \text{tr } \lambda + \det = 0.$$

The quadratic formula shows that

$$\lambda = \frac{\text{tr} \pm \sqrt{\text{tr}^2 - 4 \det}}{2}$$

Matlab can compute the trace, determinant, and eigenvalues of a matrix. For example,

`> A = [1 2; 3 4]; tr(A) cr`

gives

`ans =`

```

5
> det(A) cr
gives
ans =
-2
> eig(A) cr
gives
ans =
-0.3723
5.3723

```

These examples illustrate some of the basic operations that one uses in Matlab. There is extensive help available. For example, try

```
> help eig cr
```

and you will get a description of what the function `eig(A)` can do.

Plotting. Several methods for plotting, or visualizing, solutions and data are available in Matlab. The following have been used in the text.

```
> tt = 0 : .01 : 10; plot(tt, sin(tt)) cr
```

will plot $\sin t$ for $0 \leq t \leq 10$.

```
> plot3(tt, cos(tt), sin(tt)), rotate3don cr
```

will plot a helical curve in three dimensions and allow you to rotate the figure by clicking on the figure and holding the mouse button down as you move it.

There are two useful plotting methods to view large matrices. The first plots in three dimensions the height $A_{i,j}$ at each point (i, j) . For example,

```
> A = eye(10); mesh(A) cr
```

will plot the components of the matrix, and

```
> spy(A) cr
```

will show which components of the matrix are not zero. (See also what the function `image(A)` does.)

It is possible to make interesting animations using the MOVIE facility in Matlab.

Appendix B

Background on Random Processes

Scientists have taken an approach to describing random noise using three important ideas: samples, events, and probability. For example, consider a large population, say N , of cells. These are the samples. The events we consider are the collection of all possible aggregations of these cells: None, singletons, all possible pairs, all possible triples, etc., so there are many possible events, and the probability of each sample will be assumed to be $1/N$. The probability of an event is simply $(1/N)$ times the number of samples in the event. In this way, we have defined for this example the samples, the events of interest, and the probability of all possible events. Our notation for this is as follows: The samples are

$$\Omega = \{\text{all cells in the population}\}.$$

The events are

$$\mathcal{E} = \{\text{all possible subsets of } \Omega \text{ including } \Omega \text{ and } \emptyset\},$$

where $\emptyset$ denotes the empty set (it plays the role of the number 0 for set theory), and the probability of an event E is

$$P(E) = \frac{\text{Number of cells in } E}{N}.$$

In general, the samples Ω form a set of things, and the events $\mathcal{E}$ are a collection of subsets of samples that includes Ω , the empty set $\emptyset$, and is closed under unions and intersections. The probability function P gives for each event $E \in \mathcal{E}$ a number, $0 \leq P(E) \leq 1$, with the properties that (1) $P(\Omega) = 1$, and (2) if A and B are two disjoint events (meaning that their

intersection is empty: $A \cap B = \emptyset$), then

$$P(A) + P(B) = P(A \cup B),$$

where $A \cup B$ denotes all samples that are either in A or in B or in both. The collection $(\Omega, \mathcal{E}, P)$ is called a *probability space*, and it includes all the ideas needed to describe randomness in a way that we can carry out precise calculations.

A *discrete random variable* is a measurement of samples. It is a function, say f , that associates with a given sample a real number, say a , measuring some attribute of the sample. The function f must be defined in a way that is consistent with the events. An example is given by the function that associates with each cell in a population its dry weight. There might be many cells with the same weight, so the inverse image of a given weight a should be an event whose probability we can measure. This is the consistency we require. We write this as $f^{-1}(a) \in \mathcal{E}$ for any $a \in R$. In summary, the consistency needed for a measurement to be a random variable on $(\Omega, \mathcal{E}, P)$ is that the inverse image of any measurement value must be an event, so that its probability can be measured by P .

There are two important statistics associated with random variables that are widely used. They describe where the random values are located and how spread out they are.

The *expected value* of a random variable f (also known as its mean or average) is defined by

$$E[f] = \sum_{a \in f(\Omega)} aP(f^{-1}(a)),$$

which is the sum over the possible values of f weighted by the probability that the value is observed. The *variance* of f is defined to be

$$\text{Var}(f) = E[f^2] - E[f]^2.$$

We suppose in this section that the range of the random variable (i.e., $f(\Omega)$) is a discrete set (like the integers). Otherwise, this sum must be replaced by an integral, and a further discussion of how we evaluate P would be needed, but we do not pursue that further here.

The reason these two statistics are so important is that a basic theorem in probability, called the central limit theorem, states that if random variables are scaled correctly, then they can be approximated by a normal (or Gaussian) random variable whose probability function P is determined completely by its mean and its variance.

Specifically, we define the *cumulative distribution function* of a random variable f by the formula

$$F_f(a) = P(\{\omega \in \Omega : f(\omega) \leq a\}).$$

This function can't decrease as a increases, since the probability of successively larger sets is being measured. The derivative of this function, called

the *probability density function* of f , is denoted by

$$d_f = \frac{dF_f}{da},$$

and if we know this function, we can calculate important things about the random variable f . For example, the probability that measurements will lie in an interval $[A, B]$ is

$$\int_A^B d_f(a) da = P(\{\omega \in \Omega : A \leq f(\omega) \leq B\}),$$

and

$$E(f) = \int_{-\infty}^{\infty} a d_f(a) da.$$

A random variable f is normal (or Gaussian) if its density function is

$$d_f(a) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2}\left(\frac{a-\mu}{\sigma}\right)^2\right),$$

where μ and σ^2 are constants. In fact, as mentioned earlier, in this case $E(f) = \mu$ and $\text{Var}(f) = \sigma^2$. This shows that a normal random variable's distribution is completely determined by its mean and its variance, and this observation and the central limit theorem underlie major parts of probability theory and statistical methods. Once we know that a random variable is normal and we know its mean and variance, then we can calculate that about 68% of observations should be within distance σ of μ , about 95% should be within distance 2σ , and about 99% should be within 3σ of μ .

One last point about random variables. Each observation in Table 7.1 is a value of a random variable where the attribute being measured is the size of the population. Moreover, Table 7.1 depicts a time sequence of random variables that measures the emergence of this attribute over time. Each time we redo this experiment, we will obtain slightly different numbers, and to understand this, we consider a sequence of random variables that are attributes measured at discrete times.

Consider a sequence of random variables, say $\{f_1, f_2, f_3, \dots\}$, that denotes the measurements we would get at the first, second, third, etc., sampling times in the population size experiment. Such a sequence is referred to as a *random process*, a *random signal*, or a *stochastic process*. The term "stochastic" derives from Greek (roughly, "to aim at"). It is used in probability theory because if we aim independent shots of a projectile at a target, a random distribution (usually normal!) of hits is observed. Random processes can be simulated using a computer, and these ideas are used in the later chapters of the book.

One of the simplest, but most useful, random processes is the Markov chain, which was used in various ways in Chapters 8, 9, and 10. To use

this methodology, we change our point of view from looking at single measurements of samples to looking at how the distribution of measurements changes from time to time. For example, we consider a random variable y that can take on one of a finite number of values, say $S_1, \dots, S_M$. If $y = S_m$, we say that the system is in state m . Next, we consider a sequence of these variables in time where we know the initial state, say y_0 , but we don't know exactly what will happen to the sequence y_n at subsequent times. All we know is the probability at each step of moving from one state to another. Let $P_{j,k}$ denote the probability that the system jumps from state j to state k in one time step, and let us suppose that these numbers do not change from one measurement to the next. Note that the elements in the transition probability matrix P are non-negative and the sum across each row is 1.

It is possible to derive a formula for the probability distribution of the states after n steps. To do this, let $p_{k,n} = P\{y_n = S_k\}$. If we could calculate all of these numbers, then, while we wouldn't know exactly where the process is at time step n , we could see where it can possibly be and with what likelihood.

We can determine these numbers in the following way: First,

$$p_{j,n+1} = p_{1,n}P_{1,j} + p_{2,n}P_{2,j} + \dots + p_{M,n}P_{M,j}$$

for $j = 1, \dots, M$. That is, the probability that the process is in state j at the next time ($n+1$) is the probability that it is presently in state 1 times the probability that it jumps from state 1 to state j , plus the probability that it is in state 2 presently times the probability it jumps from 2 to j , plus the probability that it presently is in state 3 times the probability that it jumps from state 3 to state j , etc.

This can be written more concisely using vectors and matrices. Let us denote the population distribution at time n by the row vector

$$\vec{p}_n = (p_{1,n}, \dots, p_{M,n})$$

for each n . This describes the probability distribution of the measurements at time n . (It is a row vector, and the sum of its components is 1, since the process must be in one of the states.) Let us denote the transition probabilities by the matrix

$$P = \begin{pmatrix} P_{1,1} & P_{1,2} & \dots & P_{1,M} \\ P_{2,1} & P_{2,2} & \dots & P_{2,M} \\ \vdots & \vdots & \ddots & \vdots \\ P_{M,1} & P_{M,2} & \dots & P_{M,M} \end{pmatrix}.$$

This is called the *transition probability matrix*. With this notation, we can describe the whole Markov chain process with the equation

$$\vec{p}_{n+1} = \vec{p}_n P.$$

The solution can be found by successive back-substitutions:

$$\vec{p}_n = \vec{p}_0 P^n.$$

This is usually hard to evaluate in a useful form, but Matlab can be used to calculate it.

Matlab simulation of a Markov chain.

Consider a population of fruit flies occupying two neighboring trees. The first tree is less tasty than the second, so there is a probability that a fly in tree 2 will jump to tree 1, and a slightly higher probability of a fly in tree 1 jumping to tree 2. After many observations of this population's dynamics, we estimate that

$$P_{1,1} = 0.4, P_{1,2} = 0.6, P_{2,1} = 0.3, P_{2,2} = 0.7.$$

So, we write

$$> P = [0.4 \ 0.6; 0.3 \ 0.7]; \text{ cr}$$

Then if the trees are equally populated initially, so $\vec{p}_0 = (0.5 \ 0.5)$, we have that after 1 time step

$$\vec{p}_1 = \vec{p}_0 P,$$

and after 10 time steps,

$$\vec{p}_{10} = \vec{p}_0 P^{10}.$$

Matlab can be used to evaluate these formulas:

$$> P = [0.4 \ 0.6; 0.3 \ 0.7]; p0 = [0.5 \ 0.5]; p1 = p0 * P \text{ cr}$$

gives

$$p1 = 0.35 \ 0.65$$

and

$$> p10 = p0 * P^{10} \text{ cr}$$

gives

$$p10 = 0.3333 \ 0.6667$$

Therefore, after many generations, the population will be divided among the two trees in the proportion of 1:2.

Other simulations are described in Chapters 8, 9, and 10.

Index

- absolute temperature, 76
- absorbing state, 280
- accretion pattern, 311, 323
- action potential, 124, 195, 203
- afferent arteriole, 148
- age pyramid, 229
- age structure, 224
- air, alveolar, 75
- air, fresh, 79
- air, inspired, 79
- air, mixed alveolar, 86
- allele, 268
- altitude, 25, 92
- alveolar ventilation, 75, 79
- alveolar-capillary membrane, 75
- alveoli, 75, 82
- Andronov-Hopf bifurcation, 194, 213
- anemia, 25, 92, 269, 272
- angiotensin, 155
- antidiuresis, 150
- antidiuretic hormone (ADH), 150, 157
- aortic valve, 45
- areal coordinates, 274
- arrhythmia, 74
- arterial pulse, 33
- arteries, 34
 - arteries, pulmonary, 6
 - arteries, systemic, 5
 - arteriole, 10
 - arteriole, afferent, 148
 - arteriole, efferent, 149
 - atoll oscillator, 202, 210
 - attachment rate constant, 175
 - attention, 202
 - autoregulation, 10, 25, 64, 68
 - average, 344
 - Avogadro's number, 76, 92
 - axon, 195
- back-substitutions, 222, 239, 279
- bacterial growth, 219
- balanced emergences, 250
- baroreceptor, 22
- baroreceptor loop, 22
- batch culture, 217
- Beverton-Holt model, 239, 255
- bifurcation, 194, 244, 248
- bifurcation diagram, 248
- binomial distribution, 276, 279, 297
- binomial random variable, 304
- biofilm, 236, 312
- biological clock, 197, 199
- biotechnology, 253, 277, 312

- bisection method, 95
- bistable, 289
- blood, 5
- blood, arterial, 86
- blood, venous, 79
- Boltzmann's constant, 76
- boundary condition, 328
- boundary condition, absorbing, 315
- boundary condition, periodic, 314, 318
- boundary condition, reflecting, 314
- Brownian motion, 248, 312, 313, 317, 321
- canonical model, 194, 200, 224
- capillaries, pulmonary, 6
- cardiac output, 7, 19
- carrying capacity, 232, 240, 243, 262
- Central Limit Theorem, 344
- chain-branched chemical reaction, 295
- chaos and randomness, 248
- characteristic equation, 263
- chemostat, 234
- chromosome, 267
- cicadas, 262
- circadian rhythm, 199
- circulation time, 69
- circulation, controlled, 70
- circulation, steady state, 16
- circulation, uncontrolled, 20, 70
- clock, 197, 199
- clock, modulated, 199
- clock, rubber-handed, 209
- cobwebbing, 240, 248, 250
- coefficient of catchability, 254
- cohort diagram, 223, 225, 259
- collecting duct, 150
- common geometric ratio, 222
- compliance vessel, 9
- compliance, blood vessel, 9
- computer simulation, 206, 212, 231, 243, 262, 312
- concentration, 25
- continuous-time model, 219
- control law, 249, 256
- crepuscular activity, 200
- crossbridge, 173
- crossbridge cycle, 173
- current-voltage relation, 117, 118
- cuspid surface, 294
- dead space volume, 79
- DeFinetti diagram, 274, 293
- dendritic tree, 195
- density function, 318, 345
- diabetes insipidus, 150
- diastolic compliance, 23
- diastolic pressure, 34
- diffusion approximation, 317
- diffusion equation, 313
- diffusivity, 317, 321
- diploid cell, 267
- Dirac delta function, 337
- discrete-time model, 219, 312
- dissolved molecules, 77
- distal convoluted tubule, 150
- distal nephron, 150
- distribution function, 344
- diuresis, 150
- diurnal, 200
- dot product, 336
- doubling time, 218
- doubly stochastic matrix, 328
- down-regulation, 289
- drosophila, 313
- ductus arteriosus, 28
- efferent arteriole, 148
- efficiency of gas transport, 83
- effort-yield relation, 254, 256, 262
- Einstein, A., 317
- electrical conductance, 194
- electrical equilibrium, 120
- electroneutrality, 120
- entorhinal cortex, 195, 205
- entropy, 259, 327
- equilibrium of gases, 77
- equilibrium potential, 117
- Euler's renewal theory, 227
- excitatory synapse, 201
- expectation of a random variable, 300
- expected value, 344
- expired air, 79
- exponential growth phase, 217
- feedback loop (chemical), 283
- fetal circulation, 28
- Fibonacci model, 225, 263

- Fick's law, 26, 331
- filter, electronic, 200
- filter, molecular, 148
- final size of epidemic, 306
- Fisher–Wright Markov chain, 276, 292
- Fisher–Wright–Haldane model, 272
- fishery management, 255
- flow, 7
- flow resistance, 110
- flux equilibrium, 114
- FM radio, 196, 200
- focus of infestation, 311
- food chain, 236, 261
- food web, 265
- foramen ovale, 28
- force, 171
- Fourier series, 328
- frequency, 198
- frequency response, 197
- functional genomics, 269

- game of life, 330
- gamete pool, 275
- gamma rhythm, 204
- Gaussian density function, 318
- Gaussian distribution, 322, 345
- gene, 216, 268
- gene pool, 270, 292, 329
- genetic cline, 329
- genetic dominance, 271
- genome project, 268
- genotype, 268, 329
- genotype frequencies, 273
- geometric growth, 217
- geometric iteration, 240
- glomerulus, 148
- growth rate, 218
- Guttman, R., 196

- haploid cell, 267
- Hardy–Weinberg proportions, 272, 274
- harmonic oscillator, 198
- harvest, 253
- heart, left, 5
- heart, right, 6
- helminth, 308
- hemoglobin, 25

- heterosis, 272
- heterozygote, 274
- Hill coefficient, 93
- hillock region of a neuron, 195
- hippocampus, 195, 203, 204
- histidine auxotroph, 238, 323
- histidine protein kinase (HPK), 286
- histogram method, 245, 260
- horizontal transmission, 269
- hormone cycle, 200
- hyperbaric chamber, 93
- hypergeometric Markov chain, 279
- hysteresis, 233

- ideal gas law, 76, 110
- identification of parameters, 18
- inhibitory synapse, 201
- interstitial space, 159
- intrinsic growth rate, 219, 232, 238
- inverse of a matrix, 339
- ion flux, 114
- ionic channel, 199
- isometric force, 171
- iteration, 239

- Jacob–Monod model, 283, 323, 324
- juxtaglomerular apparatus, 155

- Kermack–McKendrick threshold, 306
- Kermack–McKendrick model, 301, 304
- kinase, 286
- Kronecker product of matrices, 337

- L'Hospital's rule, 132
- Laplace transform, 226, 263
- least squares estimate, ff., 220
- life table, 227
- linear independence, 338
- liter, 7
- locus of a gene, 268
- logistic equation, 232, 261
- loop of Henle, 149
- loop of Henle, ascending limb, 149
- loop of Henle, descending limb, 149

- malaria, 272
- Malthus fitness, 329
- Malthus's model, 255

- Markov chain, 276, 277, 279, 293, 312, 346
- Markov chain, periodic, 315
- mass action, 304, 307
- Matlab matrix product, 338
- Matlab program, 332
- matrix, 335
- maximum sustained yield, 253, 262
- mean, 344
- mean arterial concentration, 86
- medial septum, 195, 205
- membrane conductance, 117
- membrane potential, 201
- Mendel's laws of inheritance, 268, 270
- metabolic rate, 26
- metabolite, 290
- method of lines, 320
- method of separation of variables, 219
- mho, 145
- Michaelis-Menten approximation, 282
- microbial ecology, 264
- minimal growth media, 217
- mitral valve, 45
- mixing condition, 304
- modulator site on DNA, 286
- montane vole, 196, 199, 249
- Monte Carlo simulation, 297, 298, 316
- motor neurons, 194
- natural selection, 268
- negative feedback loop, 22
- nephron, 147
- neural engineering, 194
- neural network, 201
- neural sensors, 194
- neuron, 195, 200
- neuron, Hodgkin class 1, 194
- neuron, Hodgkin class 2, 194
- neurotransmitter, 195
- Newton's law, 193
- Newton's method, 307
- normal density function, 318, 328
- normal distribution, 312, 345
- normal resting values, 17
- novelty detection, 203
- null vector, 291, 338
- nullity, 338
- numerical analysis, 320
- Ockham's razor, 194
- oncotic pressure, 149, 153
- operon on DNA, 286
- orbit, 245
- osmotic pressure, 110, 111
- osmotic pressure, partial, 112
- osmotic pressure, relative, 112
- pacemaker, 72
- parameter, not perimeter, 15
- parasympathetic nervous system (PNS), 23
- Parkinson's disease, 195
- partial pressure of a gas, 77
- pendulum, 198
- periodical emergence, 250
- peritubular capillaries, 149, 153
- permabilities, 114
- pH, 323
- pharmaceuticals, 277
- phase locked loop (PLL), 194, 200, 205, 206
- phase locking, 206, 210
- phase locking (PLL), 196
- phase portrait, 231
- phase resetting, 209
- phase variable, 198
- phenotype, 268
- photoperiod, 199
- Picard's method, 167, 240
- pit of extinction, 243
- plasma, 147
- plasmid, 216, 268, 277
- plasmid content, 281
- plasmid incompatibility, 280
- plasmid pool, 277
- plate growth, 217, 318, 320, 321, 323
- plotting output, 341
- polymorphism, 292
- polyploid cell, 267
- population density function, 174
- population transect, 324
- positive part, 203
- pour plate, 312, 318, 321
- predator satiation, 256, 262
- pressure, 8
- pressure of chemical reaction, 282

- probability density function (pdf), 345
- probability space, 344
- promoter segment, 286
- prosthetic device, 195
- proximal convoluted tubule, 149
- pulmonary circulation, 7
- pulse pressure, 34
- pump coefficient, 12

- quiescence, 264

- radix, 316
- random genetic drift, 269, 275
- random process, 345
- random signal, 345
- random variable, 344
- random walk, 312, 326
- rank of a matrix, 339
- recurrent disease, 305
- recursion, 239
- Reed-Frost model, 297, 306
- refractory phase, 217
- regulation, Na^+ , 158
- relapse-recovery disease, 308
- renal vein, 149
- renewal equation, 227, 262, 263
- renewal theorem, 228
- renin, 155
- reproduction curve, 240
- reproduction function, 240, 243, 259
- reproduction time, 215
- residual carrying capacity, 262
- resistance vessel, 9
- resistance, blood vessel, 9
- respiration, 200
- reticular complex, 201
- Ricker's model, 244, 256
- Robinson's equation, 213
- rotation number, 197
- rubber-handed clock, 209

- Salmonella typhimurium*, 233
- sample path, 316, 319, 326
- sampling with replacement, 276
- sampling without replacement, 279
- sarcomere, 173
- saturation constant, 230, 323
- schistosomiasis, 308
- self-regulation, 14
- sexually transmitted disease, 305
- shunt, left-right, 33
- simple clock, 198, 209
- simple solution of gases, 77
- singular perturbation theory, 286
- slow selection, 272
- solubility, 77, 82
- solution, chemical, 77
- solutivory chain, 238
- spatiotemporal pattern of neural activity, 206
- speciation, 272
- spectral decomposition, 293
- stable age distribution, 229, 264
- starvation nutrition regimen, 248, 261
- stationary phase, 217
- steady state, 150
- stimulus-response experiment, 196, 203
- stochastic differential equation, 326
- stochastic matrix, 328
- stochastic process, 345
- stoichiometric matrix, 290
- strange attractor, 260
- substrate, 230
- sympathetic nervous system (SNS), 23
- synchronization of cell reproduction, 249
- synchronized emergence, 250
- synchronized reproduction, 248, 261
- systemic circulation, 7
- systolic pressure, 34

- thalamocortical model, 203
- thalamus, 195, 201
- theta rhythm, 204
- thick filaments, 173
- thin filaments, 173
- threshold for excitation, 140
- threshold, epidemic, 295, 301
- toroidal coordinates, 208
- torus knot, 208
- trace of a matrix, 340
- transition probability matrix, 346
- trapezoidal rule, 166
- triangular coordinates, 260, 274, 293
- tridiagonal matrix, 321, 322

- up-regulation, 289
- uptake rate, 230, 323
- urine, 157
- variance, 327, 344
- veins, pulmonary, 6
- veins, systemic, 6
- velocity, 171
- venous pressure, 19
- ventilation-perfusion ratio, 81, 85
- Verhulst model, 239, 255
- vertical transmission, 268
- vertical transmission of plasmids, 278
- visualization, 206, 341
- voltage-controlled oscillator, 194, 200
- Wiener-Rosenbluth automaton, 330
- winner take all in the frequency domain, 203, 211
- work, 115
- yield, bacterial growth, 230, 323
- yield, fishery, 253
- z-transform, 226

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(continued from page ii)

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